

BLACK'S RUN BRIDGE INSPECTION



PRIVATE HAUL ROAD over SR 0051 AND CSX RAILROAD

CENTER TOWNSHIP, BEAVER COUNTY, PENNSYLVANIA



Submitted to:

Ohio River Trails Council
1100 Pennsylvania Avenue
Monaca, PA 15061

Inspected By:

Larson Design Group
1000 Commerce Park Drive
Suite 201
Williamsport, PA 17701

Report Approved By:



9/2/2020

Kurt A. Brungard, P.E.

Inspection / Report:

Steve R. Haluska, P.E., C.B.S.I.
John J. Seibel, C.B.S.I.
William B. Sosko, Jr., C.B.S.I.

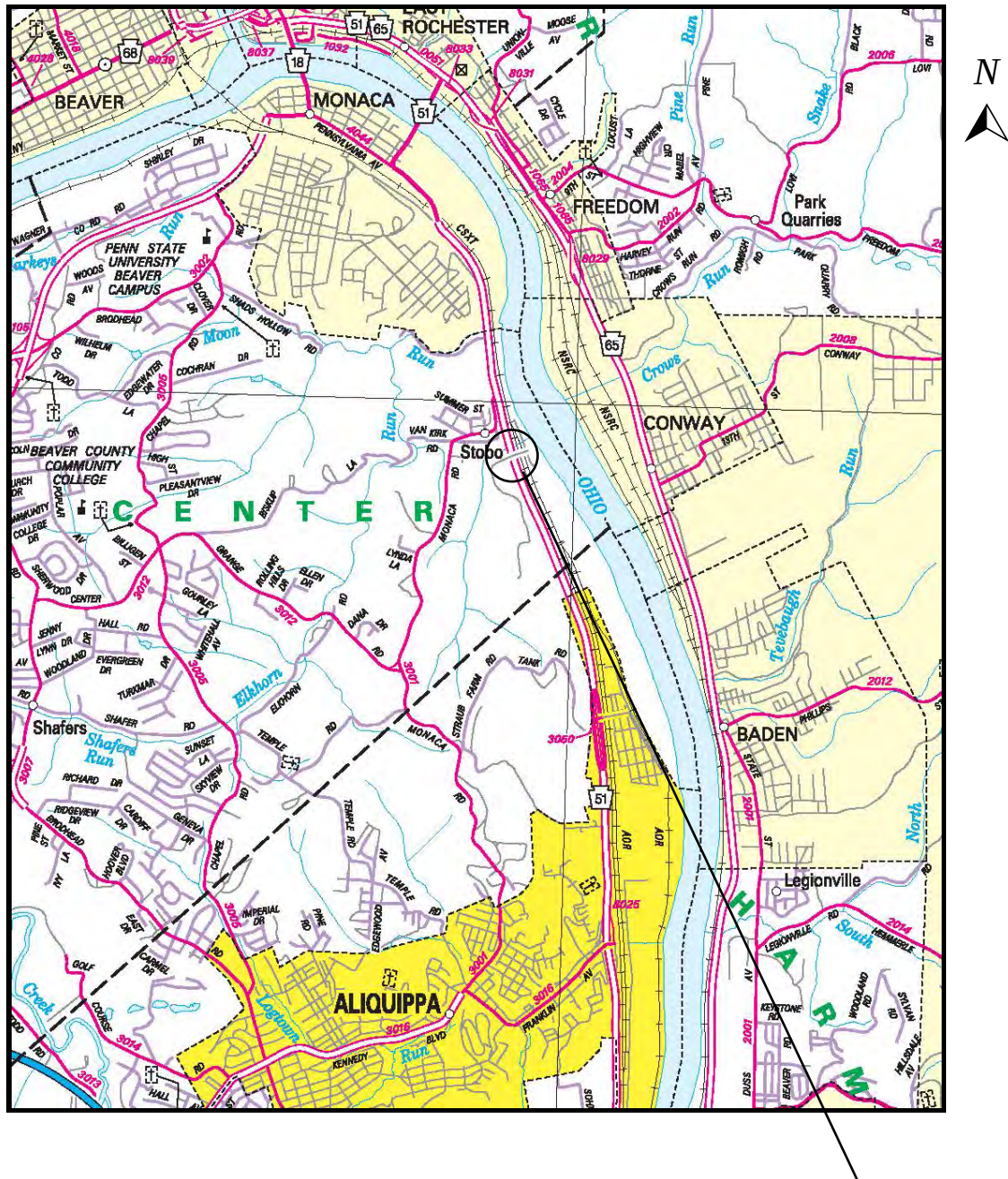
Quality Review:

Kurt A. Brungard, P.E., C.B.S.I.

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SITE LOCATION MAP



BRIDGE LOCATION

Private Haul Road over SR 0051 and CSX Railroad

INSPECTION SUMMARY

General Description

The structure carrying the Private Haul Road over SR 0051 and CSX Railroad is a four span continuous steel deck truss on reinforced concrete cellular and cantilever abutments and steel bents.

- Year built: 1943
- Total span: Span 1: 103.55'
Span 2: 114.15'
Span 3: 109.13'
Span 4: 94.0'
- Structure Length: 424'
- Clear roadway width: 18.0'
- Skew: 90°

Larson Design Group was tasked with performing a routine hands on inspection of the bridge. The inspection was performed on May 27th and July 20th through July 22nd, 2020 by Steve R. Haluska, P.E., C.B.S.I. and John J. Seibel, C.B.S.I. and William B Sosko Jr., C.B.S.I. A 60' under-bridge inspection crane, 35' bucket truck and traffic control were used to complete the inspection and was provided by Sofis Company, Inc.

Historical data was investigated for this structure. The structure is not located over a waterway and is currently gated with restricted pedestrian and vehicular access. The bridge was previously used as a haul road for a steel manufacturing facility. The bridge is currently being considered to provide pedestrian and maintenance vehicle access for a pedestrian trail.

Summary of Inspection Findings:

The overall physical condition is fair.

The bituminous wearing surface on the deck has wide cracking and areas where the wearing surface is popping off.

The underside of the concrete deck has areas of cracking, delaminations and spalls with exposed and rusted rebar.

The deck joints are heavily deteriorated and paved over.

The concrete curbs have areas of cracking and spalling with exposed and rusted rebar.

The concrete sidewalk steel access panels are bowed, deteriorated and non-functional.

The steel bridge railing panels have corrosion holes and are heavily deteriorated.

The bridge light polls are severely deteriorated.

The truss members have areas of laminated rust and section loss.

The below deck drainage systems are non-functional.

The expansion bearings at the far abutment are deteriorated and non-functional.

The concrete abutments and wingwalls have areas of cracking, delaminations and spalls with exposed and deteriorated rebar.

Approach Roadway Description:

The overall condition of the approach roadway is poor.

No transitions are provided between the bridge railings and near left and right approach railings. No transition is provided at the far left corner of the bridge and the structure mounted bridge railing and concrete curbs have blunt ends. The transition at the far right corner of the bridge consists of pipe sleeves welded to the bridge railing end post fitted with the pipe rails of the approach railing. The near left approach railing (photos 10) consists of welded steel I-beam posts and channels anchored to the abutment sidewall. The railing has a few scattered minor collision dents and is heavily rusted. Several of the railing posts have small to large corrosion holes through the webs of the I-beam posts adjacent the channel connections. The near left approach railing consists of welded steel pipe anchored to the exterior sidewalk curb. The pipe railing is heavily rusted and overgrown with vegetation. No approach rails are present at the far left shoulder. The safety feature consists of a large mound of heavily vegetated dirt and gravel along the edge of the roadway. The far right approach railing consists of welded steel pipe anchored in the ground. The pipe railing is heavily vegetated.

The near approach (photo 1) is the reinforced concrete top slab of the cellular abutment and is covered with a bituminous wearing surface. The wearing surface exhibits widespread multi-directional cracks up to 1" wide, some with vegetation growth. A gutter up to 2'-8" wide and up to 4" deep is present along the right edge of the pavement adjacent to the concrete curb and is mostly filled with dirt and vegetation growth. Heavy vegetation growth is present along the left curb. A chain link fence blocks access to the near end of the bridge roadway and sidewalk. The right panel of the fence is collision damaged and the right half of the fence is partially covered with vegetation.

The bituminous pavement at the far approach (photo 2) exhibits widespread map cracking up to 1/2" wide, a few scattered small to large shallow potholes, and two large bituminous patches. The approach transition, end of deck and deck joint were previously covered with a full roadway width bituminous patch. The patch at the transition and deck joint is severely cracked and breaking apart and has been partially covered with 10'-8" long by 6'-8" wide steel plate.

No storm drains are present at either approach.

The near left shoulder is an 18" high by 16" wide reinforced concrete curb that typically exhibits light scaling. The near right shoulder has a similar curb and a 6' wide reinforced concrete sidewalk. The near end of the curb is heavily spalled and the top of the curb near mid-length has a large spall. The sidewalk exhibits light scaling, an accumulation of dirt and debris and vegetation growth. The far left shoulder is a large heavily vegetated gravel and dirt mound. The far right shoulder is vegetated gravel and dirt. Both far shoulders appear stable.

Superstructure Description:

The overall condition of the superstructure is fair.

The bridge railings consist of structure mounted steel I-beams with welded steel channels and steel plate panels that are 11' high (photo 5). The steel plate panels are heavily deteriorated at the joints (photo 15) with the posts and along the bottom edges of the panels behind the connection angles at the sidewalk and curb. The deterioration includes section loss and large holes and is typical throughout the bridge. The panel sections are split along vertical welds at the railing posts and have separated and pushed toward the bridge roadway due to rust packing. Collision damage is present at the far left end post of the bridge railing with the top of the cap post being bent inward. The second and third right railing posts in Span 1 are bent and lean away from the bridge deck. Collision dents and scrapes are present throughout the panels. A concrete curb is present at the base of the left bridge railing and the right bridge railing is fronted by a reinforced concrete sidewalk and concrete curb.

The top of the deck is overlaid with a maximum 6" thick bituminous wearing surface (photo 18) that exhibits widespread multi-directional cracks up to 1" wide throughout the bridge. The right side of the wearing surface has random areas along the curb line that are peeling off up to 2" - 4" wide exposing the top of the deck (photo 19). This condition is present for the full length of the bridge in all spans. Vegetation growth is present along the joint with the left and right curbs for the full length of the bridge. Railroad personnel indicated that the top of the concrete deck was heavily deteriorated to the point of not driving on the bridge until the bituminous wearing surface was placed over the concrete deck.

The underside of the deck exhibits areas of hairline map cracking, transverse hairline cracks with efflorescence, pinpoint rust stains and areas of honeycombing throughout. Random small and large areas of delaminated concrete and concrete spalls with exposed and rusted rebar are present throughout the deck underside (photo 21). The left deck overhang has random transverse hairline cracks with efflorescence. The left deck fascia has random scattered small and large concrete spalls with exposed and rusted rebar. The underside of the right sidewalk has hairline map cracks, transverse hairline cracks and areas of honeycombing throughout.

Deck scuppers are present along the right curb line of each span. The scuppers vary from being open to partially or completely clogged with dirt and vegetation (photo 23).

The bridge expansion dams have been paved over (photo 24). The bituminous material over the joints is cracked full width of the bridge roadway. A 10" long hole is present through the expansion dam at the far end of Span 3 adjacent the right curb. The bituminous paving over the expansion dam at the far abutment is severely cracked and deteriorated and most of the joint has been covered with a large steel plate. The top plates of the sidewalk expansion dams are heavily rusted and have small to large corrosion holes.

A 6'-wide reinforced concrete sidewalk is present along the right side of the bridge. The sidewalk typically exhibits light scaling and very scattered transverse hairline cracks. An accumulation of dirt and debris is present on the sidewalk for the length of the bridge. A total of 23 access hatches are present in the sidewalk of Spans 1 through 4. The hatch covers are typically bowed, twisted, loose and settled into the openings (photo 26). The sidewalk support bracket angles typically exhibit paint peeling and loss of paint with heavy surface rusting and scattered laminated rust (photo 27 and 28). The support beams at the tops of the brackets are similar except at the sidewalk hatch locations where the beams have heavy laminated rust and section loss.

Concrete curbs 18" high by 16" wide are present along both sides of the bridge roadway. The curbs have very scattered full height full width hairline cracks. The corners of the left curb have small chip-type spalls for the length of the bridge. The front corners of the right curb have similar chip-type spalls and the back side of the curb along the sidewalk has scattered spalls with exposed rebar for the length of the

bridge. Large concrete spalls with exposed and corroded reinforcing are present in the top of the right curb in Spans 2, 3 and 4 (photo 29). Through drains are present in the right curb at the sidewalk level for the length of the bridge. The drains are open and functional.

One light pole is present in each span connected by four U-bolts to the right bridge railing posts. The light poles are non-functional and heavily rusted. The near side of the U-bolts is severely deteriorated with up to 100% loss. The far side of the U-bolts have varying amounts of section loss. Rust pack between the railing posts and steel plates and steel plates and poles is causing the poles to twist due to the loss of the near side of the U-bolts. The near side of the four U-bolts at the pole in Span 1 is completely sheared. The far side of the U-bolt varies from 3/8" to 1/2" remaining section. Full section is 3/4". The near side of two bolts at the pole in Span 2 is completely sheared and the remaining two bolts have 3/8" remaining section. The far side of the U-bolts vary from 3/8" to 1/2" remaining section. The bottom U-bolt for the pole in Span 3 is missing (photo 30). The near side of the three in place U-bolts is completely sheared. The far side of the three U-bolts vary from 1/2" to 5/8" remaining section. Three of the four U-bolts at the near side of the pole in Span 4 are sheared (photo 31) and the fourth bolt has 7/16" remaining section. The far side of the U-bolts vary between 3/8" and 5/8" remaining thickness.

The superstructure consists of four deck truss spans with suspended spans within Spans 2 and 4 and a rolled steel floorbeam floor system.

The rolled steel floorbeams typically exhibit areas of paint peeling and surface rust with minimal section loss throughout the bridge (photo 32). Except for the underside of the bottom flange, the end floorbeams at the abutments are not visible due to the haunch configuration of the deck for the deck joints. The underside of the bottom flanges typically exhibits heavy laminated rust. Steel channels are present at the deck haunches beneath the expansion joints at the ends of the suspended spans (photo 33). The channels typically exhibit heavy surface rusting with areas of laminated rust and because of the haunch configuration are generally inaccessible.

The left right and mid-line trusses consist of rolled steel beams and steel gusset plates with riveted connections. Generally, the truss members have areas of peeling paint with light to heavy surface rusting. The truss top chords, verticals and diagonals have scattered laminated rust at the edges of the gusset plates and fill plates with minimal section loss. Scattered areas of rust packing are present at the edges of the joints between the gusset plates and fill plates and truss members.

The bottom chords have accumulations of dirt and debris and piles of laminated rust on top of the webs (photo 42). The tops of the webs and bottoms of the flanges along the tops of the webs have areas of laminated rust with measurable section loss. The affected areas of the webs vary 1" to 3" outward from both flanges in a scalloped pattern and is scattered for the length of the member (photo 43). The affected areas of the flanges vary 1" to 3" high on the interior faces of the flanges and is in a scalloped pattern and is scattered for the length of the member. The amount of section loss varies but was generally between .0625" and .125". Three bottom chord members of the left truss in Span 1 exhibit more significant deterioration to the flanges and webs.

The truss pin connections to the pier bents (photo 44) have areas of paint peeling with light to heavy surface rust and scattered laminated rust on the plates and pins, but no significant problems were noted.

The truss sway bracing members typically have areas of paint peeling with light to heavy surface rust.

The fixed bearings at the near abutment have paint peeling and heavy surface rust and laminated rust with scattered section loss on the members (photo 45). The expansion roller bearings at the far abutment are in poor condition (photo 46). The right bearing has less than 10% remaining on the anchor bolts. The right keeper plate is broken off. The three rollers are all in different positions and appear non-functional. All the bearing members have complete loss of paint and heavy rust and areas of laminated rust. The left

bearing is partially buried in dirt and asphalt. The front anchor bolts have 100% section loss. The keeper plates are missing. All the visible members are heavily rusted, and the bearing is non-functional.

The below deck drainage systems are non-functional (photo 47 and 48). The sidewalk drain trough is filled with dirt and debris and has areas with 100% section loss. The drainpipe on the left side of the bridge has areas of 100% section loss. The drain troughs below the deck joints are filled with dirt and debris and have areas of 100% section loss. The downspout at the far end of the sidewalk drain trough is missing.

Substructure Description:

The overall condition of the substructure is fair.

The near abutment is a reinforced concrete cellular abutment (photo 7). The near backwall has scattered random vertical hairline cracks with areas of water and rust stains and heavy efflorescence. The bridge seat is covered with a heavy accumulation of dirt and debris. The front stem of the abutment has areas of hairline cracking with efflorescence, a few light cover pop-offs with exposed rebar and several large concrete spalls with exposed and deteriorated rebar (photo 49). The far abutment is a reinforced concrete cantilever abutment (photo 8). The far backwall has hairline map cracking throughout and vertical and diagonal hairline to open cracks. A 1/4" to 3/8" wide vertical crack with edge spalls is present at the end of the backwall. The bridge seat is completely covered with a heavy accumulation of dirt, debris, concrete and asphalt. A full height hairline crack with efflorescence and a full height 1/16" to 1/4" wide crack with edge spalls are present toward the left end of the stem. A full height 9' wide by up to 10" deep spall with exposed and deteriorated rebar is present on the stem (photo 50) and continues across the bridge seat under the middle of the bridge. The far right wingwall has scattered hairline map cracks and is covered with vegetation. The far left wingwall has a full height vertical hairline crack and a full height 1/16" to 1/4" wide crack with delaminations and spalls along the edges. The wingwall also has a large area of delaminated concrete and a large concrete spall with exposed and rusted rebar. The foundations are not exposed at the abutments.

The piers are steel bents on reinforced concrete foundations (photo 9). The steel members of the piers typically exhibit paint peeling and paint loss with varying degrees of surface rust and scattered laminations with minimal section loss. No significant problems were noted at the pin connections at the tops and bottoms of the piers. The top cover plate of the bottom strut at Pier 1 has a 6" diameter corrosion hole (photo 51). The foundation at Pier 1 is covered with dirt and vegetation. The foundation at Pier 2 was not accessible due to parked train cars. The foundation at Pier 3 has scattered light scaling to exposed aggregate, areas of minor honeycombing, and random chip type spalls along the corners. The Pier 3 foundation is covered with railroad supplies.

RECOMMENDATIONS AND SUMMARIES

We recommend that the following repairs be made in order of priority to open the structure to pedestrian traffic, retard further deterioration, preserve the structural integrity of the bridge and extend its useful life.

** Costs for each recommendation are an Engineer's opinion of probable construction cost based on Engineer's best judgement as an experienced and qualified professional generally familiar with the construction industry. The Engineer cannot and does not guarantee that actual construction costs will not vary from the opinions of probable construction costs noted below.

Current Recommendations:

Immediate Improvements (repairs completed prior to opening the structure to pedestrians):

1. Remove the bridge light poles (\$5,000). **
2. Clean and flush the near and far abutment bridge seats, replace the fixed and expansion bearings and repair the concrete abutment seats (\$250,000). **
3. Replace/rehabilitate the bridge expansion joints and troughs (\$25,000). **
4. Repair areas of deteriorated concrete curb (\$15,000). **
5. Mill bridge wearing surface, install waterproofing membrane and new bituminous wearing surface overlay (\$85,000). **
6. New bituminous overlay on the near and far approaches (\$10,000). **
7. Install chain link fences along the deck curb lines restricting the bridge pathway from the existing vertical wall barriers and sidewalk (\$120,000). **
8. Clean and flush all truss horizontal steel truss surfaces (\$15,000). **
9. Clean, flush and repair the under bridge drainage system (\$15,000). **

Long Term Improvements: (10 years)

1. Remove existing concrete deck, sidewalk and vertical steel plate barriers and replace with a new reinforced concrete deck (\$900,000). **
2. Sandblast clean and paint the entire structure (\$1,500,000). **
3. Repair the areas of delaminated and spalled concrete on the near and far abutments and wingwalls (\$150,000). **
4. Replace the below deck drainage systems to match the new concrete deck (\$50,000). **
5. Rehabilitate any areas of deteriorated concrete on the abutments (\$50,000). **

Load Rating Summary:

A current Load Rating Analysis has been performed for this structure to determine if it can be utilized as a pedestrian walkway. In addition, the structure was also analyzed for a F-550 maintenance truck. The structure was analyzed using inputs from the 2008 analysis performed by HGR. Section properties for the main load carrying members were gathered during this inspection and compared to the section properties used in the analysis. The section properties used in the analysis were determined to still be valid. Therefore, the bridge has adequate capacity to carry both the pedestrian loads and F-550 maintenance truck.

Signing Needs:

There are no signs required at this structure.

References Used in Preparation of this Report

Reference documents for the inspection included the following:

2008 Bridge Inspection Report Performed by HRG (Herbert, Rowland & Gruber, Inc.)



PHOTO 1 - NEAR APPROACH

Note: Pavement cracking and vegetation growth at curb lines.



PHOTO 2 - FAR APPROACH

Note: Widespread pavement map cracking, few shallow potholes, large previous patches and steel plate at transition.



PHOTO 3 - LEFT ELEVATION



PHOTO 4 - RIGHT ELEVATION



PHOTO 5 - GENERAL VIEW OF BRIDGE WEARING SURFACE AND RAILING

Note: Left railing shown.



PHOTO 6 - GENERAL UNDERSIDE

Note: Span 3 shown.



PHOTO 7 - NEAR ABUTMENT



PHOTO 8 - FAR ABUTMENT

Note: Full height wide deep spall at mid-stem and wide crack left of bridge.



PHOTO 9 - TYPICAL PIER

Note: Pier 2 shown.



PHOTO 10 - NEAR LEFT APPROACH RAILING

Note: Holes in railing posts.



PHOTO 11 - TYPICAL EXPANSION BEARING

Note: Severe rust, broken keeper plate and rollers disoriented.



PHOTO 12 - TYPICAL TRUSS CONFIGURATION

Note: Left side Span 2 shown.



PHOTO 13 - FAR RIGHT APPROACH RAILING



PHOTO 14 - NEAR RIGHT APPROACH CURB AND SIDEWALK

Note: Curb spalled, accumulation of dirt, debris and vegetation.



PHOTO 15 - LEFT BRIDGE RAILING

Note: Typical deterioration and holes through steel plate panels at railing posts.



PHOTO 16 - LEFT BRIDGE RAILING END PANEL SPAN 4

Note: Collision damaged end post and holes through panel plate at bottom and lower web channel.



PHOTO 17 - SPAN 1 RIGHT BRIDGE RAILING AT PANEL POINTS 3 AND 4

Note: Railing support posts bent outward.



PHOTO 18 - TYPICAL WIDE CRACKS IN BRIDGE DECK WEARING SURFACE



PHOTO 19 - BRIDGE DECK WEARING SURFACE ALONG RIGHT CURB

Note: Typical wearing surface peeling and exposed top of deck.



PHOTO 20 - TYPICAL UNDERSIDE OF DECK



PHOTO 21 - UNDERSIDE OF DECK SPAN 4

Note: Large area of delaminated and spalled concrete with exposed and rusted rebar.



PHOTO 22 - LEFT DECK FASCIA

Note: Typical spalls with exposed and rusted rebar.



PHOTO 23 - TYPICAL CLOGGED DECK SCUPPER



PHOTO 24 - TYPICAL PAVED OVER DECK EXPANSION DAM



PHOTO 25 - DECK EXPANSION JOINT FAR END SPAN 3

Note: Hole through paved over deck joint.



PHOTO 26 - TYPICAL BOWED AND LOOSE SIDEWALK ACCESS HATCH COVERS



PHOTO 27 - TYPICAL SIDEWALK SUPPORT BRACKETS



PHOTO 28 - TYPICAL CONDITION OF SIDEWALK SUPPORT BEAM UNDER SIDEWALK
HATCHES



PHOTO 29 - RIGHT CURB SPAN 2

Note: Large concrete spall with exposed and deteriorated rebar.



PHOTO 30 - BOTTOM OF LIGHT POLE SPAN 3

Note: Connection U-bolt missing at bottom of light pole.



PHOTO 31 - LIGHT POLE SPAN 4

Note: 100% loss to near side of connection U-bolt 2; U-bolts 1, 2 and 4 typical.



PHOTO 32 - TYPICAL CONDITION OF FLOORBEAMS

Note: Floorbeam 12 Span 2 shown.



PHOTO 33 - TYPICAL CHANNELS UNDER DECK EXPANSION JOINTS

Note: Joint at U40 Span 4 shown.



PHOTO 34 - TYPICAL CONNECTION OF MID-LINE TRUSS AT PIER SWAY BRACING



PHOTO 35 - ENDS OF MID-LINE TRUSS UNDER DECK JOINT

Note: Heavy laminated rust on top gusset plate and heavy rust on ends of truss members.



PHOTO 36 - TYPICAL CONDITION OF TRUSS VERTICALS AND DIAGONALS

Note: Left truss Span 1 shown.



PHOTO 37 - TYPICAL CONDITION OF TRUSS TOP CHORDS



PHOTO 38 - TYPICAL CONDITION OF TRUSS GUSSET PLATES



PHOTO 39 - LEFT TRUSS PANEL POINT L27 SPAN 3

Note: Rust pack between diagonal and fill plate.



PHOTO 40 - LEFT TRUSS BOTTOM CHORD L41-L43 SPAN 4

Note: Typical accumulation of laminated rust debris.



PHOTO 41 - TYPICAL DETERIORATION TO BOTTOM CHORD WEB AND FLANGES



PHOTO 42 - LEFT TRUSS BOTTOM CHORD L1-L3 SPAN 1

Note: Accumulation of laminated rust and debris and section loss to flange.



PHOTO 43 - LEFT TRUSS BOTTOM CHORD L11-L12 SPAN 2

Note: Typical scattered areas of section loss on top of web and flanges.



PHOTO 44 - TYPICAL TRUSS TO PIER PIN CONNECTION



PHOTO 45 - LEFT TRUSS FIXED BEARING AT NEAR ABUTMENT

Note: Areas of heavy laminated rust.



PHOTO 46 - LEFT TRUSS EXPANSION BEARING AT FAR ABUTMENT

Note: Partially covered, 100% loss to anchor bolts, keeper bar missing and heavy rusting.



PHOTO 47 - BELOW DECK DOWNSPOUT AT LEFT TRUSS

Note: Severe section loss and large hole.



PHOTO 48 - UNDER SIDEWALK DRAIN TROUGH

Note: Trough broken.



PHOTO 49 - FAR ABUTMENT BRIDGE SEAT

Note: Seat covered with debris and large spall in top of stem.



PHOTO 50 - RIGHT SIDE OF NEAR ABUTMENT

Note: Large spall with exposed and rusted rebar.



PHOTO 51 - BOTTOM SWAY STRUT PIER 1

Note: 6" diameter corrosion hole in top cover plate.

SAFETY FEATURES RATING SHEET

REFER TO PUB 100-A CODING GUIDE AND PDT FORM D-450-A FOR ADDITIONAL INFORMATION

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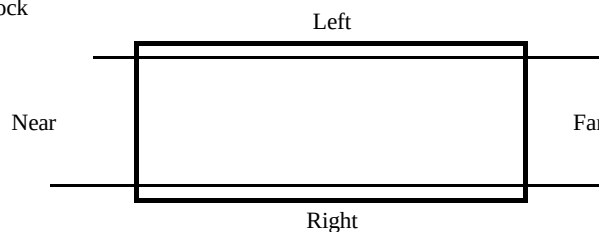
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Near Left Notes:

No trans provided betwn structure mounted panels and railing. Approach GR consists of structure mounted channel and beam railing attached to abut wall w/ blunt end. No rail ends provided. Mounded dirt and rock beyond shoulder on approach.

Far Left Notes:

No trans provided. Conc curbs and sturcture mounted panelsblunt end at abutment. No GR provided. Mounded dirt and rock beyond shoulder on approach.



Near Right Notes:

No trans provided betwn structure mounted panels and railing. Approach GR consists of structure mounted pipe railing attached to abut wall w/ blunt end behind concrete curb and sidewalk. No rail ends provided. Wide parking area at railing end.

Far Right Notes:

Trans consists of pipe sleeves welded to the end post of structure mounted panels with fitted pipes from approach rails. Approach GR consists of ground mounted pipe railing w/ blunt end. No rail ends provided. Mounded dirt and rock beyond shoulder on approach.

LEGEND:

Appr. Rdwy: N=18', F=18' to 25'

Speed Limit: 15

Measure guide rail height = guide lines ---

If between 2' 6^{15/16}" and 2' 3" code no higher than "6"

If between 2' 3" and 2' 0" code no higher than "4"

If less than 2' then code G.R. "3"

T - Trailing Edge

R - Required/Missing

D - Damaged

■ - Steel Brackets

Trans - Remember Spacer Tube

Rubrail

Distance from Roadway includes SPET Offset when applicable.

	BRIDGE RAILING RATING		
	Height	Description	Code
Left	11'	Concrete curb and structure mounted steel panel railings	3
Right	11'	Concrete curb and sidewalk and structure mounted steel panel railings	3

Controlling Rating (IA02)
3

LOCATION	TRANSITION = First 25' GR				APPROACH GUIDE RAIL (25' plus)				APPROACH RAIL ENDS			
	Length	Height	Gradual Stiffening/ Spacing	Code	Length	Height	Post Spacing	Code	Length	SPET Offset Dist.	Distance From Roadway	Code
Near Left			None Provided	2	48'-6"	50"	6'	3			None Provided	2
Near Right			None Provided	2	48'-6"	50"	6'	4			None Provided	2
Far Left			None Provided	2			None Provided	2			None Provided	2
Far Right	20'	50"	7'	3	16'-6"	50"	7'	3			None Provided	2

CONTROLLING RATINGS:

2

2

2







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JOB

Black's Run Bridge

SHEET NO.

1

OF

27

CALCULATED BY

WBS

DATE

7/22/2020

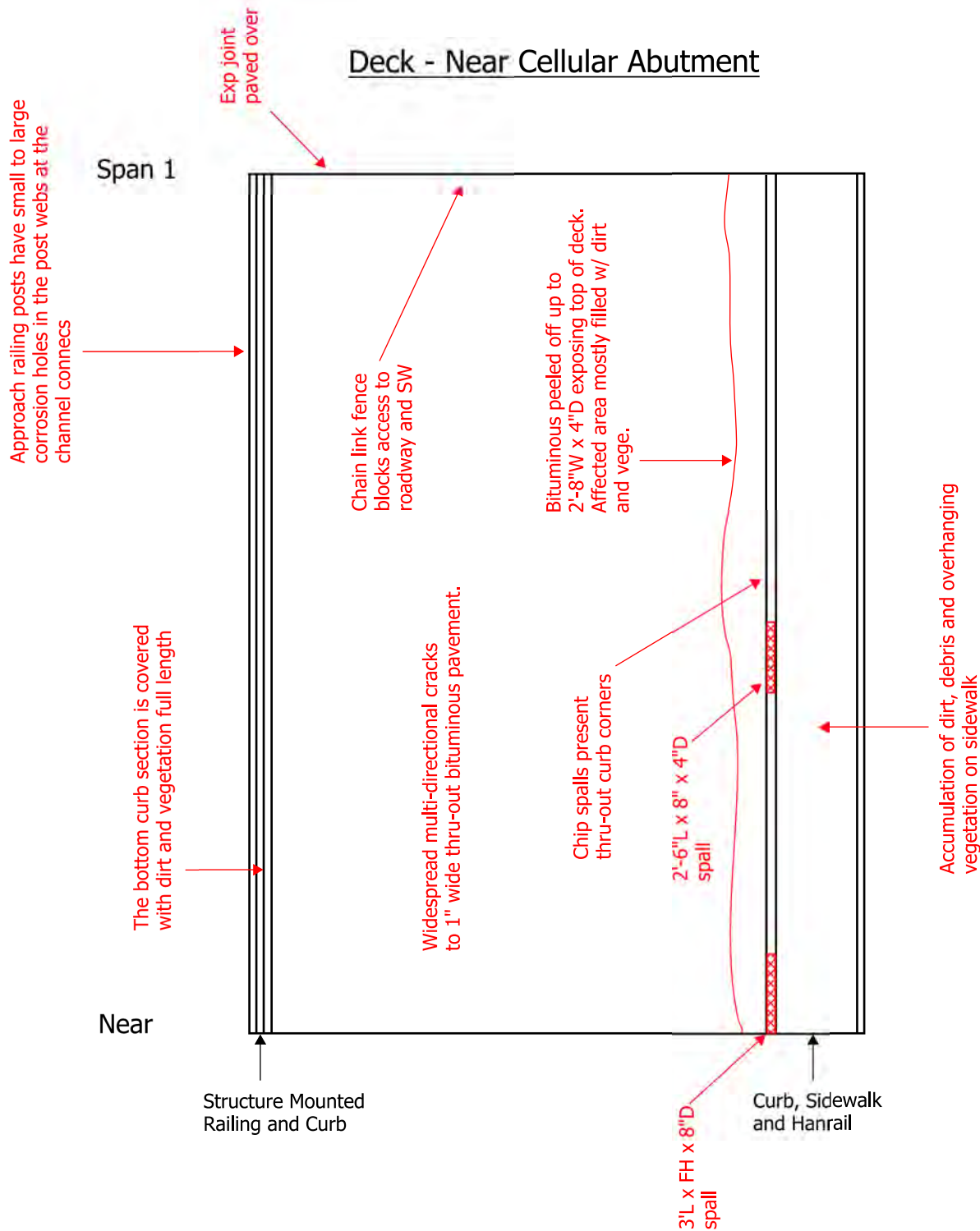
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SCALE

NTS

Deck - Near Cellular Abutment



Wearing Surface THK = 6"



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JOB Black's Run Bridge
SHEET NO. 2 OF 27
CALCULATED BY WBS DATE 7/22/2020
CHECKED BY _____ DATE _____
SCALE NTS

Deck - Span 1

Pier 1

Vegetation growth present along both curb lines for full length of bridge

Deck scuppers vary from open to partially and completely clogged with dirt and vegetation full length of bridge

Curb has scattered FH, FW HL cracks and small chip spalls at front corners

Joint paved over

Wearing Surface - Typically has widespread multi-directional cracks to 1/2"W

Joint paved over

Random areas of wearing surface peeling off up to 2'-4"W exposing top of deck along curb

Sidewalk exhibits light scaling, very scattered transverse HL cracks and an accumulation of dirt and debris

Curb has scattered FH, FW HL cracks and small chip spalls at front corners. scattered small spalls w/ exp rebar on SW side

Curb has through drain holes at SW level in all spans

23 total access hatches in Spans 1-4. Typically hatch panels are bowed, twisted, loose and settled into the openings. Hatches access a drain trough

NAB

Curb and Structure Mounted Panels

Top plate of SW exp dams typically have small to large corrosion holes

Curb, Sidewalk and Structure Mounted Panels



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SHEET NO.

3

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27

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7/22/2020

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DATE

SCALE

NTS

Deck - Span 2

Pier 2

Bridge Railing Panels (Left and Right) - Panels heavily deteriorated at joints w/ posts and bottoms behind curb angles with sec. loss and large holes. Typical thru-out bridge all spans. Collision scrapes and dents thru-out.

Curb has scattered FH, FW HL cracks and small chip spalls at front corners.

Wearing Surface - Typically has widespread multi-directional cracks to 1/2"W

Random areas of wearing surface peeling off up to 2'-4"W exposing top of deck along curb

4'L x 8"W x 8"D
spall w/ exp rebar

Sidewalk exhibits light scaling, very scattered transverse HL cracks and an accumulation of dirt and debris

Curb has scattered FH, FW HL cracks and small chip spalls at front corners. scattered small spalls w/ exp rebar on SW side

Pier 1

Curb and Structure
Mounted Panels

Curb, Sidewalk
and Structure
Mounted Panels



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Black's Run Bridge

SHEET NO.

4

OF

27

CALCULATED BY

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SCALE

NTS

Deck - Span 3

Pier 3

Bridge Light Poles - One pole right side each span. Non-functional and heavily rusted. Connected to bridge railing posts with U-bolts. Near side of U-bolts severely deteriorated with 100% loss. Sec loss to far side of U-bolts. Poles twisting away from panels. Typical all poles. Bottom U-bolt Span 3 pole missing.

Curb has scattered FH, FW HL cracks and small chip spalls at front corners.

Wearing Surface - Typically has widespread multi-directional cracks to 1/2"W

Joint paved over

10"L hole thru deck joint

2'-6"L x 8"W x 2"D spall w/ exp rebar

Random areas of wearing surface peeling off up to 2'-4"W exposing top of deck along curb

Sidewalk exhibits light scaling, very scattered transverse HL cracks and an accumulation of dirt and debris

Curb has scattered FH, FW HL cracks and small chip spalls at front corners. scattered small spalls w/ exp rebar on SW side

Pier 2

Curb and Structure Mounted Panels

Curb, Sidewalk and Structure Mounted Panels



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Black's Run Bridge

SHEET NO.

5

OF

27

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DATE

SCALE

NTS

Deck - Span 4

FAB

End Bridge Railing Panel - Collision damage w/ end post bent inward. Sec loss and holes along bottom of panel plate and at lower web channel stiffener

Joint paved over, bituminous severely cracked and deteriorated. Transition covered with 10'-8"L x 6'-8"W steel plate

3'L x 8"W x 5"D spall w/ exp rebar

Sidewalk exhibits light scaling, very scattered transverse HL cracks and an accumulation of dirt and debris

Curb has scattered FH, FW HL cracks and small chip spalls at front corners. scattered small spalls w/ exp rebar on SW side

Wearing Surface - Typically has widespread multi-directional cracks to 1/2"W

Random areas of wearing surface peeling off up to 2'-4"W exposing top of deck along curb

Curb has scattered FH, FW HL cracks and small chip spalls at front corners.

Pier 3

Curb and Structure Mounted Panels

Curb, Sidewalk and Structure Mounted Panels



Span 1 Framing Plan

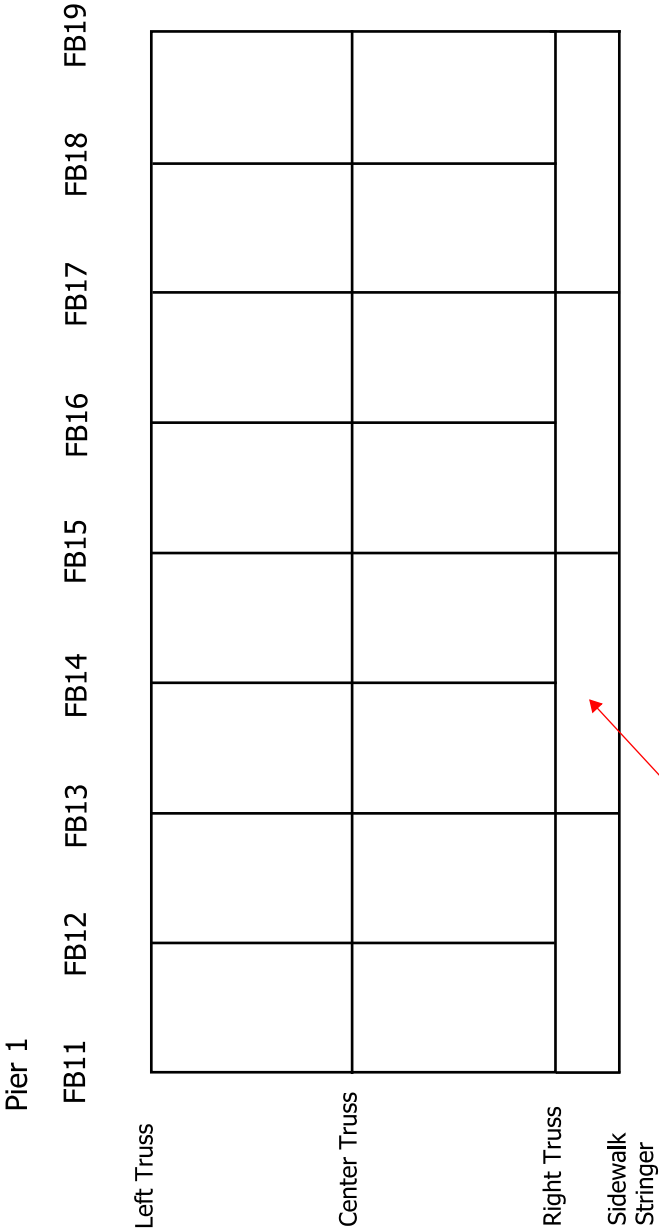
Pier 1											
NAB											
FB0	FB1	FB2	FB3	FB4	FB5	FB6	FB7	FB8	FB9	FB10	FB11
Left Truss											
Center Truss											
Right Truss											
Sidewalk Stringer											

General Notes:

- All steel members with heavy paint peel and surface corrosion with isolated areas of 1/16" deep pitting.
- Floorbeams have areas of paint peeling and surface rust with minor previous pitting with zero to minimal section loss.
- The drain trough running under the sidewalk is 100% full of debris, heavy corrosion throughout, and leaks at all joints in trough.
- Underside of deck has HL map cracks and transverse HL cracks with efflorescence, pin point rust stains and honeycombing throughout.
- The underside of deck has scattered random areas of delaminated concrete and minor spalls with significant areas noted.
- The left deck fascia has scattered random spalls with exposed and corroded rebar.
- The left deck overhang has random transverse HL cracks with efflorescence full length of the bridge.
- Underside of sidewalk has HL map cracks and transverse HL cracks with efflorescence and honeycombing.
- The sidewalk brackets typically have paint peeling and paint loss with surface rust and areas of laminated rust with minor section loss.
- The sidewalk stringer typically exhibits paint peeling and paint loss with varying degrees of surface rust and areas of heavy laminated rust.
- The center truss typically exhibits paint peeling and paint loss with varying degrees of surface rust. The gusset plates under the deck joints exhibit heavy laminated rust.



Span 2 Framing Plan (Near)



General Notes:
All steel members with heavy paint peel and surface corrosion with isolated areas of 1/16" deep pitting.
Floorbeams have areas of paint peeling and surface rust with minor previous pitting with zero to minimal section loss.
The drain trough running under the sidewalk exhibits heavy corrosion throughout and leaks at all joints in trough.
Underside of deck has HL map cracks and transverse HL cracks with efflorescence, pin point rust stains and honeycombing throughout.
The underside of deck has scattered random areas of delaminated concrete and minor spalls with significant areas noted.
The left deck fascia has scattered random spalls with exposed and corroded rebar.
The left deck overhang has random transverse HL cracks with efflorescence full length of the bridge.
Underside of sidewalk has HL map cracks and transverse HL cracks with efflorescence and honeycombing.
The sidewalk brackets typically have paint peeling and paint loss with surface rust and areas of laminated rust with minor section loss.
The sidewalk stringer typically exhibits paint peeling and paint loss with varying degrees of surface rust and areas of heavy laminated rust.
The center truss typically exhibits paint peeling and paint loss with varying degrees of surface rust. The gusset plates under the deck joints exhibit heavy laminated rust.



Span 2 Framing Plan (Far)

Pier 2

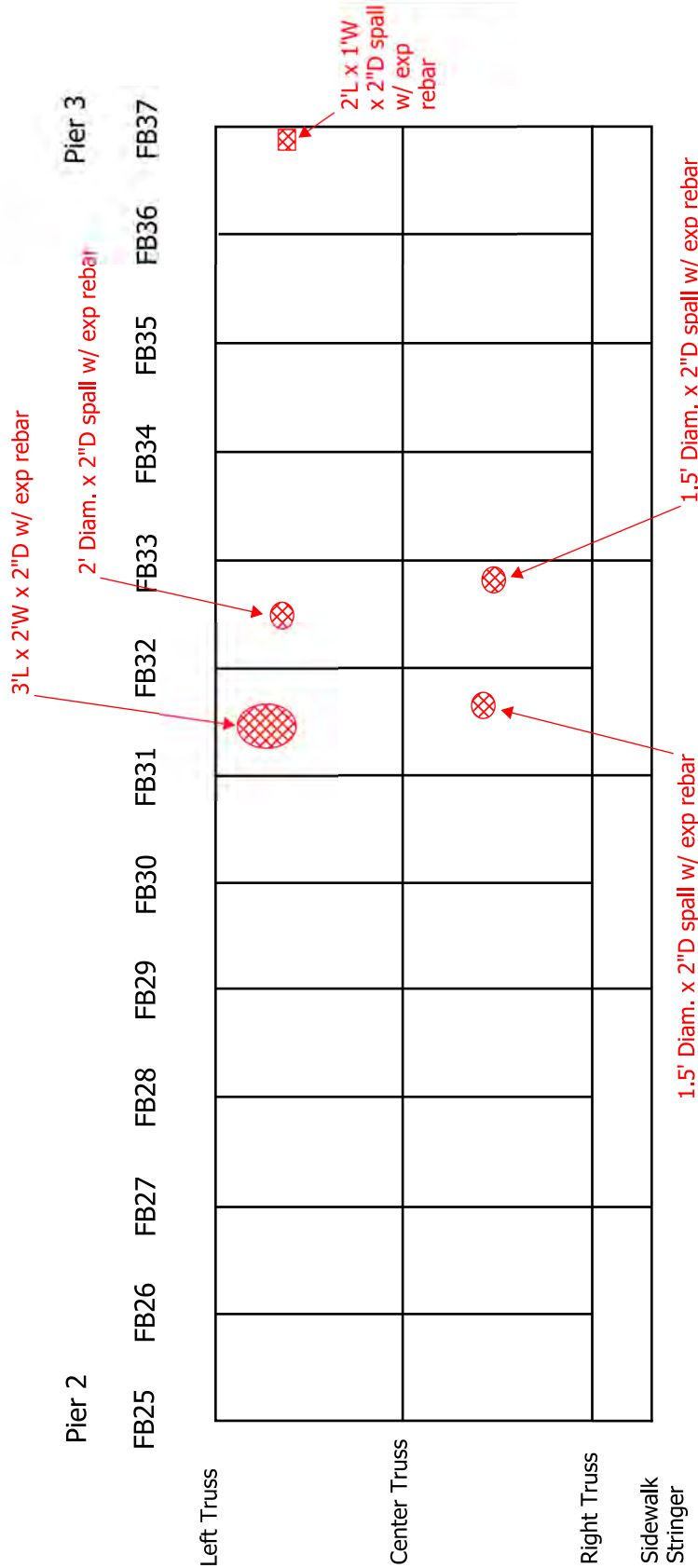
FB19 FB20 FB21 FB22 FB23 FB24 FB25

Left Truss							
Center Truss							
Right Truss							
Sidewalk Stringer							

General Notes:

- All steel members with heavy paint peel and surface corrosion with isolated areas of 1/16" deep pitting.
- Floorbeams have areas of paint peeling and surface rust with minor previous pitting with zero to minimal section loss.
- The drain trough running under the sidewalk exhibits heavy corrosion throughout and leaks at all joints in trough.
- Underside of deck has HL map cracks and transverse HL cracks with efflorescence, pin point rust stains and honeycombing throughout.
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Span 3 Framing Plan



General Notes:

All steel members with heavy paint peel and surface corrosion with isolated areas of 1/16" deep pitting. Floorbeams have areas of paint peeling and surface rust with minor previous pitting with zero to minimal section loss. The drain trough running under the sidewalk exhibits heavy corrosion throughout and leaks at all joints in trough. Underside of deck has HL map cracks and transverse HL cracks with efflorescence, pin point rust stains and honeycombing throughout. The underside of deck has scattered random areas of delaminated concrete and minor spalls with significant areas noted. The left deck fascia has scattered random spalls with exposed and corroded rebar. The left deck overhang has HL map cracks and transverse HL cracks with efflorescence full length of the bridge. Underside of sidewalk has HL map cracks and transverse HL cracks with efflorescence and honeycombing. The sidewalk brackets typically have paint peeling and paint loss with surface rust and areas of laminated rust with minor section loss. The sidewalk stringer typically exhibits paint peeling and paint loss with varying degrees of surface rust and areas of heavy laminated rust. The center truss typically exhibits paint peeling and paint loss with varying degrees of surface rust. The gusset plates under the deck joints exhibit heavy laminated rust.



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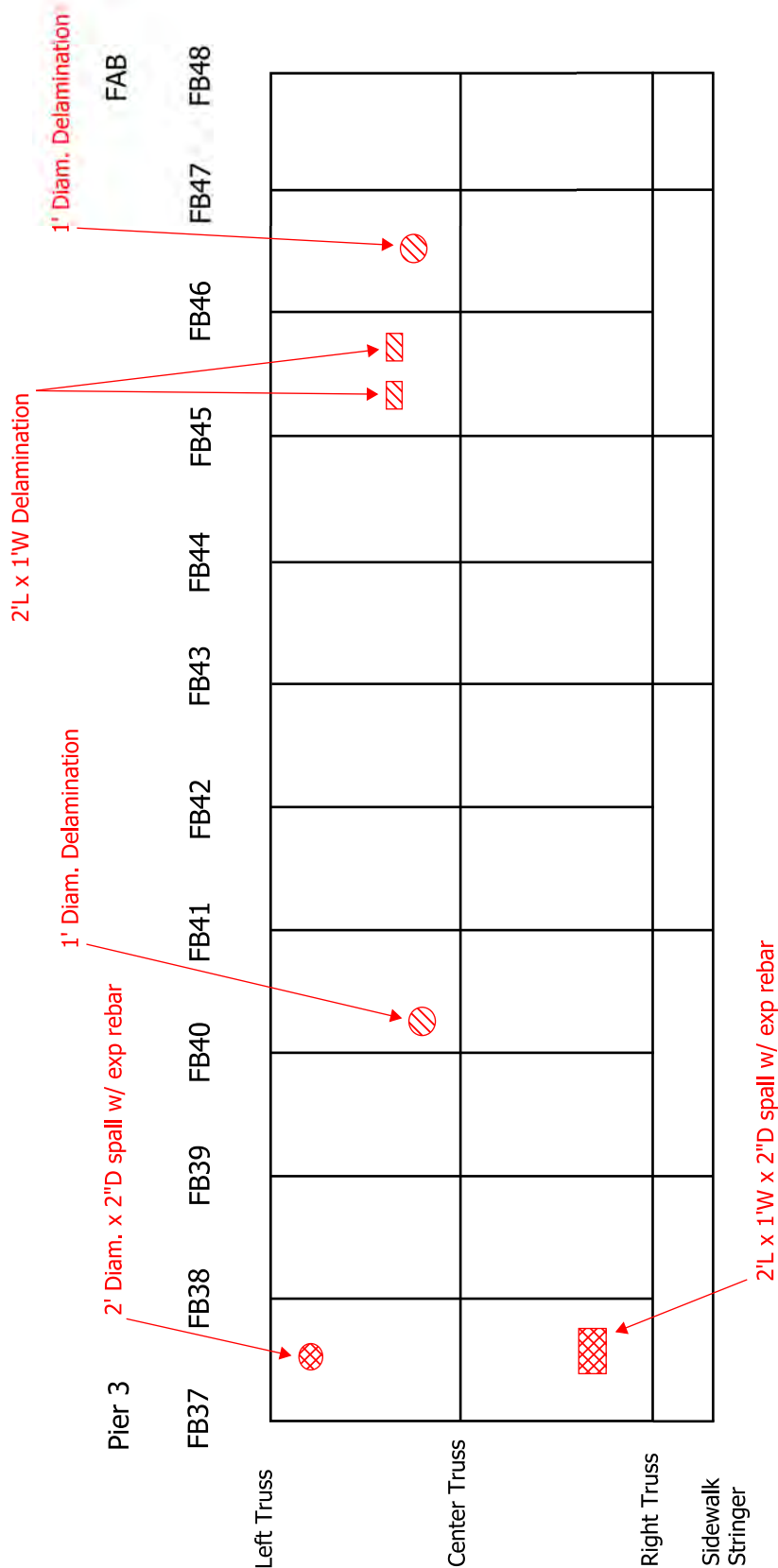
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SCALE _____ NTS _____

Span 4 Framing Plan

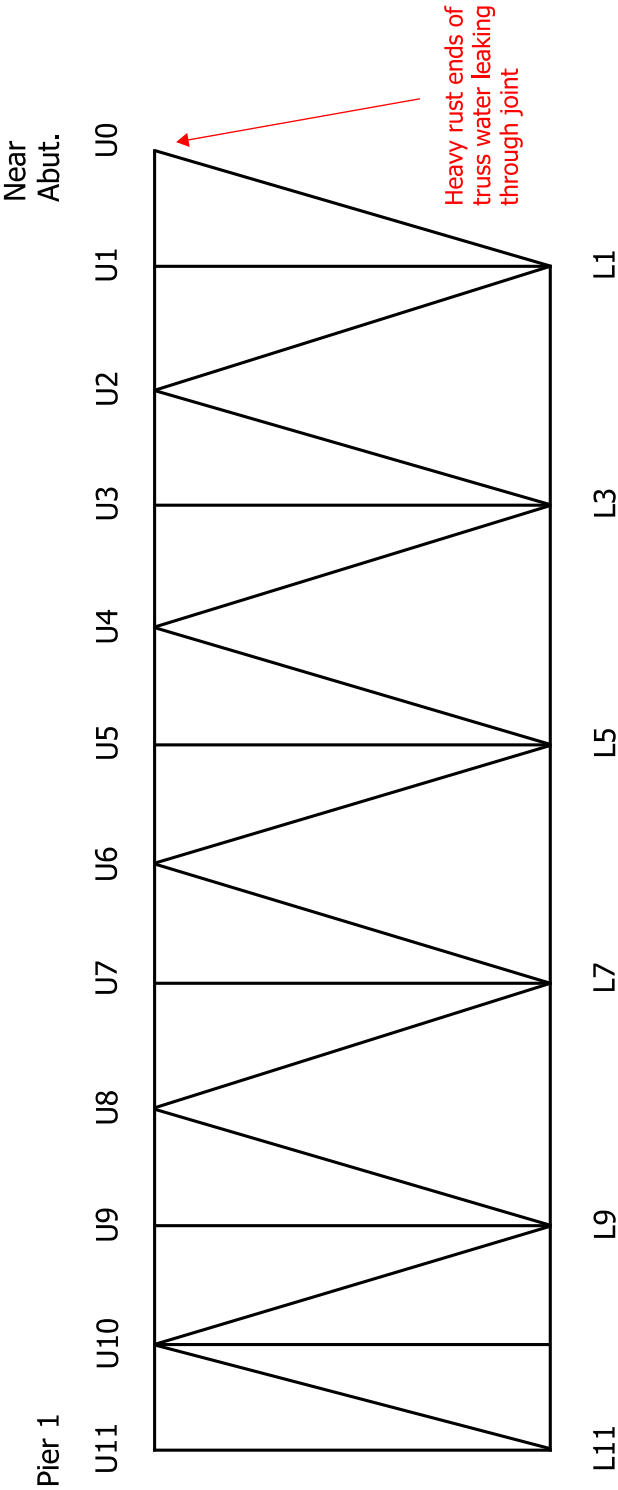


General Notes:

All steel members with heavy paint peel and surface corrosion with isolated areas of 1/16" deep pitting. Floorbeams have areas of paint peeling and surface rust with minor previous pitting with zero to minimal section loss. The drain trough running under the sidewalk exhibits heavy corrosion throughout and leaks at all joints in trough. Underside of deck has HL map cracks and transverse HL cracks with efflorescence, pin point rust stains and honeycombing throughout. The underside of deck has scattered random areas of delaminated concrete and minor spalls with significant areas noted. The left deck fascia has scattered random spalls with exposed and corroded rebar. The left deck overhang has random transverse HL cracks with efflorescence full length of the bridge. Underside of sidewalk has HL map cracks and transverse HL cracks with efflorescence and honeycombing. The sidewalk brackets typically have paint peeling and paint loss with surface rust and areas of laminated rust with minor section loss. The sidewalk stringer typically exhibits paint peeling and paint loss with varying degrees of surface rust and areas of heavy laminated rust. The center truss typically exhibits paint peeling and paint loss with varying degrees of surface rust. The gusset plates under the deck joints exhibit heavy laminated rust.



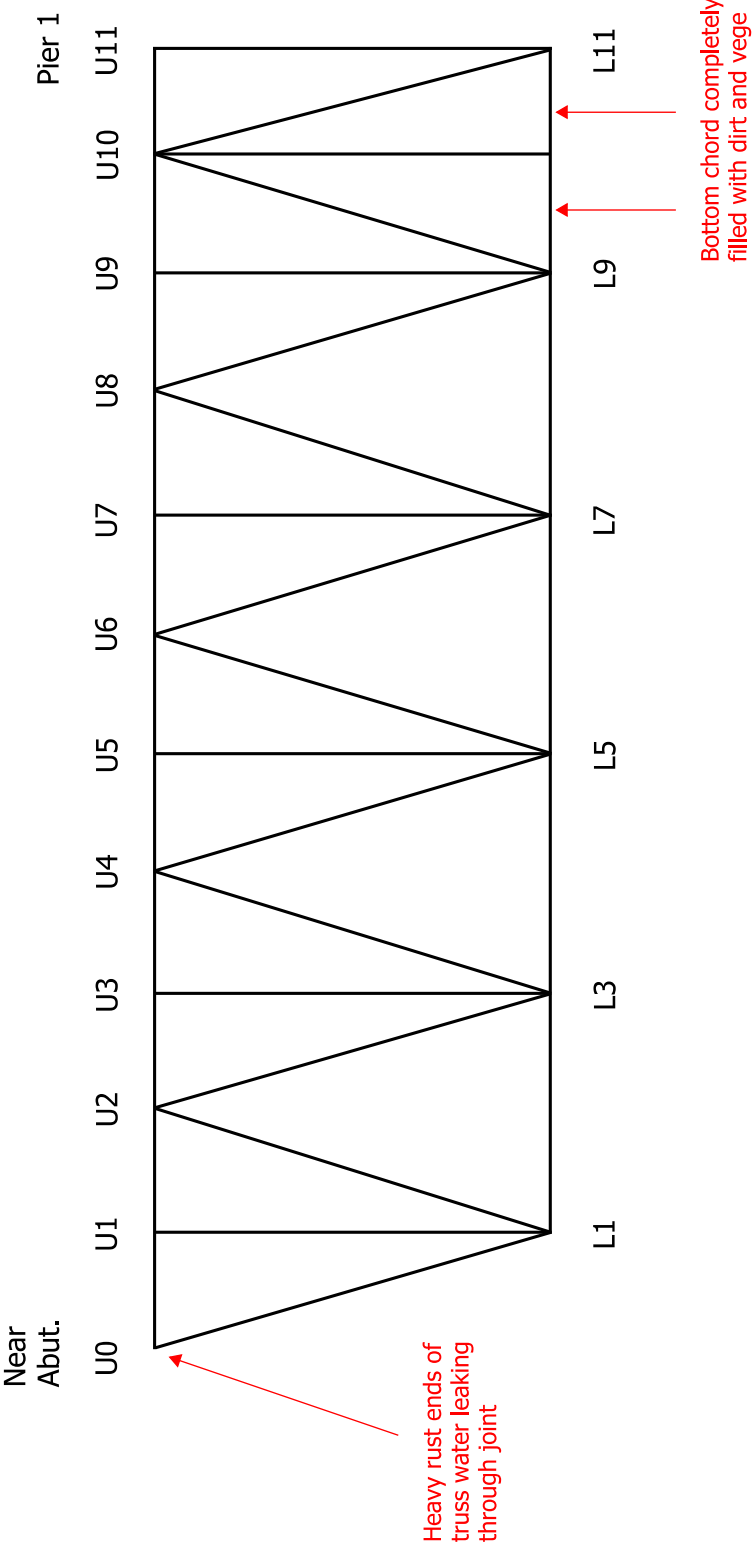
Span 1 Left Truss



General Notes:
Truss members have areas of peeling paint with light to heavy surface rusting.
Truss top chords, verticals and diagonals have scattered laminated rust at the edges of the gusset plates and fill plates with minimal section loss.
Scattered areas of rust packing are present at the edges of the joints between the gusset plates and fill plates and truss members.
Bottom chords have accumulations of dirt and debris and piles of laminated rust on top of the webs.
Tops of the webs and bottoms of the flanges along the tops of the webs have areas of laminated rust with measurable section loss.
Truss sway bracing members typically have areas of paint peeling with light to heavy surface rust.
Bottom chord members L1-L3, L3-L5 and L7-L8 have section loss on the webs and flanges.
Below deck drainage system non-functional with areas of 100% loss and holes



Span 1 Right Truss



General Notes:

- Truss members have areas of peeling paint with light to heavy surface rusting.
- Truss top chords, verticals and diagonals have scattered laminated rust at the edges of the gusset plates and fill plates with minimal section loss.
- Scattered areas of rust packing are present at the edges of the joints between the gusset plates and fill plates and truss members.
- Bottom chords have accumulations of dirt and debris and piles of laminated rust on top of the webs.
- Tops of the webs and bottoms of the flanges along the tops of the webs have areas of laminated rust with measurable section loss.
- Truss sway bracing members typically have areas of paint peeling with light to heavy surface rust.
- Below deck drainage system non-functional with areas of 100% loss and holes



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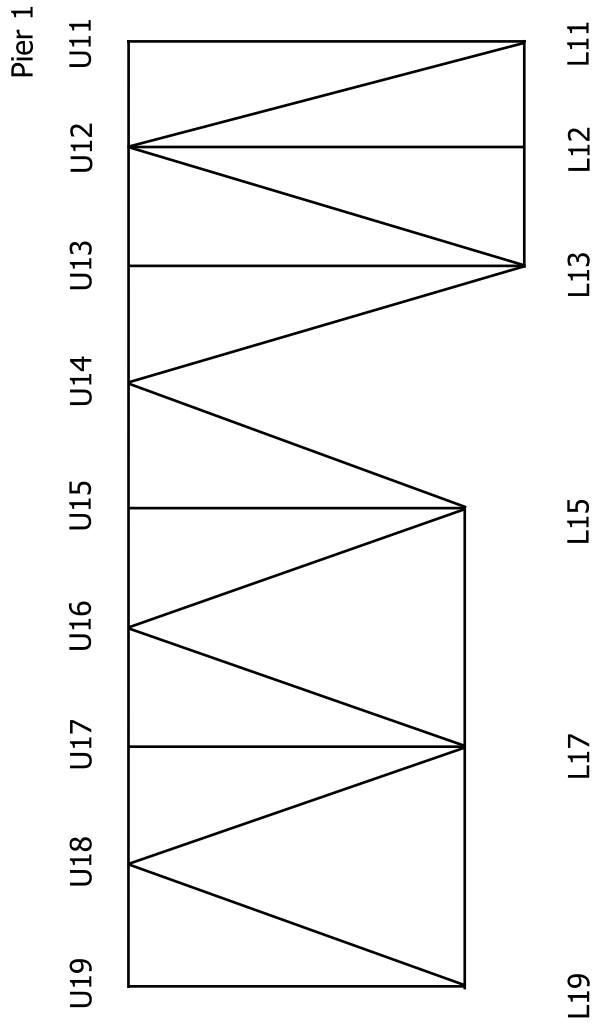
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Span 2 Left Truss

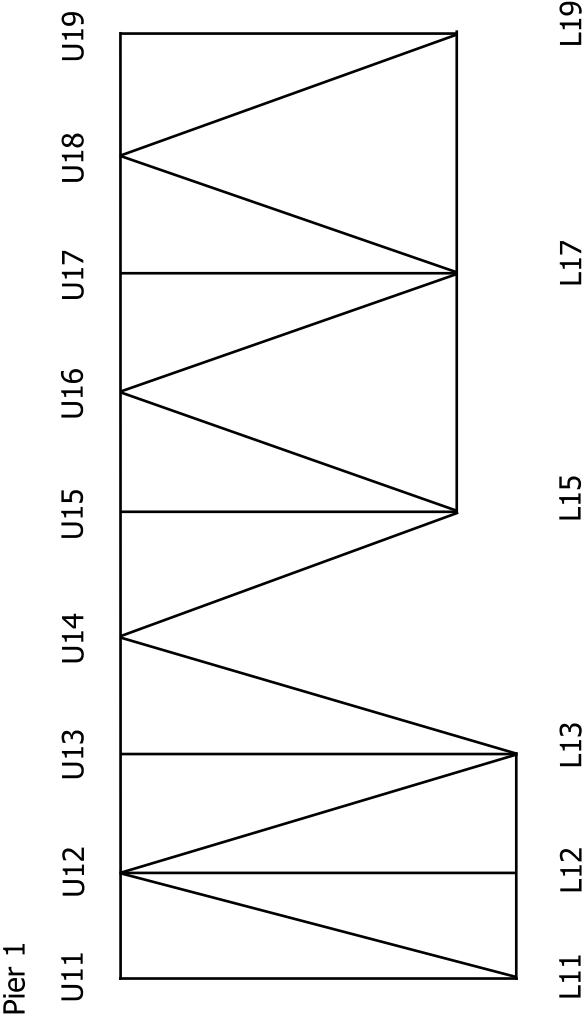


General Notes:

Truss members have areas of peeling paint with light to heavy surface rusting.
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Truss sway bracing members typically have areas of paint peeling with light to heavy surface rust.
Below deck drainage system non-functional with areas of 100% loss and holes



Span 2 Right Truss



General Notes:
Truss members have areas of peeling paint with light to heavy surface rusting.
Truss top chords, verticals and diagonals have scattered laminated rust at the edges of the gusset plates and fill plates with minimal section loss.
Scattered areas of rust packing are present at the edges of the joints between the gusset plates and fill plates and truss members.
Bottom chords have accumulations of dirt and debris and piles of laminated rust on top of the webs.
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Truss sway bracing members typically have areas of paint peeling with light to heavy surface rust.
Below deck drainage system non-functional with areas of 100% loss and holes



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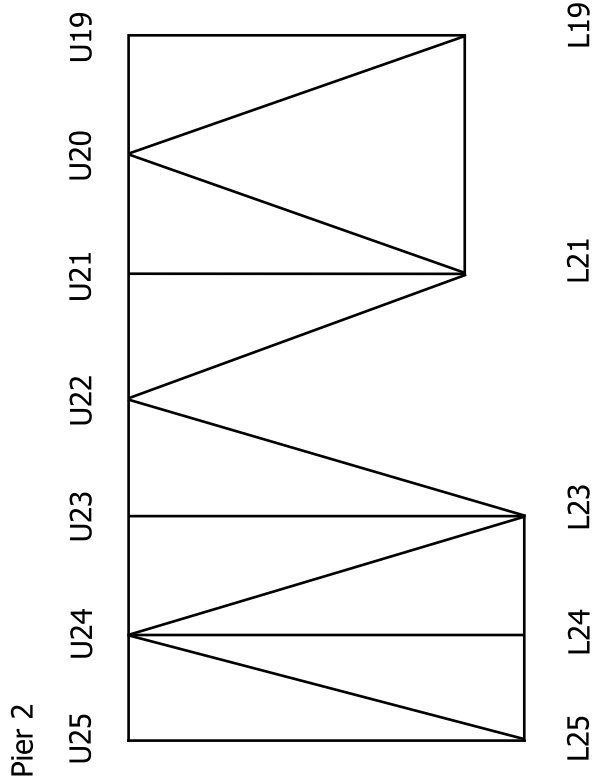
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Span 2 Left Truss

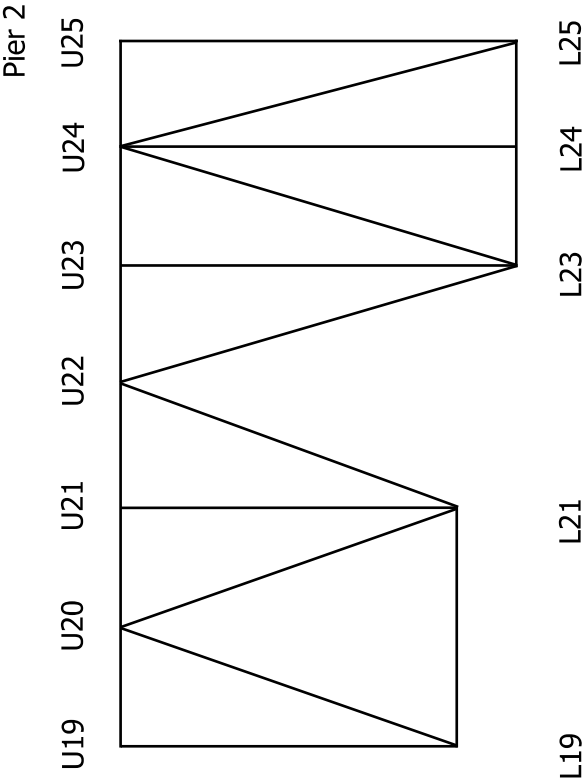


General Notes:

Truss members have areas of peeling paint with light to heavy surface rusting.
Truss top chords, verticals and diagonals have scattered laminated rust at the edges of the gusset plates and fill plates with minimal section loss.
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Truss sway bracing members typically have areas of paint peeling with light to heavy surface rust.
Below deck drainage system non-functional with areas of 100% loss and holes



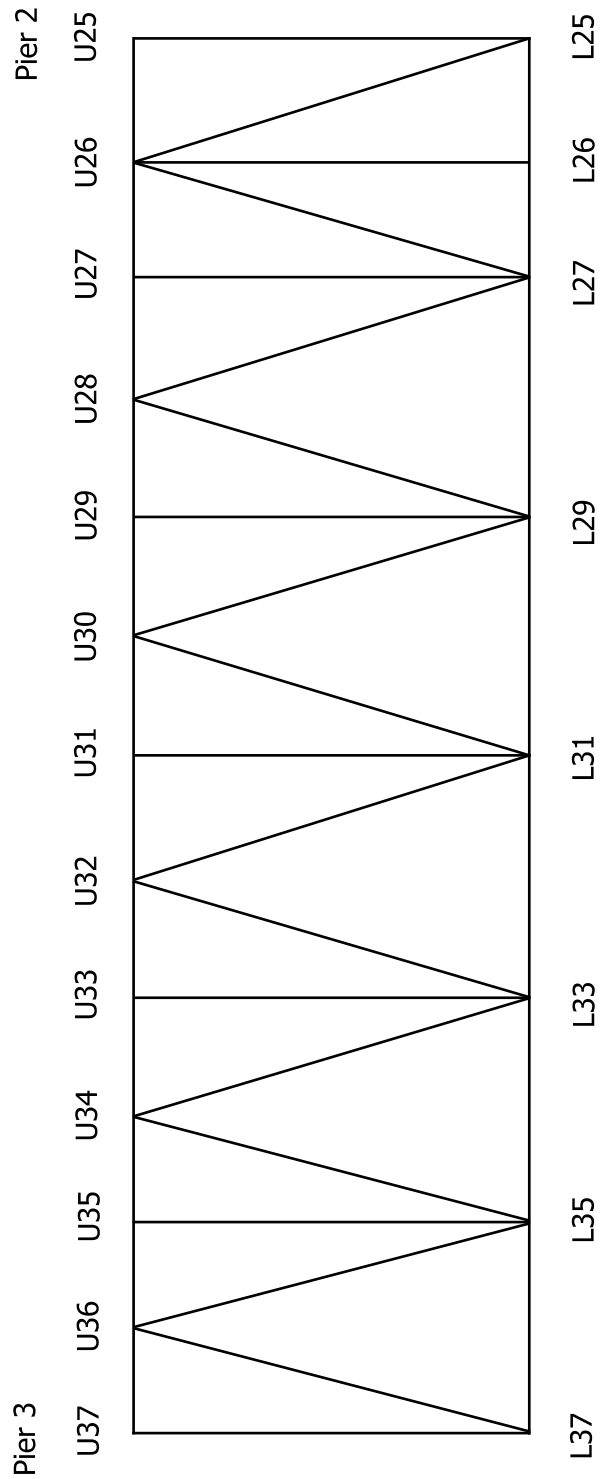
Span 2 Right Truss



General Notes:
Truss members have areas of peeling paint with light to heavy surface rusting.
Truss top chords, verticals and diagonals have scattered laminated rust at the edges of the gusset plates and fill plates with minimal section loss.
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Truss sway bracing members typically have areas of paint peeling with light to heavy surface rust.
Below deck drainage system non-functional with areas of 100% loss and holes



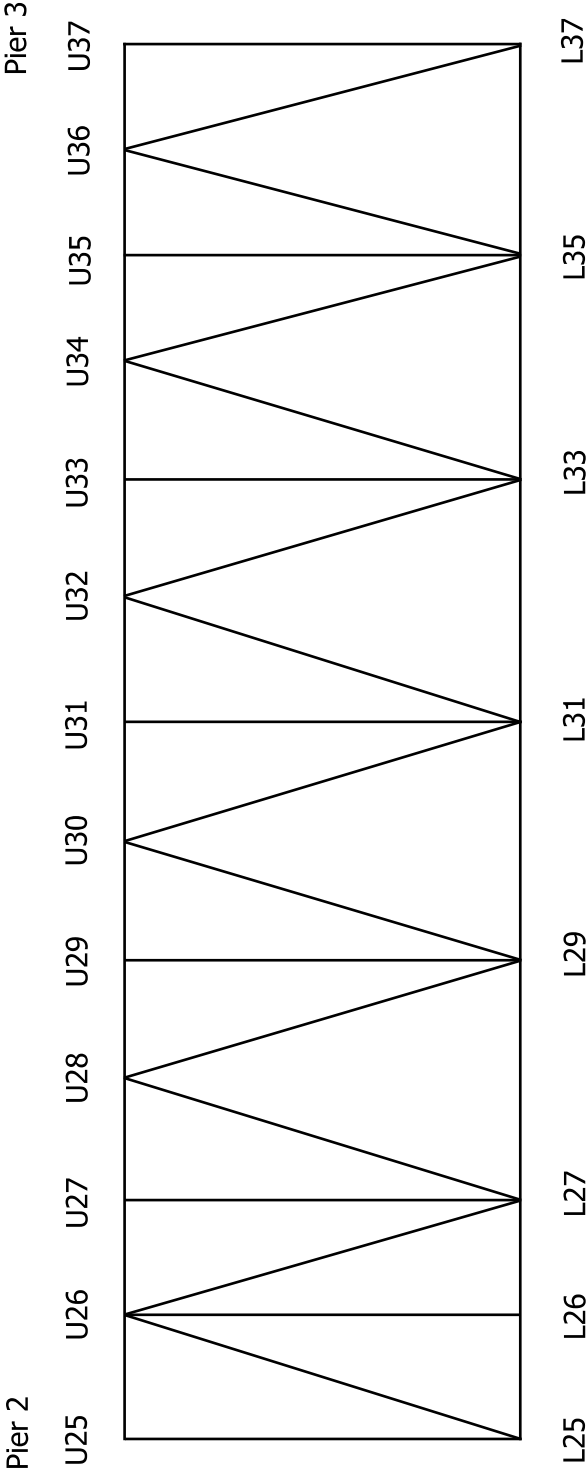
Span 3 Left Truss



General Notes:
Truss members have areas of peeling paint with light to heavy surface rusting.
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Scattered areas of rust packing are present at the edges of the joints between the gusset plates and fill plates and truss members.
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Truss sway bracing members typically have areas of paint peeling with light to heavy surface rust.
Below deck drainage system non-functional with areas of 100% loss and holes



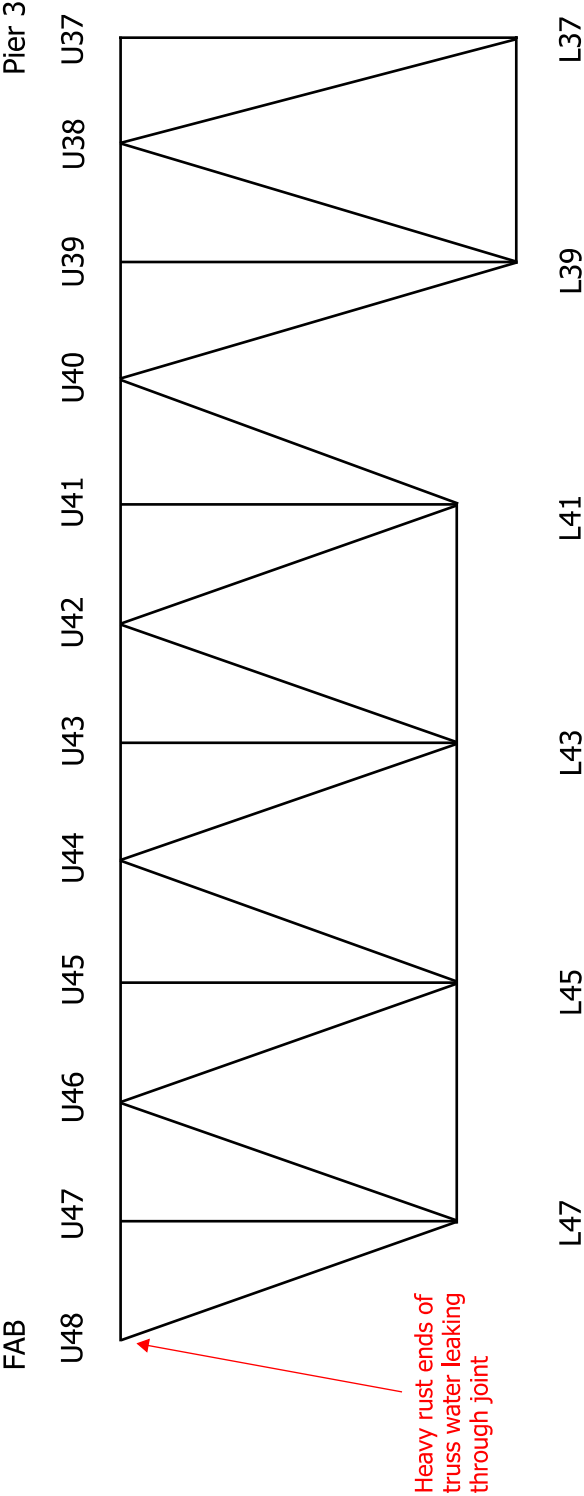
Span 3 Right Truss



General Notes:
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Truss top chords, verticals and diagonals have scattered laminated rust at the edges of the gusset plates and fill plates with minimal section loss.
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Truss sway bracing members typically have areas of paint peeling with light to heavy surface rust.
Below deck drainage system non-functional with areas of 100% loss and holes



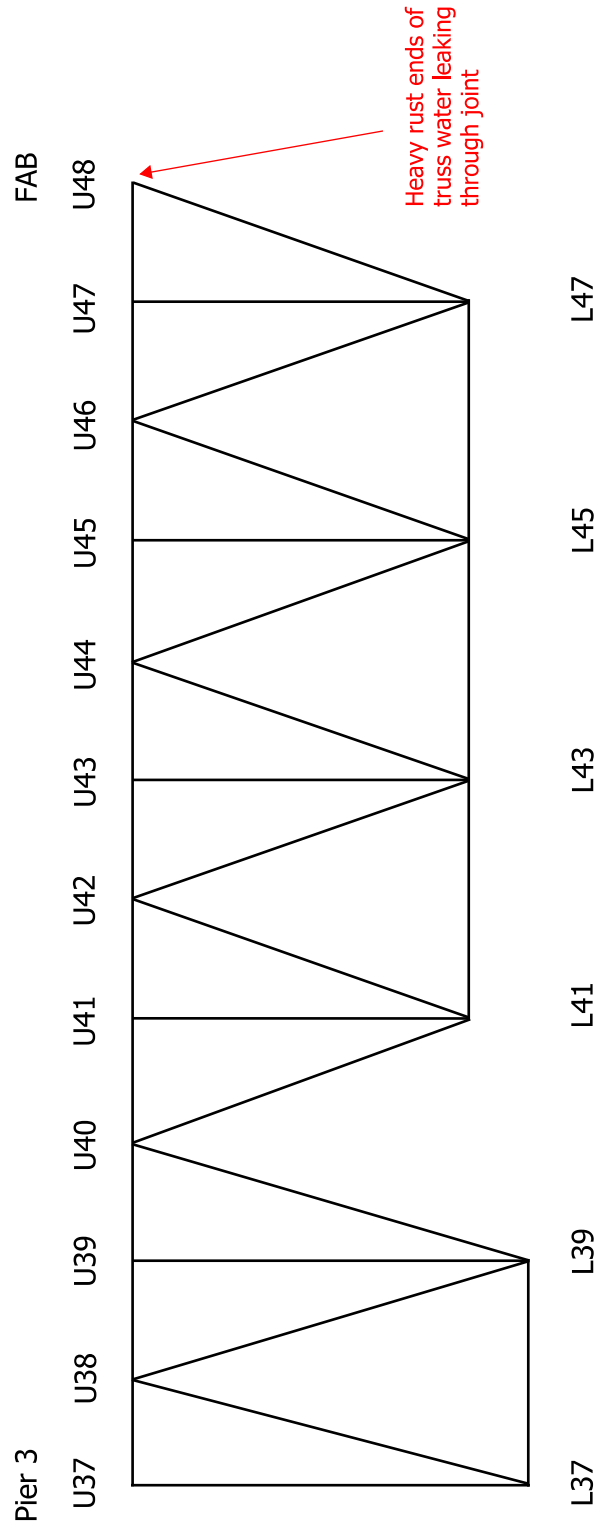
Span 4 Left Truss



General Notes:
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Truss sway bracing members typically have areas of paint peeling with light to heavy surface rust.
Below deck drainage system non-functional with areas of 100% loss and holes



Span 4 Right Truss



General Notes:

Truss members have areas of peeling paint with light to heavy surface rusting.
Truss top chords, verticals and diagonals have scattered laminated rust at the edges of the gusset plates and fill plates with minimal section loss.
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Truss sway bracing members typically have areas of paint peeling with light to heavy surface rust.
Below deck drainage system non-functional with areas of 100% loss and holes



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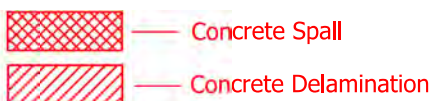
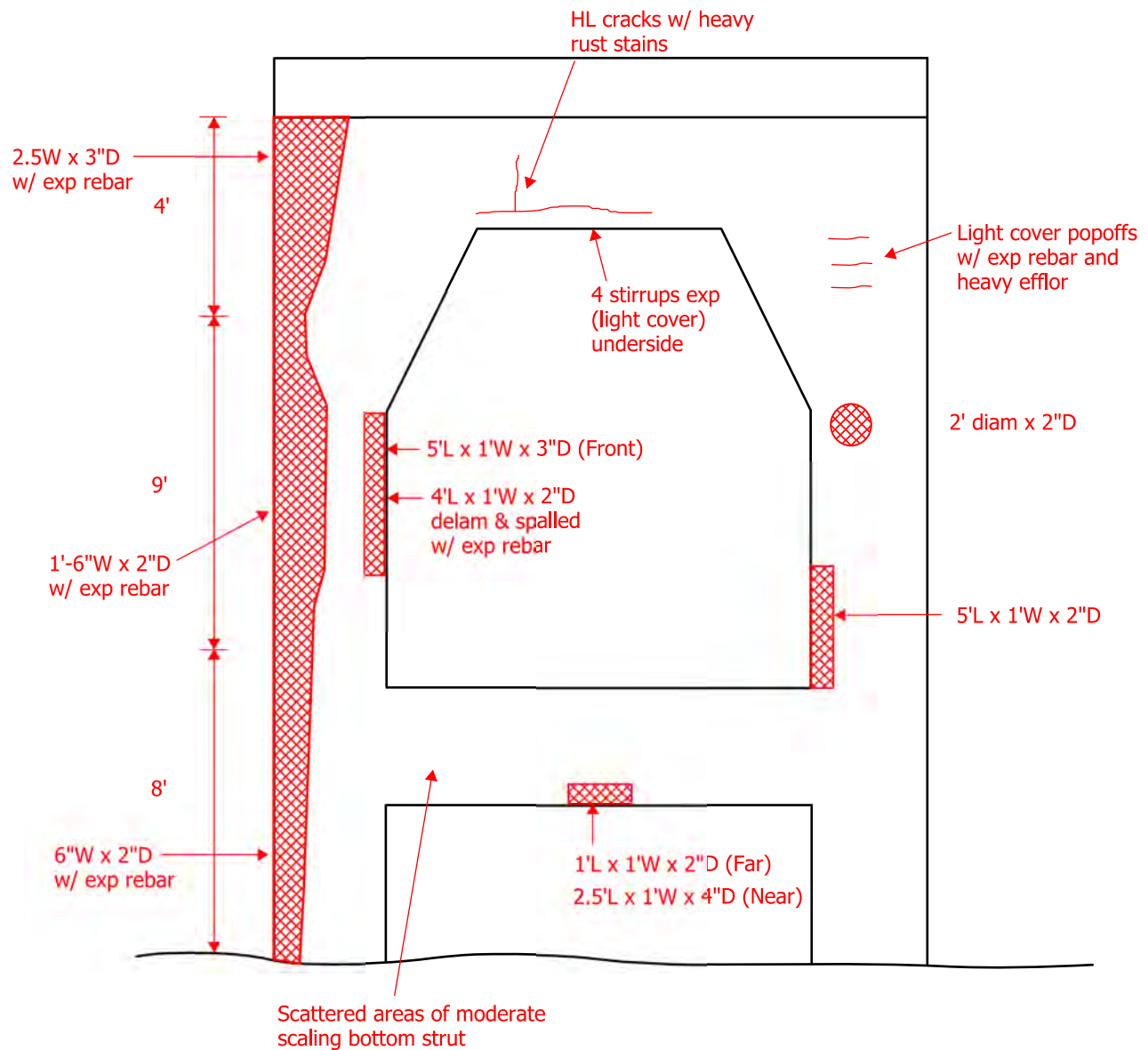
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Near Abutment Far Elevation





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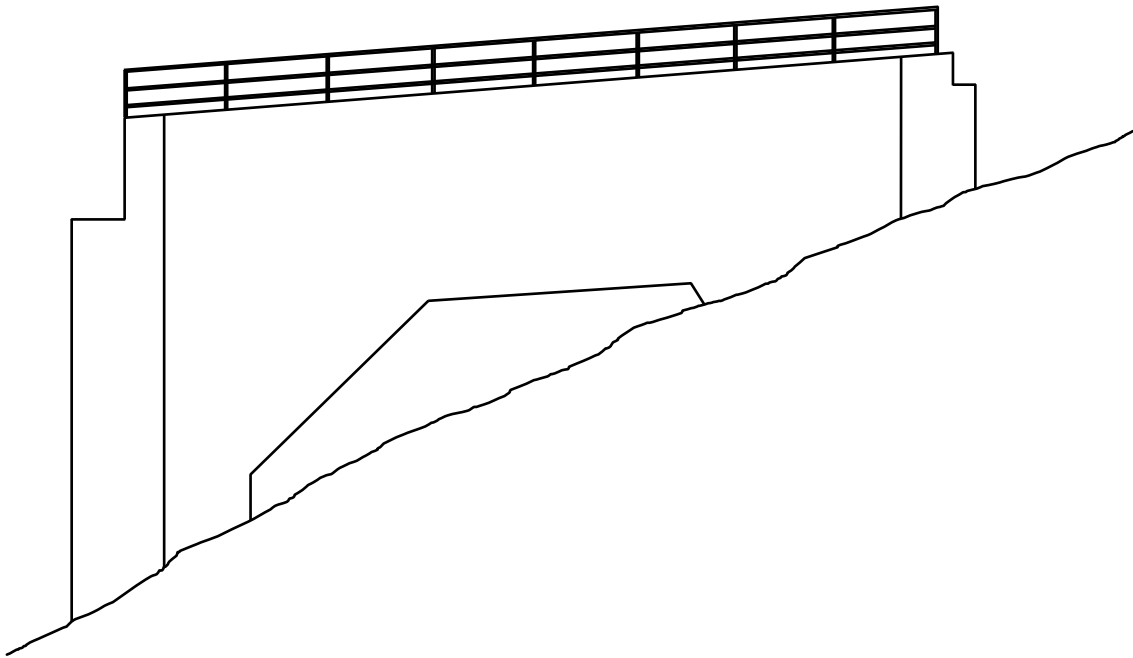
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Near Abutment
Left Elevation



No Problems Noted



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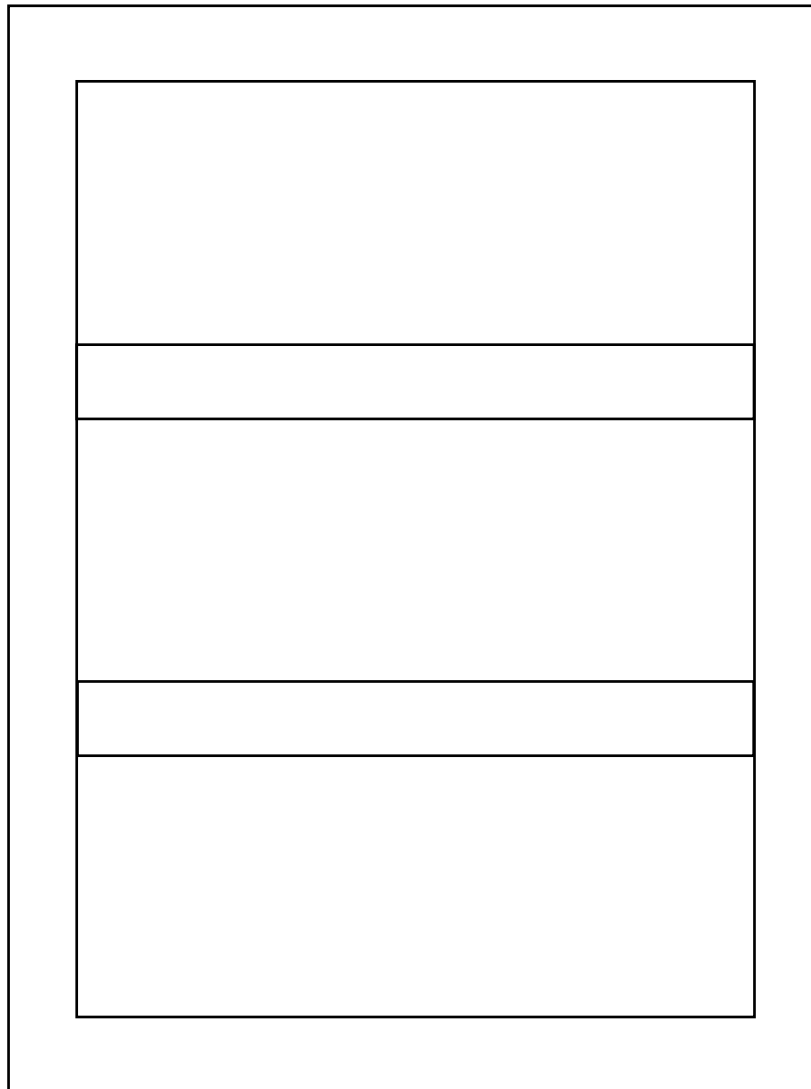
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Near Abutment
Plan View

Near



Far

Scattered light cover popoffs on underside of deck and struts.
No other problems noted.



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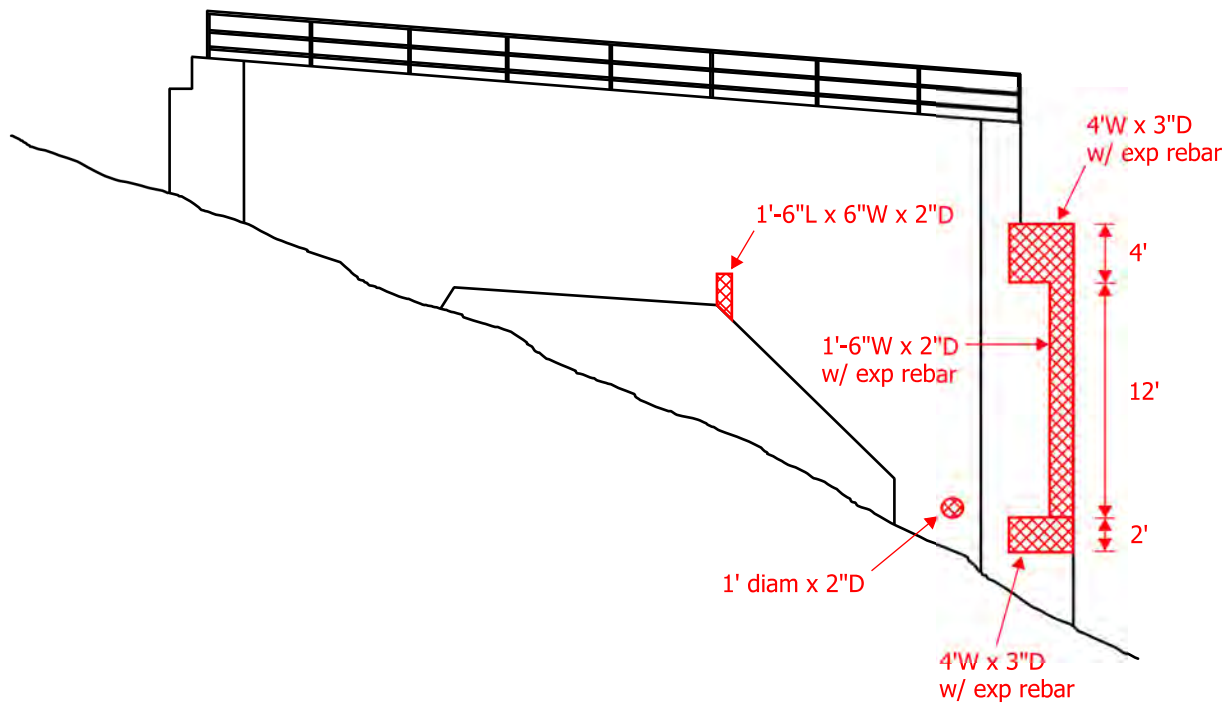
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Near Abutment
Right Elevation





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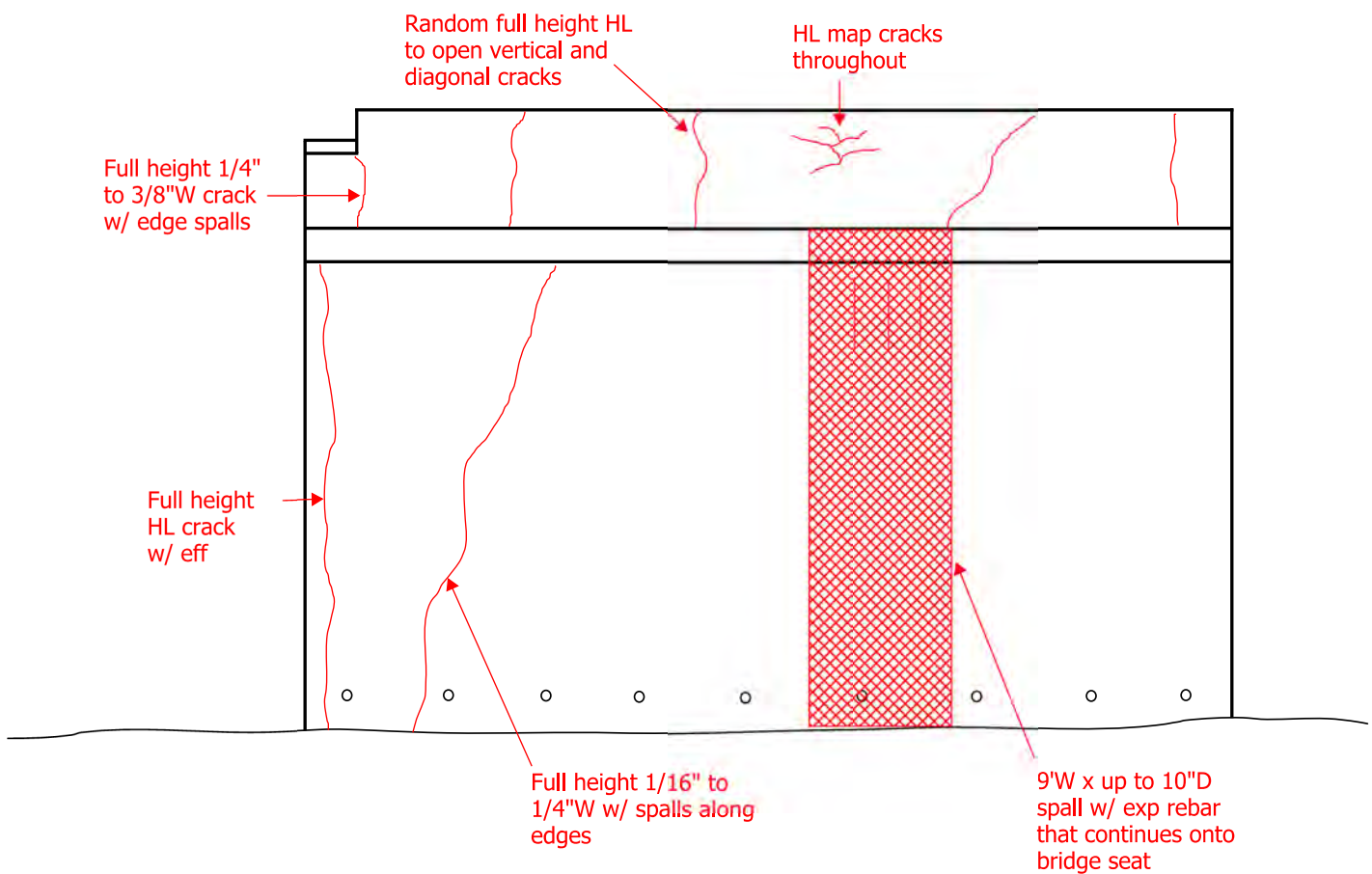
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Far Abutment





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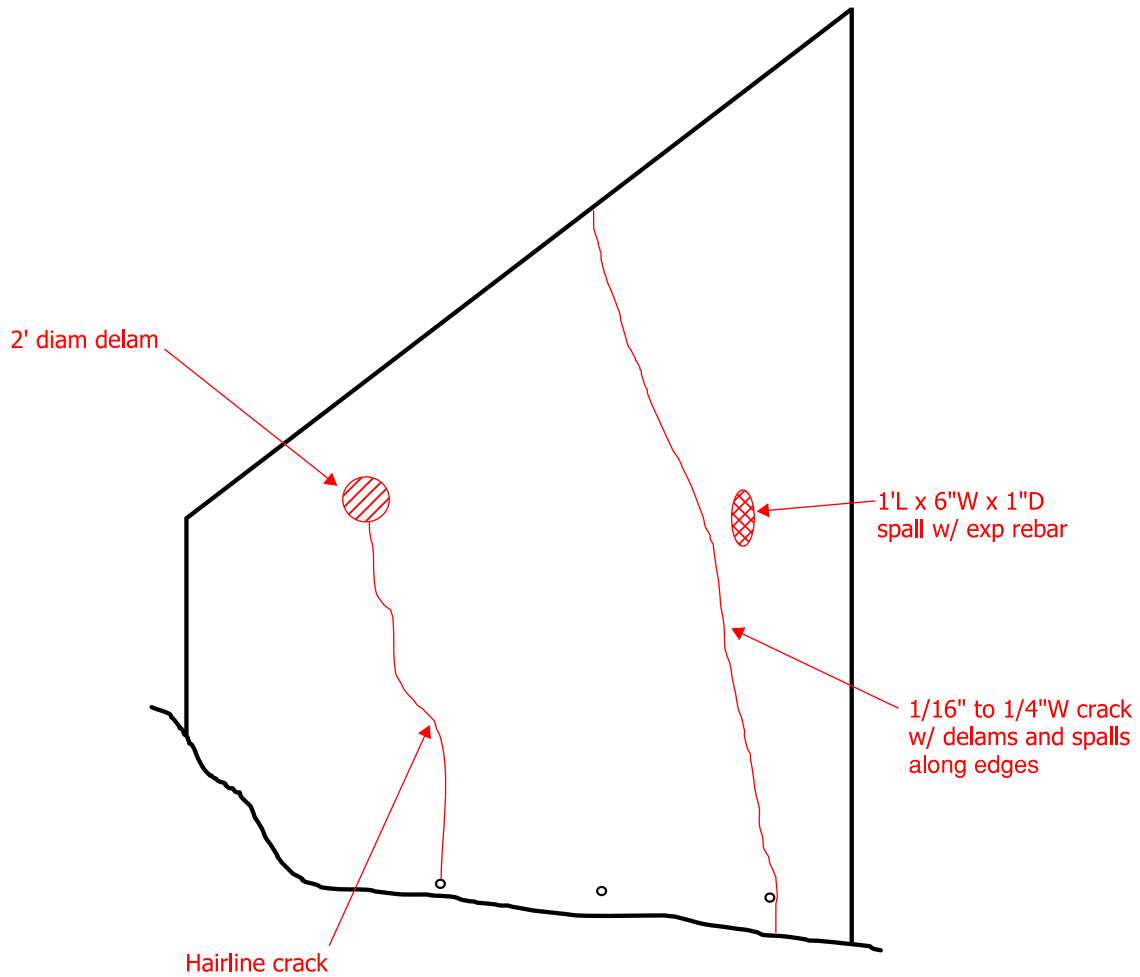
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Far Left Wingwall



Wingwall has scattered hairline
map cracks throughout



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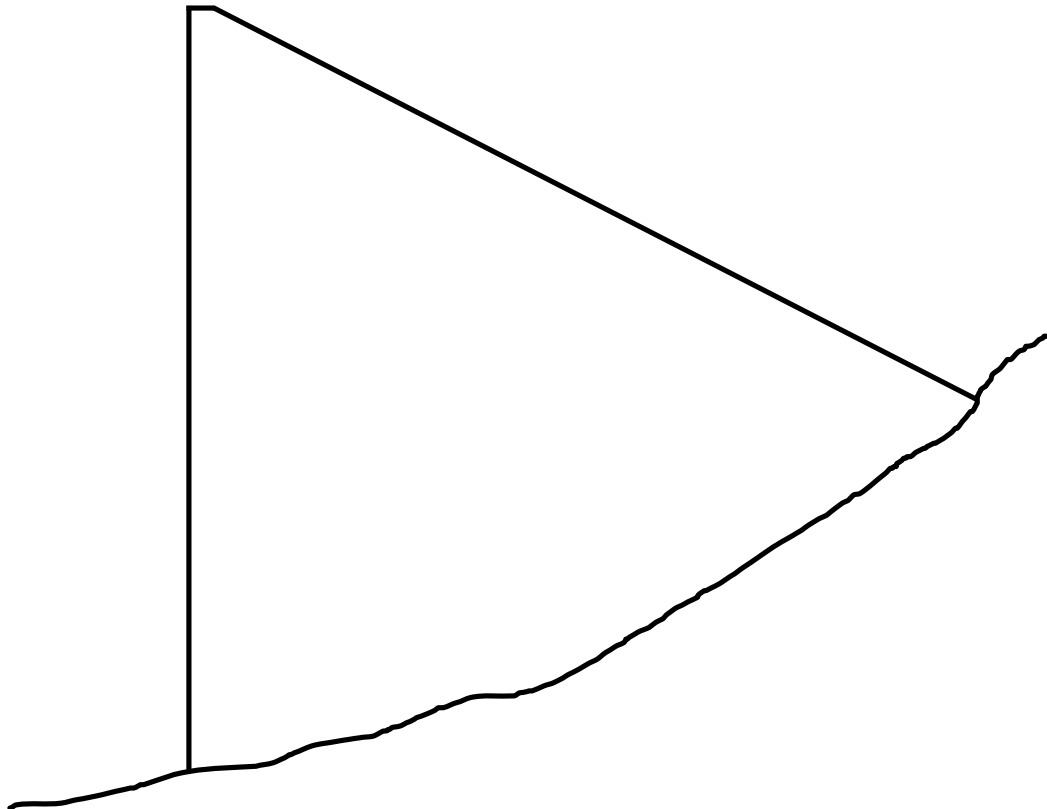
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Far Right Wingwall



Wingwall has areas of hairline map cracking throughout and is covered with vegetation.

ANALYSIS COMPUTATIONS

Black's Run Bridge Inspection

Prepared for:

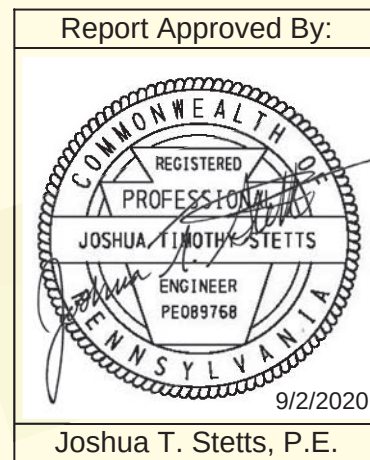
Ohio River Trail Council
1100 Pennsylvania Ave
Monaca, PA 15061

Prepared by:

LARSON DESIGN GROUP
1000 Commerce Park Drive, Suite 201
Williamsport, PA 17701

LDG Project No. 12590-001

April 2020



SUBMISSION QUALITY ASSURANCE

Prepared By: <u>ATR</u>	Date: <u>4/8/2020</u>
Checked By: <u>JTS</u>	Date: <u>4/14/2020</u>
QA/QC'd By: <u>JTS</u>	Date: <u>9/2/2020</u>



ANALYSIS TABLE OF CONTENTS

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Appendix	118
• Truss Section Data • Truck Cut Sheet • Previous Analysis • Drawings	

Section 1 – Results

SCOPE AND RESULTS

FOR

Black's Run Bridge

- The analysis was performed to determine if the structure could be utilized as a pedestrian walkway.
- In addition to the pedestrian live load, the structure was analyzed for a maintenance vehicle per AASHTO 3.14.1.4. A F-550 truck was selected and input as a Special Live Load in BAR7. A cut sheet for the F-550 is attached to this document for the loading and axle spacing.
- If any additional vehicles are required to cross the structure additional analysis will be required.
- The structure was analyzed using inputs from the previous analysis completed by HRG, dated 5/28/2008. These values were not checked and assumed to be accurate. A couple of minor invalid inputs were corrected and will be discussed later in this document.
- It was determined that the truss can service the pedestrian live load and F-550 maintenance truck with Operating Ratings above 1.0 for all analyzed components.
 - Truss
 - IR = 2.63
 - OR = 6.05
 - Stringer
 - IR = 1.97
 - OR = 3.28
 - Floorbeam
 - IR = 0.83
 - OR = 1.38

Section 2 – Analysis Notes

ANALYSIS NOTES

FOR

Black's Run Bridge

- This analysis reflects the structure's condition as detailed in the 2008 inspection report performed by HRG.
- In addition to the pedestrian live load, the structure was analyzed for a maintenance vehicle per AASHTO 3.14.1.4. A F-550 truck was selected and input as a Special Live Load in BAR7. A cut sheet for the F-550 is attached to this document for the loading and axle spacing.
- A pedestrian live load of 85 psf was used per AASHTO 3.14.1.3. The live load was input into BAR7 by increasing the DL on the floorbeam (Floorbeam DL1) and truss panel points.
- The analysis run was created from BAR7 inputs from the previous analysis (see Sheets 55 – 58 of 97). Several inputs were changed to correct mistakes or to account for additional load (Pedestrian Live Load) as follows:
 - The Stringer DL1 was increased to account for additional Pedestrian LL. See calculations later in this document for additional information.
 - The Truss DL was also increased to account for additional Pedestrian LL. The total load was divided by 48 joints (assuming each end joint counts as 1/2 of a full joint). See calculations later in this document for additional information.
 - Member ID U8L9 incorrectly input the area for Gross Area Compression as 25.52. This should be 25.56 per the Section Properties Table. This has been updated.
 - Member ID U39U39 does not exist. In sequential order, the next member should have been U38U39. This has been updated.
 - The truss rating run reduces the area of the bottom chord member by 25%. Member L35L36 did not reduce the area for Gross Area Compression or Net Area Tensions. These areas were reduced to 31.39 and 25.01, respectively. This is 75% of the numbers shown (41.85 and 33.35). This has been updated.
- The floor system acts as a two way slab spanning between the truss and stringer and between floorbeams. The structure was analyzed by applying 5/8 of the floor LL to the stringer (5/8 was used using common engineering equations for a equal spacing continuous span). The stringer acts a point load on the floorbeam. The remaining load is supported by the truss. This is modeled by increasing the truss DLs at each panel point.
- Previously, the truss was analyzed assuming the floorbeam to be supported in the middle by the center truss. Also, the stringer was not rated in the previous analysis. The DL acting on the floorbeam and truss included the weight of the center truss which would indicate the center truss was not supporting the floorbeam. This analysis assumes the floorbeam to span between outside trusses. The center truss supports its own DL only. This is conservative.
- Section properties for the main load carrying members were gathered during this inspection and compared to the section properties used in the analysis. The section properties used in the analysis were determined to still be valid.
- Analysis Program used was BAR7 version 7.15.0.0.

Section 3 - Dead Load



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BRIDGE	Black's Run Bridge		
SHEET NO.		LDG NO.	12590-001
CALCULATED BY	ATR	DATE	4/7/2020
CHECKED BY	JTS	DATE	4/15/2020
BMS NO.	-		

STRINGER DEAD LOAD

(Calculated for Spans 1 to 4)

References:

1. Standard Specifications for Highway Bridges, Sixteenth Edition, 1996 [AASHTO]
2. Steel Construction Manual, Thirteenth Edition, 2005 [AISC]

Calculations:

According to Article 3.14.1.3 [AASHTO], bridges for pedestrian and/or bicycle traffic shall be designed for a live load of 85 PSF. In order to apply that load to the stringer in BAR7, the load is entered as a linear load in Stringer DL1. The previous analysis calculate the load for the stringer as 0.020 k/ft (see Sheet 55 of 97). The stringer frames down the center of the bridge. The deck in this configuration will act like a continuous span with two equal spans on either side of the stringer. Table 3-22c [AISC] says that 5/8 of the load will be distributed to the center support. An additional 0.956 k/ft will be applied as follows:

$$0.956 \text{ k/ft} = (5/8) * 85 \text{ psf [Ped LL]} * 18.00' [\text{Roadway width}] / 1000 [\text{lbs/K}]$$

The additional DL will be added to BAR7 inputs as follows:

Input	Old	New
SDL1	0.020	0.976



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BRIDGE	Black's Run Bridge		
SHEET NO.		LDG NO.	12590-001
CALCULATED BY	ATR	DATE	4/7/2020
CHECKED BY	JTS	DATE	4/14/2020
BMS NO.	-		

TRUSS DEAD LOAD

(Calculated for Spans 1 to 4)

References:

1. Standard Specifications for Highway Bridges, Sixteenth Edition, 1996 [AASHTO]

Calculations:

According to Article 3.14.1.3 [AASHTO], bridges for pedestrian and/or bicycle traffic shall be designed for a live load of 85 PSF. In order to apply that load to the truss in BAR7, the load is entered as Truss DL at each panel point. The previous analysis calculate the load for the Floorsystem on Sheet 5 of 97. The total calculated equals 1.25K. This load gets applied to all upper joints. An additional 6.71 K should be added to account for additional Pedestrian LL as follows:

$$6.71\text{K} = 85 \text{ psf [Ped LL]} * 420.83' [\text{Length of Bridge}] * 18.00' [\text{Roadway width}] / 48 \text{ joints [assumed even dist]} / 2 [\text{ea truss}] / 1000 [\text{lbs/K}]$$

The additional DL will be added to BAR7 inputs as follows:

Joint	Old	New
U0	23.00	29.71
U1	55.00	61.71
U2	23.00	29.71
U3	57.00	63.71

Joint	Old	New
U4	23.00	29.71
U5	57.00	63.71
U6	23.00	29.71
U7	57.00	63.71

Joint	Old	New
U8	23.00	29.71
U9	57.00	63.71
U10	23.00	29.71
U11	50.00	56.71

Joint	Old	New
U12	23.00	29.71
U13	50.00	56.71
U14	20.50	27.21
U15	47.00	53.71

Joint	Old	New
U16	18.00	24.71
U17	47.00	53.71
U18	18.00	24.71
U19	47.00	53.71



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BRIDGE	<u>Black's Run Bridge</u>		
SHEET NO.		LDG NO.	<u>12590-001</u>
CALCULATED BY	<u>ATR</u>	DATE	<u>4/7/2020</u>
CHECKED BY	<u>JTS</u>	DATE	<u>4/14/2020</u>
BMS NO.	<u>-</u>		

TRUSS DEAD LOAD

(Calculated for Spans 1 to 4)

Joint	Old	New
U20	18.00	24.71
U21	47.00	53.71
U22	20.50	27.21
U23	50.00	56.71

Joint	Old	New
U24	23.00	29.71
U25	50.00	56.71
U26	23.00	29.71
U27	55.00	61.71

Joint	Old	New
U28	23.00	29.71
U29	55.00	61.71
U30	23.00	29.71
U31	55.00	61.71

Joint	Old	New
U32	23.00	29.71
U33	55.00	61.71
U34	23.00	29.71
U35	55.00	61.71

Joint	Old	New
U36	23.00	29.71
U37	50.00	56.71
U38	23.00	29.71
U39	50.00	56.71

Joint	Old	New
U40	21.50	28.21
U41	50.00	56.71
U42	20.00	26.71
U43	50.00	56.71

Joint	Old	New
U44	20.00	26.71
U45	50.00	56.71
U46	20.00	26.71
U47	50.00	56.71

Joint	Old	New
U48	20.00	26.71

Section 4 - Analysis and Ratings

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*****
*
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TFS

Calc By: ATR Date: 04/08/20
 Chk'd By: JTS Date: 04/15/20

PROGRAM P4353000
VERSION 7.15.0.0

LAST UPDATED 02/15/2018

04/15/2020 13:50
DOCUMENTATION 02/2018

INPUT: TFS.dat

PROJECT IDENTIFICATION

STRUCTURE ID -

BRG SLC	LIVE OUT-	IMP GAGE	PASS FAT-	CONC	RE-	S OVER	END
TYPE LEV LANES	LOAD PUT	FACT DIST	DIST IGUE	DECK SPEC	DIST DIR	FACTOR	PAN
TFS	D 1 0	0.00 0.0 0.0				0.00	

SKEW	
CORR	PONY
HYB FACTOR	TRUSS PDF COMPACT
0.000	Y

BRIDGE CROSS SECTION AND LOADING

DECK	OVERHANG	CL OF				
WIDTH	OR	GIRDER OR	ROADWAY	DISTRIBUTION	FACTORS	
20.67	SPACING	TRUSS TO CURB	WIDTH	SHEAR	MOMENT	DEFLECT
	0.00	-1.00	18.00	0.625	0.727	0.670

SLAB		DEAD LOADS				
THICKNESS	HAUNCH	DL1	DL2	F'C	N	SYMMETRY
10.00	0.00	0.000	0.000	0.000	0.	

LIVE LOAD	NO. OF	END		HINGE	TEMP	END
LOCATION	PANELS	COND.	CORS	AT	CHANGE	BRG.
U	48	R		L37	0.	U

STRINGER	FLOORBEAM	UNIT WEIGHT	GUSSET	PATCH LOAD
DL1	DL1	DECK CONCRETE	ANALYSIS	ANALYSIS
0.976	0.000	0.00		

UNSYMM PIER
SUPPORT

SPAN LENGTHS (CANTILEVER)

SPAN #	1	2	3	4
LENGTH	103.55	114.15	109.13	94.00

HINGE #	1	2	3
DISTANCE	128.6	192.7	351.8

TRAFFIC LANE LOCATIONS

LANE #	1	2	3	4	5	6	7
DIST							
WIDTH							
% LL							

STRINGER SPAN LENGTHS (SIMPLE)

SPAN #	1	2	3
LENGTH	9.50	9.50	9.50

STRINGER LOCATIONS

STRINGER # 1
DISTANCE 8.00

TRUSS GEOMETRY

PANEL NO.	PANEL WIDTH		VERTICAL POST			PANEL TYPE
			H1	H2	H3	
1	9.67	Y	11.67	0.00	-11.67	3
2	9.50	N	11.67	0.00	-11.67	5
3	9.50	Y	11.67	0.00	-11.67	6
4	9.50	N	11.67	0.00	-11.67	5
5	9.50	Y	11.67	0.00	-11.67	6
6	9.50	N	11.67	0.00	-11.67	5
7	9.50	Y	11.67	0.00	-11.67	6
8	9.50	N	11.67	0.00	-11.67	5
9	9.50	Y	11.67	0.00	-11.67	6
10	9.50	Y	11.67	0.00	-11.67	5
11	8.38	Y	11.67	0.00	-11.67	6
12	8.33	Y	11.67	0.00	-11.67	5
13	8.33	Y	11.67	0.00	-11.67	6
14	8.33		0.00	0.00	0.00	4
15	8.07	Y	9.00	0.00	-9.00	3
16	8.00	N	9.00	0.00	-9.00	5
17	8.00	Y	9.00	0.00	-9.00	6
18	8.01	N	9.00	0.00	-9.00	5
19	8.01	Y	9.00	0.00	-9.00	6
20	8.01	N	9.00	0.00	-9.00	5
21	8.00	Y	9.00	0.00	-9.00	6
22	8.07		0.00	0.00	0.00	4
23	8.33	Y	11.67	0.00	-11.67	3
24	8.33	Y	11.67	0.00	-11.67	5
25	8.33	Y	11.67	0.00	-11.67	6
26	8.38	Y	11.67	0.00	-11.67	5
27	9.18	Y	11.67	0.00	-11.67	6
28	9.25	N	11.67	0.00	-11.67	5
29	9.25	Y	11.67	0.00	-11.67	6
30	9.25	N	11.67	0.00	-11.67	5
31	9.25	Y	11.67	0.00	-11.67	6
32	9.25	N	11.67	0.00	-11.67	5
33	9.25	Y	11.67	0.00	-11.67	6
34	9.25	N	11.67	0.00	-11.67	5
35	9.25	Y	11.67	0.00	-11.67	6
36	9.19	N	11.67	0.00	-11.67	5
37	8.38	Y	11.67	0.00	-11.67	6
38	8.33	N	11.67	0.00	-11.67	5
39	8.33	Y	11.67	0.00	-11.67	6
40	8.33		0.00	0.00	0.00	4
41	8.63	Y	9.00	0.00	-9.00	3
42	8.63	N	9.00	0.00	-9.00	5
43	8.63	Y	9.00	0.00	-9.00	6
44	8.63	N	9.00	0.00	-9.00	5
45	8.63	Y	9.00	0.00	-9.00	6
46	8.62	N	9.00	0.00	-9.00	5
47	8.62	Y	9.00	0.00	-9.00	6
48	8.62		0.00	0.00	0.00	4

TRUSS DEAD LOADS

LOCAT	LOAD	LOCAT	LOAD	LOCAT	LOAD	LOCAT	LOAD
U 0	29.71	U 1	61.71	U 2	29.71	U 3	63.71
U 4	29.71	U 5	63.71	U 6	29.71	U 7	63.71
U 8	29.71	U 9	63.71	U 10	29.71	U 11	56.71
U 12	29.71	U 13	56.71	U 14	27.21	U 15	53.71
U 16	24.71	U 17	53.71	U 18	24.71	U 19	53.71
U 20	24.71	U 21	53.71	U 22	27.21	U 23	56.71
U 24	29.71	U 25	56.71	U 26	29.71	U 27	61.71
U 28	29.71	U 29	61.71	U 30	29.71	U 31	61.71
U 32	29.71	U 33	61.71	U 34	29.71	U 35	61.71
U 36	29.71	U 37	56.71	U 38	29.71	U 39	56.71

U 40	28.21	U 41	56.71	U 42	26.71	U 43	56.71
U 44	26.71	U 45	56.71	U 46	26.71	U 47	56.71
U 48	26.71	0	0.00	0	0.00	0	0.00

TRUSS MEMBER PROPERTIES

MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U 0U 1	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00		0.0	23.89
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U 1U 2	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00		0.0	23.89
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U 2U 3	46.47	36.97	745.0	33.0	0.00	0.00	0.00	0.00		0.0	43.94
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U 3U 4	46.47	36.97	745.0	33.0	0.00	0.00	0.00	0.00		0.0	43.94
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U 4U 5	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00		0.0	48.96
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U 5U 6	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00		0.0	48.96
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U 6U 7	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00		0.0	48.96
		UNSUPP	SECTION		MOMENT OF		TORSION		MOMENT OF		

	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U 7U 8	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00		0.0	48.96
	DEPTH	UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	0.0	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
		LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U 8U 9	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	41.85	33.35	660.1	33.0	0.00	0.00	0.00	0.00		0.0	39.63
	DEPTH	UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	0.0	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
		LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U 9U10	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	41.85	33.35	660.1	33.0	0.00	0.00	0.00	0.00		0.0	39.63
	DEPTH	UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	0.0	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
		LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U10U11	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00		0.0	48.96
	DEPTH	UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	0.0	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
		LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U11U12	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00		0.0	48.96
	DEPTH	UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	0.0	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
		LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U12U13	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00		0.0	23.89
	DEPTH	UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	0.0	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
		LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U13U14	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00		0.0	23.89
	DEPTH	UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	0.0	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
		LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					

MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U14U15	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U15U16	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U16U17	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U17U18	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U18U19	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U19U20	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U20U21	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U21U22	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30

	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.0000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U22U23	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00		0.0	23.89
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.0000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U23U24	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00		0.0	23.89
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.0000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U24U25	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00		0.0	48.96
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.0000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U25U26	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00		0.0	48.96
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.0000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U26U27	41.85	33.35	660.1	33.0	0.00	0.00	0.00	0.00		0.0	39.63
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.0000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U27U28	41.85	33.35	660.1	33.0	0.00	0.00	0.00	0.00		0.0	39.63
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.0000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U28U29	56.73	45.20	930.1	33.0	0.00	0.00	0.00	0.00		0.0	53.71
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					

	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U29U30	56.73	45.20	930.1	33.0	0.00	0.00	0.00	0.00		0.0	53.71
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U30U31	56.73	45.20	930.1	33.0	0.00	0.00	0.00	0.00		0.0	53.71
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U31U32	56.73	45.20	930.1	33.0	0.00	0.00	0.00	0.00		0.0	53.71
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U32U33	56.73	45.20	930.1	33.0	0.00	0.00	0.00	0.00		0.0	53.71
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U33U34	56.73	45.20	930.1	33.0	0.00	0.00	0.00	0.00		0.0	53.71
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U34U35	41.85	33.35	660.1	33.0	0.00	0.00	0.00	0.00		0.0	39.63
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U35U36	41.85	33.35	660.1	33.0	0.00	0.00	0.00	0.00		0.0	39.63
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER	GROSS AREA	NET AREA	MOMENT OF		UNBRACED LENGTH		FLANGE		FAT		GROSS AREA

ID	COMPR	TENS	INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS	
U36U37	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00		0.0	48.96	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	48.96
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U37U38	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00		0.0	48.96	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	48.96
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U38U39	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00		0.0	23.89	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	23.89
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U39U40	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00		0.0	23.89	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	23.89
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U40U41	34.99	27.49	491.8	33.0	0.00	0.00	0.00	0.00		0.0	32.74	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	32.74
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U41U42	34.99	27.49	491.8	33.0	0.00	0.00	0.00	0.00		0.0	32.74	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	32.74
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U42U43	34.99	27.49	491.8	33.0	0.00	0.00	0.00	0.00		0.0	32.74	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	32.74
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U43U44	34.99	27.49	491.8	33.0	0.00	0.00	0.00	0.00		0.0	32.74	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	32.74

	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U44U45	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	34.99	27.49	491.8	33.0	0.00	0.00	0.00	0.00		0.0	32.74
		UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U45U46	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	34.99	27.49	491.8	33.0	0.00	0.00	0.00	0.00		0.0	32.74
		UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U46U47	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	34.99	27.49	491.8	33.0	0.00	0.00	0.00	0.00		0.0	32.74
		UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U47U48	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	34.99	27.49	491.8	33.0	0.00	0.00	0.00	0.00		0.0	32.74
		UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U 1L 1	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
		UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U 3L 3	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
		UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U 5L 5	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
		UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					

MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U 7L 7	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U 9L 9	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U10L10	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U11L11	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U12L12	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U13L13	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U15L15	10.00	8.20	21.3	33.0	0.00	0.00	0.00	0.00		0.0	9.70
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U17L17	10.00	8.20	21.3	33.0	0.00	0.00	0.00	0.00		0.0	9.70

	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY			
	0.0	0.00	0.00	0.00		0.000		0.			
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U19L19	10.00	8.20	21.3	33.0	0.00	0.00	0.00	0.00		0.0	9.70
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY			
	0.0	0.00	0.00	0.00		0.000		0.			
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U21L21	10.00	8.20	21.3	33.0	0.00	0.00	0.00	0.00		0.0	9.70
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY			
	0.0	0.00	0.00	0.00		0.000		0.			
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U23L23	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY			
	0.0	0.00	0.00	0.00		0.000		0.			
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U24L24	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY			
	0.0	0.00	0.00	0.00		0.000		0.			
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U25L25	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY			
	0.0	0.00	0.00	0.00		0.000		0.			
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U26L26	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY			
	0.0	0.00	0.00	0.00		0.000		0.			
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U27L27	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY			

	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U29L29	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U31L31	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U33L33	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U35L35	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U37L37	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U39L39	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U41L41	10.00	8.20	21.3	33.0	0.00	0.00	0.00	0.00		0.0	9.70
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER	GROSS AREA	NET AREA	MOMENT OF		UNBRACED LENGTH		FLANGE		FAT		GROSS AREA

ID	COMPR	TENS	INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS	
U43L43	10.00	8.20	21.3	33.0	0.00	0.00	0.00	0.00		0.0	9.70	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	9.70
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U45L45	10.00	8.20	21.3	33.0	0.00	0.00	0.00	0.00		0.0	9.70	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	9.70
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U47L47	10.00	8.20	21.3	33.0	0.00	0.00	0.00	0.00		0.0	9.70	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	9.70
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U 0L 1	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	28.30
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
L 1U 2	24.71	19.40	225.5	33.0	0.00	0.00	0.00	0.00		0.0	23.11	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	23.11
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U 2L 3	17.94	15.37	107.3	33.0	0.00	0.00	0.00	0.00		0.0	17.82	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	17.82
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
L 3U 4	15.59	13.40	57.5	33.0	0.00	0.00	0.00	0.00		0.0	15.44	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	15.44
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U 4L 5	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	11.17
		UNSUPP	SECTION	MOMENT OF			TORSION		MOMENT OF			

	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
L 5U 6	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	11.17	9.17	24.6	33.0	0.00	0.00	0.00	0.00		0.0	11.17
	DEPTH	UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	0.0	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
		LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U 6L 7	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	15.59	13.40	57.5	33.0	0.00	0.00	0.00	0.00		0.0	15.44
	DEPTH	UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	0.0	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
		LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
L 7U 8	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	22.94	17.20	206.9	33.0	0.00	0.00	0.00	0.00		0.0	20.64
	DEPTH	UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	0.0	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
		LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U 8L 9	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00		0.0	23.89
	DEPTH	UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	0.0	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
		LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
L 9U10	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30
	DEPTH	UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	0.0	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
		LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U10L11	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	39.98	31.86	567.7	33.0	0.00	0.00	0.00	0.00		0.0	37.86
	DEPTH	UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	0.0	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
		LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
L11U12	COMPR	TENS	INERTIA	L	ECC		WIDTH	C	CAT	FU	TENS
	39.98	31.86	567.7	33.0	0.00	0.00	0.00	0.00		0.0	37.86
	DEPTH	UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	0.0	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
		LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					

MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U12L13	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L13U14	27.94	21.57	383.7	33.0	0.00	0.00	0.00	0.00		0.0	25.76
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U14L15	22.94	17.20	206.9	33.0	0.00	0.00	0.00	0.00		0.0	20.64
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L15U16	17.94	15.37	107.3	33.0	0.00	0.00	0.00	0.00		0.0	17.82
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U16L17	12.65	10.53	45.1	33.0	0.00	0.00	0.00	0.00		0.0	12.65
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L17U18	8.81	7.20	17.5	33.0	0.00	0.00	0.00	0.00		0.0	8.52
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U18L19	8.81	7.20	17.5	33.0	0.00	0.00	0.00	0.00		0.0	8.52
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L19U20	12.65	10.53	45.1	33.0	0.00	0.00	0.00	0.00		0.0	12.65

	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U20L21	17.94	15.37	107.3	33.0	0.00	0.00	0.00	0.00		0.0	17.82
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L21U22	22.94	17.20	206.9	33.0	0.00	0.00	0.00	0.00		0.0	20.64
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U22L23	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L23U24	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U24L25	39.98	31.86	567.7	33.0	0.00	0.00	0.00	0.00		0.0	37.86
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L25U26	34.99	27.49	491.8	33.0	0.00	0.00	0.00	0.00		0.0	32.74
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U26L27	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					

	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L27U28	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00		0.0	23.89
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U28L29	20.00	17.14	121.2	33.0	0.00	0.00	0.00	0.00		0.0	20.00
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L29U30	15.59	13.40	57.5	33.0	0.00	0.00	0.00	0.00		0.0	15.44
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U30L31	10.00	8.20	21.3	33.0	0.00	0.00	0.00	0.00		0.0	9.70
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L31U32	10.00	8.20	21.3	33.0	0.00	0.00	0.00	0.00		0.0	9.70
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
U32L33	15.59	13.40	57.5	33.0	0.00	0.00	0.00	0.00		0.0	15.44
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L33U34	20.00	17.14	121.2	33.0	0.00	0.00	0.00	0.00		0.0	20.00
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC		MOMENT OF INERTIA IYC		TORSION INERTIA JXY		MOMENT OF INERTIA IY		
	0.0	0.00	0.00		0.00		0.000		0.		
MEMBER	GROSS AREA	NET AREA	MOMENT OF		UNBRACED LENGTH		FLANGE		FAT		GROSS AREA

ID	COMPR	TENS	INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS	
U34L35	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00		0.0	23.89	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	FAT CAT	FU	GROSS AREA TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	28.30
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
L35U36	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	FAT CAT	FU	GROSS AREA TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	32.74
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U36L37	34.99	27.49	491.8	33.0	0.00	0.00	0.00	0.00		0.0	32.74	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	FAT CAT	FU	GROSS AREA TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	37.86
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
L37U38	39.98	31.86	567.7	33.0	0.00	0.00	0.00	0.00		0.0	37.86	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	FAT CAT	FU	GROSS AREA TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	28.30
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U38L39	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	FAT CAT	FU	GROSS AREA TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	20.64
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
L39U40	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00		0.0	28.30	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	FAT CAT	FU	GROSS AREA TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	17.82
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
U40L41	22.94	17.20	206.9	33.0	0.00	0.00	0.00	0.00		0.0	20.64	
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA	FY	L	ECC	WIDTH	C	FAT CAT	FU	GROSS AREA TENS
	0.0	0.00	0.00	0.00	33.0	0.00	0.00	0.00	0.00		0.0	17.82
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS	
L41U42	17.94	15.37	107.3	33.0	0.00	0.00	0.00	0.00		0.0	17.82	
		UNSUPP	SECTION	MOMENT	OF	TORSION	MOMENT	OF				

	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U42L43	COMPR	TENS	INERTIA	L	ECC		WIDTH	CAT	FU	TENS	
	12.65	10.53	45.1	33.0	0.00	0.00	0.00	0.00	0.0	12.65	
		UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
L43U44	COMPR	TENS	INERTIA	L	ECC		WIDTH	CAT	FU	TENS	
	8.81	7.20	17.5	33.0	0.00	0.00	0.00	0.00	0.0	8.52	
		UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U44L45	COMPR	TENS	INERTIA	L	ECC		WIDTH	CAT	FU	TENS	
	8.81	7.20	17.5	33.0	0.00	0.00	0.00	0.00	0.0	8.52	
		UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
L45U46	COMPR	TENS	INERTIA	L	ECC		WIDTH	CAT	FU	TENS	
	12.65	10.53	45.1	33.0	0.00	0.00	0.00	0.00	0.0	12.65	
		UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
U46L47	COMPR	TENS	INERTIA	L	ECC		WIDTH	CAT	FU	TENS	
	17.94	15.37	107.3	33.0	0.00	0.00	0.00	0.00	0.0	17.82	
		UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
L47U48	COMPR	TENS	INERTIA	L	ECC		WIDTH	CAT	FU	TENS	
	22.94	17.20	206.9	33.0	0.00	0.00	0.00	0.00	0.0	20.64	
		UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					
MEMBER	GROSS	NET	MOMENT	UNBRACED						GROSS	
ID	AREA	AREA	OF	LENGTH			FLANGE	FAT		AREA	
L 1L 2	COMPR	TENS	INERTIA	L	ECC		WIDTH	CAT	FU	TENS	
	22.69	17.82	419.7	33.0	0.00	0.00	0.00	0.00	0.0	21.23	
		UNSUPP	SECTION	MOMENT OF	TORSION	MOMENT OF					
	DEPTH	FLANGE	MODULUS	INERTIA	INERTIA	INERTIA					
	0.0	LENGTH	SXC	IYC	JXY	IY					
		0.00	0.00	0.00	0.000	0.					

MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L 2L 3	22.69	17.82	419.7	33.0	0.00	0.00	0.00	0.00		0.0	21.23
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L 3L 4	34.85	27.73	745.0	33.0	0.00	0.00	0.00	0.00		0.0	32.96
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L 4L 5	34.85	27.73	745.0	33.0	0.00	0.00	0.00	0.00		0.0	32.96
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L 5L 6	34.85	27.73	745.0	33.0	0.00	0.00	0.00	0.00		0.0	32.96
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L 6L 7	34.85	27.73	745.0	33.0	0.00	0.00	0.00	0.00		0.0	32.96
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L 7L 8	31.39	25.01	660.1	33.0	0.00	0.00	0.00	0.00		0.0	29.72
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L 8L 9	31.39	25.01	660.1	33.0	0.00	0.00	0.00	0.00		0.0	29.72
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L 9L10	31.39	25.01	660.1	33.0	0.00	0.00	0.00	0.00		0.0	29.72

	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L10L11	31.39	25.01	660.1	33.0	0.00	0.00	0.00	0.00		0.0	29.72
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L11L12	31.39	25.01	660.1	33.0	0.00	0.00	0.00	0.00		0.0	29.72
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L12L13	31.39	25.01	660.1	33.0	0.00	0.00	0.00	0.00		0.0	29.72
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L15L16	22.69	17.82	419.7	33.0	0.00	0.00	0.00	0.00		0.0	21.23
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L16L17	22.69	17.82	419.7	33.0	0.00	0.00	0.00	0.00		0.0	21.23
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L17L18	22.69	17.82	419.7	33.0	0.00	0.00	0.00	0.00		0.0	21.23
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					
	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L18L19	22.69	17.82	419.7	33.0	0.00	0.00	0.00	0.00		0.0	21.23
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY					

	0.0	0.00	0.00	0.00	0.000	0.					
MEMBER ID L19L20	GROSS AREA COMPR 22.69	NET AREA TENS 17.82	MOMENT OF INERTIA 419.7	FY 33.0	UNBRACED LENGTH L 0.00	ECC 0.00	FLANGE WIDTH 0.00	C 0.00	FAT CAT	FU 0.0	GROSS AREA TENS 21.23
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00	MOMENT OF INERTIA IYC 0.00	TORSION INERTIA JXY 0.000	MOMENT OF INERTIA IY 0.					
MEMBER ID L20L21	GROSS AREA COMPR 22.69	NET AREA TENS 17.82	MOMENT OF INERTIA 419.7	FY 33.0	UNBRACED LENGTH L 0.00	ECC 0.00	FLANGE WIDTH 0.00	C 0.00	FAT CAT	FU 0.0	GROSS AREA TENS 21.23
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00	MOMENT OF INERTIA IYC 0.00	TORSION INERTIA JXY 0.000	MOMENT OF INERTIA IY 0.					
MEMBER ID L23L24	GROSS AREA COMPR 31.39	NET AREA TENS 25.01	MOMENT OF INERTIA 660.1	FY 33.0	UNBRACED LENGTH L 0.00	ECC 0.00	FLANGE WIDTH 0.00	C 0.00	FAT CAT	FU 0.0	GROSS AREA TENS 29.72
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00	MOMENT OF INERTIA IYC 0.00	TORSION INERTIA JXY 0.000	MOMENT OF INERTIA IY 0.					
MEMBER ID L24L25	GROSS AREA COMPR 31.39	NET AREA TENS 25.01	MOMENT OF INERTIA 660.1	FY 33.0	UNBRACED LENGTH L 0.00	ECC 0.00	FLANGE WIDTH 0.00	C 0.00	FAT CAT	FU 0.0	GROSS AREA TENS 29.72
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00	MOMENT OF INERTIA IYC 0.00	TORSION INERTIA JXY 0.000	MOMENT OF INERTIA IY 0.					
MEMBER ID L25L26	GROSS AREA COMPR 31.39	NET AREA TENS 25.01	MOMENT OF INERTIA 660.1	FY 33.0	UNBRACED LENGTH L 0.00	ECC 0.00	FLANGE WIDTH 0.00	C 0.00	FAT CAT	FU 0.0	GROSS AREA TENS 29.72
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00	MOMENT OF INERTIA IYC 0.00	TORSION INERTIA JXY 0.000	MOMENT OF INERTIA IY 0.					
MEMBER ID L26L27	GROSS AREA COMPR 31.39	NET AREA TENS 25.01	MOMENT OF INERTIA 660.1	FY 33.0	UNBRACED LENGTH L 0.00	ECC 0.00	FLANGE WIDTH 0.00	C 0.00	FAT CAT	FU 0.0	GROSS AREA TENS 29.72
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00	MOMENT OF INERTIA IYC 0.00	TORSION INERTIA JXY 0.000	MOMENT OF INERTIA IY 0.					
MEMBER ID L27L28	GROSS AREA COMPR 31.39	NET AREA TENS 25.01	MOMENT OF INERTIA 660.1	FY 33.0	UNBRACED LENGTH L 0.00	ECC 0.00	FLANGE WIDTH 0.00	C 0.00	FAT CAT	FU 0.0	GROSS AREA TENS 29.72
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00	MOMENT OF INERTIA IYC 0.00	TORSION INERTIA JXY 0.000	MOMENT OF INERTIA IY 0.					
MEMBER	GROSS AREA	NET AREA	MOMENT OF		UNBRACED LENGTH		FLANGE		FAT		GROSS AREA

ID	COMPR	TENS	INERTIA	FY	L	ECC	WIDTH	C	CAT	FU	TENS
L28L29	31.39	25.01	660.1	33.0	0.00	0.00	0.00	0.00		0.0	29.72
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA FY	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY				
	0.0	0.00	0.00		0.00	0.000	0.				
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L29L30	34.85	27.73	745.0	33.0	0.00	0.00	0.00	0.00		0.0	32.96
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA FY	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY				
	0.0	0.00	0.00		0.00	0.000	0.				
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L30L31	34.85	27.73	745.0	33.0	0.00	0.00	0.00	0.00		0.0	32.96
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA FY	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY				
	0.0	0.00	0.00		0.00	0.000	0.				
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L31L32	34.85	27.73	745.0	33.0	0.00	0.00	0.00	0.00		0.0	32.96
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA FY	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY				
	0.0	0.00	0.00		0.00	0.000	0.				
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L32L33	34.85	27.73	745.0	33.0	0.00	0.00	0.00	0.00		0.0	32.96
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA FY	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY				
	0.0	0.00	0.00		0.00	0.000	0.				
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L33L34	31.39	25.01	660.1	33.0	0.00	0.00	0.00	0.00		0.0	29.72
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA FY	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY				
	0.0	0.00	0.00		0.00	0.000	0.				
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L34L35	31.39	25.01	660.1	33.0	0.00	0.00	0.00	0.00		0.0	29.72
	DEPTH	UNSUPP FLANGE LENGTH	SECTION MODULUS SXC	MOMENT OF INERTIA FY	MOMENT OF INERTIA IYC	TORSION INERTIA JXY	MOMENT OF INERTIA IY				
	0.0	0.00	0.00		0.00	0.000	0.				
MEMBER ID	GROSS AREA COMPR	NET AREA TENS	MOMENT OF INERTIA	FY	UNBRACED LENGTH L	ECC	FLANGE WIDTH	C	FAT CAT	FU	GROSS AREA TENS
L35L36	31.39	25.01	660.1	33.0	0.00	0.00	0.00	0.00		0.0	29.72
		UNSUPP	SECTION	MOMENT	OF	TORSION	MOMENT	OF			

	DEPTH 0.0	FLANGE LENGTH 0.00	MODULUS SXC 0.00		INERTIA IYC 0.00		INERTIA JXY 0.000		INERTIA IY 0.		
MEMBER ID L36L37	GROSS AREA COMPR 31.39	NET AREA TENS 25.01	MOMENT OF INERTIA 660.1	FY 33.0	UNBRACED LENGTH L 0.00	ECC 0.00	FLANGE WIDTH 0.00		FAT CAT 0.00	FU 0.0	GROSS AREA TENS 29.72
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID L37L38	GROSS AREA COMPR 31.39	NET AREA TENS 25.01	MOMENT OF INERTIA 660.1	FY 33.0	UNBRACED LENGTH L 0.00	ECC 0.00	FLANGE WIDTH 0.00		FAT CAT 0.00	FU 0.0	GROSS AREA TENS 29.72
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID L38L39	GROSS AREA COMPR 31.39	NET AREA TENS 25.01	MOMENT OF INERTIA 660.1	FY 33.0	UNBRACED LENGTH L 0.00	ECC 0.00	FLANGE WIDTH 0.00		FAT CAT 0.00	FU 0.0	GROSS AREA TENS 29.72
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID L41L42	GROSS AREA COMPR 26.24	NET AREA TENS 20.62	MOMENT OF INERTIA 491.8	FY 33.0	UNBRACED LENGTH L 0.00	ECC 0.00	FLANGE WIDTH 0.00		FAT CAT 0.00	FU 0.0	GROSS AREA TENS 24.56
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID L42L43	GROSS AREA COMPR 26.24	NET AREA TENS 20.62	MOMENT OF INERTIA 491.8	FY 33.0	UNBRACED LENGTH L 0.00	ECC 0.00	FLANGE WIDTH 0.00		FAT CAT 0.00	FU 0.0	GROSS AREA TENS 24.56
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID L43L44	GROSS AREA COMPR 26.24	NET AREA TENS 20.62	MOMENT OF INERTIA 491.8	FY 33.0	UNBRACED LENGTH L 0.00	ECC 0.00	FLANGE WIDTH 0.00		FAT CAT 0.00	FU 0.0	GROSS AREA TENS 24.56
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		
MEMBER ID L44L45	GROSS AREA COMPR 26.24	NET AREA TENS 20.62	MOMENT OF INERTIA 491.8	FY 33.0	UNBRACED LENGTH L 0.00	ECC 0.00	FLANGE WIDTH 0.00		FAT CAT 0.00	FU 0.0	GROSS AREA TENS 24.56
	DEPTH 0.0	UNSUPP FLANGE LENGTH 0.00	SECTION MODULUS SXC 0.00		MOMENT OF INERTIA IYC 0.00		TORSION INERTIA JXY 0.000		MOMENT OF INERTIA IY 0.		

STEEL MEMBER PROPERTIES													
S		T	WF BM	WF BM	FLANGE		WF BM						
G P		Y	M OF I	AREA	OR	V	OR WEB						
F A		P	OR VRT	OR HRZ	ANGLE	FLANGE	A	PLATE	WEB				
S N	RANGE	E	LEG	LEG	THICK	WIDTH	R	DEPTH	THICK	SHAPE			
F 1	8.00	W	446.30	10.59	0.4280	6.942		15.85	0.2990				
			TPW	TPT	BPW	BPT	COMP	FY	FY TOP	FY BOT	CG	TOP	CG BOT
			0.00	0.00000	0.00	0.00000	N	33.0	0.0	0.0	0.000	0.000	
S		T	WF BM	WF BM	THICK	WIDTH	V	DEPTH	THICK	SHAPE			
	RANGE	E	M OF I	AREA	THICK	WIDTH	V	DEPTH	THICK	SHAPE			
	0.00	W	446.30	10.59	0.4280	6.942		15.85	0.2990				
			TPW	TPT	BPW	BPT	COMP	FY	FY TOP	FY BOT	CG	TOP	CG BOT
			0.00	0.00000	0.00	0.00000	N	33.0	0.0	0.0	0.000	0.000	

B	OR	S			C					C					C					C				
G	OR	F			O	NO.				O	NO.				O	NO.				O	NO.			
					D	OF				D	OF				D	OF				D	OF			
CODE			SPAN		E	SPCS	SPACING			E	SPCS	SPACING			E	SPCS	SPACING			E	SPCS	SPACING		
BF			1		C	1	16.00			B	2	8.00				0	0.00				0	0.00		
						0	0.00				0	0.00				0	0.00				0	0.00		

LANE LOADING

TRUCK LOAD

DEFAULT VALUES

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U10U11	8.38	470.8	0.0	1.22	51.3	1.21	IR OR	-797.1 -994.0	881.3 1199.5	IR OR	8.001 14.204
U11U12	8.33	470.8	0.0	1.21	51.3	1.21	IR OR	-797.2 -994.1	881.3 1199.5	IR OR	8.001 14.204
U12U13	8.33	123.4	0.0	1.21	17.1	1.21	IR OR	-393.2 -490.3	430.0 585.3	IR OR	17.907 26.975
U13U14	8.33	123.4	0.0	1.21	17.1	1.21	IR OR	-393.2 -490.3	430.0 585.3	IR OR	17.907 26.975
U14U15	8.07	-128.7	-18.4	1.21	0.0	1.21	IR OR	-465.9 -581.0	509.4 693.4	IR OR	18.369 24.640
U15U16	8.00	-128.7	-18.4	1.21	0.0	1.21	IR OR	-466.0 -581.1	509.4 693.4	IR OR	18.374 24.646
U16U17	8.00	-268.3	-37.9	1.21	0.0	1.21	IR OR	-466.0 -581.1	509.4 693.4	IR OR	5.216 8.255
U17U18	8.01	-268.3	-37.9	1.21	0.0	1.21	IR OR	-465.9 -581.1	509.4 693.4	IR OR	5.216 8.254
U18U19	8.01	-268.6	-37.9	1.21	0.0	1.21	IR OR	-465.9 -581.1	509.4 693.4	IR OR	5.206 8.242
U19U20	8.01	-268.6	-37.9	1.21	0.0	1.21	IR OR	-465.9 -581.1	509.4 693.4	IR OR	5.206 8.242
U20U21	8.00	-129.3	-18.5	1.21	0.0	1.21	IR OR	-466.0 -581.1	509.4 693.4	IR OR	18.246 24.485
U21U22	8.07	-129.3	-18.5	1.21	0.0	1.21	IR OR	-465.9 -581.0	509.4 693.4	IR OR	18.241 24.479
U22U23	8.33	122.6	0.0	1.21	17.1	1.21	IR OR	-393.2 -490.3	430.0 585.3	IR OR	18.027 27.132
U23U24	8.33	122.6	0.0	1.21	17.1	1.21	IR OR	-393.2 -490.3	430.0 585.3	IR OR	18.027 27.132
U24U25	8.33	469.7	0.0	1.21	51.1	1.21	IR OR	-797.2 -994.1	881.3 1199.5	IR OR	8.047 14.268
U25U26	8.38	469.7	0.0	1.21	51.1	1.21	IR OR	-797.1 -994.0	881.3 1199.5	IR OR	8.047 14.268
U26U27	9.18	130.2	-30.9	1.21	42.9	1.21	IR OR	-643.3 -802.2	713.3 970.9	IR OR	13.588 19.590
U27U28	9.25	130.2	-30.9	1.21	42.9	1.21	IR OR	-643.2 -802.1	713.3 970.9	IR OR	13.588 19.590
U28U29	9.25	-83.7	-50.0	1.21	34.2	1.21	IR OR	-872.3 -1087.8	966.8 1315.9	IR OR	15.777 20.088
U29U30	9.25	-83.7	-50.0	1.21	34.2	1.21	IR OR	-872.3 -1087.8	966.8 1315.9	IR OR	15.777 20.088
U30U31	9.25	-152.6	-55.2	1.21	25.7	1.21	IR OR	-872.3 -1087.8	966.8 1315.9	IR OR	13.047 16.953
U31U32	9.25	-152.6	-55.2	1.21	25.9	1.22	IR OR	-872.3 -1087.8	966.8 1315.9	IR OR	13.047 16.953
U32U33	9.25	-76.6	-50.0	1.21	34.7	1.22	IR OR	-872.3 -1087.8	966.8 1315.9	IR OR	15.916 20.227
U33U34	9.25	-76.6	-50.0	1.21	34.7	1.22	IR OR	-872.3 -1087.8	966.8 1315.9	IR OR	15.916 20.227
U34U35	9.25	144.3	-30.9	1.21	43.4	1.22	IR OR	-643.2 -802.1	713.3 970.9	IR OR	13.100 19.031

U35U36	9.19	144.3	-30.9	1.21	43.4	1.22	IR OR	-643.3 -802.2	713.3 970.9	IR OR	13.100 19.031
U36U37	8.38	490.7	0.0	1.21	51.8	1.22	IR OR	-797.1 -994.0	881.3 1199.5	IR OR	7.544 13.691
U37U38	8.33	490.7	0.0	1.23	51.8	1.22	IR OR	-797.2 -994.1	881.3 1199.5	IR OR	7.544 13.691
U38U39	8.33	129.3	0.0	1.23	17.2	1.22	IR OR	-393.2 -490.3	430.0 585.3	IR OR	17.453 26.466
U39U40	8.33	129.3	0.0	1.23	17.2	1.22	IR OR	-393.2 -490.3	430.0 585.3	IR OR	17.453 26.466
U40U41	8.63	-147.5	-20.3	1.23	0.0	1.22	IR OR	-537.9 -670.8	589.3 802.1	IR OR	19.271 25.831
U41U42	8.63	-147.5	-20.3	1.23	0.0	1.22	IR OR	-537.9 -670.8	589.3 802.1	IR OR	19.271 25.831
U42U43	8.63	-307.3	-41.9	1.23	0.0	1.22	IR OR	-537.9 -670.8	589.3 802.1	IR OR	5.508 8.682
U43U44	8.63	-307.3	-41.9	1.23	0.0	1.22	IR OR	-537.9 -670.8	589.3 802.1	IR OR	5.508 8.682
U44U45	8.63	-307.1	-41.8	1.23	0.0	1.22	IR OR	-537.9 -670.8	589.3 802.1	IR OR	5.516 8.692
U45U46	8.62	-307.1	-41.8	1.23	0.0	1.22	IR OR	-537.9 -670.8	589.3 802.1	IR OR	5.517 8.693
U46U47	8.62	-147.1	-20.2	1.23	0.0	1.22	IR OR	-537.9 -670.8	589.3 802.1	IR OR	19.348 25.927
U47U48	8.62	-147.1	-20.2	1.23	0.0	1.22	IR OR	-537.9 -670.8	589.3 802.1	IR OR	19.348 25.927
L 1L 2	9.50	251.3	-9.5	1.21	32.8	1.22	IR OR	-349.2 -435.4	382.1 520.1	IR OR	3.992 8.202
L 2L 3	9.50	251.3	-9.5	1.21	32.8	1.22	IR OR	-349.2 -435.4	382.1 520.1	IR OR	3.992 8.202
L 3L 4	9.50	349.5	-18.9	1.21	49.7	1.22	IR OR	-537.1 -669.8	593.3 807.5	IR OR	4.906 9.216
L 4L 5	9.50	349.5	-18.9	1.21	49.7	1.22	IR OR	-537.1 -669.8	593.3 807.5	IR OR	4.906 9.216
L 5L 6	9.50	295.5	-28.3	1.21	52.2	1.22	IR OR	-537.1 -669.8	593.3 807.5	IR OR	5.702 9.805
L 6L 7	9.50	295.5	-28.3	1.21	52.2	1.22	IR OR	-537.1 -669.8	593.3 807.5	IR OR	5.702 9.805
L 7L 8	9.50	89.5	-37.7	1.21	42.0	1.22	IR OR	-483.7 -603.2	535.0 728.1	IR OR	10.609 15.209
L 8L 9	9.50	89.5	-37.7	1.21	42.0	1.22	IR OR	-483.7 -603.2	535.0 728.1	IR OR	10.609 15.209
L 9L10	9.50	-268.7	-47.1	1.21	16.3	1.22	IR OR	-483.7 -603.2	535.0 728.1	IR OR	4.561 7.096
L10L11	8.38	-268.7	-47.1	1.21	16.3	1.22	IR OR	-484.8 -604.6	535.0 728.1	IR OR	4.584 7.125
L11L12	8.33	-286.5	-34.2	1.21	0.0	1.21	IR OR	-484.9 -604.6	535.0 728.1	IR OR	5.798 9.299
L12L13	8.33	-286.5	-34.2	1.21	0.0	1.21	IR	-484.9	535.0	IR	5.798

							OR	-604.6	728.1	OR	9.299
L15L16	8.00	209.5	0.0	1.21	31.0	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	5.574 10.029
L16L17	8.00	209.5	0.0	1.21	31.0	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	5.574 10.029
L17L18	8.01	279.4	0.0	1.21	39.1	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	2.628 6.158
L18L19	8.01	279.4	0.0	1.21	39.1	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	2.628 6.158
L19L20	8.01	209.9	0.0	1.21	31.0	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	5.551 9.997
L20L21	8.00	209.9	0.0	1.21	31.0	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	5.551 9.997
L23L24	8.33	-285.5	-34.1	1.21	0.0	1.21	IR OR	-484.9 -604.6	535.0 728.1	IR OR	5.845 9.357
L24L25	8.33	-285.5	-34.1	1.21	0.0	1.21	IR OR	-484.9 -604.6	535.0 728.1	IR OR	5.845 9.357
L25L26	8.38	-296.5	-47.2	1.21	16.3	1.21	IR OR	-484.8 -604.6	535.0 728.1	IR OR	3.988 6.524
L26L27	9.18	-296.5	-47.2	1.21	16.3	1.21	IR OR	-484.0 -603.6	535.0 728.1	IR OR	3.971 6.504
L27L28	9.25	-11.5	-38.6	1.21	42.2	1.21	IR OR	-484.0 -603.5	535.0 728.1	IR OR	12.246 15.345
L28L29	9.25	-11.5	-38.6	1.21	42.2	1.21	IR OR	-484.0 -603.5	535.0 728.1	IR OR	12.246 15.345
L29L30	9.25	129.9	-29.9	1.21	54.3	1.21	IR OR	-537.4 -670.2	593.3 807.5	IR OR	8.532 12.477
L30L31	9.25	129.9	-29.9	1.21	54.3	1.21	IR OR	-537.4 -670.2	593.3 807.5	IR OR	8.532 12.477
L31L32	9.25	126.4	-30.3	1.22	54.3	1.21	IR OR	-537.4 -670.2	593.3 807.5	IR OR	8.596 12.541
L32L33	9.25	126.4	-30.3	1.22	54.3	1.21	IR OR	-537.4 -670.2	593.3 807.5	IR OR	8.596 12.541
L33L34	9.25	-22.1	-39.0	1.22	42.2	1.21	IR OR	-484.0 -603.5	535.0 728.1	IR OR	11.829 14.891
L34L35	9.25	-22.1	-39.0	1.22	42.2	1.21	IR OR	-484.0 -603.5	535.0 728.1	IR OR	11.829 14.891
L35L36	9.19	-314.3	-47.8	1.22	16.3	1.21	IR OR	-484.0 -603.6	535.0 728.1	IR OR	3.550 6.052
L36L37	8.38	-314.3	-47.8	1.22	16.3	1.21	IR OR	-484.8 -604.6	535.0 728.1	IR OR	3.567 6.073
L37L38	8.33	-299.4	-34.5	1.22	0.0	1.23	IR OR	-484.9 -604.6	535.0 728.1	IR OR	5.375 8.847
L38L39	8.33	-299.4	-34.5	1.22	0.0	1.23	IR OR	-484.9 -604.6	535.0 728.1	IR OR	5.375 8.847
L41L42	8.63	240.2	0.0	1.22	34.2	1.23	IR OR	-404.7 -504.6	442.1 601.7	IR OR	5.904 10.572
L42L43	8.63	240.2	0.0	1.22	34.2	1.23	IR OR	-404.7 -504.6	442.1 601.7	IR OR	5.904 10.572

L43L44	8.63	320.0	0.0	1.22	43.3	1.23	IR OR	-404.7 -504.6	442.1 601.7	IR OR	2.821 6.509
L44L45	8.63	320.0	0.0	1.22	43.3	1.23	IR OR	-404.7 -504.6	442.1 601.7	IR OR	2.821 6.509
L45L46	8.62	239.9	0.0	1.22	34.1	1.23	IR OR	-404.7 -504.6	442.1 601.7	IR OR	5.921 10.596
L46L47	8.62	239.9	0.0	1.22	34.1	1.23	IR OR	-404.7 -504.6	442.1 601.7	IR OR	5.921 10.596
U 0L 1	15.16	238.4	-7.6	1.23	29.0	1.22	IR OR	-452.8 -564.7	509.4 693.4	IR OR	9.344 15.686
L 1U 2	15.05	-157.1	-26.0	1.23	8.0	1.30	IR OR	-362.4 -451.9	416.0 566.2	IR OR	7.907 11.356
U 2L 3	15.05	118.8	-8.0	1.30	23.1	1.24	IR OR	-254.6 -317.5	320.8 436.6	IR OR	8.751 13.771
L 3U 4	15.05	-36.7	-20.2	1.25	8.3	1.30	IR OR	-207.9 -259.3	277.9 378.3	IR OR	8.494 11.042
U 4L 5	15.05	-1.6	-11.5	1.30	17.2	1.27	IR OR	-132.1 -164.8	201.1 273.7	IR OR	11.326 14.160
L 5U 6	15.05	83.8	-14.2	1.28	14.6	1.28	IR OR	-132.1 -164.8	201.1 273.7	IR OR	8.037 13.013
U 6L 7	15.05	-122.1	-17.6	1.27	11.1	1.30	IR OR	-207.9 -259.3	277.9 378.3	IR OR	4.874 7.792
L 7U 8	15.05	204.3	-7.8	1.30	20.6	1.25	IR OR	-336.2 -419.2	371.5 505.7	IR OR	8.137 14.663
U 8L 9	15.05	-242.6	-23.5	1.24	4.5	1.30	IR OR	-382.5 -477.0	430.0 585.3	IR OR	5.962 9.989
L 9U10	15.05	324.7	-1.3	1.30	26.5	1.23	IR OR	-453.1 -565.0	509.4 693.4	IR OR	6.979 13.931
U10L11	14.37	-346.6	-28.2	1.23	0.0	1.00	IR OR	-601.2 -749.7	681.5 927.6	IR OR	9.019 14.281
L11U12	14.34	-317.3	-32.2	1.23	0.0	1.00	IR OR	-601.3 -749.8	681.5 927.6	IR OR	8.807 13.414
U12L13	14.34	280.8	0.0	1.00	31.5	1.24	IR OR	-454.7 -567.1	509.4 693.4	IR OR	7.247 13.078
L13U14	14.34	-212.4	-30.4	1.25	0.0	1.00	IR OR	-419.7 -523.4	463.7 631.1	IR OR	6.814 10.221
U14L15	12.09	192.8	0.0	1.00	28.8	1.26	IR OR	-343.6 -428.5	371.5 505.7	IR OR	6.217 10.883
L15U16	12.04	-121.6	-24.6	1.28	3.1	1.30	IR OR	-263.4 -328.5	320.8 436.6	IR OR	5.760 8.404
U16L17	12.04	88.6	-6.5	1.30	20.3	1.29	IR OR	-178.2 -222.2	227.7 309.9	IR OR	6.849 10.897
L17U18	12.05	-16.7	-15.8	1.30	11.2	1.30	IR OR	-113.7 -141.8	153.4 208.7	IR OR	6.136 7.913
U18L19	12.05	-16.4	-15.8	1.30	11.2	1.30	IR OR	-113.7 -141.8	153.4 208.7	IR OR	6.171 7.952
L19U20	12.05	88.3	-6.6	1.30	20.3	1.29	IR OR	-178.2 -222.2	227.7 309.9	IR OR	6.871 10.923
U20L21	12.04	-121.3	-24.6	1.28	3.1	1.30	IR OR	-263.4 -328.5	320.8 436.6	IR OR	5.780 8.427

L21U22	12.09	193.7	0.0	1.00	28.9	1.26	IR OR	-343.6 -428.5	371.5 505.7	IR OR	6.154 10.796
U22L23	14.34	-211.0	-30.4	1.25	0.0	1.00	IR OR	-454.7 -567.1	509.4 693.4	IR OR	8.025 11.725
L23U24	14.34	280.5	0.0	1.00	31.5	1.24	IR OR	-454.7 -567.1	509.4 693.4	IR OR	7.258 13.091
U24L25	14.34	-317.0	-32.2	1.23	0.0	1.00	IR OR	-601.3 -749.8	681.5 927.6	IR OR	8.816 13.423
L25U26	14.37	-296.9	-28.4	1.23	6.9	1.23	IR OR	-526.0 -655.9	589.3 802.1	IR OR	8.055 12.624
U26L27	14.85	269.1	-7.5	1.30	26.6	1.23	IR OR	-453.5 -565.6	509.4 693.4	IR OR	9.042 15.963
L27U28	14.89	-191.1	-23.8	1.23	7.5	1.30	IR OR	-382.8 -477.4	430.0 585.3	IR OR	8.049 12.019
U28L29	14.89	153.2	-7.5	1.30	21.2	1.24	IR OR	-284.7 -355.1	360.0 490.0	IR OR	9.773 15.917
L29U30	14.89	-74.4	-18.5	1.26	10.1	1.30	IR OR	-208.6 -260.1	277.9 378.3	IR OR	7.259 10.047
U30L31	14.89	36.5	-13.0	1.29	15.8	1.27	IR OR	-117.8 -146.9	174.6 237.6	IR OR	8.758 12.757
L31U32	14.89	42.2	-13.0	1.29	15.8	1.27	IR OR	-117.8 -146.9	174.6 237.6	IR OR	8.398 12.398
U32L33	14.89	-80.1	-18.5	1.26	10.1	1.30	IR OR	-208.6 -260.1	277.9 378.3	IR OR	6.952 9.741
L33U34	14.89	158.9	-7.5	1.30	21.2	1.24	IR OR	-284.7 -355.1	360.0 490.0	IR OR	9.506 15.650
U34L35	14.89	-196.8	-23.8	1.23	7.5	1.30	IR OR	-382.8 -477.4	430.0 585.3	IR OR	7.816 11.790
L35U36	14.85	274.8	-7.5	1.30	26.5	1.23	IR OR	-453.5 -565.6	509.4 693.4	IR OR	8.859 15.806
U36L37	14.37	-302.4	-28.3	1.23	6.9	1.23	IR OR	-526.0 -655.9	589.3 802.1	IR OR	7.895 12.484
L37U38	14.34	-329.3	-32.1	1.23	0.0	1.00	IR OR	-601.3 -749.8	681.5 927.6	IR OR	8.471 13.097
U38L39	14.34	292.8	0.0	1.00	31.5	1.24	IR OR	-454.7 -567.1	509.4 693.4	IR OR	6.885 12.732
L39U40	14.34	-222.6	-30.3	1.25	0.0	1.00	IR OR	-454.7 -567.1	509.4 693.4	IR OR	7.662 11.370
U40L41	12.47	213.1	0.0	1.00	29.7	1.25	IR OR	-342.7 -427.4	371.5 505.7	IR OR	5.331 9.845
L41U42	12.47	-133.9	-25.5	1.27	3.2	1.30	IR OR	-262.3 -327.1	320.8 436.6	IR OR	5.025 7.562
U42L43	12.47	96.9	-7.0	1.30	21.1	1.29	IR OR	-176.9 -220.6	227.7 309.9	IR OR	6.194 10.089
L43U44	12.47	-18.4	-16.5	1.30	11.8	1.30	IR OR	-112.1 -139.8	153.4 208.7	IR OR	5.668 7.342
U44L45	12.47	-18.6	-16.6	1.30	11.7	1.30	IR OR	-112.1 -139.8	153.4 208.7	IR OR	5.641 7.313
L45U46	12.46	97.2	-7.0	1.30	21.1	1.29	IR	-176.9	227.7	IR	6.179

							OR	-220.6	309.9	OR	10.071
U46L47	12.46	-134.1	-25.6	1.27	3.2	1.30	IR	-262.3	320.8	IR	5.017
							OR	-327.1	436.6	OR	7.553
L47U48	12.46	212.7	0.0	1.00	29.9	1.26	IR	-342.7	371.5	IR	5.312
							OR	-427.4	505.7	OR	9.799
U 1L 1	11.67	-61.7	-18.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	4.760
							OR	-185.5	273.7	OR	6.769
U 3L 3	11.67	-63.7	-18.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	4.650
							OR	-185.5	273.7	OR	6.660
U 5L 5	11.67	-63.7	-18.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	4.650
							OR	-185.5	273.7	OR	6.660
U 7L 7	11.67	-63.7	-18.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	4.650
							OR	-185.5	273.7	OR	6.660
U 9L 9	11.67	-63.7	-18.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	4.650
							OR	-185.5	273.7	OR	6.660
U10L10	11.67	-0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	99.999
							OR	-185.5	273.7	OR	99.999
U11L11	11.67	-56.7	-18.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	5.033
							OR	-185.5	273.7	OR	7.043
U12L12	11.67	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	99.999
							OR	-185.5	273.7	OR	99.999
U13L13	11.67	-56.7	-18.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	5.033
							OR	-185.5	273.7	OR	7.043
U15L15	9.00	-53.7	-18.3	1.30	0.0	1.00	IR	-141.8	174.6	IR	4.818
							OR	-176.9	237.6	OR	6.734
U17L17	9.00	-53.7	-18.3	1.30	0.0	1.00	IR	-141.8	174.6	IR	4.818
							OR	-176.9	237.6	OR	6.734
U19L19	9.00	-53.7	-18.3	1.30	0.0	1.00	IR	-141.8	174.6	IR	4.818
							OR	-176.9	237.6	OR	6.734
U21L21	9.00	-53.7	-18.3	1.30	0.0	1.00	IR	-141.8	174.6	IR	4.818
							OR	-176.9	237.6	OR	6.734
U23L23	11.67	-56.7	-18.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	5.033
							OR	-185.5	273.7	OR	7.043
U24L24	11.67	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	99.999
							OR	-185.5	273.7	OR	99.999
U25L25	11.67	-56.7	-18.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	5.033
							OR	-185.5	273.7	OR	7.043
U26L26	11.67	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	99.999
							OR	-185.5	273.7	OR	99.999
U27L27	11.67	-61.7	-18.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	4.760
							OR	-185.5	273.7	OR	6.769
U29L29	11.67	-61.7	-18.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	4.760
							OR	-185.5	273.7	OR	6.769
U31L31	11.67	-61.7	-18.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	4.760
							OR	-185.5	273.7	OR	6.769
U33L33	11.67	-61.7	-18.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	4.760
							OR	-185.5	273.7	OR	6.769
U35L35	11.67	-61.7	-18.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	4.760
							OR	-185.5	273.7	OR	6.769

U37L37	11.67	-56.7	-18.3	1.30	0.0	1.00	IR OR	-148.8 -185.5	201.1 273.7	IR OR	5.033 7.043
U39L39	11.67	-56.7	-18.3	1.30	0.0	1.00	IR OR	-148.8 -185.5	201.1 273.7	IR OR	5.033 7.043
U41L41	9.00	-56.7	-18.3	1.30	0.0	1.00	IR OR	-141.8 -176.9	174.6 237.6	IR OR	4.654 6.570
U43L43	9.00	-56.7	-18.3	1.30	0.0	1.00	IR OR	-141.8 -176.9	174.6 237.6	IR OR	4.654 6.570
U45L45	9.00	-56.7	-18.3	1.30	0.0	1.00	IR OR	-141.8 -176.9	174.6 237.6	IR OR	4.654 6.570
U47L47	9.00	-56.7	-18.3	1.30	0.0	1.00	IR OR	-141.8 -176.9	174.6 237.6	IR OR	4.654 6.570

MEMBER L18L19 IS CRITICAL. SEE DETAILS BELOW.

DL FORCE	LL COMP	+ IMPACT I.F.	FORCE TENS	I.F.	ALLOWABLE COMP	FORCE TENS			RATING FACTOR	TONS
279.4	0.0	1.21	39.1	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	2.628 6.158	27.14 63.61

SUPPORT REACTIONS

SUPPORT	D.L.	LL+I	I.F.
U 0	213.3	24.8	1.22
L11	596.5	30.9	1.21
L25	555.9	30.6	1.21
L37	570.4	30.8	1.22
U48	180.3	24.4	1.23

PANEL POINT DEFLECTIONS

PANEL POINT	DEAD LOAD VERT	+ TEMP HOR	LIVE LOAD + IMPACT VERT+ I.F.	VERT- I.F.	HOR+ HOR-
U 0	0.00	0.61	0.00 1.22	0.00 1.21	0.05 -0.04
U 1	0.34	0.59	0.03 1.22	-0.02 1.21	0.05 -0.04
U 2	0.57	0.56	0.07 1.22	-0.03 1.21	0.05 -0.04
U 3	0.78	0.54	0.10 1.22	-0.04 1.21	0.05 -0.04
U 4	0.86	0.51	0.12 1.22	-0.05 1.21	0.05 -0.04
U 5	0.91	0.48	0.13 1.22	-0.06 1.21	0.05 -0.04
U 6	0.81	0.46	0.12 1.22	-0.07 1.21	0.05 -0.04
U 7	0.70	0.44	0.11 1.22	-0.07 1.21	0.05 -0.04
U 8	0.50	0.42	0.09 1.22	-0.06 1.21	0.05 -0.04
U 9	0.34	0.43	0.06 1.22	-0.05 1.21	0.05 -0.04
U10	0.11	0.44	0.03 1.22	-0.03 1.21	0.05 -0.04
U11	0.02	0.47	0.01 1.21	0.00 1.21	0.05 -0.04
U12	0.05	0.50	0.04 1.21	-0.02 1.21	0.05 -0.04
U13	0.19	0.51	0.08 1.21	-0.05 1.21	0.05 -0.04
U14	0.28	0.53	0.13 1.21	-0.07 1.21	0.05 -0.04
U15	0.56	0.52	0.13 1.21	-0.06 1.21	0.05 -0.04
U16	0.75	0.50	0.14 1.21	-0.05 1.21	0.05 -0.04
U17	0.94	0.47	0.15 1.21	-0.05 1.21	0.05 -0.04
U18	1.00	0.44	0.15 1.21	-0.04 1.21	0.05 -0.04
U19	1.00	0.41	0.15 1.21	-0.05 1.21	0.05 -0.04
U20	0.88	0.39	0.14 1.21	-0.06 1.21	0.05 -0.04
U21	0.74	0.37	0.13 1.21	-0.07 1.21	0.05 -0.04
U22	0.52	0.36	0.13 1.21	-0.07 1.21	0.05 -0.04
U23	0.35	0.37	0.08 1.21	-0.05 1.21	0.05 -0.04
U24	0.14	0.39	0.04 1.21	-0.02 1.21	0.05 -0.04
U25	0.02	0.42	0.01 1.21	0.00 1.21	0.05 -0.04
U26	0.03	0.45	0.03 1.21	-0.03 1.21	0.05 -0.04
U27	0.14	0.46	0.06 1.21	-0.05 1.21	0.05 -0.04
U28	0.21	0.48	0.09 1.21	-0.06 1.21	0.05 -0.04
U29	0.34	0.47	0.12 1.21	-0.07 1.21	0.05 -0.04
U30	0.38	0.47	0.13 1.21	-0.07 1.21	0.05 -0.04
U31	0.44	0.46	0.14 1.21	-0.07 1.21	0.05 -0.03

U32	0.38	0.44	0.13	1.21	-0.07	1.22	0.05	-0.03
U33	0.33	0.44	0.12	1.21	-0.07	1.22	0.05	-0.03
U34	0.20	0.43	0.09	1.21	-0.06	1.22	0.06	-0.03
U35	0.13	0.45	0.06	1.21	-0.05	1.22	0.06	-0.03
U36	0.02	0.46	0.03	1.21	-0.03	1.22	0.06	-0.02
U37	0.02	0.49	0.01	1.23	0.00	1.22	0.06	-0.02
U38	0.15	0.53	0.04	1.23	-0.03	1.22	0.06	-0.02
U39	0.38	0.54	0.09	1.23	-0.05	1.22	0.06	-0.01
U40	0.56	0.56	0.14	1.23	-0.08	1.22	0.07	-0.01
U41	0.77	0.55	0.13	1.23	-0.07	1.22	0.06	-0.01
U42	0.90	0.53	0.13	1.23	-0.06	1.22	0.06	-0.01
U43	1.01	0.50	0.14	1.23	-0.05	1.22	0.06	-0.01
U44	0.97	0.47	0.13	1.23	-0.04	1.22	0.06	-0.01
U45	0.87	0.44	0.11	1.23	-0.03	1.22	0.05	-0.01
U46	0.62	0.41	0.07	1.23	-0.02	1.22	0.05	-0.01
U47	0.35	0.39	0.04	1.23	-0.01	1.22	0.05	-0.01
U48	0.00	0.38	0.00	1.23	0.00	1.22	0.05	-0.01

```

+++++
+                               +
+   S T R I N G E R   A N A L Y S I S   +
+                               +
+++++

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THE STRINGER LOCATED AT 8.00 FT. FROM THE CENTER LINE OF GIRDER OR TRUSS IS CRITICAL.

LIVE LOAD DISTRIBUTION FACTOR FOR MOMENT: 0.727

LIVE LOAD DISTRIBUTION FACTOR FOR SHEAR: 0.625

DEAD LOADS ACTING ON THE STRINGER

STRINGER WEIGHT	SLAB WEIGHT	INPUT DL1	TOTAL DL1	TOTAL DL2
0.036	0.792	0.976	1.804	0.000

STRINGER SECTION PROPERTIES

	DEPTH	GROSS AREA	MOMENT OF INERTIA	C BOTTOM	SECTION TOP	MODULUS BOTTOM
NON-COMPOSITE	15.85	10.59	446.30	7.92	56.32	56.32

DEFLECTIONS

SPAN 1 - LIVE LOAD IMPACT FACTOR FOR DEFLECTION: 1.30

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	DEAD LOAD			LIVE LOAD + IMPACT	
X	DL1	DL2	SP-1		
0.00	0.000	0.000	0.000		
0.95	0.008	0.000	0.009		
1.90	0.015	0.000	0.017		
2.85	0.021	0.000	0.023		
3.80	0.024	0.000	0.028		
4.75	0.025	0.000	0.029		
5.70	0.024	0.000	0.028		
6.65	0.021	0.000	0.023		
7.60	0.015	0.000	0.017		
8.55	0.008	0.000	0.009		
9.50	0.000	0.000	0.000		

SPAN 2 - LIVE LOAD IMPACT FACTOR FOR DEFLECTION: 1.30

=====

	DEAD LOAD			LIVE LOAD + IMPACT	
X	DL1	DL2	SP-1		
0.00	0.000	0.000	0.000		
0.95	0.008	0.000	0.009		
1.90	0.015	0.000	0.017		
2.85	0.021	0.000	0.023		
3.80	0.024	0.000	0.028		
4.75	0.025	0.000	0.029		
5.70	0.024	0.000	0.028		

6.65	0.021	0.000	0.023
7.60	0.015	0.000	0.017
8.55	0.008	0.000	0.009
9.50	0.000	0.000	0.000

SPAN 3 - LIVE LOAD IMPACT FACTOR FOR DEFLECTION: 1.30

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	DEAD LOAD			LIVE LOAD + IMPACT
X	DL1	DL2	SP-1	
0.00	0.000	0.000	0.000	
0.95	0.008	0.000	0.009	
1.90	0.015	0.000	0.017	
2.85	0.021	0.000	0.023	
3.80	0.024	0.000	0.028	
4.75	0.025	0.000	0.029	
5.70	0.024	0.000	0.028	
6.65	0.021	0.000	0.023	
7.60	0.015	0.000	0.017	
8.55	0.008	0.000	0.009	
9.50	0.000	0.000	0.000	

 * STRINGER - LIVE LOAD SP-1 *

MAXIMUM REACTIONS

SUPPORT	DL1	DL2	+(LL+I)	-(LL+I)	REACTIONS +I.F. -I.F.	MOMENTS +I.F. -I.F.
1	8.6	0.0	13.2	0.0	1.30	
2	17.1	0.0	11.4	0.0	1.30	
3	17.1	0.0	13.2	0.0	1.30	
4	8.6	0.0	11.4	0.0	1.30	

NOTE: ALL SUPPORT REACTIONS AND END SHEARS IN EACH SPAN DUE TO A LIVE LOAD ARE CALCULATED BASED ON AASHTO ARTICLE 3.23.1 AS INTERPRETED PER SOL 431-93-05 IN PENNDOT DM4 REV 12-94, PAGE C B.3-8.

UNFACTORED MOMENTS AND SHEARS

SPAN 1 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

=====

X	DL1 MOMENT	DL2 MOMENT	+(LL+I) MOMENT	-(LL+I) MOMENT	I M	DL1 SHEAR	DL2 SHEAR	+(LL+I) SHEAR	-(LL+I) SHEAR	I.F.
0.00	0.0	0.0	0.0	0.0		8.6	0.0	13.2	0.0	1.30
	SIMULT	SHEAR	0.0	0.0		SIMULT	MOM	0.0	0.0	
0.95	7.3	0.0	11.4	0.0		6.9	0.0	12.0	-1.3	1.30
	SIMULT	SHEAR	12.0	0.0		SIMULT	MOM	11.4	11.4	
1.90	13.0	0.0	20.2	0.0		5.1	0.0	10.6	-2.7	1.30
	SIMULT	SHEAR	10.6	0.0		SIMULT	MOM	20.2	20.2	
2.85	17.1	0.0	26.5	0.0		3.4	0.0	9.3	-4.0	1.30
	SIMULT	SHEAR	9.3	0.0		SIMULT	MOM	26.5	26.5	
3.80	19.5	0.0	30.3	0.0		1.7	0.0	8.0	-5.3	1.30
	SIMULT	SHEAR	8.0	0.0		SIMULT	MOM	30.3	30.3	
4.75	20.4	0.0	31.6	0.0		-0.0	0.0	6.6	-6.6	1.30
	SIMULT	SHEAR	-6.6	0.0		SIMULT	MOM	31.6	31.6	
5.70	19.5	0.0	30.3	0.0		-1.7	0.0	5.3	-8.0	1.30
	SIMULT	SHEAR	-8.0	0.0		SIMULT	MOM	30.3	30.3	
6.65	17.1	0.0	26.5	0.0		-3.4	0.0	4.0	-9.3	1.30
	SIMULT	SHEAR	-9.3	0.0		SIMULT	MOM	26.5	26.5	
7.60	13.0	0.0	20.2	0.0		-5.1	0.0	2.7	-10.6	1.30
	SIMULT	SHEAR	-10.6	0.0		SIMULT	MOM	20.2	20.2	
8.55	7.3	0.0	11.4	0.0		-6.9	0.0	1.3	-12.0	1.30
	SIMULT	SHEAR	-12.0	0.0		SIMULT	MOM	11.4	11.4	
9.50	0.0	0.0	0.0	0.0		-8.6	0.0	0.0	-13.2	1.30
	SIMULT	SHEAR	0.0	0.0		SIMULT	MOM	0.0	0.0	

SPAN 2 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

=====

X	DL1 MOMENT	DL2 MOMENT	+(LL+I) MOMENT	-(LL+I) MOMENT	I M	DL1 SHEAR	DL2 SHEAR	+(LL+I) SHEAR	-(LL+I) SHEAR	I.F.
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0.00	0.0	0.0	0.0	0.0	8.6	0.0	13.2	0.0	1.30
	SIMULT	SHEAR	0.0	0.0	SIMULT	MOM	0.0	0.0	
0.95	7.3	0.0	11.4	0.0	6.9	0.0	12.0	-1.3	1.30
	SIMULT	SHEAR	12.0	0.0	SIMULT	MOM	11.4	11.4	
1.90	13.0	0.0	20.2	0.0	5.1	0.0	10.6	-2.7	1.30
	SIMULT	SHEAR	10.6	0.0	SIMULT	MOM	20.2	20.2	
2.85	17.1	0.0	26.5	0.0	3.4	0.0	9.3	-4.0	1.30
	SIMULT	SHEAR	9.3	0.0	SIMULT	MOM	26.5	26.5	
3.80	19.5	0.0	30.3	0.0	1.7	0.0	8.0	-5.3	1.30
	SIMULT	SHEAR	8.0	0.0	SIMULT	MOM	30.3	30.3	
4.75	20.4	0.0	31.6	0.0	0.0	0.0	6.6	-6.6	1.30
	SIMULT	SHEAR	-6.6	0.0	SIMULT	MOM	31.6	31.6	
5.70	19.5	0.0	30.3	0.0	-1.7	0.0	5.3	-8.0	1.30
	SIMULT	SHEAR	-8.0	0.0	SIMULT	MOM	30.3	30.3	
6.65	17.1	0.0	26.5	0.0	-3.4	0.0	4.0	-9.3	1.30
	SIMULT	SHEAR	-9.3	0.0	SIMULT	MOM	26.5	26.5	
7.60	13.0	0.0	20.2	0.0	-5.1	0.0	2.7	-10.6	1.30
	SIMULT	SHEAR	-10.6	0.0	SIMULT	MOM	20.2	20.2	
8.55	7.3	0.0	11.4	0.0	-6.9	0.0	1.3	-12.0	1.30
	SIMULT	SHEAR	-12.0	0.0	SIMULT	MOM	11.4	11.4	
9.50	0.0	0.0	0.0	0.0	-8.6	0.0	0.0	-11.4	1.30
	SIMULT	SHEAR	0.0	0.0	SIMULT	MOM	0.0	0.0	

SPAN 3 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

=====

X	DL1 MOMENT	DL2 MOMENT	+(LL+I) MOMENT	-(LL+I) MOMENT	I M	DL1 SHEAR	DL2 SHEAR	+(LL+I) SHEAR	-(LL+I) SHEAR	I.F.
0.00	0.0	0.0	0.0	0.0		8.6	0.0	13.2	0.0	1.30
	SIMULT	SHEAR	0.0	0.0		SIMULT	MOM	0.0	0.0	
0.95	7.3	0.0	11.4	0.0		6.9	0.0	12.0	-1.3	1.30
	SIMULT	SHEAR	12.0	0.0		SIMULT	MOM	11.4	11.4	
1.90	13.0	0.0	20.2	0.0		5.1	0.0	10.6	-2.7	1.30
	SIMULT	SHEAR	10.6	0.0		SIMULT	MOM	20.2	20.2	
2.85	17.1	0.0	26.5	0.0		3.4	0.0	9.3	-4.0	1.30
	SIMULT	SHEAR	9.3	0.0		SIMULT	MOM	26.5	26.5	
3.80	19.5	0.0	30.3	0.0		1.7	0.0	8.0	-5.3	1.30
	SIMULT	SHEAR	8.0	0.0		SIMULT	MOM	30.3	30.3	
4.75	20.4	0.0	31.6	0.0		0.0	0.0	6.6	-6.6	1.30
	SIMULT	SHEAR	-6.6	0.0		SIMULT	MOM	31.6	31.6	
5.70	19.5	0.0	30.3	0.0		-1.7	0.0	5.3	-8.0	1.30
	SIMULT	SHEAR	-8.0	0.0		SIMULT	MOM	30.3	30.3	
6.65	17.1	0.0	26.5	0.0		-3.4	0.0	4.0	-9.3	1.30
	SIMULT	SHEAR	-9.3	0.0		SIMULT	MOM	26.5	26.5	
7.60	13.0	0.0	20.2	0.0		-5.1	0.0	2.7	-10.6	1.30
	SIMULT	SHEAR	-10.6	0.0		SIMULT	MOM	20.2	20.2	
8.55	7.3	0.0	11.4	0.0		-6.9	0.0	1.3	-12.0	1.30
	SIMULT	SHEAR	-12.0	0.0		SIMULT	MOM	11.4	11.4	
9.50	0.0	0.0	0.0	0.0		-8.6	0.0	0.0	-11.4	1.30
	SIMULT	SHEAR	0.0	0.0		SIMULT	MOM	0.0	0.0	

FLEXURAL STRESSES - BEAM

SPAN 1

=====

X	TOP FIBER STEEL STRESS				BOTTOM FIBER STEEL STRESS			
	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)
0.00	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.95	-1.561	0.000	-2.423	0.000	1.561	0.000	2.423	0.000
1.90	-2.775	0.000	-4.307	0.000	2.775	0.000	4.307	0.000
2.85	-3.643	0.000	-5.653	0.000	3.643	0.000	5.653	0.000
3.80	-4.163	0.000	-6.460	0.000	4.163	0.000	6.460	0.000
4.75	-4.336	0.000	-6.730	0.000	4.336	0.000	6.730	0.000
5.70	-4.163	0.000	-6.460	0.000	4.163	0.000	6.460	0.000
6.65	-3.643	0.000	-5.653	0.000	3.643	0.000	5.653	0.000
7.60	-2.775	0.000	-4.307	0.000	2.775	0.000	4.307	0.000
8.55	-1.561	0.000	-2.423	0.000	1.561	0.000	2.423	0.000
9.50	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

SPAN 2

=====

X	TOP FIBER STEEL STRESS				BOTTOM FIBER STEEL STRESS			
	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)
0.00	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

0.95	-1.561	0.000	-2.423	0.000	1.561	0.000	2.423	0.000
1.90	-2.775	0.000	-4.307	0.000	2.775	0.000	4.307	0.000
2.85	-3.643	0.000	-5.653	0.000	3.643	0.000	5.653	0.000
3.80	-4.163	0.000	-6.460	0.000	4.163	0.000	6.460	0.000
4.75	-4.336	0.000	-6.730	0.000	4.336	0.000	6.730	0.000
5.70	-4.163	0.000	-6.460	0.000	4.163	0.000	6.460	0.000
6.65	-3.643	0.000	-5.653	0.000	3.643	0.000	5.653	0.000
7.60	-2.775	0.000	-4.307	0.000	2.775	0.000	4.307	0.000
8.55	-1.561	0.000	-2.423	0.000	1.561	0.000	2.423	0.000
9.50	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

SPAN 3

=====

TOP FIBER STEEL STRESS					BOTTOM FIBER STEEL STRESS			
X	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)
0.00	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.95	-1.561	0.000	-2.423	0.000	1.561	0.000	2.423	0.000
1.90	-2.775	0.000	-4.307	0.000	2.775	0.000	4.307	0.000
2.85	-3.643	0.000	-5.653	0.000	3.643	0.000	5.653	0.000
3.80	-4.163	0.000	-6.460	0.000	4.163	0.000	6.460	0.000
4.75	-4.336	0.000	-6.730	0.000	4.336	0.000	6.730	0.000
5.70	-4.163	0.000	-6.460	0.000	4.163	0.000	6.460	0.000
6.65	-3.643	0.000	-5.653	0.000	3.643	0.000	5.653	0.000
7.60	-2.775	0.000	-4.307	0.000	2.775	0.000	4.307	0.000
8.55	-1.561	0.000	-2.423	0.000	1.561	0.000	2.423	0.000
9.50	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

SHEAR STRESSES AND ALLOWABLE STRESS RATINGS

SPAN 1

=====

SHEAR STRESSES					ALLOW COMPR	RATING FACTORS	
X	DL1	DL2	+(LL+I)	-(LL+I)	REDUCTION	IR	OR
0.00	1.911	0.000	2.953	0.000	1.000	3.08 V	4.42 V
0.95	1.529	0.000	2.669	-0.297	1.000	3.55 V	5.03 V
1.90	1.147	0.000	2.373	-0.593	1.000	3.57 T	5.10 T
2.85	0.765	0.000	2.076	-0.890	1.000	2.57 T	3.73 T
3.80	0.382	0.000	1.780	-1.186	1.000	2.17 T	3.19 T
4.75	-0.000	0.000	1.483	-1.483	1.000	2.05 T	3.03 T
5.70	-0.382	0.000	1.186	-1.780	1.000	2.17 T	3.19 T
6.65	-0.765	0.000	0.890	-2.076	1.000	2.57 T	3.73 T
7.60	-1.147	0.000	0.593	-2.373	1.000	3.57 T	5.10 T
8.55	-1.529	0.000	0.297	-2.669	1.000	3.55 V	5.03 V
9.50	-1.911	0.000	0.000	-2.953	1.000	3.08 V	4.42 V

SPAN 2

=====

SHEAR STRESSES					ALLOW COMPR	RATING FACTORS	
X	DL1	DL2	+(LL+I)	-(LL+I)	REDUCTION	IR	OR
0.00	1.911	0.000	2.953	0.000	1.000	3.08 V	4.42 V
0.95	1.529	0.000	2.669	-0.297	1.000	3.55 V	5.03 V
1.90	1.147	0.000	2.373	-0.593	1.000	3.57 T	5.10 T
2.85	0.765	0.000	2.076	-0.890	1.000	2.57 T	3.73 T
3.80	0.382	0.000	1.780	-1.186	1.000	2.17 T	3.19 T
4.75	0.000	0.000	1.483	-1.483	1.000	2.05 T	3.03 T
5.70	-0.382	0.000	1.186	-1.780	1.000	2.17 T	3.19 T
6.65	-0.765	0.000	0.890	-2.076	1.000	2.57 T	3.73 T
7.60	-1.147	0.000	0.593	-2.373	1.000	3.57 T	5.10 T
8.55	-1.529	0.000	0.297	-2.669	1.000	3.55 V	5.03 V
9.50	-1.911	0.000	0.000	-2.550	1.000	3.56 V	5.12 V

SPAN 3

=====

SHEAR STRESSES					ALLOW COMPR	RATING FACTORS	
X	DL1	DL2	+(LL+I)	-(LL+I)	REDUCTION	IR	OR
0.00	1.911	0.000	2.953	0.000	1.000	3.08 V	4.42 V
0.95	1.529	0.000	2.669	-0.297	1.000	3.55 V	5.03 V
1.90	1.147	0.000	2.373	-0.593	1.000	3.57 T	5.10 T
2.85	0.765	0.000	2.076	-0.890	1.000	2.57 T	3.73 T
3.80	0.382	0.000	1.780	-1.186	1.000	2.17 T	3.19 T
4.75	0.000	0.000	1.483	-1.483	1.000	2.05 T	3.03 T
5.70	-0.382	0.000	1.186	-1.780	1.000	2.17 T	3.19 T
6.65	-0.765	0.000	0.890	-2.076	1.000	2.57 T	3.73 T

7.60	-1.147	0.000	0.593	-2.373	1.000	3.57 T	5.10 T
8.55	-1.529	0.000	0.297	-2.669	1.000	3.55 V	5.03 V
9.50	-1.911	0.000	0.000	-2.550	1.000	3.56 V	5.12 V

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

SPAN 1

=====

X	NON-COMP OVERLOAD			NON-COMPACT		COMPACT	COMPACT	
	MOMENT	MOMENT	SHEAR	RATING FACTORS		MOMENT	RATING FACTORS	
	STRENGTH	STRENGTH	STRENGTH	IR	OR	STRENGTH	IR	OR
0.00	154.9 B	123.9	85.8	2.60 V	4.34 V	172.2	2.60 V	4.34 V
0.95	154.9 B	123.9	85.8	2.97 V	4.94 V	172.2	2.97 V	4.94 V
1.90	154.9 B	123.9	85.8	3.15 T	5.25 T	172.2	3.29 0	5.49 0
2.85	154.9 B	123.9	85.8	2.31 T	3.85 T	172.2	2.42 0	4.03 0
3.80	154.9 B	123.9	85.8	1.97 T	3.28 T	172.2	2.07 0	3.44 0
4.75	154.9 B	123.9	85.8	1.88 T	3.13 T	172.2	1.97 0	3.28 0
5.70	154.9 B	123.9	85.8	1.97 T	3.28 T	172.2	2.07 0	3.44 0
6.65	154.9 B	123.9	85.8	2.31 T	3.85 T	172.2	2.42 0	4.03 0
7.60	154.9 B	123.9	85.8	3.15 T	5.25 T	172.2	3.29 0	5.49 0
8.55	154.9 B	123.9	85.8	2.97 V	4.94 V	172.2	2.97 V	4.94 V
9.50	154.9 B	123.9	85.8	2.60 V	4.34 V	172.2	2.60 V	4.34 V

SPAN 2

=====

X	NON-COMP OVERLOAD			NON-COMPACT		COMPACT	COMPACT	
	MOMENT	MOMENT	SHEAR	RATING FACTORS		MOMENT	RATING FACTORS	
	STRENGTH	STRENGTH	STRENGTH	IR	OR	STRENGTH	IR	OR
0.00	154.9 B	123.9	85.8	2.60 V	4.34 V	172.2	2.60 V	4.34 V
0.95	154.9 B	123.9	85.8	2.97 V	4.94 V	172.2	2.97 V	4.94 V
1.90	154.9 B	123.9	85.8	3.15 T	5.25 T	172.2	3.29 0	5.49 0
2.85	154.9 B	123.9	85.8	2.31 T	3.85 T	172.2	2.42 0	4.03 0
3.80	154.9 B	123.9	85.8	1.97 T	3.28 T	172.2	2.07 0	3.44 0
4.75	154.9 B	123.9	85.8	1.88 T	3.13 T	172.2	1.97 0	3.28 0
5.70	154.9 B	123.9	85.8	1.97 T	3.28 T	172.2	2.07 0	3.44 0
6.65	154.9 B	123.9	85.8	2.31 T	3.85 T	172.2	2.42 0	4.03 0
7.60	154.9 B	123.9	85.8	3.15 T	5.25 T	172.2	3.29 0	5.49 0
8.55	154.9 B	123.9	85.8	2.97 V	4.94 V	172.2	2.97 V	4.94 V
9.50	154.9 B	123.9	85.8	3.01 V	5.02 V	172.2	3.01 V	5.02 V

SPAN 3

=====

X	NON-COMP OVERLOAD			NON-COMPACT		COMPACT	COMPACT	
	MOMENT	MOMENT	SHEAR	RATING FACTORS		MOMENT	RATING FACTORS	
	STRENGTH	STRENGTH	STRENGTH	IR	OR	STRENGTH	IR	OR
0.00	154.9 B	123.9	85.8	2.60 V	4.34 V	172.2	2.60 V	4.34 V
0.95	154.9 B	123.9	85.8	2.97 V	4.94 V	172.2	2.97 V	4.94 V
1.90	154.9 B	123.9	85.8	3.15 T	5.25 T	172.2	3.29 0	5.49 0
2.85	154.9 B	123.9	85.8	2.31 T	3.85 T	172.2	2.42 0	4.03 0
3.80	154.9 B	123.9	85.8	1.97 T	3.28 T	172.2	2.07 0	3.44 0
4.75	154.9 B	123.9	85.8	1.88 T	3.13 T	172.2	1.97 0	3.28 0
5.70	154.9 B	123.9	85.8	1.97 T	3.28 T	172.2	2.07 0	3.44 0
6.65	154.9 B	123.9	85.8	2.31 T	3.85 T	172.2	2.42 0	4.03 0
7.60	154.9 B	123.9	85.8	3.15 T	5.25 T	172.2	3.29 0	5.49 0
8.55	154.9 B	123.9	85.8	2.97 V	4.94 V	172.2	2.97 V	4.94 V
9.50	154.9 B	123.9	85.8	3.01 V	5.02 V	172.2	3.01 V	5.02 V

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+                                     +
+   F L O O R B E A M   A N A L Y S I S   +
+                                     +
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FLOORBEAM SPAN INFORMATION
ANALYZING HALF THE FLOORBEAM

LEFT CANTILEVER: 0.00 FT, FLOORBEAM MAIN SPAN: 16.00 FT
 RIGHT CANTILEVER: 0.00 FT
 TOTAL FLOORBEAM SPAN (INCLUDING ANY EXISTING CANTILEVER): 16.00 FT

NUMBER OF TRAFFIC LANES: 2, WHERE LANE WIDTH IS BASED ON DESIGN LANE:
 DECK WIDTH: 20.67 FT, ROADWAY WIDTH: 18.00 FT
 CL OF GIRDER OR TRUSS TO CURB: -1.00 FT

MINIMAL X1 AT ROADWAY: 0.50 FT, MAXIMAL X2 AT ROADWAY: 15.50 FT

NOTE: THE WHEEL LOAD POSITIONS (X1 AND X2) MUST BE BETWEEN MINIMAL_X1_AT_ROADWAY
 AND MAXIMAL_X2_AT_ROADWAY

FLOORBEAM LIVE LOAD MOMENT AND SHEAR FACTORS

		BASED ON DESIGN LANE WHEEL LOAD POSITIONS					
		LANE 1/4/7		LANE 2/5/8		LANE 3/6/9	
X	FACTOR	X1	X2	X1	X2	X1	X2
0.00	0.000 M	0.00	0.00				
	1.000 V	0.50	6.50	9.50	15.50		
1.60	1.140 M	1.60	7.60				
	0.713 V	1.60	7.60				
3.20	1.960 M	3.20	9.20				
	0.612 V	3.20	9.20				
4.80	2.650 M	0.50	6.50	9.50	15.50		
	0.512 V	4.80	10.80				
6.40	3.450 M	0.50	6.50	9.50	15.50		
	0.500 V	0.50	6.50	9.50	15.50		
8.00	3.500 M	0.50	6.50	9.50	15.50		
	0.313 V	8.00	14.00				

MOMENT/SHEAR FACTOR CODES: M - MOMENT, V - SHEAR

NOTE: THE EFFECT OF A SET OF WHEELS PLACED TRANSVERSELY ACROSS THE
 BRIDGE (ALONG THE FLOORBEAM SPAN) TO FIND THE MAXIMUM POS AND NEG
 MOMENTS OR SHEARS EXPRESSED IN TERMS OF THE NUMBER OF AXLES IS
 REFERRED TO AS MOMENT OR SHEAR FACTORS.

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/=====\  

< FLOORBEAM 1 >  

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FLOORBEAM SECTION PROPERTIES (UNIFORM SECTION)

	DEPTH	GROSS AREA	MOMENT OF INERTIA	C BOTTOM	SECTION TOP	MODULUS BOTTOM
NON-COMPOSITE	15.85	10.59	446.30	7.92	56.32	56.32

DEAD LOADS ACTING ON THE FLOORBEAM

UNIFORM LOAD		CONCENTRATED LOADS		
FLOORBEAM	INPUT	DIST	DL1	DL2
WEIGHT	DL1FB			
0.036	0.000	8.000	8.569	0.000

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*****  

* FLOORBEAM 1 - LIVE LOAD SP-1 *  

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LIVE LOAD REACTION FROM DECK (ONE TRUCK): 14.07

LIVE LOAD IMPACT FACTORS: POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
0.00	0.0	0.0	0.0	4.6	0.0	18.3	1.30
1.60	7.3	0.0	20.9	4.5	0.0	13.0	1.30
3.20	14.4	0.0	35.8	4.5	0.0	11.2	1.30
4.80	21.5	0.0	48.5	4.4	0.0	9.4	1.30
6.40	28.5	0.0	63.1	4.3	0.0	9.1	1.30
8.00	35.4	0.0	64.0	-4.3	0.0	5.7	1.30

NOTE: SYMBOL NEXT TO X
R: IT IS A RANGE POINT
B: IT IS BOTH A RANGE POINT AND ANALYSIS POINT

FLEXURAL STRESSES - BEAM

X	TOP FIBER STEEL STRESS				BOTTOM FIBER STEEL STRESS			
	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)
0.00	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
1.60	-1.549	0.000	-4.443	0.000	1.549	0.000	4.443	0.000
3.20	-3.079	0.000	-7.639	0.000	3.079	0.000	7.639	0.000
4.80	-4.588	0.000	-10.328	0.000	4.588	0.000	10.328	0.000
6.40	-6.079	0.000	-13.446	0.000	6.079	0.000	13.446	0.000
8.00	-7.549	0.000	-13.641	0.000	7.549	0.000	13.641	0.000

SHEAR STRESSES AND ALLOWABLE STRESS RATINGS

X	SHEAR STRESSES				ALLOW COMPR REDUCTION	RATING FACTORS	
	DL1	DL2	+(LL+I)	-(LL+I)		IR	OR
0.00	1.020	0.000	4.080	0.000	1.000	2.45 V	3.42 V
1.60	1.007	0.000	2.907	0.000	1.000	3.44 V	4.80 V
3.20	0.994	0.000	2.499	0.000	1.000	1.97 T	2.84 T
4.80	0.981	0.000	2.091	0.000	1.000	1.31 T	1.95 T
6.40	0.968	0.000	2.040	0.000	1.000	0.90 T	1.39 T
8.00	-0.956	0.000	1.275	0.000	1.000	0.78 T	1.26 T

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD			NON-COMPACT		COMPACT MOMENT STRENGTH	COMPACT	
	MOMENT STRENGTH	MOMENT STRENGTH	SHEAR STRENGTH	RATING IR	FACTORS OR		RATING IR	FACTORS OR
0.00	154.9 B	123.9	85.8	2.02 V	3.36 V	172.2	2.02 V	3.36 V
1.60	154.9 B	123.9	85.8	2.83 V	4.72 V	172.2	2.83 V	4.72 V
3.20	154.9 B	123.9	85.8	1.75 T	2.92 T	172.2	1.83 0	3.05 0
4.80	154.9 B	123.9	85.8	1.21 T	2.01 T	172.2	1.27 0	2.11 0
6.40	154.9 B	123.9	85.8	0.86 T	1.44 T	172.2	0.91 0	1.51 0
8.00	154.9 B	123.9	85.8	0.78 T	1.31 T	172.2	0.83 0	1.38 0

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+           R A T I N G   S U M M A R Y           +
+
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MEMBER: TRUSS

LOAD		ALLOWABLE STRESS RATING			LOAD FACTOR RATING		
		FACTOR	TONS	MEMBER	FACTOR	TONS	MEMBER
SP-1	IR (CRITICAL)	2.63	27.1	L18L19	*****	*****	*****
	OR (CRITICAL)	6.05	62.5	L35L36	*****	*****	*****

NOTE: LOAD FACTOR RATING WAS NOT PERFORMED BECAUSE ADDITIONAL TRUSS MEMBER PROPERTIES WERE NOT INPUT BY THE USER.

MEMBER: STRINGER AT 8.000

LOAD	ALLOWABLE STRESS RATING				LOAD FACTOR RATING			
	FACTOR	TONS	X	SPAN	FACTOR	TONS	X	SPAN

		ALLOWABLE STRESS RATING				LOAD FACTOR RATING			
LOAD		FACTOR	TONS	X	FLBM	FACTOR	TONS	X	FLBM
SP-1	IR (CRITICAL)	2.05 T	21.2	4.75	3	1.97 0	20.3	4.75	3
	OR (CRITICAL)	3.03 T	31.3	4.75	3	3.28 0	33.9	4.75	3
	IR (POS MOM)	2.05 T	21.2	4.75	3	1.97 0	20.3	4.75	3
	OR (POS MOM)	3.03 T	31.3	4.75	3	3.28 0	33.9	4.75	3

OR above 1.0

NOTE: THIS RATING SUMMARY OF STRINGERS DOES FOLLOW THE ALL-OR-NONE COMPACT REQUIREMENTS. THIS STRINGER DOES HAVE ALL SECTIONS QUALIFYING AS COMPACT SECTIONS. THEREFORE, THE RATING PRINTED IN THE SUMMARY IS THAT BASED ON COMPACT SECTIONS

OR above 1.0

MEMBER: FLOORBEAM

		ALLOWABLE STRESS RATING				LOAD FACTOR RATING			
LOAD		FACTOR	TONS	X	FLBM	FACTOR	TONS	X	FLBM
SP-1	IR (CRITICAL)	0.78 T	8.0	8.00	1	0.83 0	8.6	8.00	1
	OR (CRITICAL)	1.26 T	13.0	8.00	1	1.38 0	14.3	8.00	1
	IR (POS MOM)	0.78 T	8.0	8.00	1	0.83 0	8.6	8.00	1
	OR (POS MOM)	1.26 T	13.0	8.00	1	1.38 0	14.3	8.00	1

NOTE: THIS RATING SUMMARY OF FLOORBEAMS DOES FOLLOW THE ALL-OR-NONE COMPACT REQUIREMENTS. THIS FLOORBEAM DOES HAVE ALL SECTIONS QUALIFYING AS COMPACT SECTIONS. THEREFORE, THE RATING PRINTED IN THE SUMMARY IS THAT BASED ON COMPACT SECTIONS.

RATING FACTOR CODES: FIRST CHARACTER AFTER THE RATING FACTOR

T - TOP STEEL STRESS/STRENGTH GOVERNS
 B - BOTTOM STEEL STRESS/STRENGTH GOVERNS
 C - CONCRETE STRESS/STRENGTH GOVERNS
 R - REINFORCEMENT STRESS/STRENGTH GOVERNS
 V - SHEAR STRESS/STRENGTH GOVERNS
 blank - COMPACT MOMENT STRENGTH GOVERNS
 O - OVERLOAD PROVISIONS GOVERN
 I - MOMENT-SHEAR INTERACTION GOVERNS

RATING FACTOR CODES FOR FLANGE/WEB BUCKLING: SECOND CHARACTER AFTER THE RATIN <<<<

G FACTOR

blank - SECTION DOES MEET FLANGE/WEB BUCKLING CRITERIA AND WILL NOT BUCKLE.
 F - SECTION DOES NOT MEET FLANGE PROJECTION/THICKNESS RATIO CRITERIA AND WILL BUCKLE IN FLANGE.
 W - SECTION DOES NOT MEET WEB DEPTH/THICKNESS RATIO CRITERIA AND WILL BUCKLE IN WEB.
 Z - SECTION DOES NOT MEET BOTH FLANGE PROJECTION/THICKNESS RATIO CRITERIA AND WEB DEPTH/THICKNESS RATIO CRITERIA AND WILL BUCKLE IN FLANGE OR WEB.

NON-COMPACT MOMENT STRENGTH CODES:

B - SECTION IS BRACED
 U - SECTION IS UNBRACED

NOTE: ALL RATINGS ARE BASED ON THE NUMBER OF DESIGN LANES OR THE ACTUAL TRAFFIC LANES AS DEFINED BY "D" OR "L" ENTERED FOR LANES IN THE PROJECT IDENTIFICATION.

BAR7 v7.15.0.0 PROGRAM WAS EXECUTED COMPLETELY AND SUCCESSFULLY.

Appendix

- Truss Section Data
- Truck Cut Sheet
- Previous Analysis
- Drawings



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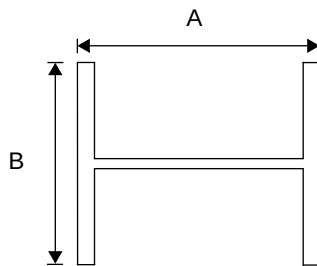
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Larson Design Group*

JOB Blacks Run Bridge
SHEET NO. 1 OF 6
CALCULATED BY WBS DATE _____
CHECKED BY _____ DATE _____
SCALE NTS

Field Measurements

(Non-Deteriorated)



C = Flange Thk.

D = Web Thk.

Span 1

L1-L3

A = 14.25"
B = 14.5"
C = .82"
D = .50"

L3-L5

A = 14.94"
B = 15.5"
C = 1.20"
D = .83"

U6-U7

A = 15.25"
B = 15.5"
C = 1.36"
D = .83"

L3-U2

A = 14.125"
B = 10.125"
C = .68"
D = .412"

L3-U3

A = 14.25"
B = 6.75"
C = .45"
D = .320"

Span 2

L17-L15

Length = 16'
A = 14.375"
B = 14.50"
C = .909"
D = .53"

U16-U17

Length = 8'
A = 14.375"
B = 14.50"
C = .917"
D = .56"

L17-U17

Length = 7'-8"
A = 14.1875"
B = 6.875"
C = .512"
D = .338"

L17-U16

Length = 9'-8"
A = 14.4375"
B = 8.0"
C = .582"
D = .363"

(Bottom and top chords appear same for suspended span)



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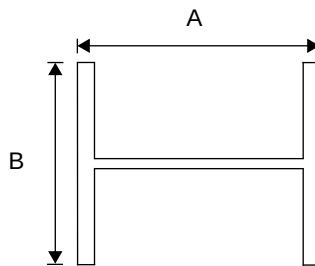
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JOB Blacks Run Bridge
SHEET NO. 2 OF 6
CALCULATED BY WBS DATE _____
CHECKED BY _____ DATE _____
SCALE _____ NTS

Field Measurements

(Non-Deteriorated to spot check truss)



C = Flange Thk.

D = Web Thk.

Span 3

L25-U25

Length = 10'-10"

A = 14.25"

B = 6.875"

C = .50"

D = .360"

L25-U26

Length = 12'-9 1/4"

A = 14.625"

B = 14.750"

C = 1.0"

D = .590"

L24-U25

Length = 11'-9 3/4"

A = 14.750"

B = 14.750"

C = 1.0"

D = .696"

L26-U26

Length = 10'-3 1/4"

A = 14.25"

B = 6.875"

C = .50"

D = .350"

L27-U27

Length = 10'-3 1/4"

A = 14.25"

B = 6.875"

C = .50"

D = .341"

L27-U26

Length = 12'-1 1/4"

A = 14.25"

B = 14.50"

C = .875"

D = .542"

L29-U29

Length = 10'-3 1/2"

A = 14.25"

B = 6.875"

C = .50"

D = .359"

L29-U30

Length = 12'-6 3/4"

A = 14"

B = 8.25"

C = .625"

D = .41"

L29-U28

Length = 12'-5"

A = 14.125"

B = 10.25"

C = .750"

D = .490"



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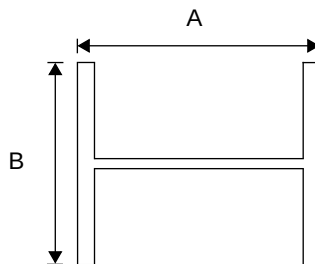
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JOB Blacks Run Bridge
SHEET NO. 3 OF 6
CALCULATED BY WBS DATE _____
CHECKED BY _____ DATE _____
SCALE _____ NTS

Field Measurements

(Non-Deteriorated to spot check truss)



C = Flange Thk.

D = Web Thk.

Span 3

L31-U30

Length = 12'-7 1/2"

A = 14.25"

B = 6.750"

C = .50"

D = .343"

L31-U32

Length = 12'-7 1/2"

A = 14.1875"

B = 6.750"

C = .50"

D = .345"

L33-U33

Length = 10'-3 1/4"

A = 14.25"

B = 6.875"

C = .50"

D = .388"

L33-U34

Length = 12'-5 1/4"

A = 14.125"

B = 10.25"

C = .750"

D = .520"

L33-U32

Length = 12'-5"

A = 14"

B = 8.25"

C = .750"

D = .414"

L35-U35

Length = 10'-3 1/4"

A = 14.25"

B = 6.875"

C = .50"

D = .363"

L35-U34

Length = 12'-1 3/4"

A = 14"

B = 14.625"

C = .750"

D = .454"

L35-U36

Length = 12'- 1 1/4"

A = 14.375"

B = 14.625"

C = .875"

D = .557"

L29-L31

A = 15.5"

B = 15.750"

C = 1.319"

D = ..445"

Bottom Chord Lengths:

L25-L26 = 8'-4 1/2"

L26-L27 = 9'-2 1/2"

L27-L29 = 18'-6"

L31-L33 = 18'-6 1/4"

L35-L37 = 17'-7"

(Top and bottom chords appear
the same throughout)



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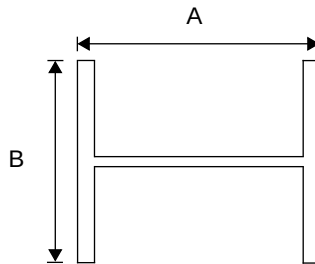
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JOB Blacks Run Bridge
SHEET NO. 4 OF 6
CALCULATED BY WBS DATE _____
CHECKED BY _____ DATE _____
SCALE _____ NTS

Field Measurements

(Non-Deteriorated to spot check truss)



C = Flange Thk.

D = Web Thk.

Span 4

L37-U37

Length = 10'-10"

A = 14.25"

B = 6.875"

C = .50"

D = .374"

L37-U38

Length = 11'-9"

A = 14.875"

B = 14.750"

C = 1.0625"

D = .754"

L37-U36

Length = 11'-9 1/4"

A = 14.750"

B = 14.625"

C = 1.0"

D = .660"

L39-U39

Length = 10'-3 1/2"

A = 14.125"

B = 6.875"

C = .50"

D = .362"

L39-U40

Length = 11'-9 1/4"

A = 14.125"

B = 14.750"

C = .875"

D = .534"

L39-U38

Length = 11'-9 3/4"

A = 14.4375"

B = 14.50"

C = .875"

D = .524"

L29-U29

Length = 10'-3 1/2"

A = 14.25"

B = 6.875"

C = .50"

D = .359"

L41-U41

Length = 7'-8"

A = 14.1875"

B = 6.750"

C = .50"

D = .323"

L41-U40

Length = 9'-8 1/4"

A = 14.25"

B = 12"

C = .750"

D = .468"

L41-U42

Length = 9'-10"

A = 14.125"

B = 10.125"

C = .750"

D = .432"

L41-U43

Length = 7'-8 1/4"

A = 14.125"

B = 6.875"

C = .50"

D = .342"

L43-U44

Length = 10'-1 1/4"

A = 14"

B = 6.875"

C = .375"

D = .372"



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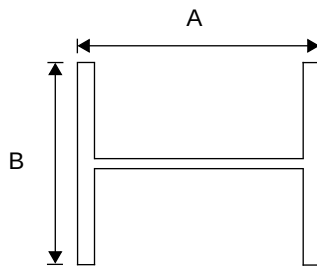
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JOB Blacks Run Bridge
SHEET NO. 5 OF 6
CALCULATED BY WBS DATE _____
CHECKED BY _____ DATE _____
SCALE _____ NTS

Field Measurements

(Non-Deteriorated to spot check truss)



C = Flange Thk.

D = Web Thk.

Span 4

L43-U42

Length = 10'-1 1/4"

A = 13.750"

B = 8.125"

C = .50"

D = .301"

L46-U46

Length = 7'-8 1/4"

A = 14.25"

B = 6.875"

C = .50"

D = .374"

L46-U45

Length = 10'-1 1/4"

A = 14"

B = 6.875"

C = .4375"

D = .341"

L46-U47

Length = 10'

A = 13.750"

B = 8.125"

C = .50"

D = .362"

L47-U47

Length = 7'-8 1/4"

A = 14.125"

B = 6.750"

C = .50"

D = .338"

L47-U46

Length = 9'-10"

A = 14"

B = 10.125"

C = .750"

D = .401"

L47-U48

Length = 9'-11 1/4"

A = 14.125"

B = 12.125"

C = .750"

D = .428"

L37-L39

Length = 17'-7"

A = 14.875"

B = 15.50"

C = 1.0625"

D = .701"

L41-L43

Length = 16'-8"

A = 14.750"

B = 15.50"

C = 1.0625"

D = .721"

L43-L45

Length = 17'-3"

A = 14.125"

B = 12.125"

C = .750"

D = .428"

L45-L47

Length = 17'-3"

A = 14.50"

B = 14.50"

C = 1.0"

D = .615"

(Assumed top and bottom chords same)



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JOB Blacks Run Bridge
SHEET NO. 6 OF 6
CALCULATED BY JJS/WBS DATE _____
CHECKED BY _____ DATE _____
SCALE _____ NTS

Field Measurements

(Deterioration)

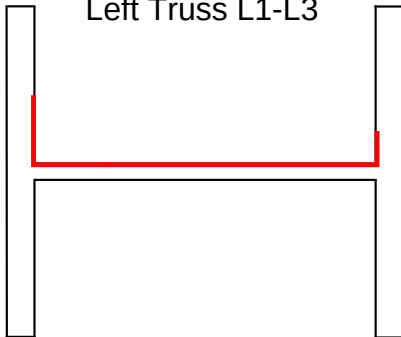
Section properties for the main load carrying members were gathered during this inspection and compared to the section properties used in the analysis. The section properties used in the analysis were determined to still be valid



Typical bottom chord deterioration:

Laminated rust present on flanges adjacent top of web and varies for height of 1" to 3" and is scattered along the length of the chord. Laminated rust present on top of webs with similar pattern of 1" to 3" inward from flanges. The loss varies from .0625" to .125" for the most part. Three bottom chord members in Span 1 have more significant deterioration and are listed below.

Left Truss L1-L3



L1-L3 Orig.

Length = ?

d = 14.25"

b = 14.50"

TF = .82"

TW = .50"

L1-L3 Deter.

Length = 0' to 17'

d = 14.25"

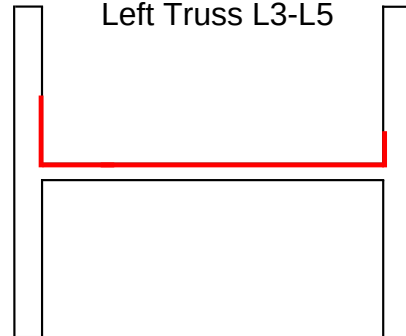
b = 14.50"

TF(L) = .50", TF(R) = .68"

(3" High) (1" High)

TW = .50"

Left Truss L3-L5



L3-L5 Orig.

Length = ?

d = 14.937"

b = 15.50"

TF = 1.20"

TW = .83"

L3-L5 Deter.

Length = 0' to 12'

d = 14.25"

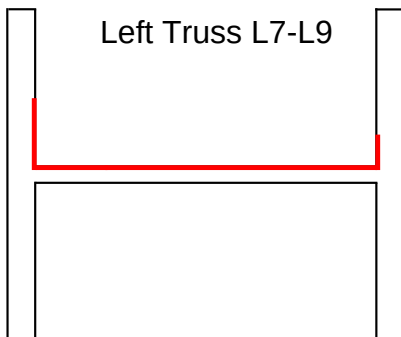
b = 14.50"

TF(L) = .80", TF(R) = .90"

(3" High) (1" High)

TW = .65"

Left Truss L7-L9



L7-L9 Orig.

Length = ?

d = 14.937"

b = 15.50"

TF = 1.20"

TW = .83"

L7-L9 Deter.

Length = Full Length

d = 14.25"

b = 14.50"

TF(L) = .80", TF(R) = .85"

(3" High) (1" High)

TW = .49"

2019 SUPER DUTY® CHASSIS CAB

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DIMENSIONS / WEIGHTS

	REGULAR CAB											
	F-350 SRW 4x2/4x4	F-350 DRW ¹ 4x2/4x4	F-350 DRW ¹ 4x2/4x4	F-450 DRW 4x2/4x4	F-450 DRW 4x2/4x4	F-450 DRW 4x2/4x4	F-450 DRW 4x2/4x4	F-550 DRW 4x2/4x4	F-550 DRW 4x2/4x4	F-550 DRW 4x2/4x4	F-550 DRW 4x2/4x4	
Wheelbase	145.3"	145.3"	169.3"	145.3"	169.3"	193.3"	205.3"	145.3"	169.3"	193.3"	205.3"	
Cab-to-axle	60"	60"	84"	60"	84"	108"	120"	60"	84"	108"	120"	
EXTERIOR												
Height												
Top of cab to ground	77.9"/81.0"	78.1"/81.0"	78.0"/80.9"	81.7"/81.5"	81.5"/81.3"	81.5"/81.4"	81.3"	81.7"/81.5"	81.7"/81.5"	81.7"/81.6"	81.5"	
Top of frame to ground	32.2"/34.3"	Shortest F550 wheel base selected		35.0"/35.1"	34.5"/34.6"	34.4"	34.4"	35.0"/35.1"	34.4"/34.5"	34.3"	34.3"	
Width												
Between frame rails	34.1"			34.2"	34.2"	34.2"	34.2"	34.2"	34.2"	34.2"	34.2"	
Axle clearance	8.6"/8.5"			8.2"	8.3"	8.3"	8.3"	8.2"	8.3"	8.3"	8.3"	
Front track	68.3"			74.8"	74.8"	74.8"	74.8"	74.8"	74.8"	74.8"	74.8"	
Rear tire outside width	79.6"	92.2"	92.2"	93.9"	93.9"	93.9"	93.9"	93.9"	93.9"	93.9"	93.9"	
Overall length	230.7"	230.7"	254.8"	230.7"	254.8"	278.8"	290.7"	230.7"	254.8"	278.8"	290.7"	
Weights (lbs.)												
Max. GVWR ²	11,100/11,500	14,000	14,000	16,500	16,500	16,500	16,500	19,000	19,500	19,500	19,500	
Max. payload ²	4,870/4,860	7,940/7,550	7,330/6,930	9,850/9,540	9,790/9,450	9,610/9,300	9,550/9,230	12,310/12,000	12,750/12,410	12,570/12,260	12,510/12,190	
	SUPERCAB					CREW CAB						
	F-350 SRW 4x2/4x4	F-350 DRW ¹ 4x2/4x4	F-450 DRW 4x2/4x4	F-450 DRW 4x2/4x4	F-550 DRW 4x2/4x4	F-550 DRW 4x2/4x4	F-350 SRW 4x2/4x4	F-350 DRW ¹ 4x2/4x4	F-450 DRW 4x2/4x4	F-450 DRW 4x2/4x4	F-550 DRW 4x2/4x4	F-550 DRW 4x2/4x4
Wheelbase	167.9"	167.9"	167.9"	191.9"	167.9"	191.9"	179.8"	179.8"	179.8"	203.8"	179.8"	203.8"
Cab-to-axle	60"	60"	60"	84"	60"	84"	60"	60"	60"	84"	60"	84"
EXTERIOR												
Height												
Top of cab to ground	78.3"/81.0"	78.5"/81.2"	81.8"	81.8"/81.6"	82.0"/81.8"	81.8"	78.4"/81.2"	78.6"/81.2"	81.9"/81.8"	81.7"/81.8"	81.9"/81.8"	81.9"/81.7"
Top of frame to ground	31.9"/33.8"	32.0"/33.7"	34.6"/34.3"	34.6"/34.4"	34.5"/34.3"	34.6"/34.3"	31.7"/33.5"	31.9"/33.7"	34.4"/34.3"	34.5"/34.3"	34.4"/34.3"	34.4"/34.3"
Width												
Between frame rails	34.1"	34.1"	34.2"	34.2"	34.2"	34.2"	34.1"	34.1"	34.2"	34.2"	34.2"	34.2"
Axle clearance	8.6"	7.9"/7.8"	8.3"	8.2"	8.3"	8.2"	8.6"	7.8"	8.2"	8.2"	8.2"	8.2"
Front track	68.3"	69.1"	74.8"	74.8"	74.8"	74.8"	68.3"	69.1"	74.8"	74.8"	74.8"	74.8"
Rear tire outside width	79.6"	92.2"	93.9"	93.9"	93.9"	93.9"	79.6"	92.2"	93.9"	93.9"	93.9"	93.9"
Overall length	253.3"	253.3"	253.3"	277.4"	253.3"	277.4"	265.2"	265.2"	265.2"	289.2"	265.2"	289.2"
Weights (lbs.)												
Max. GVWR ²	11,500	14,000	16,500	16,500	19,500	19,500	11,500	14,000	16,500	16,500	19,500	19,500
Max. payload ²	4,970/4,940	7,600/7,200	9,550/9,190	9,360/9,030	12,510/12,150	12,320/11,990	4,830/4,820	7,460/7,040	9,380/9,070	9,230/8,940	12,340/12,030	12,190/11,900
INTERIOR (front/rear)	REGULAR CAB		SUPERCAB	CREW CAB								
Head room	40.8"/ –		40.8"/40.3"	40.8"/40.4"								
Leg room (max.)	43.9"/ –		43.9"/33.5"	43.9"/43.6"								
Hip room	62.5"/ –		62.5"/64.7"	62.5"/64.7"								
Shoulder room	66.7"/ –		66.7"/65.8"	66.7"/65.9"								

Shortest F550 wheel base selected

5TH-WHEEL TOWING

Maximum Loaded Trailer Weight Ratings (lbs.)¹

Engine	Axle Ratio	GCWR	F-350 SRW 4x2	F-350 SRW 4x4	F-350 DRW 4x2	F-350 DRW 4x4	F-450 DRW 4x2	F-450 DRW 4x4	F-550 DRW 4x2	F-550 DRW 4x4
REGULAR CAB 6.2L 2-valve Gas V8 FFV	3.73	19,500	13,100	12,700	—	—	—	—	—	—
	4.30	20,000	—	—	13,200	12,800	—	—	—	—
	4.30	23,000	16,600	16,200	—	—	—	—	—	—
		23,500	—	—	16,700	16,300	—	—	—	—
6.8L 3-valve Gas V10	4.88	28,000	—	—	—	—	20,600	20,300	20,600	20,300
6.7L Diesel V8	3.73	26,500	19,200	18,700	—	—	—	—	—	—
		31,000	—	—	23,500	23,100	—	—	—	—
	4.10	32,000	—	—	24,500	24,100	23,900	23,600	23,900	23,600
	4.30	34,500	—	—	—	—	26,400	26,100	—	—
		35,000	—	—	—	—	—	—	26,900	26,600
	4.88	32,000 ²	—	—	—	—	—	—	23,900 ²	23,500 ²
		40,000 ²	—	—	—	—	—	—	31,900 ²	31,500 ²
SUPERCAB 6.2L 2-valve Gas V8 FFV	3.73	19,500	12,800	12,400	—	—	—	—	—	—
	4.30	20,000	—	—	12,900	12,500	—	—	—	—
	4.30	23,000	16,300	15,900	—	—	—	—	—	—
		23,500	—	—	16,400	16,000	—	—	—	—
6.8L 3-valve Gas V10	4.88	28,000	—	—	—	—	20,300	20,000	20,300	20,000
6.7L Diesel V8	3.73	26,500	18,800	18,400	—	—	—	—	—	—
		31,000	—	—	23,100	22,700	—	—	—	—
	4.10	32,000	—	—	24,100	23,700	23,600	23,200	23,600	23,200
	4.30	34,500	—	—	—	—	26,100	25,700	—	—
		35,000	—	—	—	—	—	—	26,600	26,200
	4.88	32,000 ²	—	—	—	—	—	—	23,500 ²	23,200 ²
		40,000 ²	—	—	—	—	—	—	31,500 ²	31,200 ²
CREW CAB 6.2L 2-valve Gas V8 FFV	3.73	19,500	12,600	12,200	—	—	—	—	—	—
	4.30	20,000	—	—	12,700	12,300	—	—	—	—
	4.30	23,000	16,100	15,700	—	—	—	—	—	—
		23,500	—	—	16,200	15,800	—	—	—	—
6.8L 3-valve Gas V10	4.88	28,000	—	—	—	—	20,200	19,900	20,200	19,900
6.7L Diesel V8	3.73	26,500	18,600	18,200	—	—	—	—	—	—
		31,000	—	—	23,000	22,600	—	—	—	—
	4.10	32,000	—	—	24,000	23,600	23,400	23,100	23,400	23,100
	4.30	34,500	—	—	—	—	25,900	25,600	—	—
		35,000	—	—	—	—	—	—	26,400	26,100
	4.88	32,000 ²	—	—	—	—	—	—	23,400 ²	23,100 ²
		40,000 ²	—	—	—	—	—	—	31,400 ²	31,100 ²

FRAMES

Number of Crossmembers

7 (145.3" WB); 8 (167.9" WB, 169.3" WB, 179.8" WB); 9 (191.9" WB, 193.3" WB, 203.8" WB); 10 (205.3" WB)

Section Modulus³

12.7 cu. in.

Frame Rail Yield Strength

50,000 psi

Max. Side Rail Section⁴ (height x width x thickness)

7.50" x 2.74" x 0.28" (F-350)

7.50" x 2.74" x 0.32" (F-450/F-550 145.3" WB, 169.3" WB, 167.9" WB, 179.8" WB and 203.8" WB)

7.50" x 2.74" x 0.60" (F-450/F-550 191.9" WB, 193.3" WB and 205.3" WB; F-550 with the Payload Plus Upgrade Package)

MECHANICAL

Max. Axle rating @ ground (lbs.)

		Front	Rear
F-350 SRW	4x2	5,250	7,230
	4x4	6,000	7,230
F-350 DRW	4x2	5,250	10,040
	4x4	6,000	10,040
F-450 DRW		7,000	13,660
F-550 DRW		7,000	13,660

Shock Absorbers

1.38" gas type

Gross Axle weights used

14.29"/14.29" (F-350); 15.39"/15.74" (F-450/F-550)

Lining area (sq. in.; front/rear) – 50.9/39.6 (F-350); 68.8/64.8 (F-450/F-550)

Wheels

Number of studs – 8 (F-350); 10 (F-450/F-550)

Bolt-circle diameter – 6.69" (F-350 SRW); 7.87" (F-350 DRW); 8.85" (F-450/F-550)

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SRW = single rear wheel DRW = dual rear wheel

¹F-350 DRW/F-450/F-550 weights shown are for 145.3" wheelbase Regular Cab, 167.9" wheelbase SuperCab or 179.8" wheelbase Crew Cab models. For 169.3," 191.9," 193.3," 203.8" and 205.3" wheelbase models, weights may be somewhat less (usually not more than 200 lbs., or 400 lbs. with 205.3" wheelbase). ²Requires High-Capacity Trailer Tow Package. ³Cross-sectional modulus calculated at back of cab. ⁴Measured to inside of metal.

5th-Wheel Towing Notes: Trailer weights shown assume 400 lbs.–800 lbs. second-unit body weight. Be sure combined weight of vehicle and trailer does not exceed listed GCWR. Trailer kingpin load weight should be 15% of total loaded trailer weight. Make sure that the vehicle payload (reduced by option weight) will accommodate trailer kingpin load weight and the weight of passengers and cargo added to the towing vehicle. The addition of trailer kingpin load weight, and the weight of passengers and cargo, cannot cause vehicle weights to exceed the rear GAWR (Gross Axle Weight Rating) or GVWR (Gross Vehicle Weight Rating). These ratings can be found on the vehicle's Safety Compliance Certification Label. See your Ford Dealer for conventional trailer weights.

LOAD RATING SUMMARY

TRUSS RATING

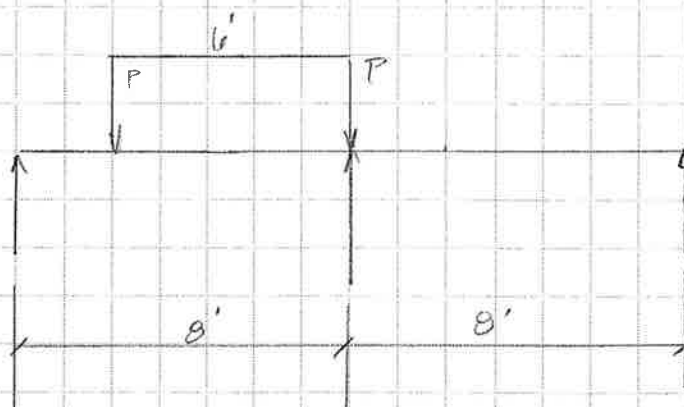
	Vehicle Rating (Tons)			
	H 20 (20)	HS 20 (36)	ML 80 (37.7)	TK527 (40)
Inventory	35	46	39	45
Operating	54	82	76	80

Load Posting Review

- A load rating analysis was performed as part of this inspection utilizing the original structure plans from Harry L. Dovell & Co. dated 1943 and field inspection findings.
- PennDOT's Computer Program BAR 7, Bridge Analysis and Rating, version 7.12.0.0 was used to perform the rating analysis, utilizing Allowable Stress Method for analysis.
- According to the bridge owner only tri-axle dump trucks utilize this structure, and no vehicles with large or unusual axle configurations. Therefore, the bridge was only rated for the above listed vehicles. If the owner would like to utilize this structure for any vehicle with a different axle configuration or loading then that of the above listed vehicles, additional analysis would be required.
- Based on inspection findings a 25% section loss was applied to the bottom chord for the full length of the bridge for the rating analysis.
- The truss bottom chord controlled the ratings for all ratings except the ML80 Operating rating which was controlled by shear in the floorbeam.
- Based on the above listed rating values that were obtained from BAR 7 it has been determined that this structure does not need to be posted at this time.
- The rating calculations can be found in the appendix of this report.

RATING ANALYSIS

DISTRIBUTION FACTORS - INPUT D.F. FOR STRINGER
FOR REGULAR LIVE LOADS - 1 LANE ONLY.



SHEAR

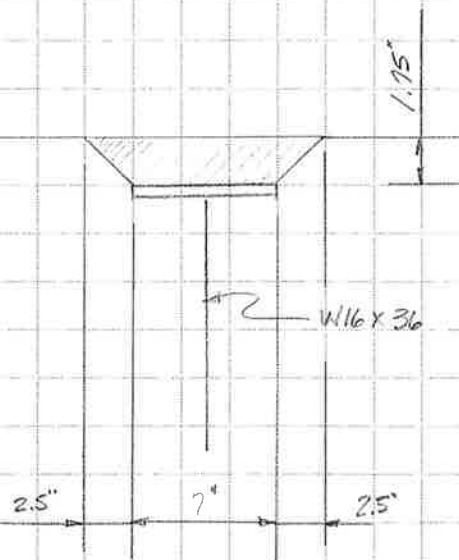
$$DF = 1 + \frac{2'}{8'} = 1.25 \text{ WH.} = 0.625 \text{ AXLES}$$

DEFLECTION

↓ INCLUDE TRUSSES

$$DF = 2 \text{ LANES} \div 3 \text{ BM.} = 0.67$$

CALCULATE STRINGER DLI



$$WT. = (7" + 2.5") \times 1.75" \div 144 \times 0.15 \text{ KCF} = 0.017 \#/\text{ft} \quad \text{SAY } 0.020 \#/\text{ft}$$

FLOORBEAM DLI

CONC. HAUNCH ON F.B. IS THE SAME AS STRINGER HAUNCH

$$WT. = 0.020 \#/\text{ft}$$

CENTER TRUSS WT & TOP CHORD BRACING

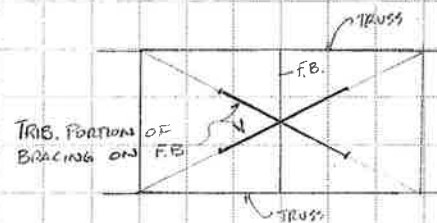
$$\begin{aligned} \text{VERT.} &= 31 \#/\text{ft} \times 4.25' \div 8.33' = 15.8 \#/\text{ft} \\ \text{DIAG.} &= 21.4 \#/\text{ft} \\ \text{BOT. CHORD} &= 2 \times 16.2 \#/\text{ft} = 32.4 \#/\text{ft} \end{aligned}$$

LENGTH OF BRACE

$$69.6 \#/\text{ft} \times 420.83' \div 48 \text{ FB'S} = 610 \#/\text{FB.}$$

$$\text{BRACING: } \angle 3 \times 3 \times \frac{3}{8} : 4 \text{ ANGLES} \times 6.2' \times 7.2 \#/\text{ft} = 178 \#$$

$$\text{CONNECTIONS: } 40\% \times (178\# + 610\#) = 315 \#$$



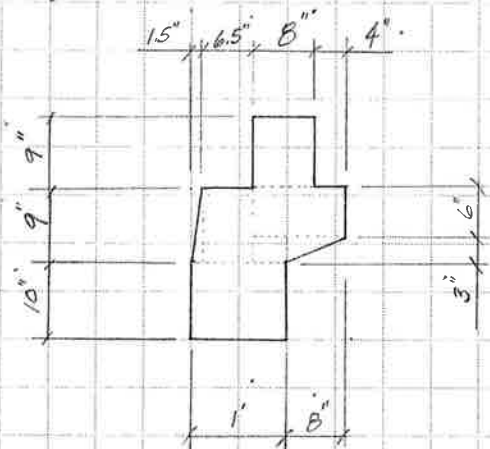
TRUSS DEAD LOADS (RIGHT SIDE TRUSS)

SLAB WT. (UPPER JOINTS)

$$\text{SLAB: } 18' \times 0.833' \times 0.15 \text{ kcf} \times \frac{1}{2} \text{ TO RT. TRUSS} = 1.13 \%$$

$$\text{X-SLOPE: } \frac{1}{2} \times 18' \times \frac{3.5}{12} \times 0.15 \text{ kcf} \times \frac{1}{2} \text{ TO RT. TRUSS} = 0.20 \%$$

CURBS



$$.83' \times 1' \times .15 \text{ kcf} = 0.12 \%$$

$$\frac{1}{2} \times 1.5' \times .75' \times .15 \text{ kcf} = 0.01 \%$$

$$6.5' \times .75' \times .15 \text{ kcf} = 0.06 \%$$

$$.75' \times .67' \times .15 \text{ kcf} = 0.08 \%$$

$$.5' \times 1' \times .15 \text{ kcf} = 0.08 \%$$

$$\frac{1}{2} \times .25' \times .67' \times .15 \text{ kcf} = 0.01 \%$$

$$.33' \times .25' \times .15 \text{ kcf} = 0.01 \%$$

$$\text{CONC IN TOP CHORD: } 1.17' \times 1.5' \times .15 \text{ kcf} = 0.26 \%$$

$$2.0 \%$$

FIND AVG. SLAB WT. PER UPPER ST.

$$2.0 \% \times 420.83' \div 48 \text{ JOINTS} = \underline{17.5' \text{ TO ALL UPPER JOINTS}}$$

← BRIDGE LENGTH

TRUSS DEAD LOAD (RIGHT TRUSS)

SIDEWALK - ATTACHED AT ODD NUMBERED JOINTS, & HINGES
DUE TO OVERHANG BRACKET, HALF OF SIDEWALK
LOAD GOES TO TOP JOINT & HALF TO EXT. JOINT.



$$Wt. = 3' \times .5' \times .15 \text{ KCF} \times 420.83' \div 29 \text{ JOINTS} = 3.3 \text{ K}$$

APPLY 3.3 K TO ODD NO. UPPER JOINTS & HINGES & END JOINTS.
APPLY 3.3 K TO ODD NO. LOWER JOINTS

SIDEWALK SUPPORT STRINGER

$\frac{1}{2}$ TO TOP & $\frac{1}{2}$ TO BOT.

$$Wt. = 36 \text{ \#/' } \div 1000 \times 7.75' \div 2 = 0.14 \text{ K TO ODD NO. UPPER JOINTS & HINGES}$$

$$0.14 \text{ K TO ODD NO. LOWER JOINTS}$$

SUPPORT BRACKET DIAG. - ODD NO. LOWER JOINTS

$$Wt. = [21.4 \text{ \#/' } \times 14' \div 1000] \times 1.4 \text{ CORN.} = 0.42 \text{ K TO ODD NO. LOWER JOINTS}$$

FENCE - ODD NO. LOWER JOINTS & HINGES

$$\text{VERT. POST. } 8" \text{ WF } 24 : .024 \text{ K' } \times 15.5' = 0.37 \text{ K}$$

$$\text{RAILS } 7 \text{ E } 9.6" : 0.010 \text{ K' } \times 4 \text{ RAILS} \times 420.83' \div 27 \text{ JOINTS} = 0.62 \text{ K}$$

$$\text{SHIELD: } .125"/12 \times .490 \text{ KCF} \times 15.5' \times 420.83' \div 27 \text{ JOINTS} = 1.23 \text{ K}$$

2.22 K TO ODD NO. LOWER
JOINTS & HINGES

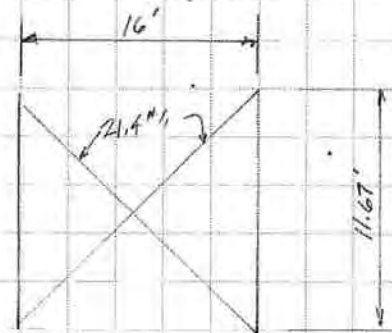
TRUSS DEAD LOAD (RIGHT TRUSS)

CROSS BRACING BTWN TRUSSES (UPPER & LOWER ODD NO. JTS.)

$$L = \sqrt{16^2 + 11.6^2} = 19.8'$$

$$\text{WT/JT} = 0.021 \frac{\text{K}}{\text{ft}} \times 19.8' \div 2 = 0.21 \text{ K}$$

CONNECTIONS SAY .10 K



.31 K TO ODD NO. LOWER & UPPER JTS.

STRINGER, CENTER TRUSS, TOP CHORD BRACING, & FLOORBEAM

STRINGER 16 WF 36 : $(0.036 \frac{\text{K}}{\text{ft}} \times 420.83' \div 48 \text{ JTS.}) \div 2 \text{ TRUSSES} = 0.16 \text{ K}$
(SEE PREV. CALCS (F.B. DL))

CENTER TRUSS : $1.10 \frac{\text{K}}{\text{FB}} \div 2 = 0.55 \text{ K}$
3/4 BRACING

FLOORBEAM 16 WF 36 : $0.036 \frac{\text{K}}{\text{ft}} \times 16' \div 2 \text{ TRUSSES} = 0.29 \text{ K}$

F.B. HAUNCH : $0.02 \frac{\text{K}}{\text{ft}} \times 16' \div 2 \text{ TRUSSES} = 0.16 \text{ K}$

STRINGER HAUNCH : $(.02 \frac{\text{K}}{\text{ft}} \times 420.83' \div 48 \text{ JTS.}) \div 2 \text{ TRUSSES} = 0.09 \text{ K}$

TOTAL = 1.25 K

APPLY TO ALL UPPER JOINTS

BOTTOM CHORD BRACING (ALL ODD NO. LOWER JTS.)

2 L's 6x4x3/8 : $16' \times 12.3 \frac{\text{K}}{\text{ft}} \times 2 \text{ L's} \div 2 \text{ SIDES} \div 1000 = 0.20 \text{ K} \times 1.4 = 0.28$

TOTAL = 0.28 K TO ODD NO. LOWER JTS.

TRUSS DEAD LOAD CONT'D

SUMMARY OF "NON-TRUSS" JOINT DEAD LOADS

- ODD NUMBERED UPPER JOINTS:

$$DL = 17.5^k + 3.3^k + 0.14^k + 3.1^k + 1.25^k = 22.5^k$$

- EVEN NUMBERED UPPER JOINTS:

$$DL = 17.5^k + 1.25^k = 18.8^k$$

- ODD NUMBERED LOWER JOINTS:

$$DL = 3.3^k + 0.14^k + 0.42^k + 2.22^k + 0.31^k + 0.28^k = 6.7^k$$

- EVEN NUMBERED LOWER JOINTS:

$$DL = 0^k$$

- HINGES & END SUPPORT JOINTS:

$$DL = 17.6^k + 3.3^k + 2.22^k + 1.14^k = 24.2^k$$

TRUSS DEAD LOAD CONT'D

TRUSS A: JOINTS $U_0 - U_{14}$, $L_1 - L_{13}$

Avg. VERTICAL MEMBER WT. = $38 \#/\text{ft} \times 1.40 = \underline{53 \#/\text{ft}}$ ^{CONN.}

Avg. TOP CHORD WT.

$87 \#/\text{ft} \times 4 \text{ MEM.}$	$= 348$	} $2004 \div 14 \text{ MEM.} = 143 \#/\text{ft}$ CONN. $\rightarrow \times 1.40$ $= \underline{200 \#/\text{ft}}$
$158 \#/\text{ft} \times 2 \text{ MEM.}$	$= 316$	
$176 \#/\text{ft} \times 6 \text{ MEM.}$	$= 1056$	
$142 \#/\text{ft} \times 2 \text{ MEM.}$	$= 284$	

Avg. BOT. CHORD WT.:

$103 \#/\text{ft} \times 2 \text{ MEM.}$	$= 206$	} $1690 \div 12 \text{ MEM.} = 141 \#/\text{ft}$ CONN. $\rightarrow \times 1.40$ $= \underline{197 \#/\text{ft}}$
$158 \#/\text{ft} \times 4 \text{ MEM.}$	$= 632$	
$142 \#/\text{ft} \times 6 \text{ MEM.}$	$= 852$	

Avg. DIAG. MEM. WT.:

$103 \#/\text{ft} \times 3 \text{ MEM.}$	$= 309$	} $1168 \div 14 = 84 \#/\text{ft}$ CONN. $\rightarrow \times 1.40$ $= \underline{118 \#/\text{ft}}$
$84 \#/\text{ft} \times 1 \text{ MEM.}$	$= 84$	
$61 \#/\text{ft} \times 1$	$= 61$	
$53 \#/\text{ft} \times 2$	$= 106$	
$38 \#/\text{ft} \times 2$	$= 76$	
$78 \#/\text{ft} \times 1$	$= 78$	
$87 \#/\text{ft} \times 1$	$= 87$	
$136 \#/\text{ft} \times 2$	$= 272$	
$95 \#/\text{ft} \times 1$	$= 95$	

Avg. LENGTH OF TOP/BOT. CHORD = $128.55' \div 14 = 9.2'$

LENGTH OF VERT. MEM. = $11.67'$

Avg. LENGTH OF DIAG. MEMBERS = $\sqrt{9.2^2 + 11.67^2} = 14.9'$

CALCULATE JOINT LOADS FOR TRUSS A USING THESE LENGTHS.

TRUSS DEAD LOAD CONT'D

TRUSS B: JOINTS $V_{14} - V_{22}$, $L_{15} - L_{21}$

Avg. VERT. MEM. WT. = $34 \#/\text{ft} \times 1.40 = \underline{48 \#/\text{ft}}$

Avg. TOP CHORD WT. = $103 \#/\text{ft} \times 1.40 = \underline{144 \#/\text{ft}}$

Avg. BOTTOM CHORD WT. = $103 \#/\text{ft} \times 1.40 = \underline{144 \#/\text{ft}}$

Avg. DIAGONAL MEM. WT. $\left. \begin{array}{l} 78 \times 2 = 156 \\ 61 \times 2 = 122 \\ 43 \times 2 = 86 \\ 30 \times 2 = 60 \end{array} \right\} 424 \div 8 \text{ MEM} = 53 \#/\text{ft}$
CONN. $\rightarrow \times 1.40$
 $= \underline{74 \#/\text{ft}}$

Avg. LENGTH OF TOP & BOT. CHORD = $64.15' \div 8 = 8.02'$

LENGTH OF VERT. MEM. = $9.0'$

Avg. LENGTH OF DIAG. MEM. = $\sqrt{8.02^2 + 9^2} = 12.1'$

CALCULATE JOINT LOADS FOR TRUSS B USING THESE LENGTHS.

TRUSS DEAD LOAD CONT'D

TRUSS C: JOINTS U22 - U40, L23 - L39

$$\text{Avg. VERT MEM. WT.} = 38 \#/\text{ft} \times 1.40^{\text{CONN.}} = \underline{53 \#/\text{ft}}$$

$$\begin{aligned} \text{Avg. TOP CHORD WT.} &= \begin{aligned} &87 \times 4 = 348 \\ &176 \times 4 = 704 \\ &142 \times 4 = 568 \\ &193 \times 6 = 1158 \end{aligned} \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} 2778 \div 18 \text{ MEM.} = 155 \#/\text{ft} \\ &\quad \text{CONN.} \times 1.40 = \underline{217 \#/\text{ft}} \end{aligned}$$

$$\begin{aligned} \text{Avg. BOTTOM CHORD WT.} &= \begin{aligned} &142 \#/\text{ft} \times 12 = 1704 \\ &158 \#/\text{ft} \times 4 = 632 \end{aligned} \left. \begin{array}{l} \\ \end{array} \right\} 2336 \div 16 \text{ MEM.} = 146 \#/\text{ft} \\ &\quad \text{CONN.} \times 1.40 = \underline{204 \#/\text{ft}} \end{aligned}$$

$$\begin{aligned} \text{Avg. DIAG. MEM. WT.} &= \begin{aligned} &103 \#/\text{ft} \times 4 = 412 \\ &136 \times 2 = 272 \\ &119 \times 2 = 238 \\ &103 \times 2 = 206 \\ &87 \times 2 = 174 \\ &68 \times 2 = 136 \\ &53 \times 2 = 106 \\ &34 \times 2 = 68 \end{aligned} \left. \begin{array}{l} \\ \\ \\ \\ \\ \\ \\ \end{array} \right\} 1612 \div 18 \text{ MEM.} = 90 \#/\text{ft} \\ &\quad \text{CONN.} \times 1.40 = \underline{126 \#/\text{ft}} \end{aligned}$$

$$\text{Avg. LENGTH OF TOP & BOT. CHORD} = 159.14' \div 18 \text{ MEM.} = 8.84'$$

$$\text{LENGTH OF VERT. MEM.} = 11.67'$$

$$\text{Avg. LENGTH OF DIAG.} = \sqrt{8.84^2 + 11.67^2} = 14.6'$$

CALCULATE JOINT LOADS FOR TRUSS C USING THESE LENGTHS.

TRUSS DEAD LOADS CONT'D

TRUSS D: JOINTS $U_{40} - U_{48}$, $L_{41} - L_{47}$

$$\text{Avg. VERT. MEM. WT.} = 34 \#/\text{ft} \times 1.40^{\text{Conn.}} = \underline{48 \#/\text{ft}}$$

$$\text{Avg. TOP CHORD WT.} = 119 \#/\text{ft} \times 1.40 = \underline{167 \#/\text{ft}}$$

$$\text{Avg. BOT. CHORD WT.} = 119 \#/\text{ft} \times 1.40 = \underline{167 \#/\text{ft}}$$

$$\begin{aligned} \text{Avg. DIAG. MEM. WT.} &= \begin{aligned} &78 \# \times 2 = 156 \\ &61 \times 2 = 122 \\ &43 \times 2 = 86 \\ &30 \times 2 = 60 \end{aligned} \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} 424 \div 8 \text{ MEM.} = 53 \#/\text{ft} \\ &\quad \text{CONN.} \rightarrow \times 1.40 \\ &\quad \quad \quad = \underline{74 \#/\text{ft}} \end{aligned}$$

$$\text{Avg. LENGTH OF TOP \& BOT. CHORD} = 69' \div 8 = 8.63'$$

$$\text{LENGTH OF VERT. MEM.} = 9.0'$$

$$\text{Avg. LENGTH OF DIAG. MEM.} = \sqrt{8.63'^2 + 9.0'^2} = 12.5'$$

CALCULATE JOINT LOADS FOR TRUSS D USING THESE LENGTHS.

TRUSS DEAD LOAD CONT'D

JOINT LOADS FOR BAR 7 INPUT

TRUSS A: JOINTS $U_0 - U_{14}$, $L_1 - L_{13}$

$$U_0 = 24.2^k + .200^k \times 9.2' \div 2 + .118^k \times 14.9' \div 2 = 26.0^k$$

SEE "NON-TRUSS" JOINT DL CALCS

$$U_{1,3,5,7,9,11,13} = 22.5^k + .200^k \times 9.2' + .053^k \times 11.67' \div 2 = 24.6^k$$

$$U_{2,4,6,8} = 18.8^k + .200^k \times 9.2' + .118^k \times 14.9' \div 2 \times 2 = 22.4^k$$

$$L_{1,13} = 6.7^k + .053^k \times \frac{11.67'}{2} + .197^k \times 9.2' + .118^k \times \frac{14.9'}{2} \times 2 = 10.6^k$$

$$L_{3,5,7,9} = 6.7^k + .053^k \times \frac{11.67'}{2} + .197^k \times 9.2' \times 2 + .118^k \times \frac{14.9'}{2} \times 2 = 12.9^k$$

$L_{2,4,6,8}$ - NO LOAD INPUT, NO VERT. OR DIAG. MEMBERS

$$U_{14} = 24.2^k + \underbrace{.200^k \times \frac{9.2'}{2}}_{\text{TRUSS A}} + .118^k \times \frac{14.9'}{2} + \underbrace{.094^k \times \frac{12.1'}{2} + .144^k \times \frac{8'}{2}}_{\text{TRUSS B}} = 27.0^k$$

$$U_{10,12} = 18.8^k + .200^k \times 9.2' + .118^k \times \frac{14.9'}{2} \times 2 + .053^k \times \frac{11.67'}{2} = 22.7^k$$

$$L_{10,12} = 0^k + .197^k \times 9.2' + .053^k \times \frac{11.67'}{2} = 2.1^k$$

$$L_{11} = 6.7^k + .053^k \times \frac{11.67'}{2} + .197^k \times 9.2' + .118^k \times \frac{14.9'}{2} \times 2 = 10.6^k$$

TRUSS DEAD LOAD CONT'D

JOINT LOADS FOR BAR 7 INPUT

TRUSS B: JOINTS $U_{14} - U_{22}$, $L_{15} - L_{21}$

U_{14} - SEE TRUSS A CALCULATION

$$U_{15, 17, 19, 21} = 22.5^k + .048^k \times \frac{9.0'}{2} + .144^k \times \frac{8.0'}{2} \times 2 = 23.9^k$$

$$U_{16, 18, 20} = 18.8^k + .144^k \times \frac{8.0'}{2} \times 2 + .074^k \times \frac{12.1'}{2} \times 2 = 20.8^k$$

$$U_{22} = 24.2^k + \underbrace{.144^k \times \frac{8.0'}{2} + .074^k \times \frac{12.1'}{2}}_{\text{TRUSS B}} + \underbrace{.217^k \times \frac{8.0'}{2} + .126^k \times \frac{14.6'}{2}}_{\text{TRUSS C}} = 27.1^k$$

$$L_{15, 21} = 6.7^k + .048^k \times \frac{9'}{2} + .144^k \times 8' + .074^k \times \frac{12.1'}{2} \times 2 = 9.0^k$$

$$L_{17, 19} = 6.7^k + .048^k \times \frac{9'}{2} + .144^k \times 8' \times 2 + .074^k \times \frac{12.1'}{2} \times 2 = 10.1^k$$

$L_{16, 18, 20}$ - NO LOAD INPUT, NO VERT. OR DIAG. MEM. PRESENT

TRUSS DEAD LOAD CONT'D

JOINT LOADS FOR BAR 7 INPUT

TRUSS C: JOINTS $U_{22} - U_{40}$, $L_{23} - L_{39}$

U_{22} - SEE TRUSS B CALCULATION.

$$U_{23, 25, 27, 29, 31, 33, 35, 37, 39} = 22.5^k + .053^k/\% \times \frac{11.67'}{2} + .217^k/\% \times \frac{8.8'}{2} \times 2 = 24.7^k$$

$$U_{24, 26} = 18.8^k + .217^k/\% \times \frac{8.8'}{2} \times 2 + .053^k/\% \times \frac{11.67'}{2} + .126^k/\% \times \frac{14.6'}{2} \times 2 = 22.9^k$$

$$U_{28, 30, 32, 34, 36, 38} = 18.8^k + .217^k/\% \times \frac{8.8'}{2} \times 2 + .126^k/\% \times \frac{14.6'}{2} \times 2 = 22.5^k$$

$$U_{40} = 24.2^k + \underbrace{.217^k/\% \times \frac{8.8'}{2} + .126^k/\% \times \frac{14.6'}{2}}_{\text{TRUSS C}} + \underbrace{.167^k/\% \times \frac{8.6'}{2} + .094^k/\% \times \frac{12.5'}{2}}_{\text{TRUSS D}} = 27.3^k$$

$$L_{23, 39} = 6.7^k + .053^k/\% \times \frac{11.67'}{2} + .204^k/\% \times \frac{8.8'}{2} + .126^k/\% \times \frac{14.6'}{2} \times 2 = 9.7^k$$

$$L_{29, 31, 33, 35, 37} = 6.7^k + .053^k/\% \times \frac{11.67'}{2} + .204^k/\% \times \frac{8.8'}{2} \times 2 + .126^k/\% \times \frac{14.6'}{2} \times 2 = 12.4^k$$

$$L_{24, 26} = 0^k + .053^k/\% \times \frac{11.67'}{2} + .204^k/\% \times \frac{8.8'}{2} \times 2 = 2.1^k$$

$L_{28, 30, 32, 34, 36, 38} = \text{NO LOAD INPUT, NO VERT. CR. DIAG. MEM.}$

$$L_{25} = 6.7^k + .053^k/\% \times \frac{11.67'}{2} + .204^k/\% \times \frac{8.8'}{2} \times 2 + .126^k/\% \times \frac{14.6'}{2} \times 2 = 10.6^k$$

$$L_{27} = 6.7^k + .053^k/\% \times \frac{11.67'}{2} + .204^k/\% \times \left(\frac{8.8'}{2} + 8.8' \right) + .126^k/\% \times \frac{14.6'}{2} \times 2 = 11.5^k$$

TRUSS DEAD LOAD CONT'D

JOINT LOADS FOR BAR 7 INPUT

TRUSS D: JOINTS $U_{40} - U_{48}$, $L_{41} - L_{47}$

U_{40} - SEE TRUSS C CALCULATIONS

$$U_{41,43,45,47} = 22.5 + .048 \text{ k/ft} \times \frac{9'}{2} + .167 \text{ k/ft} \times \frac{8.6'}{2} \times 2 = 24.2 \text{ k}$$

$$U_{42,44,46} = 18.8 \text{ k} + .167 \text{ k/ft} \times \frac{8.6'}{2} \times 2 + .074 \text{ k/ft} \times \frac{12.5'}{2} \times 2 = 21.2 \text{ k}$$

$$U_{48} = 24.2 \text{ k} + .167 \text{ k/ft} \times \frac{8.6'}{2} + .074 \text{ k/ft} \times \frac{12.5'}{2} = 25.4 \text{ k}$$

$$L_{41,47} = 6.7 \text{ k} + .048 \text{ k/ft} \times \frac{9'}{2} + .167 \text{ k/ft} \times 8.6' + .074 \text{ k/ft} \times \frac{12.5'}{2} \times 2 = 9.3 \text{ k}$$

$$L_{43,45} = 6.7 \text{ k} + .048 \text{ k/ft} \times \frac{9'}{2} + .167 \text{ k/ft} \times 8.6' \times 2 + .074 \text{ k/ft} \times \frac{12.5'}{2} \times 2 = 10.7 \text{ k}$$

$L_{42,44,46}$ = NO LOAD INPUT, NO VERT. OR DIAG. MEMBERS.

DETERMINATION OF JOINT LOADS FROM DESIGN DWG LOADS

TRUSS A

EVEN NO. UPPER JTS. :



$$\text{@ JOINT E: } 34^k \times \frac{11.67'}{15.05'} = 26^k \uparrow$$

$$4^k \times \frac{11.67'}{15.05'} = 3.1^k \downarrow$$

$$26^k - 3.1^k = 23^k \downarrow \text{ --- NET FORCE DOWN}$$

UPPER
ODD Joints : USE FORCES FROM VERT. MEMBERS

• FROM JOINT ANALYSIS, NO LOAD APPLIED TO LOWER JOINTS

TRUSS B

EVEN NO. UPPER JOINTS



$$\text{@ JT. E: } 12^k \times \frac{9'}{12.04'} = 9^k \uparrow$$

$$12^k \times \frac{9'}{12.04'} = 9^k \uparrow$$

$$9^k + 9^k = 18^k \downarrow \text{ --- NET FORCE DOWN AT JOINT}$$

* IT IS EVIDENT FROM PLAN LOADS THAT NO DEAD LOADS WERE APPLIED TO LOWER JOINTS. $\therefore$ FOR ANALYSIS THE SAME WILL BE DONE

TRUSS C

C.J. M

$$137^k \times 11.67' \div 14.89' = 107^k \downarrow$$

$$166^k \times 11.67' \div 14.89' = 130^k \uparrow$$

$$130^k - 107^k = 23^k \text{ — NET FORCE DOWN @ JOINT}$$



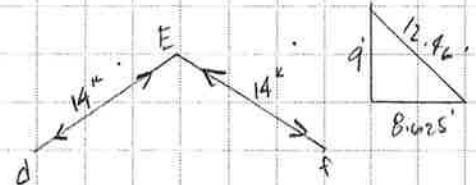
TRUSS D

C.J. E

$$14^k \times \frac{9'}{12.46'} = 10.1^k \uparrow$$

$$14^k \times \frac{9'}{12.46'} = 10.1^k \uparrow$$

$$10.1^k + 10.1^k = 20^k \text{ — NET FORCE DOWN ON JT.}$$



THE LOADS CALCULATED CONFIRM THAT THESE LOADS ARE REASONABLE, THEREFORE, IN AN EFFORT TO MATCH THE DL FORCES SHOWN ON THE PLANS APPLY THE FOLLOWING LOADS TO UPPER JTS.

TRUSS A :	EVEN JTS.	-	23 ^k
TRUSS B :	" "	-	18 ^k
TRUSS C :	" "	-	23 ^k
TRUSS D :	" "	-	20 ^k

TO ODD JOINTS APPLY LOADS MATCHING LOADS SHOWN IN VERTICAL MEMBERS ON THE DRAWINGS.

JOB Monaca Bridge Path 4
SHEET NO. 17 OF 97
CALCULATED BY AME DATE 2.20.08
CHECKED BY DFL DATE 5-13-08
SCALE 4396.001

TRUSS MEMBER PROPERTIES

	GROSS AREA COMP IN ²	NET AREA TEARS. IN ²	I IN ⁴	F _y KSI	L FT	E _c IN.	FL. WIDTH IN.	C IN.	CAT. —	F _u KSI	GROSS A. TEARS. IN ²
W14x87	25.52	20.06	349.7	33	↓	—	—	—	—	—	23.89
W14x158	46.47	36.97	745.0	—	—	—	—	—	—	—	43.94
W14x176	51.73	41.2	837.9	—	—	—	—	—	—	—	48.96
W14x142	41.85	33.35	660.1	—	—	—	—	—	—	—	39.63
W14x103	30.26	23.76	419.7	—	—	—	—	—	—	—	28.30
W14x84	24.71	19.4	225.5	—	—	—	—	—	—	—	23.11
W14x61	17.94	15.37	107.3	—	—	—	—	—	—	—	17.82
W14x53	15.59	13.4	57.5	—	—	—	—	—	—	—	15.44
W14x38	11.17	9.17	24.6	—	—	—	—	—	—	—	11.17
W14x78	22.94	17.2	206.9	—	—	—	—	—	—	—	20.64
W14x136	38.98	31.86	567.7	—	—	—	—	—	—	—	37.86
W14x95	27.94	21.57	383.7	—	—	—	—	—	—	—	25.76
W14x43	12.65	10.53	45.1	—	—	—	—	—	—	—	12.65
W14x30	8.81	7.2	17.5	—	—	—	—	—	—	—	8.52
W14x34	10.00	8.2	21.3	—	—	—	—	—	—	—	9.70
W14x119	34.99	27.49	491.8	—	—	—	—	—	—	—	32.74
W14x193	56.73	45.2	930.1	↓	—	↓	↓	↓	↓	↓	53.71
W14x68	20	17.14	121.2	33	↓	—	—	—	—	—	20

↓ FROM DECK.

TRUSS C: TOP & BOT. CHORD L = 8.0'
↓ VERT. MEM L = 11.67'
DIAG. L = 14.6'

TRUSS D: TOP & BOT. CHORD L = 8.6'
↓ VERT. MEM L = 9.0'
DIAG. L = 12.5'

TRUSS A: TOP & BOT. CHORD L = 9.2'
↓ VERT. MEM L = 11.67'
DIAG. L = 14.9'

TRUSS B: TOP & BOT. CHORD L = 8.0'
↓ VERT. MEM L = 9.0'
DIAG. L = 12.1'

TABLE 6.6.2.1-1 INVENTORY RATING ALLOWABLE STRESSES (psi)

	DATE BUILT-STEEL UNKNOWN					Silicon Steel		
	Prior to 1905	1905 to 1936	1936 to 1963	After 1963	Carbon Steel	Over 2" to 4" incl	Nickel Steel	Over 1-1/8" to 2" incl
AASHTO Designation ⁽¹⁾					M 94(1961)	M 95(1961)	M 96(1961)	
ASTM Designation ⁽¹⁾					A 7(1967)	A 94(1966)	A 8(1961)	A 94
Minimum Tensile Strength	F _t	52,000	60,000		60,000	70,000	90,000	72,000
Minimum Yield Point	F _y	26,000	30,000	33,000	36,000	45,000	55,000	47,000
Axial tension in members with no holes for high strength bolts or rivets. Use net section when member has any open holes larger than 1-1/4" diam. such as perforations	0.55F _y 0.46F _y	14,000	16,000	18,000	20,000	24,000	30,000	25,000
					NOT APPLICABLE			
Axial tension in members with holes for high strength bolts or rivets and tension in extreme fiber of rolled shapes, girders, and built-up sections subject to bending	Gross* Section 0.55F _y	14,000	16,000	18,000	20,000	24,000	30,000	25,000
• When the area of holes deducted for high strength bolts or rivets is more than 15 percent of the gross area, that area in excess of 15 percent shall be deducted from the gross area in determining stress on the gross section. In determining gross section, any open holes larger than 1-1/4" diam. such as perforations shall be deducted.	Net Section 0.50F _y Net Section 0.46F _y	26,000	30,000	30,000	30,000	35,000	45,000	36,000
					NOT APPLICABLE			
Axial tension in members without holes. Axial compression, gross section: stiffeners of plate girders. Compression in splice material, gross section	0.55F _y	14,000	16,000	18,000	20,000	24,000	30,000	25,000
Compression in extreme fibers of rolled shapes, girders and built-up sections, subject to bending, gross section, when compression flange is.	0.55F _y	14,000	16,000	18,000	20,000	24,000	30,000	25,000
(A) Supported laterally its full length by embedment in concrete								
(B) Partially supported or unsupported ⁽²⁾								

$$F_b = \frac{91 \times 10^6 C_b}{(F.S.) S_x} \left(\frac{l_x}{l} \right) \sqrt{0.772 \frac{I}{I_{yc}} + 9.87 \left(\frac{d}{l} \right)^2} \leq 0.55F_y$$

$C_b = 1.75 + 1.05 (M_1 / M_2) + 0.3 (M_1 / M_2)^2 \leq 2.3$ where M_1 is the smaller and M_2 is the larger end moment in the unbraced segment of the beams; M_1 / M_2 is positive when the moments cause reverse curvature and negative when bent in single curvature.

$C_b = 1.0$ for unbraced cantilevers and for members where the moment within a significant portion of the unbraced segment is greater than or equal to the larger of the segment end moments.

F.S. = Factor of Safety at Inventory Level = 1.82

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Do Not Use This Run For Ratings.
SEE NEXT RUN W/ REDUCED BOT. CHORD
FOR RATINGS.

BRIDGE ANALYSIS AND RATING (BAR7) 330443
PROGRAM P4353000
VERSION 7.12.0.0
LAST UPDATED 01/12/2007 05/20/2008 11:47
DOCUMENTATION 01/2007

INPUT: RATING-1.DAT

STRUCTURE ID -

BRG SLC LIVE OUT- IMP GAGE PASS FAT- CONC RE- S OVER END
TYPE LEV LANES LOAD PUT FACT DIST IGUE DECK SPEC DIST DIR FACTOR PAN HYB
TFS D H 2 0.00 N 0.00

BRIDGE CROSS SECTION AND LOADING

OVERHANG CL OF DECK OR GIRDER OR ROADWAY DISTRIBUTION FACTORS
WIDTH SPACING TRUSS TO CURB WIDTH SHEAR MOMENT DEFLECT
20.67 0.00 -1.00 18.00 0.625 0.727 0.670

SLAB DEAD LOADS
THICKNESS HAUNCH DL1 DL2 F'C N SYMMETRY
10.00 0.00 0.000 0.000 0.000 0.

LIVE LOAD NO. OF END HINGE TEMP END
LOCATION PANELS COND. CORS AT CHANGE BRG.
U 48 R L37 0. U

STRINGER FLOORBEAM UNIT WEIGHT
DL1 DL1 DECK CONCRETE
0.020 0.130 150.

SPAN LENGTHS (CANTILEVER)

SPAN # 1 2 3 4
LENGTH 103.55 114.15 109.13 94.00

HINGE # 1 2 3
DISTANCE 128.6 192.7 351.8

TRAFFIC LANE LOCATIONS

LANE # 1 2 3 4 5 6
DIST
WIDTH
LL

STRINGER SPAN LENGTHS (SIMPLE)

SPAN # 1 2 3
LENGTH 9.50 9.50 9.50

20/97

STRINGER LOCATIONS

STRINGER # 1
DISTANCE 8.00

TRUSS GEOMETRY

PANEL NO.	PANEL WIDTH	H1	H2	VERTICAL POST	H3	PANEL TYPE
1	9.67	Y	11.67	0.00	-11.67	3
2	9.50	N	11.67	0.00	-11.67	5
3	9.50	Y	11.67	0.00	-11.67	6
4	9.50	N	11.67	0.00	-11.67	5
5	9.50	Y	11.67	0.00	-11.67	6
6	9.50	N	11.67	0.00	-11.67	5
7	9.50	Y	11.67	0.00	-11.67	6
8	9.50	N	11.67	0.00	-11.67	5
9	9.50	Y	11.67	0.00	-11.67	6
10	9.50	N	11.67	0.00	-11.67	5
11	8.38	Y	11.67	0.00	-11.67	6
12	8.33	Y	11.67	0.00	-11.67	5
13	8.33	Y	11.67	0.00	-11.67	6
14	8.33	Y	0.00	0.00	0.00	4
15	8.07	Y	9.00	0.00	-9.00	3
16	8.00	N	9.00	0.00	-9.00	5
17	8.00	Y	9.00	0.00	-9.00	6
18	8.01	N	9.00	0.00	-9.00	5
19	8.01	Y	9.00	0.00	-9.00	6
20	8.01	N	9.00	0.00	-9.00	5
21	8.00	Y	9.00	0.00	-9.00	6
22	8.07	Y	0.00	0.00	0.00	4
23	8.33	Y	11.67	0.00	-11.67	3
24	8.33	Y	11.67	0.00	-11.67	5
25	8.33	Y	11.67	0.00	-11.67	6
26	8.38	Y	11.67	0.00	-11.67	5
27	9.18	Y	11.67	0.00	-11.67	6
28	9.25	N	11.67	0.00	-11.67	5
29	9.25	Y	11.67	0.00	-11.67	6
30	9.25	N	11.67	0.00	-11.67	5
31	9.25	Y	11.67	0.00	-11.67	6
32	9.25	N	11.67	0.00	-11.67	5
33	9.25	Y	11.67	0.00	-11.67	6
34	9.25	N	11.67	0.00	-11.67	5
35	9.25	Y	11.67	0.00	-11.67	6
36	9.19	N	11.67	0.00	-11.67	5
37	8.38	Y	11.67	0.00	-11.67	6
38	8.33	N	11.67	0.00	-11.67	5
39	8.33	Y	11.67	0.00	-11.67	6
40	8.33	Y	0.00	0.00	0.00	4
41	8.63	Y	9.00	0.00	-9.00	3
42	8.63	N	9.00	0.00	-9.00	5
43	8.63	Y	9.00	0.00	-9.00	6
44	8.63	N	9.00	0.00	-9.00	5
45	8.63	Y	9.00	0.00	-9.00	6
46	8.62	N	9.00	0.00	-9.00	5
47	8.62	Y	9.00	0.00	-9.00	6
48	8.62	Y	0.00	0.00	0.00	4

TRUSS DEAD LOADS

LOCAT LOAD LOCAT LOAD LOCAT LOAD LOCAT

TRUSS MEMBER PROPERTIES

MEMBER ID	GROSS AREA COMER	NET TENS	MOMENT OF INERTIA	FY	L	ECC	FLANGE WIDTH	C	CAT	FU	GROSS AREA TENS
U 0U1	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	23.89
U 1U2	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	23.89
U 2U3	46.47	36.97	745.0	33.0	0.00	0.00	0.00	0.00	0.00	0.0	43.94
U 3U4	46.47	36.97	745.0	33.0	0.00	0.00	0.00	0.00	0.00	0.0	43.94
U 4U5	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00	0.00	0.0	48.96
U 5U6	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00	0.00	0.0	48.96
U 6U7	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00	0.00	0.0	48.96
U 7U8	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00	0.00	0.0	48.96
U 8U9	41.85	33.35	660.1	33.0	0.00	0.00	0.00	0.00	0.00	0.0	39.63
U 9U10	41.85	33.35	660.1	33.0	0.00	0.00	0.00	0.00	0.00	0.0	39.63
U10U11	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00	0.00	0.0	48.96
U11U12	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00	0.00	0.0	48.96
U12U13	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	23.89
U13U14	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	23.89
U14U15	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	28.30
U15U16	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	28.30
U16U17	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	28.30
U17U18	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	28.30
U18U19	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	28.30
U19U20	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	28.30
U20U21	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	28.30
U21U22	30.26	23.76	419.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	28.30
U22U23	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	23.89
U23U24	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	23.89
U24U25	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00	0.00	0.0	48.96
U25U26	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00	0.00	0.0	48.96
U26U27	41.85	33.35	660.1	33.0	0.00	0.00	0.00	0.00	0.00	0.0	39.63
U27U28	41.85	33.35	660.1	33.0	0.00	0.00	0.00	0.00	0.00	0.0	39.63
U28U29	56.73	45.20	930.1	33.0	0.00	0.00	0.00	0.00	0.00	0.0	53.71
U29U30	56.73	45.20	930.1	33.0	0.00	0.00	0.00	0.00	0.00	0.0	53.71
U30U31	56.73	45.20	930.1	33.0	0.00	0.00	0.00	0.00	0.00	0.0	53.71
U31U32	56.73	45.20	930.1	33.0	0.00	0.00	0.00	0.00	0.00	0.0	53.71
U32U33	56.73	45.20	930.1	33.0	0.00	0.00	0.00	0.00	0.00	0.0	53.71
U33U34	56.73	45.20	930.1	33.0	0.00	0.00	0.00	0.00	0.00	0.0	53.71
U34U35	41.85	33.35	660.1	33.0	0.00	0.00	0.00	0.00	0.00	0.0	39.63
U35U36	41.85	33.35	660.1	33.0	0.00	0.00	0.00	0.00	0.00	0.0	39.63
U36U37	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00	0.00	0.0	48.96
U37U38	51.73	41.20	837.9	33.0	0.00	0.00	0.00	0.00	0.00	0.0	48.96
U38U39	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	23.89
U39U40	25.56	20.06	349.7	33.0	0.00	0.00	0.00	0.00	0.00	0.0	23.89
U40U41	34.99	27.49	491.8	33.0	0.00	0.00	0.00	0.00	0.00	0.0	32.74
U41U42	34.99	27.49	491.8	33.0	0.00	0.00	0.00	0.00	0.00	0.0	32.74
U42U43	34.99	27.49	491.8	33.0	0.00	0.00	0.00	0.00	0.00	0.0	32.74
U43U44	34.99	27.49	491.8	33.0	0.00	0.00	0.00	0.00	0.00	0.0	32.74

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NOTE: IF A SECTION DOES NOT MEET FLANGE OR WEB BUCKLING CRITERIA OF CURRENT AASHTO SPECIFICATIONS FOR LOAD FACTOR METHOD, THE RATING FACTORS ARE REPRINTED AS 88.88. THIS INDICATES THAT THERE IS A POTENTIAL FATIGUE PROBLEM.

NOTE: THE ALLOWABLE STRESS IN AN ECCENTRICALLY LOADED COMPRESSION MEMBER IS COMPUTED BY THE SECANT FORMULA. SEE ARTICLE 5.4.2 OF THE 1978 AASHTO MANUAL FOR MAINTENANCE INSPECTION OF BRIDGES.

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* TRUSS - LIVE LOAD H2O *
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MEMBER FORCES AND RATINGS
I.F. = IMPACT FACTOR

MEMBER ID	DL FORCE	LL + IMPACT FORCE		ALLOWABLE FORCE		RATING FACTORS
		COMP	I.F.	COMP	TENS	
U 0U 1	-132.0	-46.8L	1.22	11.3L 1.21	IR -391.5 430.0 OR -488.2 585.3	IR 5.542 OR 7.608
U 1U 2	-132.0	-46.8L	1.22	11.3L 1.21	IR -391.7 430.0 OR -488.5 585.3	IR 5.547 OR 7.614
U 2U 3	-283.2	-110.7L	1.22	33.5L 1.21	IR -713.8 790.9 OR -890.2 1076.5	IR 3.889 OR 5.482
U 3U 4	-283.2	-110.7L	1.22	33.5L 1.21	IR -713.8 790.9 OR -890.2 1076.5	IR 3.889 OR 5.482
U 4U 5	-304.2	-137.4L	1.22	55.7L 1.21	IR -794.8 881.3 OR -991.1 1199.5	IR 3.571 OR 5.000
U 5U 6	-304.2	-137.4L	1.22	55.7L 1.21	IR -794.8 881.3 OR -991.1 1199.5	IR 3.571 OR 5.000
U 6U 7	-194.8	-126.8L	1.22	78.0L 1.21	IR -794.8 881.3 OR -991.1 1199.5	IR 4.731 OR 6.278

LATERAL BRACE POINTS AND STIFFENER SPACINGS

B OR S G OR F	CODE	SPAN	C			C			C				
			E	SPCS	SPACING	D	OF	NO.	E	SPCS	SPACING	D	OF
BF	1	C	1	16.00	B	2	8.00	0	0.00	0	0.00	0	0.00
		C	1	16.00	B	2	8.00	0	0.00	0	0.00	0	0.00
BF	2	C	1	16.00	B	2	8.00	0	0.00	0	0.00	0	0.00
		C	1	16.00	B	2	8.00	0	0.00	0	0.00	0	0.00
BF	3	C	1	16.00	B	2	8.00	0	0.00	0	0.00	0	0.00
		C	1	16.00	B	2	8.00	0	0.00	0	0.00	0	0.00

DEFAULT VALUES

[illegible]

LIVE LOAD DISTRIBUTION FACTORS

TRUSS	TRUSS
FORCE	DEFLECTION
1.000 (2)	1.000 (2)

U15U16	-107.8	-36.3L 1.21	0.0	1.21	IR	-581.0	693.4	OR	13.042
					IR	-466.0	509.4	IR	9.872
					OR	-581.1	693.4	OR	13.045
U16U17	-223.5	-77.7L 1.21	0.0	1.21	IR	-466.0	509.4	IR	3.120
					OR	-581.1	693.4	OR	4.601
U17U18	-223.5	-77.7L 1.21	0.0	1.21	IR	-465.9	509.4	IR	3.120
					OR	-581.1	693.4	OR	4.601
U18U19	-223.7	-77.8L 1.21	0.0	1.21	IR	-465.9	509.4	IR	3.114
					OR	-581.1	693.4	OR	4.594
U19U20	-223.7	-77.8L 1.21	0.0	1.21	IR	-465.9	509.4	IR	3.114
					OR	-581.1	693.4	OR	4.594
U20U21	-108.3	-36.5L 1.21	0.0	1.21	IR	-466.0	509.4	IR	9.810
					OR	-581.1	693.4	OR	12.967
U21U22	-108.3	-36.5L 1.21	0.0	1.21	IR	-465.9	509.4	IR	9.807
					OR	-581.0	693.4	OR	12.964
U22U23	101.0	0.0 1.21	35.6L 1.21	IR	-393.2	430.0	IR	9.234	
					OR	-490.3	585.3	OR	13.592
U23U24	101.0	0.0 1.21	35.6L 1.21	IR	-393.2	430.0	IR	9.234	
					OR	-490.3	585.3	OR	13.592
U24U25	390.7	0.0 1.21	120.7L 1.21	IR	-797.2	881.3	IR	4.065	
					OR	-994.1	1199.5	OR	6.702
U25U26	390.7	0.0 1.21	120.7L 1.21	IR	-797.1	881.3	IR	4.065	
					OR	-994.0	1199.5	OR	6.702
U26U27	101.4	-81.1L 1.21	121.3L 1.21	IR	-643.3	713.3	IR	5.044	
					OR	-802.2	970.9	OR	7.168
U27U28	101.4	-81.1L 1.21	121.3L 1.21	IR	-643.2	713.3	IR	5.044	
					OR	-802.1	970.9	OR	7.168
U28U29	-80.5	-132.9L 1.21	122.0L 1.21	IR	-872.3	966.8	IR	5.959	
					OR	-1087.8	1315.9	OR	7.581
U29U30	-80.5	-132.9L 1.21	122.0L 1.21	IR	-872.3	966.8	IR	5.959	
					OR	-1087.8	1315.9	OR	7.581
U30U31	-138.8	-150.1L 1.21	122.6L 1.21	IR	-872.3	966.8	IR	4.885	
					OR	-1087.8	1315.9	OR	6.321
U31U32	-138.8	-150.1L 1.21	123.6L 1.22	IR	-872.3	966.8	IR	4.885	
					OR	-1087.8	1315.9	OR	6.321
U32U33	-73.4	-132.9L 1.21	124.3L 1.22	IR	-872.3	966.8	IR	6.012	
					OR	-1087.8	1315.9	OR	7.633
U33U34	-73.4	-132.9L 1.21	124.3L 1.22	IR	-872.3	966.8	IR	6.012	
					OR	-1087.8	1315.9	OR	7.633
U34U35	115.6	-81.1L 1.21	124.9L 1.22	IR	-643.2	713.3	IR	4.785	
					OR	-802.1	970.9	OR	6.847
U35U36	115.6	-81.1L 1.21	124.9L 1.22	IR	-643.3	713.3	IR	4.785	
					OR	-802.2	970.9	OR	6.847

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U36U37	411.8	0.0	1.21	125.6L 1.22	IR	-797.1	881.3	IR	3.739
					OR	-994.0	1199.5	OR	6.274
U37U38	411.8	0.0	1.23	125.6L 1.22	IR	-797.2	881.3	IR	3.739
					OR	-994.1	1199.5	OR	6.274
U39U39	107.8	0.0	1.23	37.1L 1.22	IR	-393.2	430.0	IR	8.673
					OR	-490.3	585.3	OR	12.854
U39U40	107.8	0.0	1.23	37.1L 1.22	IR	-393.2	430.0	IR	8.673
					OR	-490.3	585.3	OR	12.854
U40U41	-124.9	-41.4L 1.23	0.0	1.22	IR	-537.9	589.3	IR	9.975
					OR	-670.8	802.1	OR	13.185
U41U42	-124.9	-41.4L 1.23	0.0	1.22	IR	-537.9	589.3	IR	9.975
					OR	-670.8	802.1	OR	13.185
U42U43	-259.0	-88.6L 1.23	0.0	1.22	IR	-537.9	589.3	IR	3.150
					OR	-670.8	802.1	OR	4.650
U43U44	-259.0	-88.6L 1.23	0.0	1.22	IR	-537.9	589.3	IR	3.150
					OR	-670.8	802.1	OR	4.650
U44U45	-258.8	-88.5L 1.23	0.0	1.22	IR	-537.9	589.3	IR	3.154
					OR	-670.8	802.1	OR	4.656
U45U46	-258.8	-88.5L 1.23	0.0	1.22	IR	-537.9	589.3	IR	3.154
					OR	-670.8	802.1	OR	4.656
U46U47	-124.6	-41.3L 1.23	0.0	1.22	IR	-537.9	589.3	IR	10.014
					OR	-670.8	802.1	OR	13.234
U47U48	-124.6	-41.3L 1.23	0.0	1.22	IR	-537.9	589.3	IR	10.014
					OR	-670.8	802.1	OR	13.234
L 1 L 2	217.0	-22.4L 1.21	83.4L 1.22	IR	-463.9	509.4	IR	3.505	
					OR	-578.5	693.4	OR	5.710
L 2 L 3	217.0	-22.4L 1.21	83.4L 1.22	IR	-463.9	509.4	IR	3.505	
					OR	-578.5	693.4	OR	5.710
L 3 L 4	303.0	-44.6L 1.21	128.7L 1.22	IR	-713.8	790.9	IR	3.791	
					OR	-890.2	1076.5	OR	6.010
L 4 L 5	303.0	-44.6L 1.21	128.7L 1.22	IR	-713.8	790.9	IR	3.791	
					OR	-890.2	1076.5	OR	6.010
L 5 L 6	258.9	-66.9L 1.21	136.7L 1.22	IR	-713.8	790.9	IR	3.891	
					OR	-890.2	1076.5	OR	5.979
L 6 L 7	258.9	-66.9L 1.21	136.7L 1.22	IR	-713.8	790.9	IR	3.891	
					OR	-890.2	1076.5	OR	5.979
L 7 L 8	84.4	-89.1L 1.21	107.6L 1.22	IR	-642.7	713.3	IR	5.847	
					OR	-801.5	970.9	OR	8.242
L 8 L 9	84.4	-89.1L 1.21	107.6L 1.22	IR	-642.7	713.3	IR	5.847	
					OR	-801.5	970.9	OR	8.242
L 9L10	-220.3	-111.3L 1.21	41.1L 1.22	IR	-642.7	713.3	IR	3.796	
					OR	-801.5	970.9	OR	5.223

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U12L13	235.4	0.0	1.00	83.3L 1.24	IR	-454.7	509.4	IR	3.289	OR	5.498	L33U34	137.5	-25.3L 1.30	65.7L 1.24	IR	-284.7	360.0	IR	3.387	OR	5.365		
L13U14	-175.0	-76.0L 1.25	0.0	1.00	IR	-419.7	463.7	IR	3.220	OR	4.585	U34L35	-166.8	-75.8L 1.23	22.4L 1.30	IR	-382.8	430.0	IR	2.849	OR	4.097		
U14L15	161.4	0.0	1.00	68.6L 1.26	IR	-343.6	371.5	IR	3.061	OR	5.016	L35U36	236.4	-21.0L 1.30	86.8L 1.23	IR	-453.5	509.4	IR	3.144	OR	5.262		
L15U16	-99.1	-55.9L 1.28	7.0	1.30	IR	-263.4	320.8	IR	2.939	OR	4.103	U36L37	-257.0	-95.5L 1.23	19.0L 1.23	IR	-526.0	589.3	IR	2.818	OR	4.179		
U16L17	75.1	-14.3	1.30	44.0L 1.29	IR	-178.2	227.7	IR	3.470	OR	5.340	L37U38	-275.7	-92.7L 1.23	0.0	1.00	IR	-601.3	681.5	IR	3.513	OR	5.116	
L17U18	-12.2	-32.9L 1.30	23.0	1.30	IR	-113.7	153.4	IR	3.090	OR	3.945	U38L39	247.5	0.0	1.00	85.2L 1.24	IR	-454.7	509.4	IR	3.073	OR	5.231	
U18L19	-11.9	-32.8L 1.30	23.1	1.30	IR	-113.7	153.4	IR	3.106	OR	3.963	L39U40	-185.6	-77.5L 1.25	0.0	1.00	IR	-454.7	509.4	IR	3.471	OR	4.919	
L19U20	74.8	-14.4	1.30	43.9L 1.29	IR	-178.2	227.7	IR	3.480	OR	5.352	U40L41	180.5	0.0	1.00	72.9L 1.25	IR	-342.7	371.5	IR	2.622	OR	4.464	
U20L21	-98.9	-55.8L 1.28	7.0	1.30	IR	-263.4	320.8	IR	2.948	OR	4.114	L41U42	-110.7	-59.3L 1.27	7.2	1.30	IR	-262.3	320.8	IR	2.555	OR	3.647	
L21U22	162.2	0.0	1.00	69.0L 1.26	IR	-343.6	371.5	IR	3.035	OR	4.980	U42L43	83.0	-15.1	1.30	46.5L 1.29	IR	-176.9	227.7	IR	3.110	OR	4.878	
U22L23	-173.9	-75.8L 1.25	0.0	1.00	IR	-454.7	509.4	IR	3.707	OR	5.189	L43U44	-13.7	-34.7L 1.30	24.1	1.30	IR	-112.1	153.4	IR	2.830	OR	3.627	
L23U24	235.2	0.0	1.00	83.3L 1.24	IR	-454.7	509.4	IR	3.294	OR	5.504	U44L45	-14.0	-34.8L 1.30	24.1	1.30	IR	-112.1	153.4	IR	2.818	OR	3.613	
U24L25	-263.4	-90.7L 1.23	0.0	1.00	IR	-601.3	681.5	IR	3.725	OR	5.363	L45U46	83.2	-15.1	1.30	46.6L 1.29	IR	-176.9	227.7	IR	3.103	OR	4.869	
L25U26	-251.5	-95.3L 1.23	19.4L 1.23	IR	-526.0	589.3	IR	2.880	OR	4.243	U46L47	-110.9	-59.4L 1.27	7.2	1.30	IR	-262.3	320.8	IR	2.551	OR	3.643		
U26L27	230.6	-21.6L 1.30	86.6L 1.23	IR	-453.5	509.4	IR	3.218	OR	5.342	L47U48	180.1	0.0	1.00	73.3L 1.26	IR	-342.7	371.5	IR	2.611	OR	4.442		
L27U28	-161.1	-75.3L 1.23	22.9L 1.30	IR	-382.8	430.0	IR	2.943	OR	4.199	U 1L 1	-55.0	-41.8L 1.30	0.0	1.00	IR	-148.8	201.1	IR	2.245	OR	3.124		
U28L29	131.8	-25.8L 1.30	65.2L 1.24	IR	-284.7	360.0	IR	3.503	OR	5.498	U 3L 3	-57.0	-41.7L 1.30	0.0	1.00	IR	-148.8	201.1	IR	2.200	OR	3.082		
L29U30	-61.6	-55.8L 1.26	32.6L 1.30	IR	-208.6	277.9	IR	2.636	OR	3.559	U 5L 5	-57.0	-41.7L 1.30	0.0	1.00	IR	-148.8	201.1	IR	2.200	OR	3.082		
U30L31	32.2	-39.8L 1.29	47.1L 1.27	IR	-117.8	174.6	IR	3.020	OR	4.357	U 7L 7	-57.0	-41.7L 1.30	0.0	1.00	IR	-148.8	201.1	IR	2.200	OR	3.082		
L31U32	37.9	-39.3L 1.29	47.7L 1.27	IR	-117.8	174.6	IR	2.865	OR	4.186	U 9L 9	-57.0	-41.7L 1.30	0.0	1.00	IR	-148.8	201.1	IR	2.200	OR	3.082		
U32L33	-67.3	-56.3L 1.26	32.0L 1.30	IR	-208.6	277.9	IR	2.508	OR	3.423	U10L10	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999	OR		
												U11L11	-50.0	-41.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.374	OR	3.258

26/97

DL FORCE	LL + IMPACT FORCE		ALLOWABLE FORCE		RATING	
	COMP	I.F.	TENS	I.F.	COMP	TENS
-101.4	-52.9L	1.27	23.3L	1.30	IR -207.9	277.9
					OR -259.3	378.3
					IR	2.015
					OR	2.987
					IR	40.29
					OR	59.73
SUPPORT REACTIONS						
	SUPPORT	D.L.	LL+I	I.F.		
	U 0	182.4	72.1	L 1.22		
	L11	503.2	132.1	L 1.21		
	L25	468.7	133.1	L 1.21		
	L37	483.2	136.5	L 1.22		
	U48	150.1	59.1	L 1.23		

* TRUSS - LIVE LOAD HS20 *						

MEMBER FORCES AND RATINGS						
I.F. = IMPACT FACTOR						
MEMBER ID	DL FORCE	LL + IMPACT FORCE		ALLOWABLE FORCE		RATING FACTORS
		COMP	I.F.	TENS	I.F.	
U 0U 1	-132.0	-59.4	1.22	15.0	1.21	IR -391.5
						OR -488.2
						IR 4.371
						OR 6.000
U 1U 2	-132.0	-59.4	1.22	15.0	1.21	IR -391.7
						OR -488.5
						IR 4.374
						OR 6.004
U 2U 3	-283.2	-136.5	1.22	44.3	1.21	IR -713.8
						OR -890.2
						IR 3.156
						OR 4.448
U 3U 4	-283.2	-136.5	1.22	44.3	1.21	IR -713.8
						OR -890.2
						IR 3.156
						OR 4.448
U 4U 5	-304.2	-165.6	1.22	73.7	1.21	IR -794.8
						OR -991.1
						IR 2.963
						OR 4.149
U 5U 6	-304.2	-165.6	1.22	73.7	1.21	IR -794.8
						OR -991.1
						IR 2.963
						OR 4.149
U 6U 7	-194.8	-154.4	1.22	103.1	1.21	IR -794.8
						OR -991.1
						IR 3.887
						OR 5.159
U 7U 8	-194.8	-154.4	1.22	103.1	1.21	IR -794.8
						OR -991.1
						IR 3.887
						OR 5.159
U 8U 9	44.7	-99.1	1.22	132.5	1.21	IR -642.7
						OR -801.5
						IR 5.045
						OR 6.989
U 9U10	44.7	-99.1	1.22	132.5	1.21	IR -642.7
						OR -801.5
						IR 5.045
						OR 6.989
U10U11	391.6	0.0	1.22	160.2	1.21	IR -797.1
						OR -994.0
						IR 3.057
						OR 5.044
U11U12	391.6	0.0	1.21	160.2	1.21	IR -797.2
						OR -994.1
						IR 3.057
						OR 5.044
U12U13	101.7	0.0	1.21	53.5	1.21	IR -393.2
						IR 6.135

MEMBER U 6L 7 IS CRITICAL. SEE DETAILS BELOW.

U13U14	101.7	0.0	1.21	53.5	1.21	IR	-490.3	585.3	OR	9.037	U34U35	115.6	-99.1	1.21	136.4	1.22	IR	-643.2	713.3	IR	4.388
						IR	-393.2	430.0	IR	6.135						OR	-802.1	970.9	OR	6.271	
U14U15	-107.8	-56.5	1.21	0.0	1.21	IR	-490.3	585.3	OR	9.037	U35U36	115.6	-99.1	1.21	136.4	1.22	IR	-643.3	713.3	IR	4.382
						IR	-465.9	509.4	IR	6.336						OR	-802.2	970.9	OR	6.271	
U15U16	-107.8	-56.5	1.21	0.0	1.21	IR	-581.0	693.4	OR	8.373	U36U37	411.8	0.0	1.21	162.6	1.22	IR	-797.1	881.3	IR	2.888
						IR	-466.0	509.4	IR	6.338						OR	-994.0	1199.5	OR	4.845	
U16U17	-223.5	-113.3	1.21	0.0	1.21	IR	-581.1	693.4	OR	8.375	U37U38	411.8	0.0	1.23	162.6	1.22	IR	-797.2	881.3	IR	2.888
						IR	-466.0	509.4	IR	2.140						OR	-994.1	1199.5	OR	4.845	
U17U18	-223.5	-113.3	1.21	0.0	1.21	IR	-581.1	693.4	OR	3.156	U39U39	107.8	0.0	1.23	54.1	1.22	IR	-393.2	430.0	IR	5.955
						IR	-465.9	509.4	IR	2.140						OR	-490.3	585.3	OR	8.825	
U18U19	-223.7	-113.4	1.21	0.0	1.21	IR	-581.1	693.4	OR	3.151	U39U40	107.8	0.0	1.23	54.1	1.22	IR	-393.2	430.0	IR	5.955
						IR	-465.9	509.4	IR	2.136						OR	-490.3	585.3	OR	8.825	
U19U20	-223.7	-113.4	1.21	0.0	1.21	IR	-581.1	693.4	OR	3.151	U40U41	-124.9	-62.9	1.23	0.0	1.22	IR	-537.9	589.3	IR	6.571
						IR	-465.9	509.4	IR	2.136						OR	-670.8	802.1	OR	8.685	
U20U21	-108.3	-56.8	1.21	0.0	1.21	IR	-581.1	693.4	OR	3.151	U41U42	-124.9	-62.9	1.23	0.0	1.22	IR	-537.9	589.3	IR	6.571
						IR	-465.9	509.4	IR	6.297						OR	-670.8	802.1	OR	8.685	
U21U22	-108.3	-56.8	1.21	0.0	1.21	IR	-581.1	693.4	OR	8.323	U42U43	-259.0	-126.5	1.23	0.0	1.22	IR	-537.9	589.3	IR	2.204
						IR	-465.9	509.4	IR	6.295						OR	-670.8	802.1	OR	3.254	
U22U23	101.0	0.0	1.21	53.2	1.21	IR	-581.0	693.4	OR	8.321	U43U44	-259.0	-126.5	1.23	0.0	1.22	IR	-537.9	589.3	IR	2.204
						IR	-393.2	430.0	IR	6.181						OR	-670.8	802.1	OR	3.254	
U23U24	101.0	0.0	1.21	53.2	1.21	IR	-490.3	585.3	OR	9.099	U44U45	-258.8	-126.5	1.23	0.0	1.22	IR	-537.9	589.3	IR	2.207
						IR	-393.2	430.0	IR	6.181						OR	-670.8	802.1	OR	3.258	
U24U25	390.7	0.0	1.21	159.6	1.21	IR	-490.3	585.3	OR	9.099	U45U46	-258.8	-126.5	1.23	0.0	1.22	IR	-537.9	589.3	IR	2.207
						IR	-797.2	881.3	IR	3.074						OR	-670.8	802.1	OR	3.258	
U25U26	390.7	0.0	1.21	159.6	1.21	IR	-994.1	1199.5	OR	5.068	U46U47	-124.6	-62.7	1.23	0.0	1.22	IR	-537.9	589.3	IR	6.595
						IR	-797.1	881.3	IR	3.074						OR	-670.8	802.1	OR	8.716	
U26U27	101.4	-99.1	1.21	133.9	1.21	IR	-994.0	1199.5	OR	5.068	U47U48	-124.6	-62.7	1.23	0.0	1.22	IR	-537.9	589.3	IR	6.595
						IR	-643.3	713.3	IR	4.569						OR	-670.8	802.1	OR	8.716	
U27U28	101.4	-99.1	1.21	133.9	1.21	IR	-802.2	970.9	OR	6.492	L 1L 2	217.0	-29.7	1.21	104.5	1.22	IR	-463.9	509.4	IR	2.799
						IR	-643.2	713.3	IR	4.569						OR	-578.5	693.4	OR	4.560	
U28U29	-80.5	-157.7	1.21	122.0L	1.21	IR	-802.1	970.9	OR	6.492	L 2L 3	217.0	-29.7	1.21	104.5	1.22	IR	-463.9	509.4	IR	2.799
						IR	-872.3	966.8	IR	5.021						OR	-578.5	693.4	OR	4.560	
U29U30	-80.5	-157.7	1.21	122.0L	1.21	IR	-1087.8	1315.9	OR	6.388	L 3L 4	303.0	-59.0	1.21	156.6	1.22	IR	-713.8	790.9	IR	3.116
						IR	-872.3	966.8	IR	5.021						OR	-890.2	1076.5	OR	4.940	
U30U31	-138.8	-175.2	1.21	122.6L	1.21	IR	-1087.8	1315.9	OR	6.388	L 4L 5	303.0	-59.0	1.21	156.6	1.22	IR	-713.8	790.9	IR	3.116
						IR	-872.3	966.8	IR	4.188						OR	-890.2	1076.5	OR	4.940	
U31U32	-138.8	-175.2	1.21	123.6L	1.22	IR	-1087.8	1315.9	OR	5.418	L 5L 6	258.9	-88.4	1.21	165.1	1.22	IR	-713.8	790.9	IR	3.222
						IR	-872.3	966.8	IR	4.188						OR	-890.2	1076.5	OR	4.952	
U32U33	-73.4	-157.7	1.21	124.3L	1.22	IR	-1087.8	1315.9	OR	5.418	L 6L 7	258.9	-88.4	1.21	165.1	1.22	IR	-713.8	790.9	IR	3.222
						IR	-872.3	966.8	IR	5.066						OR	-890.2	1076.5	OR	4.952	
U33U34	-73.4	-157.7	1.21	124.3L	1.22	IR	-1087.8	1315.9	OR	6.432	L 7L 8	84.4	-117.8	1.21	132.9	1.22	IR	-642.7	713.3	IR	4.733
						IR	-872.3	966.8	IR	5.066						OR	-801.5	970.9	OR	6.671	

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L 8L 9	84.4	-117.8	1.21	132.9	1.22	IR	-642.7	713.3	IR	4.733
						OR	-801.5	970.9	OR	6.671
L 9L10	-220.3	-147.2	1.21	52.2	1.22	IR	-642.7	713.3	IR	2.870
						OR	-801.5	970.9	OR	3.948
L10L11	-220.3	-147.2	1.21	52.2	1.22	IR	-644.7	713.3	IR	2.883
						OR	-803.9	970.9	OR	3.965
L11L12	-238.5	-106.8	1.21	0.0	1.21	IR	-644.7	713.3	IR	3.802
						OR	-804.0	970.9	OR	5.293
L12L13	-238.5	-106.8	1.21	0.0	1.21	IR	-644.7	713.3	IR	3.802
						OR	-804.0	970.9	OR	5.293
L15L16	173.7	0.0	1.21	93.6	1.21	IR	-466.0	509.4	IR	3.586
						OR	-581.1	693.4	OR	5.550
L16L17	173.7	0.0	1.21	93.6	1.21	IR	-466.0	509.4	IR	3.586
						OR	-581.1	693.4	OR	5.550
L17L18	231.6	0.0	1.21	117.4	1.21	IR	-465.9	509.4	IR	2.366
						OR	-581.1	693.4	OR	3.933
L18L19	231.6	0.0	1.21	117.4	1.21	IR	-465.9	509.4	IR	2.366
						OR	-581.1	693.4	OR	3.933
L19L20	174.0	0.0	1.21	93.8	1.21	IR	-465.9	509.4	IR	3.576
						OR	-581.1	693.4	OR	5.537
L20L21	174.0	0.0	1.21	93.8	1.21	IR	-466.0	509.4	IR	3.576
						OR	-581.1	693.4	OR	5.537
L23L24	-237.7	-106.4	1.21	0.0	1.21	IR	-644.7	713.3	IR	3.825
						OR	-804.0	970.9	OR	5.322
L24L25	-237.7	-106.4	1.21	0.0	1.21	IR	-644.7	713.3	IR	3.825
						OR	-804.0	970.9	OR	5.322
L25L26	-244.0	-147.4	1.21	52.6	1.21	IR	-644.7	713.3	IR	2.719
						OR	-803.9	970.9	OR	3.800
L26L27	-244.0	-147.4	1.21	52.6	1.21	IR	-643.3	713.3	IR	2.710
						OR	-802.2	970.9	OR	3.788
L27L28	-1.3	-121.6L	1.21	134.3	1.21	IR	-643.2	713.3	IR	5.277
						OR	-802.1	970.9	OR	6.583
L28L29	-1.3	-121.6L	1.21	134.3	1.21	IR	-643.2	713.3	IR	5.277
						OR	-802.1	970.9	OR	6.583
L29L30	118.8	-122.3L	1.21	172.2	1.21	IR	-714.3	790.9	IR	3.902
						OR	-890.8	1076.5	OR	5.561
L30L31	118.8	-122.3L	1.21	172.2	1.21	IR	-714.3	790.9	IR	3.902
						OR	-890.8	1076.5	OR	5.561
L31L32	115.2	-123.9L	1.22	172.2	1.21	IR	-714.3	790.9	IR	3.923
						OR	-890.8	1076.5	OR	5.581
L32L33	115.2	-123.9L	1.22	172.2	1.21	IR	-714.3	790.9	IR	3.923
						OR	-890.8	1076.5	OR	5.581
L33L34	-12.0	-124.6L	1.22	134.3	1.21	IR	-643.2	713.3	IR	5.066

U10L11	-293.8	-93.0L 1.23	0.0	1.00	IR OR	-601.2 -749.7	681.5 927.6	IR OR	3.305 4.902	L31U32	37.9	-39.3L 1.29	48.4	1.27	IR OR	-117.8 -146.9	174.6 237.6	IR OR	2.824 4.127
L11U12	-263.7	-107.0 1.23	0.0	1.00	IR OR	-601.3 -749.8	681.5 927.6	IR OR	3.156 4.545	U32L33	-67.3	-57.6 1.26	32.0L 1.30	IR OR	-208.6 -260.1	277.9 378.3	IR OR	2.452 3.346	
U12L13	235.4	0.0 1.00	101.9	1.24	IR OR	-454.7 -567.1	509.4 693.4	IR OR	2.688 4.493	L33U34	137.5	-25.3L 1.30	66.7 1.24	IR OR	-284.7 -355.1	360.0 490.0	IR OR	3.334 5.281	
L13U14	-175.0	-95.1 1.25	0.0	1.00	IR OR	-419.7 -523.4	463.7 631.1	IR OR	2.573 3.663	U34L35	-166.8	-75.8L 1.23	23.4 1.30	IR OR	-382.8 -477.4	430.0 585.3	IR OR	2.849 4.097	
U14L15	161.4	0.0 1.00	88.5	1.26	IR OR	-343.6 -428.5	371.5 505.7	IR OR	2.373 3.889	L35U36	236.4	-23.3 1.30	86.8L 1.23	IR OR	-453.5 -565.6	509.4 693.4	IR OR	3.144 5.262	
L15U16	-99.1	-74.4 1.28	7.0	1.30	IR OR	-263.4 -328.5	320.8 436.6	IR OR	2.208 3.082	U36L37	-257.0	-95.5L 1.23	21.4 1.23	IR OR	-526.0 -655.9	589.3 802.1	IR OR	2.818 4.179	
U16L17	75.1	-15.7 1.30	59.8	1.29	IR OR	-178.2 -222.2	227.7 309.9	IR OR	2.554 3.931	L37U38	-275.7	-106.7 1.23	0.0 1.00	IR OR	-601.3 -749.8	681.5 927.6	IR OR	3.052 4.445	
L17U18	-12.2	-44.5 1.30	29.6	1.30	IR OR	-113.7 -141.8	153.4 208.7	IR OR	2.284 2.916	U38L39	247.5	0.0 1.00	102.0 1.24	IR OR	-454.7 -567.1	509.4 693.4	IR OR	2.569 4.372	
U18L19	-11.9	-44.4 1.30	29.7	1.30	IR OR	-113.7 -141.8	153.4 208.7	IR OR	2.296 2.929	L39U40	-185.6	-95.1 1.25	0.0 1.00	IR OR	-454.7 -567.1	509.4 693.4	IR OR	2.829 4.010	
L19U20	74.8	-15.7 1.30	59.7	1.29	IR OR	-178.2 -222.2	227.7 309.9	IR OR	2.562 3.940	U40L41	180.5	0.0 1.00	92.2 1.25	IR OR	-342.7 -427.4	371.5 505.7	IR OR	2.072 3.527	
U20L21	-98.9	-74.3 1.28	7.0	1.30	IR OR	-263.4 -328.5	320.8 436.6	IR OR	2.214 3.090	L41U42	-110.7	-77.9 1.27	7.2 1.30	IR OR	-262.3 -327.1	320.8 436.6	IR OR	1.945 2.776	
L21U22	162.2	0.0 1.00	89.0	1.26	IR OR	-343.6 -428.5	371.5 505.7	IR OR	2.353 3.860	U42L43	83.0	-17.2 1.30	62.8 1.29	IR OR	-176.9 -220.6	227.7 309.9	IR OR	2.303 3.611	
U22L23	-173.9	-94.8 1.25	0.0	1.00	IR OR	-454.7 -567.1	509.4 693.4	IR OR	2.963 4.148	L43U44	-13.7	-47.3 1.30	31.6 1.30	IR OR	-112.1 -139.8	153.4 208.7	IR OR	2.081 2.667	
L23U24	235.2	0.0 1.00	101.9	1.24	IR OR	-454.7 -567.1	509.4 693.4	IR OR	2.692 4.497	U44L45	-14.0	-47.4 1.30	31.5 1.30	IR OR	-112.1 -139.8	153.4 208.7	IR OR	2.072 2.657	
U24L25	-263.4	-107.0 1.23	0.0	1.00	IR OR	-601.3 -749.8	681.5 927.6	IR OR	3.159 4.547	L45U46	83.2	-17.1 1.30	62.9 1.29	IR OR	-176.9 -220.6	227.7 309.9	IR OR	2.298 3.605	
L25U26	-251.5	-95.3L 1.23	21.5	1.23	IR OR	-526.0 -655.9	589.3 802.1	IR OR	2.880 4.243	U46L47	-110.9	-78.0 1.27	7.2 1.30	IR OR	-262.3 -327.1	320.8 436.6	IR OR	1.942 2.774	
U26L27	230.6	-23.6 1.30	86.6L 1.23	IR	-453.5	509.4	IR	3.218	L47U48	180.1	0.0 1.00	92.8 1.26	IR	-342.7	371.5	IR	2.063		
					OR	-565.6	693.4	OR	5.342					OR	-427.4	505.7	OR	3.509	
L27U28	-161.1	-75.8 1.23	23.6	1.30	IR OR	-382.8 -477.4	430.0 585.3	IR OR	2.925 4.173	U 1L 1	-55.0	-41.8L 1.30	0.0 1.00	IR OR	-148.8 -185.5	201.1 273.7	IR OR	2.245 3.124	
U28L29	131.8	-25.8L 1.30	66.8 1.24	IR OR	-284.7 -355.1	360.0 490.0	IR OR	3.419 5.366	U 3L 3	-57.0	-41.7L 1.30	0.0 1.00	IR OR	-148.8 -185.5	201.1 273.7	IR OR	2.200 3.082		
L29U30	-61.6	-57.6 1.26	32.6L 1.30	IR OR	-208.6 -260.1	277.9 378.3	IR OR	2.550 3.444	U 5L 5	-57.0	-41.7L 1.30	0.0 1.00	IR OR	-148.8 -185.5	201.1 273.7	IR OR	2.200 3.082		
U30L31	32.2	-39.8L 1.29	48.4 1.27	IR OR	-117.8 -146.9	174.6 237.6	IR OR	2.942 4.244	U 7L 7	-57.0	-41.7L 1.30	0.0 1.00	IR OR	-148.8 -185.5	201.1 273.7	IR OR	2.200 3.082		
									U 9L 9	-57.0	-41.7L 1.30	0.0 1.00	IR OR	-148.8 -185.5	201.1 273.7	IR OR	2.200 3.082		

U10L10	0.0	0.0	1.00	0.0	1.00	IR	-185.5	273.7	OR	3.082
U11L11	-50.0	-41.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	999.999
U12L12	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	2.374
U13L13	-50.0	-41.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	3.258
U15L15	-47.0	-41.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	999.999
U17L17	-47.0	-41.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	2.280
U19L19	-47.0	-41.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	3.122
U21L21	-47.0	-41.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	2.280
U23L23	-50.0	-41.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	3.122
U24L24	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	2.280
U25L25	-50.0	-41.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	3.122
U26L26	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	2.280
U27L27	-55.0	-41.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	3.122
U29L29	-55.0	-41.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.254
U31L31	-55.0	-41.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	3.138
U33L33	-55.0	-41.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.254
U35L35	-55.0	-41.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	3.138
U37L37	-50.0	-41.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.254
U39L39	-50.0	-41.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	3.138
U41L41	-50.0	-41.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	2.280
U43L43	-50.0	-41.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	3.050
U45L45	-50.0	-41.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	2.208
U47L47	-50.0	-41.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	3.050

U47L47	-50.0	-41.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	2.208
						OR	-176.9	237.6	OR	3.050

MEMBER U46L47 IS CRITICAL. SEE DETAILS BELOW.

DL FORCE	LL + IMPACT COMP	LL + IMPACT I.F.	TENS	I.F.	ALLOWABLE FORCE COMP	TENS	RATING FACTOR	TONS		
-110.9	-78.0	1.27	7.2	1.30	IR	-262.3	320.8	IR	1.942	69.92
					OR	-327.1	436.6	OR	2.774	99.85

SUPPORT REACTIONS

SUPPORT	D.L.	LL+I	I.F.
U 0	182.4	79.8	1.22
L11	503.2	132.1	L 1.21
L25	468.7	133.1	L 1.21
L37	483.2	136.5	L 1.22
U48	150.1	76.5	1.23

* TRUSS - LIVE LOAD TK527 *

MEMBER FORCES AND RATINGS
I.F. = IMPACT FACTOR

MEMBER ID	DL FORCE	LL + IMPACT COMP	LL + IMPACT I.F.	TENS	I.F.	ALLOWABLE FORCE COMP	TENS	RATING FACTOR		
U 0U 1	-132.0	-67.4	1.22	17.0	1.21	IR	-391.5	430.0	IR	3.849
						OR	-488.2	585.3	OR	5.284
U 1U 2	-132.0	-67.4	1.22	17.0	1.21	IR	-391.7	430.0	IR	3.853
						OR	-488.5	585.3	OR	5.288
U 2U 3	-283.2	-155.5	1.22	50.5	1.21	IR	-713.8	790.9	IR	2.769
						OR	-890.2	1076.5	OR	3.903
U 3U 4	-283.2	-155.5	1.22	50.5	1.21	IR	-713.8	790.9	IR	2.769
						OR	-890.2	1076.5	OR	3.903
U 4U 5	-304.2	-189.0	1.22	84.0	1.21	IR	-794.8	881.3	IR	2.596
						OR	-991.1	1199.5	OR	3.635
U 5U 6	-304.2	-189.0	1.22	84.0	1.21	IR	-794.8	881.3	IR	2.596
						OR	-991.1	1199.5	OR	3.635
U 6U 7	-194.8	-177.3	1.22	117.5	1.21	IR	-794.8	881.3	IR	3.383
						OR	-991.1	1199.5	OR	4.490
U 7U 8	-194.8	-177.3	1.22	117.5	1.21	IR	-794.8	881.3	IR	3.383
						OR	-991.1	1199.5	OR	4.490
U 8U 9	44.7	-112.4	1.22	151.0	1.21	IR	-642.7	713.3	IR	4.429
						OR	-801.5	970.9	OR	6.135
U 9U10	44.7	-112.4	1.22	151.0	1.21	IR	-642.7	713.3	IR	4.429
						OR	-801.5	970.9	OR	6.135
U10U11	391.6	0.0	1.22	182.5	1.21	IR	-797.1	881.3	IR	2.683

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U11U12	391.6	0.0	1.21	182.5	1.21	IR	-994.0	1199.5	OR	4.427
U12U13	101.7	0.0	1.21	60.5	1.21	IR	-797.2	881.3	IR	2.683
U13U14	101.7	0.0	1.21	60.5	1.21	IR	-994.1	1199.5	OR	4.427
U14U15	-107.8	-63.8	1.21	0.0	1.21	IR	-393.2	430.0	IR	5.423
U15U16	-107.8	-63.8	1.21	0.0	1.21	IR	-490.3	585.3	OR	7.989
U16U17	-223.5	-130.3	1.21	0.0	1.21	IR	-393.2	430.0	IR	5.423
U17U18	-223.5	-130.3	1.21	0.0	1.21	IR	-490.3	585.3	OR	7.989
U18U19	-223.7	-130.4	1.21	0.0	1.21	IR	-465.9	509.4	IR	5.616
U19U20	-223.7	-130.4	1.21	0.0	1.21	IR	-581.0	693.4	OR	7.422
U20U21	-108.3	-64.1	1.21	0.0	1.21	IR	-466.0	509.4	IR	5.618
U21U22	-108.3	-64.1	1.21	0.0	1.21	IR	-581.1	693.4	OR	7.423
U22U23	101.0	0.0	1.21	60.2	1.21	IR	-466.0	509.4	IR	1.861
U23U24	101.0	0.0	1.21	60.2	1.21	IR	-581.1	693.4	OR	2.744
U24U25	390.7	0.0	1.21	181.9	1.21	IR	-465.9	509.4	IR	1.858
U25U26	390.7	0.0	1.21	181.9	1.21	IR	-581.1	693.4	OR	2.741
U26U27	101.4	-112.5	1.21	152.6	1.21	IR	-465.9	509.4	IR	1.858
U27U28	101.4	-112.5	1.21	152.6	1.21	IR	-581.1	693.4	OR	2.741
U28U29	-80.5	-181.3	1.21	121.8	1.21	IR	-466.0	509.4	IR	5.581
U29U30	-80.5	-181.3	1.21	121.8	1.21	IR	-581.1	693.4	OR	7.378
U30U31	-138.8	-199.5	1.21	91.7	1.21	IR	-465.9	509.4	IR	5.580
U31U32	-138.8	-199.5	1.21	92.4	1.22	IR	-581.0	693.4	OR	7.376

U32U33	-73.4	-181.4	1.21	123.7	1.22	IR	-872.3	966.8	IR	4.405
U33U34	-73.4	-181.4	1.21	123.7	1.22	IR	-1087.8	1315.9	OR	5.593
U34U35	115.6	-112.5	1.21	155.0	1.22	IR	-872.3	966.8	IR	4.405
U35U36	115.6	-112.5	1.21	155.0	1.22	IR	-1087.8	1315.9	OR	5.593
U36U37	411.8	0.0	1.21	184.7	1.22	IR	-643.2	713.3	IR	3.856
U37U38	411.8	0.0	1.23	184.7	1.22	IR	-802.1	970.9	OR	5.518
U39U39	107.8	0.0	1.23	61.2	1.22	IR	-643.3	713.3	IR	3.856
U39U40	107.8	0.0	1.23	61.2	1.22	IR	-802.2	970.9	OR	5.518
U40U41	-124.9	-71.0	1.23	0.0	1.22	IR	-797.1	881.3	IR	2.541
U41U42	-124.9	-71.0	1.23	0.0	1.22	IR	-994.0	1199.5	OR	4.264
U42U43	-259.0	-145.4	1.23	0.0	1.22	IR	-797.2	881.3	IR	2.541
U43U44	-259.0	-145.4	1.23	0.0	1.22	IR	-994.1	1199.5	OR	4.264
U44U45	-258.8	-145.3	1.23	0.0	1.22	IR	-393.2	430.0	IR	5.262
U45U46	-258.8	-145.3	1.23	0.0	1.22	IR	-490.3	585.3	OR	7.798
U46U47	-124.6	-70.8	1.23	0.0	1.22	IR	-537.9	589.3	IR	5.817
U47U48	-124.6	-70.8	1.23	0.0	1.22	IR	-670.8	802.1	OR	7.689
L 1L 2	217.0	-33.8	1.21	118.5	1.22	IR	-537.9	589.3	IR	5.817
L 2L 3	217.0	-33.8	1.21	118.5	1.22	IR	-670.8	802.1	OR	7.689
L 3L 4	303.0	-67.3	1.21	179.8	1.22	IR	-537.9	589.3	IR	1.918
L 4L 5	303.0	-67.3	1.21	179.8	1.22	IR	-670.8	802.1	OR	2.832
L 5L 6	258.9	-100.7	1.21	188.6	1.22	IR	-537.9	589.3	IR	1.918

L 6L 7	258.9	-100.7	1.21	188.6	1.22	IR OR	-713.8 -890.2	790.9 1076.5	IR OR	2.821 4.335
L 7L 8	84.4	-134.2	1.21	151.1	1.22	IR OR	-642.7 -801.5	713.3 970.9	IR OR	4.163 5.869
L 8L 9	84.4	-134.2	1.21	151.1	1.22	IR OR	-642.7 -801.5	713.3 970.9	IR OR	4.163 5.869
L 9L10	-220.3	-167.7	1.21	59.3	1.22	IR OR	-642.7 -801.5	713.3 970.9	IR OR	2.519 3.466
L10L11	-220.3	-167.7	1.21	59.3	1.22	IR OR	-644.7 -803.9	713.3 970.9	IR OR	2.530 3.480
L11L12	-238.5	-120.8	1.21	0.0	1.21	IR OR	-644.7 -804.0	713.3 970.9	IR OR	3.362 4.681
L12L13	-238.5	-120.8	1.21	0.0	1.21	IR OR	-644.7 -804.0	713.3 970.9	IR OR	3.362 4.681
L15L16	173.7	0.0	1.21	105.3	1.21	IR OR	-466.0 -581.1	509.4 693.4	IR OR	3.188 4.935
L16L17	173.7	0.0	1.21	105.3	1.21	IR OR	-466.0 -581.1	509.4 693.4	IR OR	3.188 4.935
L17L18	231.6	0.0	1.21	133.1	1.21	IR OR	-465.9 -581.1	509.4 693.4	IR OR	2.086 3.468
L18L19	231.6	0.0	1.21	133.1	1.21	IR OR	-465.9 -581.1	509.4 693.4	IR OR	2.086 3.468
L19L20	174.0	0.0	1.21	105.5	1.21	IR OR	-465.9 -581.1	509.4 693.4	IR OR	3.179 4.922
L20L21	174.0	0.0	1.21	105.5	1.21	IR OR	-466.0 -581.1	509.4 693.4	IR OR	3.179 4.922
L23L24	-237.7	-120.3	1.21	0.0	1.21	IR OR	-644.7 -804.0	713.3 970.9	IR OR	3.384 4.708
L24L25	-237.7	-120.3	1.21	0.0	1.21	IR OR	-644.7 -804.0	713.3 970.9	IR OR	3.384 4.708
L25L26	-244.0	-167.9	1.21	59.7	1.21	IR OR	-644.7 -803.9	713.3 970.9	IR OR	2.386 3.335
L26L27	-244.0	-167.9	1.21	59.7	1.21	IR OR	-643.3 -802.2	713.3 970.9	IR OR	2.378 3.324
L27L28	-1.3	-137.2	1.21	152.2	1.21	IR OR	-643.2 -802.1	713.3 970.9	IR OR	4.678 5.836
L28L29	-1.3	-137.2	1.21	152.2	1.21	IR OR	-643.2 -802.1	713.3 970.9	IR OR	4.678 5.836
L29L30	118.8	-106.4	1.21	197.1	1.21	IR OR	-714.3 -890.8	790.9 1076.5	IR OR	3.410 4.858
L30L31	118.8	-106.4	1.21	197.1	1.21	IR OR	-714.3 -890.8	790.9 1076.5	IR OR	3.410 4.858
L31L32	115.2	-108.0	1.22	197.1	1.21	IR OR	-714.3 -890.8	790.9 1076.5	IR OR	3.427 4.876
L32L33	115.2	-108.0	1.22	197.1	1.21	IR OR	-714.3 -890.8	790.9 1076.5	IR OR	3.427 4.876
L33L34	-12.0	-139.3	1.22	152.3	1.21	IR OR	-643.2 -802.1	713.3 970.9	IR OR	4.530 5.670
L34L35	-12.0	-139.3	1.22	152.3	1.21	IR OR	-643.2 -802.1	713.3 970.9	IR OR	4.530 5.670
L35L36	-261.9	-170.6	1.22	59.7	1.21	IR OR	-643.3 -802.2	713.3 970.9	IR OR	2.236 3.168
L36L37	-261.9	-170.6	1.22	59.7	1.21	IR OR	-644.7 -803.9	713.3 970.9	IR OR	2.244 3.178
L37L38	-251.6	-122.6	1.22	0.0	1.23	IR OR	-644.7 -804.0	713.3 970.9	IR OR	3.207 4.506
L38L39	-251.6	-122.6	1.22	0.0	1.23	IR OR	-644.7 -804.0	713.3 970.9	IR OR	3.207 4.506
L41L42	201.5	0.0	1.22	117.5	1.23	IR OR	-537.9 -670.8	589.3 802.1	IR OR	3.299 5.109
L42L43	201.5	0.0	1.22	117.5	1.23	IR OR	-537.9 -670.8	589.3 802.1	IR OR	3.299 5.109
L43L44	268.5	0.0	1.22	149.1	1.23	IR OR	-537.9 -670.8	589.3 802.1	IR OR	2.151 3.579
L44L45	268.5	0.0	1.22	149.1	1.23	IR OR	-537.9 -670.8	589.3 802.1	IR OR	2.151 3.579
L45L46	201.3	0.0	1.22	117.4	1.23	IR OR	-537.9 -670.8	589.3 802.1	IR OR	3.306 5.119
L46L47	201.3	0.0	1.22	117.4	1.23	IR OR	-537.9 -670.8	589.3 802.1	IR OR	3.306 5.119
U 0L 1	207.0	-27.1	1.23	105.6	1.22	IR OR	-452.8 -564.7	509.4 693.4	IR OR	2.863 4.604
L 1U 2	-134.6	-93.8	1.23	28.4	1.30	IR OR	-362.4 -451.9	416.0 566.2	IR OR	2.428 3.362
U 2L 3	104.9	-28.4	1.30	82.6	1.24	IR OR	-254.6 -317.5	320.8 436.6	IR OR	2.613 4.015
L 3U 4	-31.4	-71.2	1.25	28.4	1.30	IR OR	-207.9 -259.3	277.9 378.3	IR OR	2.477 3.198
U 4L 5	1.7	-37.6	1.30	59.7	1.27	IR OR	-132.1 -164.8	201.1 273.7	IR OR	3.339 4.434
L 5U 6	71.8	-47.9	1.28	49.5	1.28	IR OR	-132.1 -164.8	201.1 273.7	IR OR	2.610 4.076
U 6L 7	-101.4	-61.3	1.27	35.9	1.30	IR OR	-207.9 -259.3	277.9 378.3	IR OR	1.738 2.576
L 7U 8	174.9	-23.2	1.30	72.8	1.25	IR OR	-336.2 -419.2	371.5 505.7	IR OR	2.701 4.543

U 8L 9	-204.6	-84.1	1.24	10.5	1.30	IR	-381.9	430.0	IR	2.108	L29U30	-61.6	-65.2	1.26	32.6	1.30	IR	-208.6	277.9	IR	2.256
						OR	-476.3	585.3	OR	3.229							OR	-260.1	378.3	OR	3.047
L 9U10	278.1	0.0	1.30	95.8	1.23	IR	-453.1	509.4	IR	2.416	U30L31	32.2	-43.8	1.29	54.6	1.27	IR	-117.8	174.6	IR	2.609
						OR	-565.0	693.4	OR	4.336							OR	-146.9	237.6	OR	3.764
U10L11	-293.8	-102.9	1.23	0.0	1.00	IR	-601.2	681.5	IR	2.987	L31U32	37.9	-43.8	1.29	54.6	1.27	IR	-117.8	174.6	IR	2.505
						OR	-749.7	927.6	OR	4.430							OR	-146.9	237.6	OR	3.660
L11U12	-263.7	-120.9	1.23	0.0	1.00	IR	-601.3	681.5	IR	2.794	U32L33	-67.3	-65.2	1.26	32.7	1.30	IR	-208.6	277.9	IR	2.169
						OR	-749.8	927.6	OR	4.023							OR	-260.1	378.3	OR	2.960
U12L13	235.4	0.0	1.00	116.3	1.24	IR	-454.7	509.4	IR	2.356	L33U34	137.5	-26.6	1.30	75.6	1.24	IR	-284.7	360.0	IR	2.944
						OR	-567.1	693.4	OR	3.937							OR	-355.1	490.0	OR	4.664
L13U14	-175.0	-107.6	1.25	0.0	1.00	IR	-419.7	463.7	IR	2.274	U34L35	-166.8	-85.9	1.23	26.6	1.30	IR	-382.8	430.0	IR	2.515
						OR	-523.4	631.1	OR	3.238							OR	-477.4	585.3	OR	3.616
U14L15	161.4	0.0	1.00	99.9	1.26	IR	-343.6	371.5	IR	2.104	L35U36	236.4	-26.6	1.30	96.3	1.23	IR	-453.5	509.4	IR	2.835
						OR	-428.5	505.7	OR	3.447							OR	-565.6	693.4	OR	4.746
L15U16	-99.1	-83.7	1.28	6.8	1.30	IR	-263.4	320.8	IR	1.963	U36L37	-257.0	-103.6	1.23	24.4	1.23	IR	-526.0	589.3	IR	2.596
						OR	-328.5	436.6	OR	2.740							OR	-655.9	802.1	OR	3.850
U16L17	75.1	-18.4	1.30	66.9	1.29	IR	-178.2	227.7	IR	2.282	L37U38	-275.7	-120.6	1.23	0.0	1.00	IR	-601.3	681.5	IR	2.699
						OR	-222.2	309.9	OR	3.511							OR	-749.8	927.6	OR	3.931
L17U18	-12.2	-49.8	1.30	33.6	1.30	IR	-113.7	153.4	IR	2.038	U38L39	247.5	0.0	1.00	116.4	1.24	IR	-454.7	509.4	IR	2.250
						OR	-141.8	208.7	OR	2.602							OR	-567.1	693.4	OR	3.830
U18L19	-11.9	-49.7	1.30	33.7	1.30	IR	-113.7	153.4	IR	2.049	L39U40	-185.6	-107.7	1.25	0.0	1.00	IR	-454.7	509.4	IR	2.500
						OR	-141.8	208.7	OR	2.614							OR	-567.1	693.4	OR	3.543
L19U20	74.8	-18.5	1.30	66.8	1.29	IR	-178.2	227.7	IR	2.288	U40L41	180.5	0.0	1.00	104.1	1.25	IR	-342.7	371.5	IR	1.834
						OR	-222.2	309.9	OR	3.519							OR	-427.4	505.7	OR	3.123
U20L21	-98.9	-83.6	1.28	6.8	1.30	IR	-263.4	320.8	IR	1.969	L41U42	-110.7	-87.8	1.27	7.3	1.30	IR	-262.3	320.8	IR	1.727
						OR	-328.5	436.6	OR	2.747							OR	-327.1	436.6	OR	2.465
L21U22	162.2	0.0	1.00	100.4	1.26	IR	-343.6	371.5	IR	2.086	U42L43	83.0	-19.8	1.30	70.5	1.29	IR	-176.9	227.7	IR	2.053
						OR	-428.5	505.7	OR	3.422							OR	-220.6	309.9	OR	3.219
U22L23	-173.9	-107.2	1.25	0.0	1.00	IR	-454.7	509.4	IR	2.621	L43U44	-13.7	-52.7	1.30	35.9	1.30	IR	-112.1	153.4	IR	1.868
						OR	-567.1	693.4	OR	3.669							OR	-139.8	208.7	OR	2.394
L23U24	235.2	0.0	1.00	116.3	1.24	IR	-454.7	509.4	IR	2.358	U44L45	-14.0	-52.8	1.30	35.8	1.30	IR	-112.1	153.4	IR	1.860
						OR	-567.1	693.4	OR	3.940							OR	-139.8	208.7	OR	2.384
U24L25	-263.4	-120.8	1.23	0.0	1.00	IR	-601.3	681.5	IR	2.796	L45U46	83.2	-19.8	1.30	70.6	1.29	IR	-176.9	227.7	IR	2.048
						OR	-749.8	927.6	OR	4.025							OR	-220.6	309.9	OR	3.214
L25U26	-251.5	-104.1	1.23	24.5	1.23	IR	-526.0	589.3	IR	2.638	U46L47	-110.9	-87.8	1.27	7.2	1.30	IR	-262.3	320.8	IR	1.724
						OR	-655.9	802.1	OR	3.887							OR	-327.1	436.6	OR	2.462
U26L27	230.6	-26.8	1.30	96.7	1.23	IR	-453.5	509.4	IR	2.884	L47U48	180.1	0.0	1.00	104.8	1.26	IR	-342.7	371.5	IR	1.826
						OR	-565.6	693.4	OR	4.787							OR	-427.4	505.7	OR	3.107
L27U28	-161.1	-85.9	1.23	26.8	1.30	IR	-382.8	430.0	IR	2.579	U 1L 1	-55.0	-50.4	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.860
						OR	-477.4	585.3	OR	3.680							OR	-185.5	273.7	OR	2.589
U28L29	131.8	-26.8	1.30	75.6	1.24	IR	-284.7	360.0	IR	3.019	U 3L 3	-57.0	-50.2	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.829
						OR	-355.1	490.0	OR	4.739							OR	-185.5	273.7	OR	2.561
											U 5L 5	-57.0	-50.2	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.829

U 7L 7	-57.0	-50.2	1.30	0.0	1.00	IR	-185.5	273.7	OR	2.561
										IR 1.829
U 9L 9	-57.0	-50.2	1.30	0.0	1.00	IR	-185.5	273.7	OR	2.561
										IR 1.829
U10L10	0.0	0.0	1.00	0.0	1.00	IR	-185.5	273.7	OR	2.561
										IR 999.999
U11L11	-50.0	-46.8	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.111
										OR 2.897
U12L12	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U13L13	-50.0	-46.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.115
										OR 2.902
U15L15	-47.0	-45.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	2.077
										OR 2.844
U17L17	-47.0	-45.5	1.30	0.0	1.00	IR	-141.8	174.6	IR	2.082
										OR 2.852
U19L19	-47.0	-45.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	2.082
										OR 2.851
U21L21	-47.0	-45.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	2.077
										OR 2.844
U23L23	-50.0	-46.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.115
										OR 2.902
U24L24	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U25L25	-50.0	-46.8	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.111
										OR 2.897
U26L26	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U27L27	-55.0	-49.4	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.897
										OR 2.641
U29L29	-55.0	-49.5	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.894
										OR 2.636
U31L31	-55.0	-49.5	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.894
										OR 2.636
U33L33	-55.0	-49.5	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.894
										OR 2.636
U35L35	-55.0	-49.4	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.897
										OR 2.641
U37L37	-50.0	-46.8	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.111
										OR 2.897
U39L39	-50.0	-46.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.115
										OR 2.902
U41L41	-50.0	-47.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.926
										OR 2.661

U43L43	-50.0	-47.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.926
										OR 2.661
U45L45	-50.0	-47.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.927
										OR 2.662
U47L47	-50.0	-47.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.928
										OR 2.663

MEMBER U46L47 IS CRITICAL. SEE DETAILS BELOW.

DL FORCE	LL + IMPACT COMP I.F.	FORCE TENS	ALLOWABLE I.F.	FORCE COMP TENS	RATING FACTOR	TONS
-110.9	-87.8	1.27	7.2	1.30	IR	262.3
					OR	320.8
					IR	1.724
					OR	2.462
					OR	68.97
					OR	98.49

SUPPORT REACTIONS

SUPPORT	D.L.	LL+I	I.F.
U 0	182.4	90.7	1.22
L11	503.2	115.5	1.21
L25	468.7	114.4	1.21
L37	483.2	115.3	1.22
U48	150.1	86.6	1.23

 * TRUSS - LIVE LOAD ML80 *

MEMBER FORCES AND RATINGS I.F. = IMPACT FACTOR

MEMBER ID	DL FORCE	LL + IMPACT COMP I.F.	FORCE TENS	ALLOWABLE I.F.	FORCE COMP TENS	RATING FACTOR
U 0U 1	-132.0	-64.3	1.22	16.5	1.21	IR -391.5 430.0
					OR -488.2 585.3	OR 5.540
U 1U 2	-132.0	-64.3	1.22	16.5	1.21	IR -391.7 430.0
					OR -488.5 585.3	OR 5.544
U 2U 3	-283.2	-149.3	1.22	48.9	1.21	IR -713.8 790.9
					OR -890.2 1076.5	OR 4.066
U 3U 4	-283.2	-149.3	1.22	48.9	1.21	IR -713.8 790.9
					OR -890.2 1076.5	OR 4.066
U 4U 5	-304.2	-185.0	1.22	81.3	1.21	IR -794.8 881.3
					OR -991.1 1199.5	OR 3.714
U 5U 6	-304.2	-185.0	1.22	81.3	1.21	IR -794.8 881.3
					OR -991.1 1199.5	OR 3.714
U 6U 7	-194.8	-171.4	1.22	113.8	1.21	IR -794.8 881.3
					OR -991.1 1199.5	OR 4.644
U 7U 8	-194.8	-171.4	1.22	113.8	1.21	IR -794.8 881.3
					OR -991.1 1199.5	OR 4.644
U 8U 9	44.7	-107.7	1.22	146.2	1.21	IR -642.7 713.3
						IR 4.574

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U 9U10	44.7	-107.7	1.22	146.2	1.21	IR	-801.5	970.9	OR	6.336	U30U31	-138.8	-195.6	1.21	88.5	1.21	IR	-872.3	966.8	IR	3.750
						OR	-642.7	713.3	IR	4.574						OR	-1087.8	1315.9	OR	4.851	
U10U11	391.6	0.0	1.22	176.7	1.21	IR	-797.1	881.3	IR	2.771	U31U32	-138.8	-195.6	1.21	89.2	1.22	IR	-872.3	966.8	IR	3.750
						OR	-994.0	1199.5	OR	4.572						OR	-1087.8	1315.9	OR	4.851	
U11U12	391.6	0.0	1.21	176.7	1.21	IR	-797.2	881.3	IR	2.771	U32U33	-73.4	-174.4	1.21	119.4	1.22	IR	-872.3	966.8	IR	4.581
						OR	-994.1	1199.5	OR	4.572						OR	-1087.8	1315.9	OR	5.817	
U12U13	101.7	0.0	1.21	59.0	1.21	IR	-393.2	430.0	IR	5.566	U33U34	-73.4	-174.4	1.21	119.4	1.22	IR	-872.3	966.8	IR	4.581
						OR	-490.3	585.3	OR	8.198						OR	-1087.8	1315.9	OR	5.817	
U13U14	101.7	0.0	1.21	59.0	1.21	IR	-393.2	430.0	IR	5.566	U34U35	115.6	-107.4	1.21	149.7	1.22	IR	-643.2	713.3	IR	3.993
						OR	-490.3	585.3	OR	8.198						OR	-802.1	970.9	OR	5.714	
U14U15	-107.8	-62.8	1.21	0.0	1.21	IR	-465.9	509.4	IR	5.703	U35U36	115.6	-107.4	1.21	149.7	1.22	IR	-643.3	713.3	IR	3.993
						OR	-581.0	693.4	OR	7.536						OR	-802.2	970.9	OR	5.714	
U15U16	-107.8	-62.8	1.21	0.0	1.21	IR	-466.0	509.4	IR	5.704	U36U37	411.8	0.0	1.21	178.4	1.22	IR	-797.1	881.3	IR	2.631
						OR	-581.1	693.4	OR	7.538						OR	-994.0	1199.5	OR	4.415	
U16U17	-223.5	-131.6	1.21	0.0	1.21	IR	-466.0	509.4	IR	1.843	U37U38	411.8	0.0	1.23	178.4	1.22	IR	-797.2	881.3	IR	2.631
						OR	-581.1	693.4	OR	2.718						OR	-994.1	1199.5	OR	4.415	
U17U18	-223.5	-131.6	1.21	0.0	1.21	IR	-465.9	509.4	IR	1.843	U39U39	107.8	0.0	1.23	59.4	1.22	IR	-393.2	430.0	IR	5.426
						OR	-581.1	693.4	OR	2.718						OR	-490.3	585.3	OR	8.042	
U18U19	-223.7	-131.7	1.21	0.0	1.21	IR	-465.9	509.4	IR	1.839	U39U40	107.8	0.0	1.23	59.4	1.22	IR	-393.2	430.0	IR	5.426
						OR	-581.1	693.4	OR	2.714						OR	-490.3	585.3	OR	8.042	
U19U20	-223.7	-131.7	1.21	0.0	1.21	IR	-465.9	509.4	IR	1.839	U40U41	-124.9	-69.5	1.23	0.0	1.22	IR	-537.9	589.3	IR	5.942
						OR	-581.1	693.4	OR	2.714						OR	-670.8	802.1	OR	7.854	
U20U21	-108.3	-63.1	1.21	0.0	1.21	IR	-466.0	509.4	IR	5.667	U41U42	-124.9	-69.5	1.23	0.0	1.22	IR	-537.9	589.3	IR	5.942
						OR	-581.1	693.4	OR	7.491						OR	-670.8	802.1	OR	7.854	
U21U22	-108.3	-63.1	1.21	0.0	1.21	IR	-465.9	509.4	IR	5.666	U42U43	-259.0	-145.7	1.23	0.0	1.22	IR	-537.9	589.3	IR	1.915
						OR	-581.0	693.4	OR	7.489						OR	-670.8	802.1	OR	2.827	
U22U23	101.0	0.0	1.21	58.6	1.21	IR	-393.2	430.0	IR	5.609	U43U44	-259.0	-145.7	1.23	0.0	1.22	IR	-537.9	589.3	IR	1.915
						OR	-490.3	585.3	OR	8.257						OR	-670.8	802.1	OR	2.827	
U23U24	101.0	0.0	1.21	58.6	1.21	IR	-393.2	430.0	IR	5.609	U44U45	-258.8	-145.6	1.23	0.0	1.22	IR	-537.9	589.3	IR	1.917
						OR	-490.3	585.3	OR	8.257						OR	-670.8	802.1	OR	2.830	
U24U25	390.7	0.0	1.21	176.2	1.21	IR	-797.2	881.3	IR	2.785	U45U46	-258.8	-145.6	1.23	0.0	1.22	IR	-537.9	589.3	IR	1.917
						OR	-994.1	1199.5	OR	4.591						OR	-670.8	802.1	OR	2.830	
U25U26	390.7	0.0	1.21	176.2	1.21	IR	-797.1	881.3	IR	2.785	U46U47	-124.6	-69.3	1.23	0.0	1.22	IR	-537.9	589.3	IR	5.965
						OR	-994.0	1199.5	OR	4.591						OR	-670.8	802.1	OR	7.882	
U26U27	101.4	-107.4	1.21	147.8	1.21	IR	-643.3	713.3	IR	4.139	U47U48	-124.6	-69.3	1.23	0.0	1.22	IR	-537.9	589.3	IR	5.965
						OR	-802.2	970.9	OR	5.882						OR	-670.8	802.1	OR	7.882	
U27U28	101.4	-107.4	1.21	147.8	1.21	IR	-643.2	713.3	IR	4.139	L 1L 2	217.0	-32.7	1.21	113.6	1.22	IR	-463.9	509.4	IR	2.574
						OR	-802.1	970.9	OR	5.882						OR	-578.5	693.4	OR	4.193	
U28U29	-80.5	-174.4	1.21	118.0	1.21	IR	-872.3	966.8	IR	4.541	L 2L 3	217.0	-32.7	1.21	113.6	1.22	IR	-463.9	509.4	IR	2.574
						OR	-1087.8	1315.9	OR	5.777						OR	-578.5	693.4	OR	4.193	
U29U30	-80.5	-174.4	1.21	118.0	1.21	IR	-872.3	966.8	IR	4.541	L 3L 4	303.0	-65.1	1.21	174.0	1.22	IR	-713.8	790.9	IR	2.804
						OR	-1087.8	1315.9	OR	5.777						OR	-890.2	1076.5	OR	4.445	

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U 6L 7	-101.4	-60.3	1.27	37.1	1.30	IR	1.767	IR	277.9	-207.9	277.9	IR	1.767	L27U28	-161.1	-82.5	1.23	25.9	1.30	IR	-382.8	430.0	IR	2.687
L 7U 8	174.9	-25.5	1.30	70.8	1.25	IR	2.778	OR	371.5	-336.2	371.5	IR	2.620	U28L29	131.8	-25.9	1.30	73.1	1.24	IR	-284.7	360.0	OR	3.834
U 8L 9	-204.6	-81.1	1.24	13.9	1.30	IR	2.778	OR	505.7	-419.2	505.7	OR	4.674	L29U30	-61.6	-63.5	1.26	33.9	1.30	IR	-355.1	490.0	OR	3.124
L 9U10	278.1	-2.3	1.30	91.7	1.23	IR	2.186	OR	430.0	-476.3	430.0	IR	3.350	U30L31	32.2	-44.0	1.29	53.9	1.27	IR	-260.1	378.3	OR	4.904
U10L11	-293.8	-98.1	1.23	0.0	1.00	IR	2.522	OR	693.4	-565.0	693.4	IR	4.527	L31U32	37.9	-44.0	1.29	53.9	1.27	IR	-117.8	174.6	OR	2.314
L11U12	-263.7	-114.0	1.23	0.0	1.00	IR	3.133	OR	927.6	-749.7	927.6	IR	4.647	U32L33	-67.3	-63.5	1.26	33.9	1.30	IR	-146.9	237.6	OR	3.125
U12L13	235.4	0.0	1.00	112.1	1.24	IR	2.963	OR	681.5	-601.3	681.5	IR	4.266	L33U34	137.5	-25.8	1.30	73.0	1.24	IR	-284.7	360.0	OR	2.643
L13U14	-175.0	-104.8	1.25	0.0	1.00	IR	2.443	OR	509.4	-454.7	509.4	IR	4.084	U34L35	-166.8	-82.4	1.23	25.8	1.30	IR	-355.1	490.0	OR	3.813
U14L15	161.4	0.0	1.00	98.3	1.26	IR	2.334	OR	371.5	-343.6	371.5	IR	3.323	L35U36	236.4	-25.7	1.30	91.9	1.23	IR	-477.4	585.3	OR	2.537
L15U16	-99.1	-83.7	1.28	6.8	1.30	IR	2.136	OR	505.7	-428.5	505.7	IR	3.500	U36L37	-257.0	-98.5	1.23	23.6	1.23	IR	-453.5	509.4	OR	3.708
U16L17	75.1	-20.1	1.30	68.3	1.29	IR	1.964	OR	320.8	-263.4	320.8	IR	2.742	L37U38	-275.7	-113.5	1.23	0.0	1.00	IR	-565.6	693.4	OR	2.969
L17U18	-12.2	-52.3	1.30	35.8	1.30	IR	1.940	OR	208.7	-141.8	208.7	IR	2.477	U38L39	247.5	0.0	1.00	111.8	1.24	IR	-601.3	681.5	OR	4.970
U18L19	-11.9	-52.2	1.30	35.9	1.30	IR	1.950	OR	153.4	-113.7	153.4	IR	2.488	L39U40	-185.6	-104.4	1.25	0.0	1.00	IR	-454.7	509.4	OR	2.868
L19U20	74.8	-20.2	1.30	68.3	1.29	IR	2.240	OR	309.9	-222.2	309.9	IR	3.445	U40L41	180.5	0.0	1.00	102.0	1.25	IR	-567.1	693.4	OR	4.177
U20L21	-98.9	-83.6	1.28	6.8	1.30	IR	1.970	OR	436.6	-328.5	436.6	IR	2.749	L41U42	-110.7	-87.1	1.27	7.5	1.30	IR	-454.7	509.4	OR	2.343
L21U22	162.2	0.0	1.00	98.9	1.26	IR	2.117	OR	371.5	-343.6	371.5	IR	3.474	U42L43	83.0	-21.4	1.30	71.3	1.29	IR	-454.7	509.4	OR	2.578
U22L23	-173.9	-104.4	1.25	0.0	1.00	IR	2.689	OR	509.4	-454.7	509.4	IR	3.765	L43U44	-13.7	-55.0	1.30	38.2	1.30	IR	-567.1	693.4	OR	3.654
L23U24	235.2	0.0	1.00	112.1	1.24	IR	2.446	OR	693.4	-567.1	693.4	IR	4.087	U44L45	-14.0	-55.1	1.30	38.1	1.30	IR	-454.7	509.4	OR	1.874
U24L25	-263.4	-114.0	1.23	0.0	1.00	IR	2.965	OR	681.5	-601.3	681.5	IR	4.269	L45U46	83.2	-21.3	1.30	71.4	1.29	IR	-427.4	505.7	OR	3.190
L25U26	-251.5	-99.0	1.23	23.6	1.23	IR	2.774	OR	589.3	-526.0	589.3	IR	4.087	U46L47	-110.9	-87.1	1.27	7.4	1.30	IR	-262.3	320.8	OR	1.741
U26L27	230.6	-25.9	1.30	92.3	1.23	IR	3.020	OR	509.4	-453.5	509.4	IR	5.013	L47U48	180.1	0.0	1.00	102.6	1.26	IR	-327.1	436.6	OR	2.485
U 1L 1	-55.0	-58.0	1.30	0.0	1.00	IR	1.617	OR	201.1	-148.8	201.1	IR	1.617	U 1L 1	-55.0	-58.0	1.30	0.0	1.00	IR	-148.8	201.1	OR	2.025

U 3L 3	-57.0	-57.8	1.30	0.0	1.00	IR	-185.5	273.7	OR	2.251
							OR			
U 5L 5	-57.0	-57.8	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.588
							OR			
U 7L 7	-57.0	-57.8	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.588
							OR			
U 9L 9	-57.0	-57.8	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.588
							OR			
U10L10	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U11L11	-50.0	-54.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.806
							OR			
U12L12	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U13L13	-50.0	-54.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.808
							OR			
U15L15	-47.0	-53.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.767
							OR			
U17L17	-47.0	-53.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.770
							OR			
U19L19	-47.0	-53.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.770
							OR			
U21L21	-47.0	-53.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.767
							OR			
U23L23	-50.0	-54.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.808
							OR			
U24L24	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U25L25	-50.0	-54.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.806
							OR			
U26L26	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U27L27	-55.0	-57.1	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.642
							OR			
U29L29	-55.0	-57.2	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.640
							OR			
U31L31	-55.0	-57.2	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.640
							OR			
U33L33	-55.0	-57.2	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.640
							OR			
U35L35	-55.0	-57.1	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.642
							OR			
U37L37	-50.0	-54.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.806
							OR			

MEMBER U 9L 9 IS CRITICAL. SEE DETAILS BELOW.

DL FORCE	LL + IMPACT FORCE		ALLOWABLE FORCE		RATING	
	COMP	I.F.	TENS	I.F.	COMP	TENS
-57.0	-57.8	1.30	0.0	1.00	IR	-148.8 201.1
					OR	-185.5 273.7

SUPPORT REACTIONS

SUPPORT	D.L.	LL+I	I.F.
U 0	182.4	86.2	1.22
L11	503.2	108.6	1.21
L25	468.7	107.4	1.21
L37	483.2	108.3	1.22
U48	150.1	83.9	1.23

 * TRUSS - LIVE LOAD P-82 *

MEMBER FORCES AND RATINGS I.F. = IMPACT FACTOR

MEMBER ID	DL + IMPACT FORCE		ALLOWABLE FORCE		RATING FACTORS	
	COMP	I.F.	TENS	I.F.	COMP	TENS
U 0U 1	-132.0	-140.8	1.22	33.5	1.21	IR -391.5 430.0
					OR	-488.2 585.3
U 1U 2	-132.0	-140.8	1.22	33.5	1.21	IR -391.7 430.0
					OR	-488.5 585.3
U 2U 3	-283.2	-318.8	1.22	99.4	1.21	IR -713.8 790.9
					OR	-890.2 1076.5
U 3U 4	-283.2	-318.8	1.22	99.4	1.21	IR -713.8 790.9
					OR	-890.2 1076.5
U 4U 5	-304.2	-371.5	1.22	165.3	1.21	IR -794.8 881.3
					OR	-991.1 1199.5
U 5U 6	-304.2	-371.5	1.22	165.3	1.21	IR -794.8 881.3
					OR	-991.1 1199.5
U 6U 7	-194.8	-357.4	1.22	231.2	1.21	IR -794.8 881.3
						IR 1.678

U 7U 8	-194.8	-357.4	1.22	231.2	1.21	IR	-794.8	881.3	IR	1.678	OR	2.228	U28U29	-80.5	-372.6	1.21	239.6	1.21	IR	-872.3	966.8	IR	2.135
U 8U 9	44.7	-233.6	1.22	297.1	1.21	IR	-642.7	713.3	IR	2.251	OR	3.118	U29U30	-80.5	-372.6	1.21	239.6	1.21	IR	-872.3	966.8	IR	2.125
U 9U10	44.7	-233.6	1.22	297.1	1.21	IR	-642.7	713.3	IR	2.251	OR	3.118	U30U31	-138.8	-393.6	1.21	183.6	1.21	IR	-872.3	966.8	IR	2.703
U10U11	391.6	0.0	1.22	359.1	1.21	IR	-797.1	881.3	IR	1.364	OR	2.250	U31U32	-138.8	-393.6	1.21	185.0	1.22	IR	-872.3	966.8	IR	1.864
U11U12	391.6	0.0	1.21	359.1	1.21	IR	-797.2	881.3	IR	1.364	OR	2.250	U32U33	-73.4	-372.7	1.21	247.7	1.22	IR	-872.3	966.8	IR	2.411
U12U13	101.7	0.0	1.21	113.7	1.21	IR	-393.2	430.0	IR	2.887	OR	4.252	U33U34	-73.4	-372.7	1.21	247.7	1.22	IR	-872.3	966.8	IR	1.864
U13U14	101.7	0.0	1.21	113.7	1.21	IR	-393.2	430.0	IR	2.887	OR	4.252	U34U35	115.6	-236.2	1.21	310.5	1.22	IR	-872.3	966.8	IR	2.411
U14U15	-107.8	-113.0	1.21	0.0	1.21	IR	-465.9	509.4	IR	3.170	OR	4.188	U35U36	115.6	-236.2	1.21	310.5	1.22	IR	-872.3	966.8	IR	2.144
U15U16	-107.8	-113.0	1.21	0.0	1.21	IR	-466.0	509.4	IR	3.170	OR	4.189	U36U37	411.8	0.0	1.21	370.1	1.22	IR	-872.3	966.8	IR	2.722
U16U17	-223.5	-218.2	1.21	0.0	1.21	IR	-466.0	509.4	IR	1.111	OR	1.639	U37U38	411.8	0.0	1.23	370.1	1.22	IR	-872.3	966.8	IR	2.144
U17U18	-223.5	-218.2	1.21	0.0	1.21	IR	-465.9	509.4	IR	1.111	OR	1.639	U39U39	107.8	0.0	1.23	117.9	1.22	IR	-872.3	966.8	IR	2.722
U18U19	-223.7	-218.3	1.21	0.0	1.21	IR	-465.9	509.4	IR	1.110	OR	1.637	U39U40	107.8	0.0	1.23	117.9	1.22	IR	-872.3	966.8	IR	4.049
U19U20	-223.7	-218.3	1.21	0.0	1.21	IR	-465.9	509.4	IR	1.110	OR	1.637	U40U41	-124.9	-130.1	1.23	0.0	1.22	IR	-872.3	966.8	IR	2.732
U20U21	-108.3	-113.5	1.21	0.0	1.21	IR	-466.0	509.4	IR	3.151	OR	4.165	U41U42	-124.9	-130.1	1.23	0.0	1.22	IR	-872.3	966.8	IR	4.049
U21U22	-108.3	-113.5	1.21	0.0	1.21	IR	-465.9	509.4	IR	3.150	OR	4.164	U42U43	-259.0	-251.5	1.23	0.0	1.22	IR	-872.3	966.8	IR	3.176
U22U23	101.0	0.0	1.21	112.9	1.21	IR	-393.2	430.0	IR	2.913	OR	4.288	U43U44	-259.0	-251.5	1.23	0.0	1.22	IR	-872.3	966.8	IR	4.198
U23U24	101.0	0.0	1.21	112.9	1.21	IR	-393.2	430.0	IR	2.913	OR	4.288	U44U45	-258.8	-251.5	1.23	0.0	1.22	IR	-872.3	966.8	IR	1.109
U24U25	390.7	0.0	1.21	357.8	1.21	IR	-797.2	881.3	IR	1.371	OR	2.260	U45U46	-258.8	-251.5	1.23	0.0	1.22	IR	-872.3	966.8	IR	1.637
U25U26	390.7	0.0	1.21	357.8	1.21	IR	-797.1	881.3	IR	1.371	OR	2.260	U46U47	-124.6	-129.7	1.23	0.0	1.22	IR	-872.3	966.8	IR	1.110
U26U27	101.4	-236.1	1.21	300.3	1.21	IR	-643.3	713.3	IR	2.038	OR	2.896	U47U48	-124.6	-129.7	1.23	0.0	1.22	IR	-872.3	966.8	IR	1.638
U27U28	101.4	-236.1	1.21	300.3	1.21	IR	-643.2	713.3	IR	2.038	OR	2.896	L 1L 2	217.0	-66.5	1.21	246.2	1.22	IR	-872.3	966.8	IR	1.110

L 2L 3	217.0	-66.5	1.21	246.2	1.22	IR	1.188	-463.9	509.4	IR	1.188	OR	2.967
						OR	1.935	-578.5	693.4	OR	1.935	OR	2.967
L 3L 4	303.0	-132.4	1.21	361.0	1.22	IR	1.351	-713.8	790.9	IR	1.351	OR	2.246
						OR	2.143	-890.2	1076.5	OR	2.143	OR	2.967
L 4L 5	303.0	-132.4	1.21	361.0	1.22	IR	1.351	-713.8	790.9	IR	1.351	OR	1.693
						OR	2.143	-890.2	1076.5	OR	2.143	OR	2.413
L 5L 6	258.9	-198.3	1.21	371.9	1.22	IR	1.431	-713.8	790.9	IR	1.431	OR	1.693
						OR	2.199	-890.2	1076.5	OR	2.199	OR	2.413
L 6L 7	258.9	-198.3	1.21	371.9	1.22	IR	1.431	-713.8	790.9	IR	1.431	OR	1.702
						OR	2.199	-890.2	1076.5	OR	2.199	OR	2.422
L 7L 8	84.4	-264.1	1.21	310.2	1.22	IR	2.028	-642.7	713.3	IR	2.028	OR	1.702
						OR	2.858	-801.5	970.9	OR	2.858	OR	2.422
L 8L 9	84.4	-264.1	1.21	310.2	1.22	IR	2.028	-642.7	713.3	IR	2.028	OR	1.702
						OR	2.858	-801.5	970.9	OR	2.858	OR	2.422
L 9L10	-220.3	-330.0	1.21	124.3	1.22	IR	1.280	-642.7	713.3	IR	1.280	OR	1.702
						OR	1.761	-801.5	970.9	OR	1.761	OR	2.422
L10L11	-220.3	-330.0	1.21	124.3	1.22	IR	1.286	-644.7	713.3	IR	1.286	OR	1.702
						OR	1.768	-803.9	970.9	OR	1.768	OR	2.422
L11L12	-238.5	-234.7	1.21	0.0	1.21	IR	1.731	-644.7	713.3	IR	1.731	OR	1.702
						OR	2.410	-804.0	970.9	OR	2.410	OR	2.422
L12L13	-238.5	-234.7	1.21	0.0	1.21	IR	1.731	-644.7	713.3	IR	1.731	OR	1.702
						OR	2.410	-804.0	970.9	OR	2.410	OR	2.422
L13L16	173.7	0.0	1.21	185.4	1.21	IR	1.811	-466.0	509.4	IR	1.811	OR	1.620
						OR	2.804	-581.1	693.4	OR	2.804	OR	2.276
L16L17	173.7	0.0	1.21	185.4	1.21	IR	1.811	-466.0	509.4	IR	1.811	OR	1.822
						OR	2.804	-581.1	693.4	OR	2.804	OR	2.822
L17L18	231.6	0.0	1.21	211.1	1.21	IR	1.316	-465.9	589.3	IR	1.316	OR	1.822
						OR	2.187	-581.1	693.4	OR	2.187	OR	2.822
L18L19	231.6	0.0	1.21	211.1	1.21	IR	1.316	-465.9	589.3	IR	1.316	OR	1.303
						OR	2.187	-581.1	693.4	OR	2.187	OR	2.167
L19L20	174.0	0.0	1.21	185.7	1.21	IR	1.807	-465.9	589.3	IR	1.807	OR	1.826
						OR	2.797	-581.1	693.4	OR	2.797	OR	2.827
L20L21	174.0	0.0	1.21	185.7	1.21	IR	1.807	-466.0	589.3	IR	1.807	OR	1.826
						OR	2.797	-581.1	693.4	OR	2.797	OR	2.827
L23L24	-237.7	-233.6	1.21	0.0	1.21	IR	1.743	-644.7	589.3	IR	1.743	OR	1.826
						OR	2.424	-804.0	802.1	OR	2.424	OR	2.827
L24L25	-237.7	-233.6	1.21	0.0	1.21	IR	1.743	-644.7	589.3	IR	1.743	OR	1.826
						OR	2.424	-804.0	802.1	OR	2.424	OR	2.827
L25L26	-244.0	-330.4	1.21	126.5	1.21	IR	1.213	-644.7	509.4	IR	1.213	OR	1.190
						OR	1.695	-803.9	566.2	OR	1.695	OR	1.658
L26L27	-244.0	-330.4	1.21	126.5	1.21	IR	1.209	-643.3	320.8	IR	1.209	OR	1.322
						OR	1.690	-802.2	436.6	OR	1.690	OR	2.032
L 27L28	-1.3	-269.9	1.21	318.2	1.21	IR	1.311	-643.2	277.9	IR	1.311	OR	1.311
						OR	1.692	-802.1	378.3	OR	1.692	OR	1.692

U 4L 5	1.7	-57.2	1.30	105.6	1.27	IR	-132.1	201.1	IR	1.888	L25U26	-251.5	-220.4	1.23	49.0	1.23	IR	-526.0	589.3	IR	1.245
						OR	-164.8	273.7	OR	2.575							OR	-655.9	802.1	OR	1.835
L 5U 6	71.8	-75.9	1.28	81.7	1.28	IR	-132.1	201.1	IR	1.582	U26L27	230.6	-53.6	1.30	200.9	1.23	IR	-453.5	509.4	IR	1.388
						OR	-164.8	273.7	OR	2.470							OR	-565.6	693.4	OR	2.304
U 6L 7	-101.4	-109.6	1.27	45.5	1.30	IR	-207.9	277.9	IR	0.972	L27U28	-161.1	-174.2	1.23	53.8	1.30	IR	-382.8	430.0	IR	1.272
						OR	-259.3	378.3	OR	1.441							OR	-477.4	585.3	OR	1.815
L 7U 8	174.9	-15.0	1.30	138.6	1.25	IR	-336.2	371.5	IR	1.419	U28L29	131.8	-53.8	1.30	148.3	1.24	IR	-284.7	360.0	IR	1.539
						OR	-419.2	505.7	OR	2.387							OR	-355.1	490.0	OR	2.416
U 8L 9	-204.6	-167.1	1.24	0.0	1.30	IR	-381.9	430.0	IR	1.062	L29U30	-61.6	-122.1	1.26	53.8	1.30	IR	-208.6	277.9	IR	1.205
						OR	-476.3	585.3	OR	1.626							OR	-260.1	378.3	OR	1.627
L 9U10	278.1	0.0	1.30	196.0	1.23	IR	-453.1	509.4	IR	1.180	U30L31	32.2	-68.1	1.29	95.4	1.27	IR	-117.8	174.6	IR	1.493
						OR	-565.0	693.4	OR	2.118							OR	-146.9	237.6	OR	2.154
U10L11	-293.8	-215.6	1.23	0.0	1.00	IR	-601.2	681.5	IR	1.426	L31U32	37.9	-68.2	1.29	95.4	1.27	IR	-117.8	174.6	IR	1.433
						OR	-749.7	927.6	OR	2.115							OR	-146.9	237.6	OR	2.094
L11U12	-263.7	-251.0	1.23	0.0	1.00	IR	-601.3	681.5	IR	1.345	U32L33	-67.3	-122.0	1.26	52.4	1.30	IR	-208.6	277.9	IR	1.158
						OR	-749.8	927.6	OR	1.937							OR	-260.1	378.3	OR	1.580
U12L13	235.4	0.0	1.00	231.5	1.24	IR	-454.7	509.4	IR	1.183	L33U34	137.5	-52.4	1.30	148.3	1.24	IR	-284.7	360.0	IR	1.501
						OR	-567.1	693.4	OR	1.978							OR	-355.1	490.0	OR	2.378
L13U14	-175.0	-202.1	1.25	0.0	1.00	IR	-419.7	463.7	IR	1.211	U34L35	-166.8	-174.1	1.23	52.4	1.30	IR	-382.8	430.0	IR	1.241
						OR	-523.4	631.1	OR	1.724							OR	-477.4	585.3	OR	1.784
U14L15	161.4	0.0	1.00	177.0	1.26	IR	-343.6	371.5	IR	1.187	L35U36	236.4	-52.3	1.30	200.1	1.23	IR	-453.5	509.4	IR	1.364
						OR	-428.5	505.7	OR	1.945							OR	-565.6	693.4	OR	2.284
L15U16	-99.1	-138.7	1.28	8.9	1.30	IR	-263.4	320.8	IR	1.184	U36L37	-257.0	-219.5	1.23	48.0	1.23	IR	-526.0	589.3	IR	1.226
						OR	-328.5	436.6	OR	1.654							OR	-655.9	802.1	OR	1.818
U16L17	75.1	-29.4	1.30	102.4	1.29	IR	-178.2	227.7	IR	1.491	L37U38	-275.7	-253.9	1.23	0.0	1.00	IR	-601.3	681.5	IR	1.282
						OR	-222.2	309.9	OR	2.294							OR	-749.8	927.6	OR	1.867
L17U18	-12.2	-76.5	1.30	52.8	1.30	IR	-113.7	153.4	IR	1.328	U38L39	247.5	0.0	1.00	236.0	1.24	IR	-454.7	509.4	IR	1.110
						OR	-141.8	208.7	OR	1.695							OR	-567.1	693.4	OR	1.890
U18L19	-11.9	-76.3	1.30	53.0	1.30	IR	-113.7	153.4	IR	1.335	L39U40	-185.6	-207.4	1.25	0.0	1.00	IR	-454.7	509.4	IR	1.298
						OR	-141.8	208.7	OR	1.703							OR	-567.1	693.4	OR	1.840
L19U20	74.8	-29.5	1.30	102.2	1.29	IR	-178.2	227.7	IR	1.495	U40L41	180.5	0.0	1.00	190.8	1.25	IR	-342.7	371.5	IR	1.001
						OR	-222.2	309.9	OR	2.299							OR	-427.4	505.7	OR	1.705
U20L21	-98.9	-138.5	1.28	8.9	1.30	IR	-263.4	320.8	IR	1.188	L41U42	-110.7	-150.4	1.27	9.8	1.30	IR	-262.3	320.8	IR	1.008
						OR	-328.5	436.6	OR	1.658							OR	-327.1	436.6	OR	1.439
L21U22	162.2	0.0	1.00	177.8	1.26	IR	-343.6	371.5	IR	1.177	U42L43	83.0	-31.8	1.30	110.7	1.29	IR	-176.9	227.7	IR	1.307
						OR	-428.5	505.7	OR	1.932							OR	-220.6	309.9	OR	2.050
U22L23	-173.9	-201.1	1.25	0.0	1.00	IR	-454.7	509.4	IR	1.397	L43U44	-13.7	-80.3	1.30	56.1	1.30	IR	-112.1	153.4	IR	1.225
						OR	-567.1	693.4	OR	1.955							OR	-139.8	208.7	OR	1.570
L23U24	235.2	0.0	1.00	231.4	1.24	IR	-454.7	509.4	IR	1.185	U44L45	-14.0	-80.4	1.30	56.0	1.30	IR	-112.1	153.4	IR	1.220
						OR	-567.1	693.4	OR	1.980							OR	-139.8	208.7	OR	1.564
U24L25	-263.4	-250.9	1.23	0.0	1.00	IR	-601.3	681.5	IR	1.347	L45U46	83.2	-31.7	1.30	110.8	1.29	IR	-176.9	227.7	IR	1.304
						OR	-749.8	927.6	OR	1.939							OR	-220.6	309.9	OR	2.046
											U46L47	-110.9	-150.5	1.27	9.8	1.30	IR	-262.3	320.8	IR	1.006

U35L35	-55.0	-79.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.178
						OR	-185.5	273.7	OR	1.640
U37L37	-50.0	-73.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.348
						OR	-185.5	273.7	OR	1.849
U39L39	-50.0	-73.0	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.353
						OR	-185.5	273.7	OR	1.857
U41L41	-50.0	-75.3	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.219
						OR	-176.9	237.6	OR	1.684
U43L43	-50.0	-75.3	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.219
						OR	-176.9	237.6	OR	1.684
U45L45	-50.0	-75.3	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.230
						OR	-176.9	237.6	OR	1.685
U47L47	-50.0	-75.2	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.220
						OR	-176.9	237.6	OR	1.686

MEMBER U 6L 7 IS CRITICAL. SEE DETAILS BELOW.

DL FORCE	LL + IMPACT FORCE		ALLOWABLE FORCE		RATING
	COMP	I.F.	COMP	TENS	
-101.4	-109.6	1.27	45.5	1.30	IR
			-207.9	277.9	IR
			OR -259.3	378.3	OR
					1.441
					146.95

SUPPORT REACTIONS

SUPPORT	D.L.	LL+I	I.F.
U 0	182.4	193.2	1.22
L11	503.2	259.9	1.21
L25	468.7	258.4	1.21
L37	483.2	260.9	1.22
U48	150.1	166.7	1.23

+++++
+
+ S T R I N G E R A N A L Y S I S +
+
+++++

THE STRINGER LOCATED AT 8.00 FT. FROM THE CENTER LINE OF GIRDER OR TRUSS IS CRITICAL.

DEAD LOADS ACTING ON STRINGER

STRINGER	SLAB	INPUT	TOTAL
WEIGHT	DL1	DL1	DL2
0.036	0.792	0.020	0.848
			0.000

* STRINGER - LIVE LOAD H20 *

MAXIMUM REACTIONS

REACTIONS

MOMENTS

L47U48	180.1	0.0	1.00	192.0	1.26	IR	-327.1	436.6	OR	1.437
						IR	-342.7	371.5	IR	0.997
						OR	-427.4	505.7	OR	1.696
U 1L 1	-55.0	-82.1	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.143
						OR	-185.5	273.7	OR	1.591
U 3L 3	-57.0	-81.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.129
						OR	-185.5	273.7	OR	1.581
U 5L 5	-57.0	-81.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.129
						OR	-185.5	273.7	OR	1.581
U 7L 7	-57.0	-81.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.129
						OR	-185.5	273.7	OR	1.581
U 9L 9	-57.0	-81.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.129
						OR	-185.5	273.7	OR	1.581
U10L10	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U11L11	-50.0	-73.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.348
						OR	-185.5	273.7	OR	1.849
U12L12	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U13L13	-50.0	-73.0	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.353
						OR	-185.5	273.7	OR	1.857
U15L15	-47.0	-70.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.342
						OR	-176.9	237.6	OR	1.838
U17L17	-47.0	-70.3	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.350
						OR	-176.9	237.6	OR	1.848
U19L19	-47.0	-70.3	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.349
						OR	-176.9	237.6	OR	1.848
U21L21	-47.0	-70.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.342
						OR	-176.9	237.6	OR	1.838
U23L23	-50.0	-73.0	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.353
						OR	-185.5	273.7	OR	1.857
U24L24	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U25L25	-50.0	-73.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.348
						OR	-185.5	273.7	OR	1.849
U26L26	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U27L27	-55.0	-79.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.178
						OR	-185.5	273.7	OR	1.640
U29L29	-55.0	-79.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.177
						OR	-185.5	273.7	OR	1.638
U31L31	-55.0	-79.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.177
						OR	-185.5	273.7	OR	1.638
U33L33	-55.0	-79.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.177
						OR	-185.5	273.7	OR	1.638

SUPPORT	DL1	DL2	+(LL+I)	- (LL+I)	+I.F. -I.F.	+I.F. -I.F.
1	4.0	0.0	29.0	0.0	1.30	0.0
2	8.1	0.0	26.9 L	0.0	1.30	0.0
3	8.1	0.0	26.9 L	0.0	1.30	0.0
4	4.0	0.0	29.0	0.0	1.30	0.0

NOTE: ALL SUPPORT REACTIONS AND END SHEARS IN EACH SPAN DUE TO A LIVE LOAD ARE CALCULATED BASED ON AASHTO ARTICLE 3.23.1 AS INTERPRETED IN SOL 431-93-05.

UNFACTORED MOMENTS AND SHEARS

SPAN 1 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X	DL1	DL2	+(LL+I)	- (LL+I)	DL1	DL2	+(LL+I)	- (LL+I)	SHEAR	I.F.
0.00	0.0	0.0	0.0	0.0	0.0	0.0	29.0	0.0	0.0	1.30
0.95	3.4	0.0	25.9	0.0	3.2	0.0	27.2	0.0	0.0	1.30
1.90	6.1	0.0	46.0	0.0	2.4	0.0	24.2	0.0	0.0	1.30
2.85	8.0	0.0	60.3	0.0	1.6	0.0	21.2	0.0	0.0	1.30
3.80	9.2	0.0	69.0	0.0	0.8	0.0	18.1	0.0	0.0	1.30
4.75	9.6	0.0	71.8	0.0	0.0	0.0	15.1	0.0	0.0	1.30
5.70	9.2	0.0	69.0	0.0	-0.8	0.0	12.1	0.0	0.0	1.30
6.65	8.0	0.0	60.3	0.0	-1.6	0.0	9.1	0.0	0.0	1.30
7.60	6.1	0.0	46.0	0.0	-2.4	0.0	6.0	0.0	0.0	1.30
8.55	3.4	0.0	25.9	0.0	-3.2	0.0	3.0	0.0	0.0	1.30
9.50	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	0.0	0.0	1.30

SPAN 2 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X	DL1	DL2	+(LL+I)	- (LL+I)	DL1	DL2	+(LL+I)	- (LL+I)	SHEAR	I.F.
0.00	0.0	0.0	0.0	0.0	0.0	0.0	29.0	0.0	0.0	1.30
0.95	3.4	0.0	25.9	0.0	3.2	0.0	27.2	0.0	0.0	1.30
1.90	6.1	0.0	46.0	0.0	2.4	0.0	24.2	0.0	0.0	1.30
2.85	8.0	0.0	60.3	0.0	1.6	0.0	21.2	0.0	0.0	1.30
3.80	9.2	0.0	69.0	0.0	0.8	0.0	18.1	0.0	0.0	1.30
4.75	9.6	0.0	71.8	0.0	0.0	0.0	15.1	0.0	0.0	1.30
5.70	9.2	0.0	69.0	0.0	-0.8	0.0	12.1	0.0	0.0	1.30
6.65	8.0	0.0	60.3	0.0	-1.6	0.0	9.1	0.0	0.0	1.30
7.60	6.1	0.0	46.0	0.0	-2.4	0.0	6.0	0.0	0.0	1.30
8.55	3.4	0.0	25.9	0.0	-3.2	0.0	3.0	0.0	0.0	1.30
9.50	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	0.0	0.0	1.30

9.50	SIMULT SHEAR	-27.2	0.0	SIMULT MOM	25.9	25.9
		0.0	0.0	-4.0	0.0	-29.0
	SIMULT SHEAR	0.0	0.0	SIMULT MOM	0.0	0.0

SPAN 3 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X	DL1	DL2	+(LL+I)	- (LL+I)	DL1	DL2	+(LL+I)	- (LL+I)	SHEAR	I.F.
0.00	0.0	0.0	0.0	0.0	0.0	0.0	29.0	0.0	0.0	1.30
0.95	3.4	0.0	25.9	0.0	3.2	0.0	27.2	0.0	0.0	1.30
1.90	6.1	0.0	46.0	0.0	2.4	0.0	24.2	0.0	0.0	1.30
2.85	8.0	0.0	60.3	0.0	1.6	0.0	21.2	0.0	0.0	1.30
3.80	9.2	0.0	69.0	0.0	0.8	0.0	18.1	0.0	0.0	1.30
4.75	9.6	0.0	71.8	0.0	0.0	0.0	15.1	0.0	0.0	1.30
5.70	9.2	0.0	69.0	0.0	-0.8	0.0	12.1	0.0	0.0	1.30
6.65	8.0	0.0	60.3	0.0	-1.6	0.0	9.1	0.0	0.0	1.30
7.60	6.1	0.0	46.0	0.0	-2.4	0.0	6.0	0.0	0.0	1.30
8.55	3.4	0.0	25.9	0.0	-3.2	0.0	3.0	0.0	0.0	1.30
9.50	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	0.0	0.0	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

SPAN 1										SPAN 2																			
NON-COMP OVERLOAD					NON-COMPACT					COMPACT					NON-COMP OVERLOAD					NON-COMPACT					COMPACT				
MOMENT					RATING FACTORS					MOMENT					RATING FACTORS					MOMENT					RATING FACTORS				
STRENGTH					IR					STRENGTH					IR					STRENGTH					IR				
X	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)					
0.00	154.9 B	123.9	85.8	85.8	1.28 V	2.14 V	172.2	172.2	1.28 V	2.14 V	172.2	172.2	1.28 V	2.14 V	172.2	172.2	1.28 V	2.14 V	172.2	172.2	1.28 V	2.14 V	172.2	172.2					
0.95	154.9 B	123.9	85.8	85.8	1.38 V	2.31 V	172.2	172.2	1.38 V	2.31 V	172.2	172.2	1.38 V	2.31 V	172.2	172.2	1.38 V	2.31 V	172.2	172.2	1.38 V	2.31 V	172.2	172.2					
1.90	154.9 B	123.9	85.8	85.8	1.47 T	2.46 T	172.2	172.2	1.47 T	2.46 T	172.2	172.2	1.47 T	2.46 T	172.2	172.2	1.47 T	2.46 T	172.2	172.2	1.47 T	2.46 T	172.2	172.2					
2.85	154.9 B	123.9	85.8	85.8	1.10 T	1.84 T	172.2	172.2	1.10 T	1.84 T	172.2	172.2	1.10 T	1.84 T	172.2	172.2	1.10 T	1.84 T	172.2	172.2	1.10 T	1.84 T	172.2	172.2					
3.80	154.9 B	123.9	85.8	85.8	0.96 T	1.59 T	172.2	172.2	0.96 T	1.59 T	172.2	172.2	0.96 T	1.59 T	172.2	172.2	0.96 T	1.59 T	172.2	172.2	0.96 T	1.59 T	172.2	172.2					
4.75	154.9 B	123.9	85.8	85.8	0.92 T	1.53 T	172.2	172.2	0.92 T	1.53 T	172.2	172.2	0.92 T	1.53 T	172.2	172.2	0.92 T	1.53 T	172.2	172.2	0.92 T	1.53 T	172.2	172.2					
5.70	154.9 B	123.9	85.8	85.8	0.96 T	1.59 T	172.2	172.2	0.96 T	1.59 T	172.2	172.2	0.96 T	1.59 T	172.2	172.2	0.96 T	1.59 T	172.2	172.2	0.96 T	1.59 T	172.2	172.2					
6.65	154.9 B	123.9	85.8	85.8	1.10 T	1.84 T	172.2	172.2	1.10 T	1.84 T	172.2	172.2	1.10 T	1.84 T	172.2	172.2	1.10 T	1.84 T	172.2	172.2	1.10 T	1.84 T	172.2	172.2					
7.60	154.9 B	123.9	85.8	85.8	1.47 T	2.46 T	172.2	172.2	1.47 T	2.46 T	172.2	172.2	1.47 T	2.46 T	172.2	172.2	1.47 T	2.46 T	172.2	172.2	1.47 T	2.46 T	172.2	172.2					
8.55	154.9 B	123.9	85.8	85.8	1.38 V	2.31 V	172.2	172.2	1.38 V	2.31 V	172.2	172.2	1.38 V	2.31 V	172.2	172.2	1.38 V	2.31 V	172.2	172.2	1.38 V	2.31 V	172.2	172.2					
9.50	154.9 B	123.9	85.8	85.8	1.28 V	2.14 V	172.2	172.2	1.28 V	2.14 V	172.2	172.2	1.28 V	2.14 V	172.2	172.2	1.28 V	2.14 V	172.2	172.2	1.28 V	2.14 V	172.2	172.2					

2.85	154.9 B	123.9	85.8	1.10 T	1.84 T	172.2	1.15 O	1.92 O	5.70	9.2	0.0	69.0	0.0	-0.8	0.0	12.1	-18.1	1.30
3.80	154.9 B	123.9	85.8	0.96 T	1.59 T	172.2	1.00 O	1.66 O		SIMULT SHEAR		-18.1	0.0	SIMULT	MOM	69.0	69.0	
4.75	154.9 B	123.9	85.8	0.92 T	1.53 T	172.2	0.96 O	1.59 O	6.65	8.0	0.0	60.3	0.0	-1.6	0.0	9.1	-21.2	1.30
5.70	154.9 B	123.9	85.8	0.96 T	1.59 T	172.2	1.00 O	1.66 O		SIMULT SHEAR		20.1	0.0	SIMULT	MOM	60.3	60.3	
6.65	154.9 B	123.9	85.8	1.10 T	1.84 T	172.2	1.15 O	1.92 O	7.60	6.1	0.0	46.0	0.0	-2.4	0.0	6.0	-24.2	1.30
7.60	154.9 B	123.9	85.8	1.47 T	2.46 T	172.2	1.54 O	2.56 O		SIMULT SHEAR		-24.2	0.0	SIMULT	MOM	46.0	46.0	
8.55	154.9 B	123.9	85.8	1.38 V	2.31 V	172.2	1.38 V	2.31 V	8.55	3.4	0.0	25.9	0.0	-3.2	0.0	3.0	-27.2	1.30
9.50	154.9 B	123.9	85.8	1.28 V	2.14 V	172.2	1.28 V	2.14 V		SIMULT SHEAR		-27.2	0.0	SIMULT	MOM	25.9	25.9	
									9.50	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	-30.0	1.30
										SIMULT SHEAR		0.0	0.0	SIMULT	MOM	0.0	0.0	

SPAN 2 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X	DL1		DL2		+(LL+I)		-(LL+I)		DL1		DL2		+(LL+I)		-(LL+I)	
	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4.0	0.0	0.0	0.0	30.0	0.0	0.0	1.30
0.95	SIMULT	SHEAR	0.0	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	0.0	0.0	0.0	0.0	0.0	0.0
	3.4	0.0	25.9	0.0	0.0	0.0	0.0	0.0	3.2	0.0	27.2	0.0	25.9	0.0	0.0	1.30
1.90	SIMULT	SHEAR	27.2	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	25.9	0.0	25.9	0.0	0.0	0.0
	6.1	0.0	46.0	0.0	0.0	0.0	0.0	0.0	2.4	0.0	24.2	0.0	24.2	0.0	0.0	1.30
2.85	SIMULT	SHEAR	24.2	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	46.0	0.0	46.0	0.0	0.0	0.0
	8.0	0.0	60.3	0.0	0.0	0.0	0.0	0.0	1.6	0.0	21.2	0.0	21.2	0.0	0.0	1.30
3.80	SIMULT	SHEAR	21.2	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	60.3	0.0	60.3	0.0	0.0	0.0
	9.2	0.0	69.0	0.0	0.0	0.0	0.0	0.0	0.8	0.0	18.1	0.0	18.1	0.0	0.0	1.30
4.75	SIMULT	SHEAR	18.1	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	69.0	0.0	69.0	0.0	0.0	0.0
	9.6	0.0	71.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	15.1	0.0	15.1	0.0	0.0	1.30
5.70	SIMULT	SHEAR	-15.1	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	71.8	0.0	71.8	0.0	0.0	0.0
	9.2	0.0	69.0	0.0	0.0	0.0	0.0	0.0	-0.8	0.0	12.1	0.0	12.1	0.0	0.0	1.30
6.65	SIMULT	SHEAR	-18.1	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	69.0	0.0	69.0	0.0	0.0	0.0
	8.0	0.0	60.3	0.0	0.0	0.0	0.0	0.0	-1.6	0.0	9.1	0.0	9.1	0.0	0.0	1.30
7.60	SIMULT	SHEAR	-21.2	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	60.3	0.0	60.3	0.0	0.0	0.0
	9.6	0.0	46.0	0.0	0.0	0.0	0.0	0.0	-2.4	0.0	6.0	0.0	6.0	0.0	0.0	1.30
8.55	SIMULT	SHEAR	-24.2	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	46.0	0.0	46.0	0.0	0.0	0.0
	3.4	0.0	25.9	0.0	0.0	0.0	0.0	0.0	-3.2	0.0	3.0	0.0	3.0	0.0	0.0	1.30
9.50	SIMULT	SHEAR	-27.2	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	25.9	0.0	25.9	0.0	0.0	0.0
	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	0.0	0.0	0.0	0.0	1.30
	SIMULT	SHEAR	0.0	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	0.0	0.0	0.0	0.0	0.0	0.0

SPAN 3 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X	DL1		DL2		+(LL+I)		-(LL+I)		DL1		DL2		+(LL+I)		-(LL+I)		I.F.
	MOMENT	0.0	MOMENT	0.0	MOMENT	0.0	MOMENT	0.0	SHEAR	4.0	SHEAR	MOM	SHEAR	MOM	SHEAR	0.0	
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	SIMULT	SHEAR	0.0	SIMULT	MOM	0.0	0.0	1.30	
0.95	3.4	0.0	25.9	27.2	0.0	0.0	0.0	0.0	3.2	0.0	27.2	0.0	27.2	-3.0	-3.0	1.30	
1.90	6.1	0.0	46.0	0.0	0.0	0.0	0.0	0.0	SIMULT	SHEAR	25.9	SIMULT	MOM	25.9	25.9	1.30	
2.85	SIMULT	SHEAR	24.2	0.0	0.0	0.0	0.0	0.0	2.4	0.0	24.2	0.0	24.2	-6.0	-6.0	1.30	
3.80	8.0	0.0	60.3	0.0	0.0	0.0	0.0	0.0	SIMULT	SHEAR	46.0	SIMULT	MOM	46.0	46.0	1.30	
4.75	SIMULT	SHEAR	21.2	0.0	0.0	0.0	0.0	0.0	1.6	0.0	21.2	0.0	21.2	-9.1	-9.1	1.30	
5.70	9.2	0.0	69.0	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	60.3	SIMULT	MOM	60.3	60.3	1.30	
6.65	SIMULT	SHEAR	18.1	0.0	0.0	0.0	0.0	0.0	0.8	0.0	18.1	0.0	18.1	-12.1	-12.1	1.30	
7.60	9.6	0.0	71.8	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	69.0	SIMULT	MOM	69.0	69.0	1.30	
8.55	SIMULT	SHEAR	-15.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	15.1	0.0	15.1	-15.1	-15.1	1.30	
9.50	9.2	0.0	69.0	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	71.8	SIMULT	MOM	71.8	71.8	1.30	
	SIMULT	SHEAR	18.1	0.0	0.0	0.0	0.0	0.0	-0.8	0.0	12.1	0.0	12.1	-18.1	-18.1	1.30	
	8.0	0.0	60.3	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	69.0	SIMULT	MOM	69.0	69.0	1.30	
	SIMULT	SHEAR	-21.2	0.0	0.0	0.0	0.0	0.0	-1.6	0.0	9.1	0.0	9.1	-21.2	-21.2	1.30	
	6.1	0.0	46.0	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	60.3	SIMULT	MOM	60.3	60.3	1.30	
	SIMULT	SHEAR	-24.2	0.0	0.0	0.0	0.0	0.0	-2.4	0.0	6.0	0.0	6.0	-24.2	-24.2	1.30	
	3.4	0.0	25.9	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	46.0	SIMULT	MOM	46.0	46.0	1.30	
	SIMULT	SHEAR	-27.2	0.0	0.0	0.0	0.0	0.0	-3.2	0.0	3.0	0.0	3.0	-27.2	-27.2	1.30	
	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	25.9	SIMULT	MOM	25.9	25.9	1.30	
	SIMULT	SHEAR	0.0	0.0	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	0.0	0.0	-26.0	-26.0	1.30	
	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	0.0	SIMULT	MOM	0.0	0.0	1.30	

NOTE: ALL SUPPORT REACTIONS AND END SHEARS IN EACH SPAN DUE TO A LIVE LOAD ARE CALCULATED BASED ON AASHTO ARTICLE 3.23.1 AS INTERPRETED IN SOL 431-93-05

UNFACTORED MOMENTS AND SHEARS

SPAN 1 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30												
X	DL1		DL2		-(LL+I)		DL1		DL2		+(LL+I)	
	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR
0.00	0.00	0.0	0.0	0.0	0.0	0.0	4.0	0.0	30.0	0.0	0.0	0.0
	SIMULT	SHEAR	0.0	0.0	SIMULT	MOM	0.0	0.0	0.0	0.0	0.0	0.0
0.95		3.4	0.0	25.9	0.0	27.2	3.2	0.0	27.2	3.0	0.0	0.0
	SIMULT	SHEAR	27.2	0.0	SIMULT	MOM	25.9	0.0	25.9	25.9	0.0	0.0
1.90		6.1	0.0	46.0	0.0	24.2	2.4	0.0	24.2	6.0	0.0	0.0
	SIMULT	SHEAR	24.2	0.0	SIMULT	MOM	46.0	0.0	46.0	46.0	0.0	0.0
2.85		8.0	0.0	60.3	0.0	21.2	1.6	0.0	21.2	9.1	0.0	0.0
	SIMULT	SHEAR	21.2	0.0	SIMULT	MOM	60.3	0.0	60.3	60.3	0.0	0.0
3.80		9.2	0.0	69.0	0.0	18.1	0.8	0.0	18.1	12.1	0.0	0.0
	SIMULT	SHEAR	18.1	0.0	SIMULT	MOM	69.0	0.0	69.0	69.0	0.0	0.0
4.75		9.6	0.0	71.8	0.0	15.1	0.0	0.0	15.1	15.1	0.0	0.0
	SIMULT	SHEAR	-15.1	0.0	SIMULT	MOM	71.8	0.0	71.8	71.8	0.0	0.0

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NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

SPAN 1

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	RATING FACTORS
			IR	OR	IR	OR
0.00	154.9 B	123.9	85.8	1.24 V	172.2	1.24 V
0.95	154.9 B	123.9	85.8	1.38 V	172.2	1.38 V
1.90	154.9 B	123.9	85.8	1.47 T	172.2	1.54 O
2.85	154.9 B	123.9	85.8	1.10 T	172.2	1.15 O
3.80	154.9 B	123.9	85.8	0.96 T	172.2	1.00 O
4.75	154.9 B	123.9	85.8	0.92 T	172.2	0.96 O
5.70	154.9 B	123.9	85.8	0.96 T	172.2	1.00 O
6.65	154.9 B	123.9	85.8	1.10 T	172.2	1.15 O
7.60	154.9 B	123.9	85.8	1.47 T	172.2	1.54 O
8.55	154.9 B	123.9	85.8	1.38 V	172.2	1.38 V
9.50	154.9 B	123.9	85.8	1.24 V	172.2	1.24 V

SPAN 2

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	RATING FACTORS
			IR	OR	IR	OR
0.00	154.9 B	123.9	85.8	1.24 V	172.2	1.24 V
0.95	154.9 B	123.9	85.8	1.38 V	172.2	1.38 V
1.90	154.9 B	123.9	85.8	1.47 T	172.2	1.54 O
2.85	154.9 B	123.9	85.8	1.10 T	172.2	1.15 O
3.80	154.9 B	123.9	85.8	0.96 T	172.2	1.00 O
4.75	154.9 B	123.9	85.8	0.92 T	172.2	0.96 O
5.70	154.9 B	123.9	85.8	0.96 T	172.2	1.00 O
6.65	154.9 B	123.9	85.8	1.10 T	172.2	1.15 O
7.60	154.9 B	123.9	85.8	1.47 T	172.2	1.54 O
8.55	154.9 B	123.9	85.8	1.38 V	172.2	1.38 V
9.50	154.9 B	123.9	85.8	1.25 V	172.2	1.25 V

SPAN 3

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	RATING FACTORS
			IR	OR	IR	OR
0.00	154.9 B	123.9	85.8	1.24 V	172.2	1.24 V
0.95	154.9 B	123.9	85.8	1.38 V	172.2	1.38 V
1.90	154.9 B	123.9	85.8	1.47 T	172.2	1.54 O
2.85	154.9 B	123.9	85.8	1.10 T	172.2	1.15 O
3.80	154.9 B	123.9	85.8	0.96 T	172.2	1.00 O
4.75	154.9 B	123.9	85.8	0.92 T	172.2	0.96 O
5.70	154.9 B	123.9	85.8	0.96 T	172.2	1.00 O
6.65	154.9 B	123.9	85.8	1.10 T	172.2	1.15 O
7.60	154.9 B	123.9	85.8	1.47 T	172.2	1.54 O
8.55	154.9 B	123.9	85.8	1.38 V	172.2	1.38 V
9.50	154.9 B	123.9	85.8	1.43 V	172.2	1.43 V

* STRINGER - LIVE LOAD TK527 *

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MAXIMUM REACTIONS

SUPPORT	DL1	DL2	+(LL+I)	-(LL+I)	+I.F. -I.F.	MOMENTS
						+I.F. -I.F.
1	4.0	0.0	29.2	0.0	1.30	
2	8.1	0.0	33.7	0.0	1.30	
3	8.1	0.0	33.7	0.0	1.30	
4	4.0	0.0	29.2	0.0	1.30	

NOTE: ALL SUPPORT REACTIONS AND END SHEARS IN EACH SPAN DUE TO A LIVE LOAD ARE CALCULATED BASED ON AASHTO ARTICLE 3.23.1 AS INTERPRETED IN SOL 431-93-05.

UNFACTORED MOMENTS AND SHEARS

SPAN 1 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X	DL1		DL2		+(LL+I)		-(LL+I)		DL1		DL2		+(LL+I)		-(LL+I)	
	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4.0	0.0	4.0	0.0	29.2	0.0	0.0	0.0
0.95	3.4	0.0	25.9	0.0	0.0	0.0	0.0	0.0	3.2	0.0	27.3	0.0	0.0	0.0	0.0	0.0
1.90	6.1	0.0	43.6	0.0	0.0	0.0	0.0	0.0	2.4	0.0	23.0	0.0	0.0	0.0	0.0	0.0
2.85	8.0	0.0	54.3	0.0	0.0	0.0	0.0	0.0	1.6	0.0	19.1	0.0	0.0	0.0	0.0	0.0
3.80	9.2	0.0	57.6	0.0	0.0	0.0	0.0	0.0	0.8	0.0	15.2	0.0	0.0	0.0	0.0	0.0
4.75	9.6	0.0	56.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5.70	9.2	0.0	57.6	0.0	0.0	0.0	0.0	0.0	-0.8	0.0	7.8	0.0	0.0	0.0	0.0	0.0
6.65	8.0	0.0	54.3	0.0	0.0	0.0	0.0	0.0	-1.6	0.0	5.8	0.0	0.0	0.0	0.0	0.0
7.60	6.1	0.0	43.6	0.0	0.0	0.0	0.0	0.0	-2.4	0.0	3.9	0.0	0.0	0.0	0.0	0.0
8.55	3.4	0.0	25.9	0.0	0.0	0.0	0.0	0.0	-3.2	0.0	1.9	0.0	0.0	0.0	0.0	0.0
9.50	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

SPAN 2 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X	DL1		DL2		+(LL+I)		-(LL+I)		DL1		DL2		+(LL+I)		-(LL+I)	
	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4.0	0.0	4.0	0.0	29.2	0.0	0.0	0.0
0.95	3.4	0.0	25.9	0.0	0.0	0.0	0.0	0.0	3.2	0.0	27.3	0.0	0.0	0.0	0.0	0.0
1.90	6.1	0.0	43.6	0.0	0.0	0.0	0.0	0.0	2.4	0.0	23.0	0.0	0.0	0.0	0.0	0.0
2.85	8.0	0.0	54.3	0.0	0.0	0.0	0.0	0.0	1.6	0.0	19.1	0.0	0.0	0.0	0.0	0.0
3.80	9.2	0.0	57.6	0.0	0.0	0.0	0.0	0.0	0.8	0.0	15.2	0.0	0.0	0.0	0.0	0.0
4.75	9.6	0.0	56.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5.70	9.2	0.0	57.6	0.0	0.0	0.0	0.0	0.0	-0.8	0.0	7.8	0.0	0.0	0.0	0.0	0.0
6.65	8.0	0.0	54.3	0.0	0.0	0.0	0.0	0.0	-1.6	0.0	5.8	0.0	0.0	0.0	0.0	0.0

	SIMULT SHEAR	-19.1	0.0	SIMULT MOM	38.8	54.3
7.60	6.1	0.0	0.0	-2.4	0.0	-23.0
	SIMULT SHEAR	-23.0	0.0	SIMULT MOM	29.6	43.6
8.55	3.4	0.0	0.0	-3.2	0.0	-27.3
	SIMULT SHEAR	-27.3	0.0	SIMULT MOM	16.6	25.9
9.50	0.0	0.0	0.0	-4.0	0.0	-29.2
	SIMULT SHEAR	0.0	0.0	SIMULT MOM	0.0	0.0

SPAN 3 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)
X	MOMENT	MOMENT	MOMENT	MOMENT	SHEAR	SHEAR	SHEAR	SHEAR	SHEAR	SHEAR	SHEAR	SHEAR
0.00	0.0	0.0	0.0	0.0	4.0	0.0	29.2	0.0	0.0	0.0	0.0	0.0
	SIMULT SHEAR	0.0	0.0	0.0	SIMULT MOM	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.95	3.4	0.0	25.9	0.0	3.2	0.0	27.3	-1.9	1.30			
	SIMULT SHEAR	27.3	0.0	0.0	SIMULT MOM	25.9	16.6					
1.90	6.1	0.0	43.6	0.0	2.4	0.0	23.0	-3.9	1.30			
	SIMULT SHEAR	23.0	0.0	0.0	SIMULT MOM	43.6	29.6					
2.85	8.0	0.0	54.3	0.0	1.6	0.0	19.1	-5.8	1.30			
	SIMULT SHEAR	19.1	0.0	0.0	SIMULT MOM	54.3	38.8					
3.80	9.2	0.0	57.6	0.0	0.8	0.0	15.2	-7.8	1.30			
	SIMULT SHEAR	15.2	0.0	0.0	SIMULT MOM	57.6	44.4					
4.75	9.6	0.0	56.5	0.0	0.0	0.0	11.3	-11.3	1.30			
	SIMULT SHEAR	-10.7	0.0	0.0	SIMULT MOM	53.5	53.5					
5.70	9.2	0.0	57.6	0.0	-0.8	0.0	7.8	-15.2	1.30			
	SIMULT SHEAR	-15.2	0.0	0.0	SIMULT MOM	44.4	57.6					
6.65	8.0	0.0	54.3	0.0	-1.6	0.0	5.8	-19.1	1.30			
	SIMULT SHEAR	-19.1	0.0	0.0	SIMULT MOM	38.8	54.3					
7.60	6.1	0.0	43.6	0.0	-2.4	0.0	3.9	-23.0	1.30			
	SIMULT SHEAR	-23.0	0.0	0.0	SIMULT MOM	29.6	43.6					
8.55	3.4	0.0	25.9	0.0	-3.2	0.0	1.9	-27.3	1.30			
	SIMULT SHEAR	-27.3	0.0	0.0	SIMULT MOM	16.6	25.9					
9.50	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	-29.2	1.30			
	SIMULT SHEAR	0.0	0.0	0.0	SIMULT MOM	0.0	0.0					

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

	NON-COMP OVERLOAD	NON-COMPACT	NON-COMPACT	COMPACT	COMPACT	COMPACT
X	MOMENT	RATING FACTORS	RATING FACTORS	MOMENT	RATING FACTORS	RATING FACTORS
	STRENGTH	STRENGTH	IR	STRENGTH	IR	OR
0.00	154.9 B	123.9	85.8	1.27 V	1.27 V	2.12 V
0.95	154.9 B	123.9	85.8	1.38 V	1.38 V	2.30 V
1.90	154.9 B	123.9	85.8	1.55 T	1.55 T	2.70 O
2.85	154.9 B	123.9	85.8	1.23 T	1.23 T	2.13 O
3.80	154.9 B	123.9	85.8	1.14 T	1.14 T	1.99 O
4.75	154.9 B	123.9	85.8	1.16 T	1.16 T	2.02 O
5.70	154.9 B	123.9	85.8	1.14 T	1.14 T	1.99 O
6.65	154.9 B	123.9	85.8	1.23 T	1.23 T	2.13 O
7.60	154.9 B	123.9	85.8	1.55 T	1.55 T	2.70 O
8.55	154.9 B	123.9	85.8	1.38 V	1.38 V	2.30 V
9.50	154.9 B	123.9	85.8	1.27 V	1.27 V	2.12 V

	NON-COMP OVERLOAD	NON-COMPACT	NON-COMPACT	COMPACT	COMPACT	COMPACT
X	MOMENT	RATING FACTORS	RATING FACTORS	MOMENT	RATING FACTORS	RATING FACTORS
	STRENGTH	STRENGTH	IR	STRENGTH	IR	OR
0.00	154.9 B	123.9	85.8	1.27 V	1.27 V	2.12 V
0.95	154.9 B	123.9	85.8	1.38 V	1.38 V	2.30 V
1.90	154.9 B	123.9	85.8	1.55 T	1.55 T	2.70 O
2.85	154.9 B	123.9	85.8	1.23 T	1.23 T	2.13 O
3.80	154.9 B	123.9	85.8	1.14 T	1.14 T	1.99 O
4.75	154.9 B	123.9	85.8	1.16 T	1.16 T	2.02 O
5.70	154.9 B	123.9	85.8	1.14 T	1.14 T	1.99 O
6.65	154.9 B	123.9	85.8	1.23 T	1.23 T	2.13 O
7.60	154.9 B	123.9	85.8	1.55 T	1.55 T	2.70 O
8.55	154.9 B	123.9	85.8	1.38 V	1.38 V	2.30 V
9.50	154.9 B	123.9	85.8	1.27 V	1.27 V	2.12 V

SPAN 2

	STRENGTH	STRENGTH	STRENGTH	IR	OR	STRENGTH	IR	OR
X	MOMENT	MOMENT	MOMENT	RATING FACTORS	RATING FACTORS	MOMENT	RATING FACTORS	RATING FACTORS
	STRENGTH	STRENGTH	STRENGTH	IR	OR	STRENGTH	IR	OR
0.00	154.9 B	123.9	85.8	1.27 V	1.27 V	172.2	1.27 V	2.12 V
0.95	154.9 B	123.9	85.8	1.38 V	1.38 V	172.2	1.38 V	2.30 V
1.90	154.9 B	123.9	85.8	1.55 T	1.55 T	172.2	1.62 O	2.70 O
2.85	154.9 B	123.9	85.8	1.23 T	1.23 T	172.2	1.28 O	2.13 O
3.80	154.9 B	123.9	85.8	1.14 T	1.14 T	172.2	1.19 O	1.99 O
4.75	154.9 B	123.9	85.8	1.16 T	1.16 T	172.2	1.21 O	2.02 O
5.70	154.9 B	123.9	85.8	1.14 T	1.14 T	172.2	1.19 O	1.99 O
6.65	154.9 B	123.9	85.8	1.23 T	1.23 T	172.2	1.28 O	2.13 O
7.60	154.9 B	123.9	85.8	1.55 T	1.55 T	172.2	1.62 O	2.70 O
8.55	154.9 B	123.9	85.8	1.38 V	1.38 V	172.2	1.38 V	2.30 V
9.50	154.9 B	123.9	85.8	1.27 V	1.27 V	172.2	1.27 V	2.12 V

SPAN 3

	NON-COMP OVERLOAD	NON-COMPACT	NON-COMPACT	COMPACT	COMPACT	COMPACT
X	MOMENT	RATING FACTORS	RATING FACTORS	MOMENT	RATING FACTORS	RATING FACTORS
	STRENGTH	STRENGTH	IR	STRENGTH	IR	OR
0.00	154.9 B	123.9	85.8	1.27 V	1.27 V	2.12 V
0.95	154.9 B	123.9	85.8	1.38 V	1.38 V	2.30 V
1.90	154.9 B	123.9	85.8	1.55 T	1.55 T	2.70 O
2.85	154.9 B	123.9	85.8	1.23 T	1.23 T	2.13 O
3.80	154.9 B	123.9	85.8	1.14 T	1.14 T	1.99 O
4.75	154.9 B	123.9	85.8	1.16 T	1.16 T	2.02 O
5.70	154.9 B	123.9	85.8	1.14 T	1.14 T	1.99 O
6.65	154.9 B	123.9	85.8	1.23 T	1.23 T	2.13 O
7.60	154.9 B	123.9	85.8	1.55 T	1.55 T	2.70 O
8.55	154.9 B	123.9	85.8	1.38 V	1.38 V	2.30 V
9.50	154.9 B	123.9	85.8	1.27 V	1.27 V	2.12 V

* STRINGER - LIVE LOAD ML80 *

MAXIMUM REACTIONS

	SUPPORT	DL1	DL2	+(LL+I)	-(LL+I)	REACTIONS	MOMENTS
						+I.F. -I.F.	+I.F. -I.F.
1	4.0	0.0	31.1	0.0	1.30		
2	8.1	0.0	39.3	0.0	1.30		
3	8.1	0.0	39.3	0.0	1.30		
4	4.0	0.0	31.1	0.0	1.30		

NOTE: ALL SUPPORT REACTIONS AND END SHEARS IN EACH SPAN DUE TO A LIVE LOAD ARE CALCULATED BASED ON AASHTO ARTICLE 3.23.1 AS INTERPRETED IN SOL 431-93-05.

UNFACTORED MOMENTS AND SHEARS

SPAN 1 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)	I.F.
X	MOMENT	MOMENT	MOMENT	MOMENT	SHEAR	SHEAR	SHEAR	SHEAR	I.F.
0.00	0.0	0.0	0.0	0.0	4.0	0.0	31.1	0.0	1.30
	SIMULT SHEAR	0.0	0.0	0.0	SIMULT MOM	0.0	0.0	0.0	
0.95	3.4	0.0	26.6	0.0	3.2	0.0	28.0	-1.9	1.30
	SIMULT SHEAR	28.0	0.0	0.0	SIMULT MOM	26.6	16.6		
1.90	6.1	0.0	43.6	0.0	2.4	0.0	23.0	-3.9	1.30
	SIMULT SHEAR	23.0	0.0	0.0	SIMULT MOM	43.6	29.6		
2.85	8.0	0.0	54.3	0.0	1.6	0.0	19.1	-5.8	1.30
	SIMULT SHEAR	19.1	0.0	0.0	SIMULT MOM	54.3	38.8		

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3.80	9.2	0.0	57.6	0.0	0.8	0.0	15.2	-7.8	1.30
	SIMULT SHEAR	15.2	0.0	SIMULT MOM	57.6	44.4			
4.75	9.6	0.0	60.8	0.0	0.0	11.3	-11.3	1.30	
	SIMULT SHEAR	9.7	0.0	SIMULT MOM	53.5	53.5			
5.70	9.2	0.0	57.6	0.0	-0.8	0.0	-15.2	1.30	
	SIMULT SHEAR	-15.2	0.0	SIMULT MOM	44.4	57.6			
6.65	8.0	0.0	54.3	0.0	-1.6	0.0	-19.1	1.30	
	SIMULT SHEAR	-19.1	0.0	SIMULT MOM	38.8	54.3			
7.60	6.1	0.0	43.6	0.0	-2.4	0.0	-23.0	1.30	
	SIMULT SHEAR	-23.0	0.0	SIMULT MOM	29.6	43.6			
8.55	3.4	0.0	26.6	0.0	-3.2	0.0	-28.0	1.30	
	SIMULT SHEAR	-28.0	0.0	SIMULT MOM	16.6	26.6			
9.50	0.0	0.0	0.0	0.0	-4.0	0.0	-31.1	1.30	
	SIMULT SHEAR	0.0	0.0	SIMULT MOM	0.0	0.0			

SPAN 2 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X	DL1 MOMENT	DL2 MOMENT	+(LL+I) MOMENT	-(LL+I) MOMENT	DL1 SHEAR	DL2 SHEAR	+(LL+I) SHEAR	-(LL+I) SHEAR	I.F.
0.00	0.0	0.0	0.0	0.0	4.0	0.0	31.1	0.0	1.30
0.95	3.4	0.0	26.6	0.0	3.2	0.0	28.0	0.0	1.30
1.90	6.1	0.0	43.6	0.0	2.4	0.0	23.0	-3.9	1.30
2.85	8.0	0.0	54.3	0.0	1.6	0.0	19.1	-5.8	1.30
3.80	9.2	0.0	57.6	0.0	0.8	0.0	15.2	-7.8	1.30
4.75	9.6	0.0	60.8	0.0	0.0	0.0	11.3	-11.3	1.30
5.70	9.2	0.0	57.6	0.0	-0.8	0.0	7.8	-15.2	1.30
6.65	8.0	0.0	54.3	0.0	-1.6	0.0	5.8	-19.1	1.30
7.60	6.1	0.0	43.6	0.0	-2.4	0.0	3.9	-23.0	1.30
8.55	3.4	0.0	26.6	0.0	-3.2	0.0	1.9	-28.0	1.30
9.50	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	-31.1	1.30

SPAN 3 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X	DL1 MOMENT	DL2 MOMENT	+(LL+I) MOMENT	-(LL+I) MOMENT	DL1 SHEAR	DL2 SHEAR	+(LL+I) SHEAR	-(LL+I) SHEAR	I.F.
0.00	0.0	0.0	0.0	0.0	4.0	0.0	31.1	0.0	1.30
0.95	3.4	0.0	26.6	0.0	3.2	0.0	28.0	0.0	1.30
1.90	6.1	0.0	43.6	0.0	2.4	0.0	23.0	-3.9	1.30
2.85	8.0	0.0	54.3	0.0	1.6	0.0	19.1	-5.8	1.30
3.80	9.2	0.0	57.6	0.0	0.8	0.0	15.2	-7.8	1.30
4.75	9.6	0.0	60.8	0.0	0.0	0.0	11.3	-11.3	1.30
5.70	9.2	0.0	57.6	0.0	-0.8	0.0	7.8	-15.2	1.30
6.65	8.0	0.0	54.3	0.0	-1.6	0.0	5.8	-19.1	1.30
7.60	6.1	0.0	43.6	0.0	-2.4	0.0	3.9	-23.0	1.30

8.55	3.4	0.0	26.6	0.0	-3.2	0.0	1.9	-28.0	1.30
	SIMULT SHEAR	-28.0	0.0	SIMULT MOM	16.6	26.6			
9.50	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	-31.1	1.30
	SIMULT SHEAR	0.0	0.0	SIMULT MOM	0.0	0.0			

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

SPAN 1

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT		COMPACT	
	MOMENT	STRENGTH	MOMENT	RATING	MOMENT	RATING	MOMENT	RATING
0.00	154.9 B	123.9	85.8	1.20 V	1.99 V	172.2	1.20 V	1.99 V
0.95	154.9 B	123.9	85.8	1.35 V	2.24 V	172.2	1.35 V	2.24 V
1.90	154.9 B	123.9	85.8	1.55 T	2.59 T	172.2	1.62 O	2.70 O
2.85	154.9 B	123.9	85.8	1.23 T	2.05 T	172.2	1.28 O	2.13 O
3.80	154.9 B	123.9	85.8	1.14 T	1.91 T	172.2	1.19 O	1.99 O
4.75	154.9 B	123.9	85.8	1.08 T	1.80 T	172.2	1.13 O	1.88 O
5.70	154.9 B	123.9	85.8	1.14 T	1.91 T	172.2	1.19 O	1.99 O
6.65	154.9 B	123.9	85.8	1.23 T	2.05 T	172.2	1.28 O	2.13 O
7.60	154.9 B	123.9	85.8	1.55 T	2.59 T	172.2	1.62 O	2.70 O
8.55	154.9 B	123.9	85.8	1.35 V	2.24 V	172.2	1.35 V	2.24 V
9.50	154.9 B	123.9	85.8	1.20 V	1.99 V	172.2	1.20 V	1.99 V

SPAN 2

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT		COMPACT	
	MOMENT	STRENGTH	MOMENT	RATING	MOMENT	RATING	MOMENT	RATING
0.00	154.9 B	123.9	85.8	1.20 V	1.99 V	172.2	1.20 V	1.99 V
0.95	154.9 B	123.9	85.8	1.35 V	2.24 V	172.2	1.35 V	2.24 V
1.90	154.9 B	123.9	85.8	1.55 T	2.59 T	172.2	1.62 O	2.70 O
2.85	154.9 B	123.9	85.8	1.23 T	2.05 T	172.2	1.28 O	2.13 O
3.80	154.9 B	123.9	85.8	1.14 T	1.91 T	172.2	1.19 O	1.99 O
4.75	154.9 B	123.9	85.8	1.08 T	1.80 T	172.2	1.13 O	1.88 O
5.70	154.9 B	123.9	85.8	1.14 T	1.91 T	172.2	1.19 O	1.99 O
6.65	154.9 B	123.9	85.8	1.23 T	2.05 T	172.2	1.28 O	2.13 O
7.60	154.9 B	123.9	85.8	1.55 T	2.59 T	172.2	1.62 O	2.70 O
8.55	154.9 B	123.9	85.8	1.35 V	2.24 V	172.2	1.35 V	2.24 V
9.50	154.9 B	123.9	85.8	1.20 V	1.99 V	172.2	1.20 V	1.99 V

SPAN 3

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT		COMPACT	
	MOMENT	STRENGTH	MOMENT	RATING	MOMENT	RATING	MOMENT	RATING
0.00	154.9 B	123.9	85.8	1.20 V	1.99 V	172.2	1.20 V	1.99 V
0.95	154.9 B	123.9	85.8	1.35 V	2.24 V	172.2	1.35 V	2.24 V
1.90	154.9 B	123.9	85.8	1.55 T	2.59 T	172.2	1.62 O	2.70 O
2.85	154.9 B	123.9	85.8	1.23 T	2.05 T	172.2	1.28 O	2.13 O
3.80	154.9 B	123.9	85.8	1.14 T	1.91 T	172.2	1.19 O	1.99 O
4.75	154.9 B	123.9	85.8	1.08 T	1.80 T	172.2	1.13 O	1.88 O
5.70	154.9 B	123.9	85.8	1.14 T	1.91 T	172.2	1.19 O	1.99 O
6.65	154.9 B	123.9	85.8	1.23 T	2.05 T	172.2	1.28 O	2.13 O
7.60	154.9 B	123.9	85.8	1.55 T	2.59 T	172.2	1.62 O	2.70 O
8.55	154.9 B	123.9	85.8	1.35 V	2.24 V	172.2	1.35 V	2.24 V
9.50	154.9 B	123.9	85.8	1.20 V	1.99 V	172.2	1.20 V	1.99 V

SPAN 2

X	NON-COMP OVERLOAD		SHEAR	NON-COMPACT		COMPACT	
	MOMENT	STRENGTH		RATING FACTORS	IR	OR	RATING FACTORS
0.00	154.9 B	123.9	85.8	0.91 V	1.52 V	172.2	0.91 V 1.52 V
0.95	154.9 B	123.9	85.8	1.03 V	1.71 V	172.2	1.03 V 1.71 V
1.90	154.9 B	123.9	85.8	1.19 T	1.98 T	172.2	1.24 O 2.06 O
2.85	154.9 B	123.9	85.8	0.94 T	1.56 T	172.2	0.98 O 1.63 O
3.80	154.9 B	123.9	85.8	0.87 T	1.46 T	172.2	0.91 O 1.52 O
4.75	154.9 B	123.9	85.8	0.82 T	1.37 T	172.2	0.86 O 1.43 O
5.70	154.9 B	123.9	85.8	0.87 T	1.46 T	172.2	0.91 O 1.52 O
6.65	154.9 B	123.9	85.8	0.94 T	1.56 T	172.2	0.98 O 1.63 O
7.60	154.9 B	123.9	85.8	1.19 T	1.98 T	172.2	1.24 O 2.06 O
8.55	154.9 B	123.9	85.8	1.03 V	1.71 V	172.2	1.03 V 1.71 V
9.50	154.9 B	123.9	85.8	0.91 V	1.52 V	172.2	0.91 V 1.52 V

SPAN 3

X	NON-COMP OVERLOAD		SHEAR	NON-COMPACT		COMPACT	
	MOMENT	STRENGTH		RATING FACTORS	IR	OR	RATING FACTORS
0.00	154.9 B	123.9	85.8	0.91 V	1.52 V	172.2	0.91 V 1.52 V
0.95	154.9 B	123.9	85.8	1.03 V	1.71 V	172.2	1.03 V 1.71 V
1.90	154.9 B	123.9	85.8	1.19 T	1.98 T	172.2	1.24 O 2.06 O
2.85	154.9 B	123.9	85.8	0.94 T	1.56 T	172.2	0.98 O 1.63 O
3.80	154.9 B	123.9	85.8	0.87 T	1.46 T	172.2	0.91 O 1.52 O
4.75	154.9 B	123.9	85.8	0.82 T	1.37 T	172.2	0.86 O 1.43 O
5.70	154.9 B	123.9	85.8	0.87 T	1.46 T	172.2	0.91 O 1.52 O
6.65	154.9 B	123.9	85.8	0.94 T	1.56 T	172.2	0.98 O 1.63 O
7.60	154.9 B	123.9	85.8	1.19 T	1.98 T	172.2	1.24 O 2.06 O
8.55	154.9 B	123.9	85.8	1.03 V	1.71 V	172.2	1.03 V 1.71 V
9.50	154.9 B	123.9	85.8	0.91 V	1.52 V	172.2	0.91 V 1.52 V

+++++ FLOORBEAM ANALYSIS +++++
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+
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FLOORBEAM SPAN: 16.00 CANTILEVER: 0.00

FLOORBEAM LIVE LOAD MOMENT AND SHEAR FACTORS

X	FACTOR	WHEEL LOAD POSITIONS		LANE 2/5/8		LANE 3/6	
		LANE 1/4/7	LANE 2/5/8	LANE 2/5/8	LANE 3/6	LANE 2/5/8	LANE 3/6
0.00	0.000 M	0.00	0.00	0.00	0.00	0.00	0.00
1.60	1.000 V	0.50	6.50	9.50	15.50	0.50	6.50
3.20	1.960 M	3.20	9.20	9.50	15.50	3.20	9.20
4.80	2.650 M	0.50	6.50	9.50	15.50	0.50	6.50
6.40	3.450 M	0.50	6.50	9.50	15.50	0.50	6.50
8.00	3.500 M	0.50	6.50	9.50	15.50	0.50	6.50
9.50	0.313 V	8.00	14.00	9.50	15.50	8.00	14.00

FACTOR CODES: M - MOMENT, V - SHEAR

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< FLOORBEAM 1 >
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DEAD LOADS ACTING ON FLOORBEAM

UNIFORM LOAD		CONCENTRATED LOADS	
FLOORBEAM	INPUT	DIST	DL1 DL2
WEIGHT	DL1	DL1	DL2
0.036	0.130	8.000	4.028 0.000

* FLOORBEAM - LIVE LOAD H20 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 32.00
LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1		DL2		LL+I		LL+I	
	MOMENT	STRENGTH	MOMENT	STRENGTH	MOMENT	STRENGTH	MOMENT	STRENGTH
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.60	5.1	0.0	47.4	3.1	0.0	0.0	29.6	1.30
3.20	9.8	0.0	81.5	2.8	0.0	0.0	25.5	1.30
4.80	14.1	0.0	110.2	2.5	0.0	0.0	21.3	1.30
6.40	18.0	0.0	143.5	2.3	0.0	0.0	20.8	1.30
8.00	21.4	0.0	145.6	-2.0	0.0	0.0	13.0	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	RATING FACTORS
0.00	154.9 B	123.9	85.8	0.90 V	172.2	0.90 V
1.60	154.9 B	123.9	85.8	1.27 V	172.2	1.27 V
3.20	154.9 B	123.9	85.8	0.80 T	1.34 T	0.84 O
4.80	154.9 B	123.9	85.8	0.57 T	0.95 T	0.60 O
6.40	154.9 B	123.9	85.8	0.42 T	0.70 T	0.44 O
8.00	154.9 B	123.9	85.8	0.40 T	0.67 T	0.42 O

* FLOORBEAM - LIVE LOAD HS20 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 32.00
LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1		DL2		LL+I		LL+I	
	MOMENT	STRENGTH	MOMENT	STRENGTH	MOMENT	STRENGTH	MOMENT	STRENGTH
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.60	5.1	0.0	47.4	3.1	0.0	0.0	29.6	1.30

3.20 9.8 0.0 81.5 2.8 0.0 25.5 1.30
 4.80 14.1 0.0 110.2 2.5 0.0 21.3 1.30
 6.40 18.0 0.0 143.5 2.3 0.0 20.8 1.30
 8.00 21.4 0.0 145.6 -2.0 0.0 13.0 1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
 UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
 SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT		COMPACT	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	STRENGTH	RATING FACTORS	OR
0.00	154.9 B	123.9	85.8	0.90 V	1.51 V	172.2	0.90 V	1.51 V
1.60	154.9 B	123.9	85.8	1.27 V	2.12 V	172.2	1.27 V	2.12 V
3.20	154.9 B	123.9	85.8	0.80 T	1.34 T	172.2	0.84 O	1.40 O
4.80	154.9 B	123.9	85.8	0.57 T	0.95 T	172.2	0.60 O	1.00 O
6.40	154.9 B	123.9	85.8	0.42 T	0.70 T	172.2	0.44 O	0.74 O
8.00	154.9 B	123.9	85.8	0.40 T	0.67 T	172.2	0.42 O	0.70 O

 * FLOORBEAM - LIVE LOAD TK527 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 33.83
 LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1		DL2		LL+I		LL+I	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	STRENGTH	SHEAR	I.F.
0.00	0.0	0.0	0.0	3.3	0.0	44.0	0.0	1.30
1.60	5.1	0.0	50.1	3.1	0.0	31.3	0.0	1.30
3.20	9.8	0.0	86.2	2.8	0.0	26.9	0.0	1.30
4.80	14.1	0.0	116.5	2.5	0.0	22.5	0.0	1.30
6.40	18.0	0.0	151.7	2.3	0.0	22.0	0.0	1.30
8.00	21.4	0.0	153.9	-2.0	0.0	13.7	0.0	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
 UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
 SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT		COMPACT	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	STRENGTH	RATING FACTORS	OR
0.00	154.9 B	123.9	85.8	0.85 V	1.42 V	172.2	0.85 V	1.42 V
1.60	154.9 B	123.9	85.8	1.21 V	2.01 V	172.2	1.21 V	2.01 V
3.20	154.9 B	123.9	85.8	0.76 T	1.32 O	172.2	0.79 O	1.32 O
4.80	154.9 B	123.9	85.8	0.54 T	0.90 T	172.2	0.57 O	0.94 O
6.40	154.9 B	123.9	85.8	0.40 T	0.67 T	172.2	0.42 O	0.70 O
8.00	154.9 B	123.9	85.8	0.38 T	0.63 T	172.2	0.40 O	0.67 O

 * FLOORBEAM - LIVE LOAD ML80 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 35.78
 LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1		DL2		LL+I		LL+I	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	STRENGTH	SHEAR	I.F.
0.00	0.0	0.0	0.0	3.3	0.0	46.5	0.0	1.30
1.60	5.1	0.0	53.0	3.1	0.0	33.1	0.0	1.30
3.20	9.8	0.0	91.2	2.8	0.0	28.5	0.0	1.30
4.80	14.1	0.0	123.3	2.5	0.0	23.8	0.0	1.30
6.40	18.0	0.0	160.5	2.3	0.0	23.3	0.0	1.30
8.00	21.4	0.0	162.8	-2.0	0.0	14.5	0.0	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
 UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
 SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT		COMPACT	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	STRENGTH	RATING FACTORS	OR
0.00	154.9 B	123.9	85.8	0.81 V	1.35 V	172.2	0.81 V	1.35 V
1.60	154.9 B	123.9	85.8	1.14 V	1.90 V	172.2	1.14 V	1.90 V
3.20	154.9 B	123.9	85.8	0.72 T	1.20 T	172.2	0.75 O	1.25 O
4.80	154.9 B	123.9	85.8	0.51 T	0.85 T	172.2	0.53 O	0.89 O
6.40	154.9 B	123.9	85.8	0.38 T	0.63 T	172.2	0.40 O	0.66 O
8.00	154.9 B	123.9	85.8	0.36 T	0.60 T	172.2	0.38 O	0.63 O

 * FLOORBEAM - LIVE LOAD P-82 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 46.89
 LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1		DL2		LL+I		LL+I	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	STRENGTH	SHEAR	I.F.
0.00	0.0	0.0	0.0	3.3	0.0	61.0	0.0	1.30
1.60	5.1	0.0	69.5	3.1	0.0	43.4	0.0	1.30
3.20	9.8	0.0	119.5	2.8	0.0	37.3	0.0	1.30
4.80	14.1	0.0	161.6	2.5	0.0	31.2	0.0	1.30
6.40	18.0	0.0	210.3	2.3	0.0	30.5	0.0	1.30
8.00	21.4	0.0	213.4	-2.0	0.0	19.1	0.0	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
 UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
 SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT		COMPACT	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	STRENGTH	RATING FACTORS	OR
0.00	154.9 B	123.9	85.8	0.62 V	1.03 V	172.2	0.62 V	1.03 V

1.60 154.9 B 123.9 85.8 0.87 V 1.45 V 172.2 0.87 V 1.45 V
 3.20 154.9 B 123.9 85.8 0.55 T 0.91 T 172.2 0.57 O 0.95 O
 4.80 154.9 B 123.9 85.8 0.39 T 0.65 T 172.2 0.41 O 0.68 O
 6.40 154.9 B 123.9 85.8 0.29 T 0.48 T 172.2 0.30 O 0.50 O
 8.00 154.9 B 123.9 85.8 0.27 T 0.46 T 172.2 0.29 O 0.48 O

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 < FLOORBEAM 2 >
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DEAD LOADS ACTING ON FLOORBEAM

UNIFORM LOAD FLOORBEAM INPUT CONCENTRATED LOADS
 WEIGHT DL1 DIST DL1 DL2
 0.036 0.130 8.000 8.055 0.000

 * FLOORBEAM - LIVE LOAD H20 *

LIVE LOAD REACTION FROM DECK (ONE LANE) : 32.08
 LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	41.7	1.30
1.60	8.4	0.0	47.5	5.1	0.0	29.7	1.30
3.20	16.3	0.0	81.7	4.8	0.0	25.5	1.30
4.80	23.8	0.0	110.5	4.6	0.0	21.4	1.30
6.40	30.9	0.0	143.9	4.3	0.0	20.9	1.30
8.00	37.5	0.0	146.0	-4.0	0.0	13.0	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
 UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
 SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD			NON-COMPACT		COMPACT		RATING FACTORS
	MOMENT	STRENGTH	STRENGTH	SHEAR	RATING FACTORS	MOMENT	IR	
0.00	154.9 B	123.9	85.8	0.87 V	1.45 V	172.2	0.87 V	1.45 V
1.60	154.9 B	123.9	85.8	1.23 V	2.05 V	172.2	1.23 V	2.05 V
3.20	154.9 B	123.9	85.8	0.75 T	1.26 T	172.2	0.79 O	1.32 O
4.80	154.9 B	123.9	85.8	0.52 T	0.86 T	172.2	0.54 O	0.91 O
6.40	154.9 B	123.9	85.8	0.37 T	0.61 T	172.2	0.39 O	0.65 O
8.00	154.9 B	123.9	85.8	0.34 T	0.56 T	172.2	0.35 O	0.59 O

 * FLOORBEAM - LIVE LOAD HS20 *

LIVE LOAD REACTION FROM DECK (ONE LANE) : 32.08
 LIVE LOAD IMPACT FACTORS : POS MOM 1.30

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UNFACTORED MOMENTS AND SHEARS

X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	41.7	1.30
1.60	8.4	0.0	47.5	5.1	0.0	29.7	1.30
3.20	16.3	0.0	81.7	4.8	0.0	25.5	1.30
4.80	23.8	0.0	110.5	4.6	0.0	21.4	1.30
6.40	30.9	0.0	143.9	4.3	0.0	20.9	1.30
8.00	37.5	0.0	146.0	-4.0	0.0	13.0	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
 UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
 SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD			NON-COMPACT		COMPACT		RATING FACTORS
	MOMENT	STRENGTH	STRENGTH	SHEAR	RATING FACTORS	MOMENT	IR	
0.00	154.9 B	123.9	85.8	0.87 V	1.45 V	172.2	0.87 V	1.45 V
1.60	154.9 B	123.9	85.8	1.23 V	2.05 V	172.2	1.23 V	2.05 V
3.20	154.9 B	123.9	85.8	0.75 T	1.26 T	172.2	0.79 O	1.32 O
4.80	154.9 B	123.9	85.8	0.52 T	0.86 T	172.2	0.54 O	0.91 O
6.40	154.9 B	123.9	85.8	0.37 T	0.61 T	172.2	0.39 O	0.65 O
8.00	154.9 B	123.9	85.8	0.34 T	0.56 T	172.2	0.35 O	0.59 O

 * FLOORBEAM - LIVE LOAD TK527 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 38.60
 LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	50.2	1.30
1.60	8.4	0.0	57.2	5.1	0.0	35.8	1.30
3.20	16.3	0.0	98.3	4.8	0.0	30.7	1.30
4.80	23.8	0.0	133.0	4.6	0.0	25.7	1.30
6.40	30.9	0.0	173.1	4.3	0.0	25.1	1.30
8.00	37.5	0.0	175.6	-4.0	0.0	15.7	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
 UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
 SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD			NON-COMPACT		COMPACT		RATING FACTORS
	MOMENT	STRENGTH	STRENGTH	SHEAR	RATING FACTORS	MOMENT	IR	
0.00	154.9 B	123.9	85.8	0.73 V	1.21 V	172.2	0.73 V	1.21 V
1.60	154.9 B	123.9	85.8	1.02 V	1.70 V	172.2	1.02 V	1.70 V
3.20	154.9 B	123.9	85.8	0.63 T	1.05 T	172.2	0.66 O	1.09 O
4.80	154.9 B	123.9	85.8	0.43 T	0.72 T	172.2	0.45 O	0.75 O
6.40	154.9 B	123.9	85.8	0.31 T	0.51 T	172.2	0.32 O	0.54 O

8.00	154.9 B	123.9	85.8	0.28 T	0.46 T	172.2	0.30 O	0.49 O
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X	NON-COMP OVERLOAD		SHEAR	NON-COMPACT		COMPACT							
	MOMENT	STRENGTH		RATING FACTORS	IR	OR	RATING FACTORS	IR	OR				
0.00	154.9	B	123.9	85.8	0.45	V	0.75	V	172.2	0.45	V	0.75	V
1.60	154.9	B	123.9	85.8	0.63	T	1.05	V	172.2	0.63	V	1.05	V
3.20	154.9	B	123.9	85.8	0.39	T	0.65	T	172.2	0.41	O	0.68	O
4.80	154.9	B	123.9	85.8	0.27	T	0.44	T	172.2	0.28	O	0.46	O
6.40	154.9	B	123.9	85.8	0.19	T	0.31	T	172.2	0.20	O	0.33	O
8.00	154.9	B	123.9	85.8	0.17	T	0.29	T	172.2	0.18	O	0.30	O

FLOORBEAM 3 >

DEAD LOADS ACTING ON FLOORBEAM

UNIFORM LOAD		CONCENTRATED LOADS	
FLOORBEAM	INPUT	DIST	DL1
WEIGHT	DL1	DL2	
0.036	0.130	8.000	8.055
			0.000

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* FLOORBEAM - LIVE LOAD H2O *
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LIVE LOAD REACTION FROM DECK (ONE LANE) : 32.08
LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

	X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
1	0.00	0.0	0.0	0.0	5.4	0.0	41.7	1.30
2	1.60	8.4	0.0	47.5	5.1	0.0	29.7	1.30
3	3.20	16.3	0.0	81.7	4.8	0.0	25.5	1.30
4	4.80	23.8	0.0	110.5	4.6	0.0	21.4	1.30
5	6.40	30.9	0.0	143.9	4.3	0.0	20.9	1.30
6	8.00	37.5	0.0	146.0	-4.0	0.0	13.0	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD		SHEAR		NON-COMPACT		COMPACT	
	MOMENT	STRENGTH	MOMENT	STRENGTH	IR	OR	IR	OR
0.00	154.9	B	123.9	85.8	0.87	V	1.45	V
1.60	154.9	B	123.9	85.8	1.23	V	2.05	V
3.20	154.9	B	123.9	85.8	0.75	T	1.26	T
4.80	154.9	B	123.9	85.8	0.52	T	0.86	T
6.40	154.9	B	123.9	85.8	0.37	T	0.61	T
8.00	154.9	B	123.9	85.8	0.34	T	0.56	T

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STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP		OVERLOAD		SHEAR	NON-COMPACT		COMPACT					
	MOMENT	STRENGTH	MOMENT	STRENGTH		RATING FACTORS	IR	OR	RATING FACTORS	IR	OR		
0.00	154.9	B	123.9	85.8	85.8	0.63	V	1.05	V	0.63	V	1.05	V
0.60	154.9	B	123.9	85.8	85.8	0.89	V	1.48	V	0.89	V	1.48	V
1.20	154.9	B	123.9	85.8	85.8	0.54	T	0.91	T	0.57	O	0.95	O
1.80	154.9	B	123.9	85.8	85.8	0.37	T	0.62	T	0.39	O	0.65	O
2.40	154.9	B	123.9	85.8	85.8	0.27	T	0.44	T	0.28	O	0.47	O
3.00	154.9	B	123.9	85.8	85.8	0.24	T	0.40	T	0.26	O	0.43	O

* FLOORBEAM - LIVE LOAD P-82 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 62.53
LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	81.3	1.30
0.20	8.4	0.0	92.7	5.1	0.0	57.9	1.30
1.60	16.3	0.0	159.3	4.8	0.0	49.8	1.30
3.20	23.8	0.0	215.4	4.6	0.0	41.7	1.30
4.80	30.9	0.0	280.4	4.3	0.0	40.6	1.30
6.40	37.5	0.0	284.5	4.0	0.0	25.4	1.30
8.00	37.5	0.0	284.5	4.0	0.0	25.4	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA

* FLOORBEAM - LIVE LOAD HS20 *

LIVE LOAD REACTION FROM DECK (ONE LANE) : 32.08
LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	41.7	1.30
1.60	8.4	0.0	47.5	5.1	0.0	29.7	1.30
3.20	16.3	0.0	81.7	4.8	0.0	25.5	1.30
4.80	23.8	0.0	110.5	4.6	0.0	21.4	1.30
6.40	30.9	0.0	143.9	4.3	0.0	20.9	1.30
8.00	37.5	0.0	146.0	-4.0	0.0	13.0	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD MOMENT	STRENGTH	SHEAR	NON-COMPACT RATING FACTORS	COMPACT MOMENT	COMPACT RATING FACTORS	IR	OR
0.00	154.9 B	123.9	85.8	0.87 V	1.45 V	1.72.2	0.87 V	1.45 V
1.60	154.9 B	123.9	85.8	1.23 V	2.05 V	1.72.2	1.23 V	2.05 V
3.20	154.9 B	123.9	85.8	0.75 T	1.26 T	1.72.2	0.79 O	1.32 O
4.80	154.9 B	123.9	85.8	0.52 T	0.86 T	1.72.2	0.54 O	0.91 O
6.40	154.9 B	123.9	85.8	0.37 T	0.61 T	1.72.2	0.39 O	0.65 O
8.00	154.9 B	123.9	85.8	0.34 T	0.56 T	1.72.2	0.35 O	0.59 O

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 36.60
LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	50.2	1.30
1.60	8.4	0.0	57.2	5.1	0.0	35.8	1.30
3.20	16.3	0.0	98.3	4.8	0.0	30.7	1.30
4.80	23.8	0.0	133.0	4.6	0.0	25.7	1.30
6.40	30.9	0.0	173.1	4.3	0.0	25.1	1.30
8.00	37.5	0.0	175.6	-4.0	0.0	15.7	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

NON-COMP OVERLOAD MOMENT	STRENGTH	SHEAR	NON-COMPACT RATING FACTORS	COMPACT MOMENT	COMPACT RATING FACTORS
--------------------------------	----------	-------	-------------------------------	-------------------	---------------------------

X	STRENGTH	STRENGTH	STRENGTH	IR	OR	STRENGTH	IR	OR
0.00	154.9 B	123.9	85.8	0.73 V	1.21 V	172.2	0.73 V	1.21 V
1.60	154.9 B	123.9	85.8	1.02 V	1.70 V	172.2	1.02 V	1.70 V
3.20	154.9 B	123.9	85.8	0.63 T	1.05 T	172.2	0.66 O	1.09 O
4.80	154.9 B	123.9	85.8	0.43 T	0.72 T	172.2	0.45 O	0.75 O
6.40	154.9 B	123.9	85.8	0.31 T	0.51 T	172.2	0.32 O	0.54 O
8.00	154.9 B	123.9	85.8	0.28 T	0.46 T	172.2	0.30 O	0.49 O

* FLOORBEAM - LIVE LOAD ML80 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 44.45
LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	57.8	1.30
1.60	8.4	0.0	65.9	5.1	0.0	41.2	1.30
3.20	16.3	0.0	113.3	4.8	0.0	35.4	1.30
4.80	23.8	0.0	153.1	4.6	0.0	29.6	1.30
6.40	30.9	0.0	199.4	4.3	0.0	28.9	1.30
8.00	37.5	0.0	202.3	-4.0	0.0	18.1	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD MOMENT	STRENGTH	SHEAR	NON-COMPACT RATING FACTORS	COMPACT MOMENT	COMPACT RATING FACTORS	IR	OR
0.00	154.9 B	123.9	85.8	0.63 V	1.05 V	172.2	0.63 V	1.05 V
1.60	154.9 B	123.9	85.8	0.89 V	1.48 V	172.2	0.89 V	1.48 V
3.20	154.9 B	123.9	85.8	0.54 T	0.91 T	172.2	0.57 O	0.95 O
4.80	154.9 B	123.9	85.8	0.37 T	0.62 T	172.2	0.39 O	0.65 O
6.40	154.9 B	123.9	85.8	0.27 T	0.44 T	172.2	0.28 O	0.47 O
8.00	154.9 B	123.9	85.8	0.24 T	0.40 T	172.2	0.26 O	0.43 O

* FLOORBEAM - LIVE LOAD P-82 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 62.53
LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	81.3	1.30
1.60	8.4	0.0	92.7	5.1	0.0	57.9	1.30
3.20	16.3	0.0	159.3	4.8	0.0	49.8	1.30
4.80	23.8	0.0	215.4	4.6	0.0	41.7	1.30
6.40	30.9	0.0	280.4	4.3	0.0	40.6	1.30
8.00	37.5	0.0	284.5	-4.0	0.0	25.4	1.30

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP		OVERLOAD		NON-COMPACT		COMPACT	
	MOMENT	STRENGTH	MOMENT	STRENGTH	RATING FACTORS	IR	RATING FACTORS	IR
0.00	154.9 B	123.9	85.8	85.8	0.45 V	0.75 V	172.2	0.45 V
0.60	154.9 B	123.9	85.8	85.8	0.63 V	1.05 V	172.2	0.63 V
1.20	154.9 B	123.9	85.8	85.8	0.39 T	0.65 T	172.2	0.41 O
1.80	154.9 B	123.9	85.8	85.8	0.27 T	0.44 T	172.2	0.28 O
2.40	154.9 B	123.9	85.8	85.8	0.19 T	0.31 T	172.2	0.20 O
3.00	154.9 B	123.9	85.8	85.8	0.17 T	0.29 T	172.2	0.18 O

+ + + + + R A T I N G S U M M A R Y + + + + +

+ + +

LOAD	ALLOWABLE STRESS RATING		
	IR (CRITICAL)	FACTOR	TONS
420	IR (CRITICAL)	2.01	40.3
	OR (CRITICAL)	2.99	59.7
420	IR (CRITICAL)	1.94	69.9
	OR (CRITICAL)	2.66	95.6
4527	IR (CRITICAL)	1.72	69.0
	OR (CRITICAL)	2.38	95.4
480	IR (CRITICAL)	1.59	58.2
	OR (CRITICAL)	2.22	81.5
8-82	IR (CRITICAL)	0.97	99.1
	OR (CRITICAL)	1.44	146.6

see next page

LOAD		FACTOR		LOAD FACTOR		RATING	
				TONS	X	SPAN	
H20	IR (CRITICAL)	0.96	0	19.1	4.75	3	
	OR (CRITICAL)	1.59	0	31.8	4.75	3	
	IR (POS MOM)	0.96	0	19.1	4.75	3	
	OR (POS MOM)	1.59	0	31.8	4.75	3	
HS20	IR (CRITICAL)	0.96	0	34.4	4.75	3	
	OR (CRITICAL)	1.59	0	57.3	4.75	3	
	IR (POS MOM)	0.96	0	34.4	4.75	3	
	OR (POS MOM)	1.59	0	57.3	4.75	3	
TK527	IR (CRITICAL)	1.19	0	47.8	5.70	3	
	OR (CRITICAL)	1.99	0	79.6	5.70	3	
	IR (POS MOM)	1.19	0	47.8	5.70	3	
	OR (POS MOM)	1.99	0	79.6	5.70	3	
ML80	IR (CRITICAL)	1.13	0	41.3	4.75	3	
	OR (CRITICAL)	1.88	0	68.9	4.75	3	
	IR (POS MOM)	1.13	0	41.3	4.75	3	
	OR (POS MOM)	1.88	0	68.9	4.75	3	
P-82	IR (CRITICAL)	0.86	0	87.7	4.75	3	
	OR (CRITICAL)	1.43	0	146.2	4.75	3	
	IR (POS MOM)	0.86	0	87.7	4.75	3	
	OR (POS MOM)	1.43	0	146.2	4.75	3	

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SEE HAND OUTS
FOR FLOORPLAN
PAGES.

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LOAD	IR	CRITICAL	LOAD FACTOR RATING		X	FLEM
			FACTOR	TONS		
H20	IR	(CRITICAL)	0.35	0	7.1	8.00
	OR	(CRITICAL)	0.59	0	11.8	8.00
	IR	(POS MOM)	0.35	0	7.1	8.00
HS20	OR	(POS MOM)	0.59	0	11.8	8.00
	IR	(CRITICAL)	0.35	0	12.8	8.00
	OR	(CRITICAL)	0.59	0	21.3	8.00
K527	IR	(POS MOM)	0.35	0	12.8	8.00
	OR	(POS MOM)	0.59	0	21.3	8.00
	IR	(CRITICAL)	0.30	0	11.8	8.00
ML80	OR	(CRITICAL)	0.49	0	19.7	8.00
	IR	(POS MOM)	0.30	0	11.8	8.00
	OR	(POS MOM)	0.49	0	19.7	8.00
P-82	IR	(CRITICAL)	0.26	0	9.4	8.00
	OR	(CRITICAL)	0.43	0	15.6	8.00
	IR	(POS MOM)	0.26	0	9.4	8.00
	OR	(POS MOM)	0.43	0	15.6	8.00
	IR	(CRITICAL)	0.18	0	18.6	8.00
	OR	(CRITICAL)	0.30	0	31.0	8.00
	IR	(POS MOM)	0.18	0	18.6	8.00
	OR	(POS MOM)	0.30	0	31.0	8.00
	OR	(POS MOM)	0.30	0	31.0	8.00

RATING FACTOR CODES:

T - TOP STEEL STRESS/STRENGTH GOVERNS
B - BOTTOM STEEL STRESS/STRENGTH GOVERNS
C - CONCRETE STRESS/STRENGTH GOVERNS
R - REINFORCEMENT STRESS/STRENGTH GOVERNS
V - SHEAR STRESS/STRENGTH GOVERNS
b - blank - COMPACT MOMENT STRENGTH GOVERNS
O - OVERLOAD PROVISIONS GOVERN
I - MOMENT-SHEAR INTERACTION GOVERNS
F - SECTION DOES NOT MEET FLANGE PROJECTION
W - SECTION DOES NOT MEET WEB DEPTH/THICKNESS

NON-COMPACT MOMENT STRENGTH CODES:

B - SECTION IS BRACED

U - SECTION IS UNBRACED

NOTE: ALL RATINGS ARE BASED ON THE NUMBER OF DESIGN LANES OR THE ACTUAL TRAFFIC LANES AS DEFINED BY "D" OR "L" ENTERED FOR LANES IN THE PROJECT IDENTIFICATION.

JOB Monaca BRIDGE RATING
SHEET NO. OF
CALCULATED BY Amik DATE
CHECKED BY DFL DATE 5-20-08
DESCRIPTION

BRIDGE ANALYSIS AND RATING (BAR7) 330443
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USE THIS RUN FOR TRUSS RATINGS

BRIDGE ANALYSIS AND RATING (BAR7) 330443
LAST UPDATED 01/12/2007 05/20/2008 11:48
DOCUMENTATION 01/2007

INPUT: RATING-2.DAT

Special LL ran

STRUCTURE ID -
BRG SLC LIVE OUT- IMP GAGE PASS FAT- CONC RE- S OVER END
TYPE LEV LINES LOAD PUT FACT DIST DIST IGUE DECK SPEC DIST DIR FACTOR PAN HYB
TFS D 2 0.00 N 0.00

BRIDGE CROSS SECTION AND LOADING

OVERHANG CL OF
OR GIRDER OR ROADWAY DISTRIBUTION FACTORS
WIDTH SPACING TRUSS TO CURB WIDTH SHEAR MOMENT DEFLECT
20.67 0.00 -1.00 18.00 0.625 0.727 0.670

SLAB THICKNESS HAUNCH DL1 DL2 F'C N SYMMETRY
10.00 0.00 0.000 0.000 0.000 0.

LIVE LOAD NO. OF END HINGE TEMP
LOCATION PANELS COND. CORRS AT CHANGE BRG.
U 48 R L37 0. U

STRINGER FLOORBEAM
DL1 DL1 UNIT WEIGHT
0.020 0.130 DECK CONCRETE
150.

SPAN LENGTHS (CANTILEVER)

SPAN # 1 2 3 4
LENGTH 103.55 114.15 109.13 94.00

HINGE # 1 2 3
DISTANCE 128.6 192.7 351.8

TRAFFIC LANE LOCATIONS

LANE # 1 2 3 4 5 6
DIST
WIDTH
% LL

STRINGER SPAN LENGTHS (SIMPLE)

SPAN # 1 2 3
LENGTH 9.50 9.50 9.50

DL increased to
account for
Pedestrian LL

55/97

Invalid input:
Should be 25.56 per
steel member prop table

LATERAL BRACE POINTS AND STIFFENER SPACINGS

DEFAULT VALUES

```
+ + + + + T R U S S   A N A L Y S I S + + + + +
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LIVE LOAD DISTRIBUTION FACTORS

TRUSS	TRUSS
FORCE	DEFLECTION
1.000 (2)	1.000 (2)

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* TRUSS - LIVE LOAD H20 *

I.F. = IMPACT FACTOR

MEMBER ID	DL FORCE	LL + IMPACT FORCE		ALLOWABLE FORCE		RATING FACTORS
		COMP	I.F.	COMP	TENS	
U 0U 1	-132.0	-46.8L	1.22	11.3L 1.21	IR -391.5 430.0 OR -488.2 585.3	IR 5.542 OR 7.608
U 1U 2	-132.0	-46.8L	1.22	11.3L 1.21	IR -391.7 430.0 OR -488.5 585.3	IR 5.547 OR 7.614
U 2U 3	-283.2	-110.7L	1.22	33.5L 1.21	IR -713.8 790.9 OR -890.2 1076.5	IR 3.889 OR 5.482
U 3U 4	-283.2	-110.7L	1.22	33.5L 1.21	IR -713.8 790.9 OR -890.2 1076.5	IR 3.889 OR 5.482
U 4U 5	-304.2	-137.4L	1.22	55.7L 1.21	IR -794.8 881.3 OR -991.1 1199.5	IR 3.571 OR 5.000
U 5U 6	-304.2	-137.4L	1.22	55.7L 1.21	IR -794.8 881.3 OR -991.1 1199.5	IR 3.571 OR 5.000
U 6U 7	-194.8	-126.8L	1.22	78.0L 1.21	IR -794.8 881.3 OR -991.1 1199.5	IR 4.731 OR 6.279
U 7U 8	-194.8	-126.8L	1.22	78.0L 1.21	IR -794.8 881.3 OR -991.1 1199.5	IR 4.731 OR 6.279
U 8U 9	44.7	-79.0L	1.22	100.2L 1.21	IR -642.7 713.3 OR -801.5 970.9	IR 6.674 OR 9.245
U 9U10	44.7	-79.0L	1.22	100.2L 1.21	IR -642.7 713.3 OR -801.5 970.9	IR 6.674 OR 9.245
U10U11	391.6	0.0	1.22	121.1L 1.21	IR -797.1 881.3 OR -994.0 1199.5	IR 4.043 OR 6.671
U11U12	391.6	0.0	1.21	121.1L 1.21	IR -797.2 881.3 OR -994.1 1199.5	IR 4.043 OR 6.671
U12U13	101.7	0.0	1.21	35.8L 1.21	IR -393.2 430.0 OR -490.3 585.3	IR 9.166 OR 13.501
U13U14	101.7	0.0	1.21	35.8L 1.21	IR -393.2 430.0 OR -490.3 585.3	IR 9.166 OR 13.501
U14U15	-107.8	-36.3L	1.21	0.0 1.21	IR -465.9 509.4	IR 9.870

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U15U16	-107.8	-36.3L 1.21	0.0	1.21	IR	-466.0	509.4	IR	9.872	OR	13.042
					OR	-581.1	693.4	OR	13.045		
U16U17	-223.5	-77.7L 1.21	0.0	1.21	IR	-466.0	509.4	IR	3.120	OR	4.601
					OR	-581.1	693.4	OR	4.601		
U17U18	-223.5	-77.7L 1.21	0.0	1.21	IR	-465.9	509.4	IR	3.120	OR	4.601
					OR	-581.1	693.4	OR	4.601		
U18U19	-223.7	-77.8L 1.21	0.0	1.21	IR	-465.9	509.4	IR	3.114	OR	4.594
					OR	-581.1	693.4	OR	4.594		
U19U20	-223.7	-77.8L 1.21	0.0	1.21	IR	-465.9	509.4	IR	3.114	OR	4.594
					OR	-581.1	693.4	OR	4.594		
U20U21	-108.3	-36.5L 1.21	0.0	1.21	IR	-466.0	509.4	IR	9.810	OR	12.967
					OR	-581.1	693.4	OR	12.967		
U21U22	-108.3	-36.5L 1.21	0.0	1.21	IR	-465.9	509.4	IR	9.807	OR	12.964
					OR	-581.0	693.4	OR	12.964		
U22U23	101.0	0.0 1.21	35.6L 1.21	IR	-393.2	430.0	IR	9.234	OR	13.592	
					OR	-490.3	585.3	OR	13.592		
U23U24	101.0	0.0 1.21	35.6L 1.21	IR	-393.2	430.0	IR	9.234	OR	13.592	
					OR	-490.3	585.3	OR	13.592		
U24U25	390.7	0.0 1.21	120.7L 1.21	IR	-797.2	881.3	IR	4.065	OR	6.702	
					OR	-994.1	1199.5	OR	6.702		
U25U26	390.7	0.0 1.21	120.7L 1.21	IR	-797.1	881.3	IR	4.065	OR	6.702	
					OR	-994.0	1199.5	OR	6.702		
U26U27	101.4	-81.1L 1.21	121.3L 1.21	IR	-643.3	713.3	IR	5.044	OR	7.168	
					OR	-802.2	970.9	OR	7.168		
U27U28	101.4	-81.1L 1.21	121.3L 1.21	IR	-643.2	713.3	IR	5.044	OR	7.168	
					OR	-802.1	970.9	OR	7.168		
U28U29	-80.5	-132.9L 1.21	122.0L 1.21	IR	-872.3	966.8	IR	5.959	OR	7.581	
					OR	-1087.8	1315.9	OR	7.581		
U29U30	-80.5	-132.9L 1.21	122.0L 1.21	IR	-872.3	966.8	IR	5.959	OR	7.581	
					OR	-1087.8	1315.9	OR	7.581		
U30U31	-138.8	-150.1L 1.21	122.6L 1.21	IR	-872.3	966.8	IR	4.885	OR	6.321	
					OR	-1087.8	1315.9	OR	6.321		
U31U32	-138.8	-150.1L 1.21	123.6L 1.21	IR	-872.3	966.8	IR	4.885	OR	6.321	
					OR	-1087.8	1315.9	OR	6.321		
U32U33	-73.4	-132.9L 1.21	124.3L 1.21	IR	-872.3	966.8	IR	6.012	OR	7.633	
					OR	-1087.8	1315.9	OR	7.633		
U33U34	-73.4	-132.9L 1.21	124.3L 1.21	IR	-872.3	966.8	IR	6.012	OR	7.633	
					OR	-1087.8	1315.9	OR	7.633		
U34U35	115.6	-81.1L 1.21	124.9L 1.21	IR	-643.2	713.3	IR	4.785	OR	6.847	
					OR	-802.1	970.9	OR	6.847		
U35U36	115.6	-81.1L 1.21	124.9L 1.21	IR	-643.3	713.3	IR	4.785	OR	6.847	
					OR	-802.2	970.9	OR	6.847		

U36U37	411.8	0.0 1.21	125.6L 1.22	IR	-797.1	881.3	IR	3.739	OR	6.274	
				OR	-994.0	1199.5	OR	6.274			
U37U38	411.8	0.0 1.23	125.6L 1.22	IR	-797.2	881.3	IR	3.739	OR	6.274	
				OR	-994.1	1199.5	OR	6.274			
U39U39	107.8	0.0 1.23	37.1L 1.22	IR	-393.2	430.0	IR	8.673	OR	12.854	
				OR	-490.3	585.3	OR	12.854			
U39U40	107.8	0.0 1.23	37.1L 1.22	IR	-393.2	430.0	IR	8.673	OR	12.854	
				OR	-490.3	585.3	OR	12.854			
U40U41	-124.9	-41.4L 1.23	0.0 1.22	IR	-537.9	589.3	IR	9.975	OR	13.185	
				OR	-670.8	802.1	OR	13.185			
U41U42	-124.9	-41.4L 1.23	0.0 1.22	IR	-537.9	589.3	IR	9.975	OR	13.185	
				OR	-670.8	802.1	OR	13.185			
U42U43	-259.0	-88.6L 1.23	0.0 1.22	IR	-537.9	589.3	IR	3.150	OR	4.650	
				OR	-670.8	802.1	OR	4.650			
U43U44	-259.0	-88.6L 1.23	0.0 1.22	IR	-537.9	589.3	IR	3.150	OR	4.650	
				OR	-670.8	802.1	OR	4.650			
U44U45	-258.8	-88.5L 1.23	0.0 1.22	IR	-537.9	589.3	IR	3.154	OR	4.656	
				OR	-670.8	802.1	OR	4.656			
U45U46	-258.8	-88.5L 1.23	0.0 1.22	IR	-537.9	589.3	IR	3.154	OR	4.656	
				OR	-670.8	802.1	OR	4.656			
U46U47	-124.6	-41.3L 1.23	0.0 1.22	IR	-537.9	589.3	IR	10.014	OR	13.234	
				OR	-670.8	802.1	OR	13.234			
U47U48	-124.6	-41.3L 1.23	0.0 1.22	IR	-537.9	589.3	IR	10.014	OR	13.234	
				OR	-670.8	802.1	OR	13.234			
L 1L 2	217.0	-22.4L 1.21	83.4L 1.22	IR	-349.2	382.1	IR	1.980	OR	3.634	
				OR	-435.4	520.1	OR	3.634			
L 2L 3	217.0	-22.4L 1.21	83.4L 1.22	IR	-349.2	382.1	IR	1.980	OR	3.634	
				OR	-435.4	520.1	OR	3.634			
L 3L 4	303.0	-44.6L 1.21	128.7L 1.22	IR	-537.1	593.3	IR	2.255	OR	3.920	
				OR	-669.8	807.5	OR	3.920			
L 4L 5	303.0	-44.6L 1.21	128.7L 1.22	IR	-537.1	593.3	IR	2.255	OR	3.920	
				OR	-669.8	807.5	OR	3.920			
L 5L 6	258.9	-66.9L 1.21	136.7L 1.22	IR	-537.1	593.3	IR	2.446	OR	4.012	
				OR	-669.8	807.5	OR	4.012			
L 6L 7	258.9	-66.9L 1.21	136.7L 1.22	IR	-537.1	593.3	IR	2.446	OR	4.012	
				OR	-669.8	807.5	OR	4.012			
L 7L 8	84.4	-89.1L 1.21	107.6L 1.22	IR	-483.7	535.0	IR	4.189	OR	5.985	
				OR	-603.2	728.1	OR	5.985			
L 8L 9	84.4	-89.1L 1.21	107.6L 1.22	IR	-483.7	535.0	IR	4.189	OR	5.985	
				OR	-603.2	728.1	OR	5.985			
L 9L10	-220.3	-111.3L 1.21	41.1L 1.22	IR	-483.7	535.0	IR	2.367	OR	3.441	
				OR	-603.2	728.1	OR	3.441			

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L10L11	-220.3	-111.3L 1.21	41.1L 1.22	IR -484.8 OR -604.6	535.0 728.1	IR 2.377 OR 3.453
L11L12	-238.5	-76.1L 1.21	0.0 1.21	IR -484.9 OR -604.6	535.0 728.1	IR 3.236 OR 4.809
L12L13	-238.5	-76.1L 1.21	0.0 1.21	IR -484.9 OR -604.6	535.0 728.1	IR 3.236 OR 4.809
L15L16	173.7	0.0 1.21	62.2L 1.21	IR -350.3 OR -436.9	382.1 520.1	IR 3.354 OR 5.574
L16L17	173.7	0.0 1.21	62.2L 1.21	IR -350.3 OR -436.9	382.1 520.1	IR 3.354 OR 5.574
L17L18	231.6	0.0 1.21	82.9L 1.21	IR -350.3 OR -436.9	382.1 520.1	IR 1.815 OR 3.479
L18L19	231.6	0.0 1.21	82.9L 1.21	IR -350.3 OR -436.9	382.1 520.1	IR 1.815 OR 3.479
L19L20	174.0	0.0 1.21	62.3L 1.21	IR -350.3 OR -436.9	382.1 520.1	IR 3.343 OR 5.558
L20L21	174.0	0.0 1.21	62.3L 1.21	IR -350.3 OR -436.9	382.1 520.1	IR 3.343 OR 5.558
L23L24	-237.7	-75.8L 1.21	0.0 1.21	IR -484.9 OR -604.6	535.0 728.1	IR 3.259 OR 4.838
L24L25	-237.7	-75.8L 1.21	0.0 1.21	IR -484.9 OR -604.6	535.0 728.1	IR 3.259 OR 4.838
L25L26	-244.0	-121.0L 1.21	42.6L 1.21	IR -484.8 OR -604.6	535.0 728.1	IR 1.990 OR 2.981
L26L27	-244.0	-121.0L 1.21	42.6L 1.21	IR -484.0 OR -603.6	535.0 728.1	IR 1.984 OR 2.973
L27L28	-1.3	-121.6L 1.21	111.3L 1.21	IR -484.0 OR -603.5	535.0 728.1	IR 3.968 OR 4.951
L28L29	-1.3	-121.6L 1.21	111.3L 1.21	IR -484.0 OR -603.5	535.0 728.1	IR 3.968 OR 4.951
L29L30	118.8	-122.3L 1.21	145.8L 1.21	IR -537.4 OR -670.2	593.3 807.5	IR 3.254 OR 4.723
L30L31	118.8	-122.3L 1.21	145.8L 1.21	IR -537.4 OR -670.2	593.3 807.5	IR 3.254 OR 4.723
L31L32	115.2	-123.9L 1.22	145.8L 1.21	IR -537.4 OR -670.2	593.3 807.5	IR 3.278 OR 4.747
L32L33	115.2	-123.9L 1.22	145.8L 1.21	IR -537.4 OR -670.2	593.3 807.5	IR 3.278 OR 4.747
L33L34	-12.0	-124.6L 1.22	111.3L 1.21	IR -484.0 OR -603.5	535.0 728.1	IR 3.788 OR 4.748
L34L35	-12.0	-124.6L 1.22	111.3L 1.21	IR -484.0 OR -603.5	535.0 728.1	IR 3.788 OR 4.748
L35L36	-261.9	-125.3L 1.22	42.6L 1.21	IR -643.3 OR -749.8	535.0 927.6	IR 3.045 OR 5.358

U12L13	235.4	0.0	1.00	83.3L 1.24	IR	-454.7	509.4	IR	3.289	L33U34	137.5	-25.3L 1.30	65.7L 1.24	IR	-284.7	360.0	IR	3.387	
					OR	-567.1	693.4	OR	5.498					OR	-355.1	490.0	OR	5.365	
L13U14	-175.0	-76.0L 1.25	0.0	1.00	IR	-419.7	463.7	IR	3.220	U34L35	-166.8	-75.8L 1.23	22.4L 1.30	IR	-382.8	430.0	IR	2.849	
					OR	-523.4	631.1	OR	4.585					OR	-477.4	585.3	OR	4.097	
U14L15	161.4	0.0	1.00	68.6L 1.26	IR	-343.6	371.5	IR	3.061	L35U36	236.4	-21.0L 1.30	86.8L 1.23	IR	-453.5	509.4	IR	3.144	
					OR	-428.5	505.7	OR	5.016					OR	-565.6	693.4	OR	5.262	
L15U16	-99.1	-55.9L 1.28	7.0	1.30	IR	-263.4	320.8	IR	2.939	U36L37	-257.0	-95.5L 1.23	19.0L 1.23	IR	-526.0	589.3	IR	2.818	
					OR	-328.5	436.6	OR	4.103					OR	-655.9	802.1	OR	4.179	
U16L17	75.1	-14.3	1.30	44.0L 1.29	IR	-178.2	227.7	IR	3.470	L37U38	-275.7	-92.7L 1.23	0.0	1.00	IR	-601.3	681.5	IR	3.513
					OR	-222.2	309.9	OR	5.340					OR	-749.8	927.6	OR	5.116	
L17U18	-12.2	-32.9L 1.30	23.0	1.30	IR	-113.7	153.4	IR	3.090	U38L39	247.5	0.0	1.00	85.2L 1.24	IR	-454.7	509.4	IR	3.073
					OR	-141.8	208.7	OR	3.945					OR	-567.1	693.4	OR	5.231	
U18L19	-11.9	-32.8L 1.30	23.1	1.30	IR	-113.7	153.4	IR	3.106	L39U40	-185.6	-77.5L 1.25	0.0	1.00	IR	-454.7	509.4	IR	3.471
					OR	-141.8	208.7	OR	3.963					OR	-567.1	693.4	OR	4.919	
L19U20	74.8	-14.4	1.30	43.9L 1.29	IR	-178.2	227.7	IR	3.480	U40L41	180.5	0.0	1.00	72.9L 1.25	IR	-342.7	371.5	IR	2.622
					OR	-222.2	309.9	OR	5.352					OR	-427.4	505.7	OR	4.464	
U20L21	-98.9	-55.8L 1.28	7.0	1.30	IR	-263.4	320.8	IR	2.948	L41U42	-110.7	-59.3L 1.27	7.2	1.30	IR	-262.3	320.8	IR	2.555
					OR	-328.5	436.6	OR	4.114					OR	-327.1	436.6	OR	3.647	
L21U22	162.2	0.0	1.00	69.0L 1.26	IR	-343.6	371.5	IR	3.035	U42L43	83.0	-15.1	1.30	46.5L 1.29	IR	-176.9	227.7	IR	3.110
					OR	-428.5	505.7	OR	4.980					OR	-220.6	309.9	OR	4.878	
U22L23	-173.9	-75.8L 1.25	0.0	1.00	IR	-454.7	509.4	IR	3.707	L43U44	-13.7	-34.7L 1.30	24.1	1.30	IR	-112.1	153.4	IR	2.830
					OR	-567.1	693.4	OR	5.189					OR	-139.8	208.7	OR	3.627	
L23U24	235.2	0.0	1.00	83.3L 1.24	IR	-454.7	509.4	IR	3.294	U44L45	-14.0	-34.8L 1.30	24.1	1.30	IR	-112.1	153.4	IR	2.818
					OR	-567.1	693.4	OR	5.504					OR	-139.8	208.7	OR	3.613	
U24L25	-263.4	-90.7L 1.23	0.0	1.00	IR	-601.3	681.5	IR	3.725	L45U46	83.2	-15.1	1.30	46.6L 1.29	IR	-176.9	227.7	IR	3.103
					OR	-749.8	927.6	OR	5.363					OR	-220.6	309.9			

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DL FORCE	LL + IMPACT FORCE		ALLOWABLE FORCE		RATING	
	COMP	I.F.	TENS	I.F.	FACTOR	TONS
-261.9	-125.3L	1.22	42.6L	1.21	IR	1.780
					OR	2.736
					IR	35.59
					OR	54.72
SUPPORT REACTIONS						
	SUPPORT	D.L.	LL+I	I.F.		
	U 0	182.4	72.1	L	1.22	
	L11	503.2	132.1	L	1.21	
	L25	468.7	133.1	L	1.21	
	L37	483.2	136.5	L	1.22	
	U48	150.1	59.1	L	1.23	
***** * TRUSS - LIVE LOAD HS20 * *****						
MEMBER FORCES AND RATINGS I.F. = IMPACT FACTOR						
MEMBER ID	DL FORCE	LL + IMPACT FORCE		ALLOWABLE FORCE		RATING FACTORS
		COMP	I.F.	TENS	I.F.	
U 0U 1	-132.0	-59.4	1.22	15.0	1.21	IR 4.371
					OR	6.000
U 1U 2	-132.0	-59.4	1.22	15.0	1.21	IR 4.374
					OR	6.004
U 2U 3	-283.2	-136.5	1.22	44.3	1.21	IR 3.156
					OR	4.448
U 3U 4	-283.2	-136.5	1.22	44.3	1.21	IR 3.156
					OR	4.448
U 4U 5	-304.2	-165.6	1.22	73.7	1.21	IR 2.963
					OR	4.149
U 5U 6	-304.2	-165.6	1.22	73.7	1.21	IR 2.963
					OR	4.149
U 6U 7	-194.8	-154.4	1.22	103.1	1.21	IR 3.887
					OR	5.159
U 7U 8	-194.8	-154.4	1.22	103.1	1.21	IR 3.887
					OR	5.159
U 8U 9	44.7	-99.1	1.22	132.5	1.21	IR 5.045
					OR	6.989
U 9U10	44.7	-99.1	1.22	132.5	1.21	IR 5.045
					OR	6.989
U10U11	391.6	0.0	1.22	160.2	1.21	IR 3.057
					OR	5.044
U11U12	391.6	0.0	1.21	160.2	1.21	IR 3.057
					OR	5.044
U12U13	101.7	0.0	1.21	53.5	1.21	IR 6.135
					OR	6.135

MEMBER L36L37 IS CRITICAL. SEE DETAILS BELOW.

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U13U14	101.7	0.0	1.21	53.5	1.21	IR	-490.3	585.3	OR	9.037	U34U35	115.6	-99.1	1.21	136.4	1.22	IR	-643.2	713.3	IR	4.382
						OR	-393.2	430.0	IR	6.135						OR	-802.1	970.9	OR	6.271	
U14U15	-107.8	-56.5	1.21	0.0	1.21	IR	-490.3	585.3	OR	9.037	U35U36	115.6	-99.1	1.21	136.4	1.22	IR	-643.3	713.3	IR	4.382
						OR	-465.9	509.4	IR	6.336						OR	-802.2	970.9	OR	6.271	
U15U16	-107.8	-56.5	1.21	0.0	1.21	IR	-581.0	693.4	OR	8.373	U36U37	411.8	0.0	1.21	162.6	1.22	IR	-797.1	881.3	IR	2.888
						OR	-466.0	509.4	IR	6.338						OR	-994.0	1199.5	OR	4.845	
U16U17	-223.5	-113.3	1.21	0.0	1.21	IR	-581.1	693.4	OR	8.375	U37U38	411.8	0.0	1.23	162.6	1.22	IR	-797.2	881.3	IR	2.888
						OR	-466.0	509.4	IR	2.140						OR	-994.1	1199.5	OR	4.845	
U17U18	-223.5	-113.3	1.21	0.0	1.21	IR	-581.1	693.4	OR	3.156	U39U39	107.8	0.0	1.23	54.1	1.22	IR	-393.2	430.0	IR	5.955
						OR	-465.9	509.4	IR	2.140						OR	-490.3	585.3	OR	8.625	
U18U19	-223.7	-113.4	1.21	0.0	1.21	IR	-581.1	693.4	OR	3.151	U39U40	107.8	0.0	1.23	54.1	1.22	IR	-393.2	430.0	IR	5.955
						OR	-465.9	509.4	IR	2.136						OR	-490.3	585.3	OR	8.625	
U19U20	-223.7	-113.4	1.21	0.0	1.21	IR	-581.1	693.4	OR	3.151	U40U41	-124.9	-62.9	1.23	0.0	1.22	IR	-537.9	589.3	IR	6.571
						OR	-465.9	509.4	IR	2.136						OR	-670.8	802.1	OR	8.685	
U20U21	-108.3	-56.8	1.21	0.0	1.21	IR	-581.1	693.4	OR	8.323	U41U42	-124.9	-62.9	1.23	0.0	1.22	IR	-537.9	589.3	IR	6.571
						OR	-466.0	509.4	IR	6.297						OR	-670.8	802.1	OR	8.685	
U21U22	-108.3	-56.8	1.21	0.0	1.21	IR	-581.1	693.4	OR	8.323	U42U43	-259.0	-126.5	1.23	0.0	1.22	IR	-537.9	589.3	IR	2.204
						OR	-465.9	509.4	IR	6.295						OR	-670.8	802.1	OR	3.254	
U22U23	101.0	0.0	1.21	53.2	1.21	IR	-581.0	693.4	OR	8.321	U43U44	-259.0	-126.5	1.23	0.0	1.22	IR	-537.9	589.3	IR	2.204
						OR	-393.2	430.0	IR	6.181						OR	-670.8	802.1	OR	3.254	
U23U24	101.0	0.0	1.21	53.2	1.21	IR	-490.3	585.3	OR	9.099	U44U45	-258.8	-126.5	1.23	0.0	1.22	IR	-537.9	589.3	IR	2.207
						OR	-393.2	430.0	IR	6.181						OR	-670.8	802.1	OR	3.258	
U24U25	390.7	0.0	1.21	159.6	1.21	IR	-490.3	585.3	OR	9.099	U45U46	-258.8	-126.5	1.23	0.0	1.22	IR	-537.9	589.3	IR	2.207
						OR	-797.2	881.3	IR	3.074						OR	-670.8	802.1	OR	3.258	
U25U26	390.7	0.0	1.21	159.6	1.21	IR	-994.1	1199.5	OR	5.068	U46U47	-124.6	-62.7	1.23	0.0	1.22	IR	-537.9	589.3	IR	6.595
						OR	-797.1	881.3	IR	3.074						OR	-670.8	802.1	OR	8.716	
U26U27	101.4	-99.1	1.21	133.9	1.21	IR	-994.0	1199.5	OR	5.068	U47U48	-124.6	-62.7	1.23	0.0	1.22	IR	-537.9	589.3	IR	6.595
						OR	-643.3	713.3	IR	4.569						OR	-670.8	802.1	OR	8.716	
U27U28	101.4	-99.1	1.21	133.9	1.21	IR	-802.2	970.9	OR	6.492	L 1L 2	217.0	-29.7	1.21	104.5	1.22	IR	-349.2	382.1	IR	1.581
						OR	-643.2	713.3	IR	4.569						OR	-435.4	520.1	OR	2.902	
U28U29	-80.5	-157.7	1.21	122.0L	1.21	IR	-802.1	970.9	OR	6.492	L 2L 3	217.0	-29.7	1.21	104.5	1.22	IR	-349.2	382.1	IR	1.581
						OR	-872.3	966.8	IR	5.021						OR	-435.4	520.1	OR	2.902	
U29U30	-80.5	-157.7	1.21	122.0L	1.21	IR	-1087.8	1315.9	OR	6.388	L 3L 4	303.0	-59.0	1.21	156.6	1.22	IR	-537.1	593.3	IR	1.853
						OR	-872.3	966.8	IR	5.021						OR	-669.8	807.5	OR	3.222	
U30U31	-138.8	-175.2	1.21	122.6L	1.21	IR	-1087.8	1315.9	OR	6.388	L 4L 5	303.0	-59.0	1.21	156.6	1.22	IR	-537.1	593.3	IR	1.853
						OR	-872.3	966.8	IR	4.188						OR	-669.8	807.5	OR	3.222	
U31U32	-138.8	-175.2	1.21	123.6L	1.22	IR	-1087.8	1315.9	OR	5.418	L 5L 6	258.9	-88.4	1.21	165.1	1.22	IR	-537.1	593.3	IR	2.025
						OR	-872.3	966.8	IR	4.188						OR	-669.8	807.5	OR	3.323	
U32U33	-73.4	-157.7	1.21	124.3L	1.22	IR	-1087.8	1315.9	OR	5.418	L 6L 7	258.9	-88.4	1.21	165.1	1.22	IR	-537.1	593.3	IR	2.025
						OR	-872.3	966.8	IR	5.066						OR	-669.8	807.5	OR	3.323	
U33U34	-73.4	-157.7	1.21	124.3L	1.22	IR	-1087.8	1315.9	OR	6.432	L 7L 8	84.4	-117.8	1.21	132.9	1.22	IR	-483.7	535.0	IR	3.390
						OR	-872.3	966.8	IR	5.066						OR	-603.2	728.1	OR	4.844	

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U10L11	-293.8	-93.0L 1.23	0.0	1.00	IR	-601.2	681.5	IR	3.305	L31U32	37.9	-39.3L 1.29	48.4	1.27	IR	-117.8	174.6	IR	2.824
					OR	-749.7	927.6	OR	4.902						OR	-146.9	237.6	OR	4.127
L11U12	-263.7	-107.0 1.23	0.0	1.00	IR	-601.3	681.5	IR	3.156	U32L33	-67.3	-57.6 1.26	32.0L 1.30	IR	-208.6	277.9	IR	2.452	
					OR	-749.8	927.6	OR	4.545						OR	-260.1	378.3	OR	3.346
U12L13	235.4	0.0 1.00	101.9	1.24	IR	-454.7	509.4	IR	2.688	L33U34	137.5	-25.3L 1.30	66.7 1.24	IR	-284.7	360.0	IR	3.334	
					OR	-567.1	693.4	OR	4.493						OR	-355.1	490.0	OR	5.281
L13U14	-175.0	-95.1 1.25	0.0	1.00	IR	-419.7	463.7	IR	2.573	U34L35	-166.8	-75.8L 1.23	23.4 1.30	IR	-382.8	430.0	IR	2.849	
					OR	-523.4	631.1	OR	3.663						OR	-477.4	585.3	OR	4.097
U14L15	161.4	0.0 1.00	88.5	1.26	IR	-343.6	371.5	IR	2.373	L35U36	236.4	-23.3 1.30	86.8L 1.23	IR	-453.5	509.4	IR	3.144	
					OR	-428.5	505.7	OR	3.889						OR	-565.6	693.4	OR	5.262
L15U16	-99.1	-74.4 1.28	7.0	1.30	IR	-263.4	320.8	IR	2.208	U36L37	-257.0	-95.5L 1.23	21.4 1.23	IR	-526.0	589.3	IR	2.818	
					OR	-328.5	436.6	OR	3.082						OR	-655.9	802.1	OR	4.179
U16L17	75.1	-15.7 1.30	59.8	1.29	IR	-178.2	227.7	IR	2.554	L37U38	-275.7	-106.7 1.23	0.0 1.00	IR	-601.3	681.5	IR	3.052	
					OR	-222.2	309.9	OR	3.931						OR	-749.8	927.6	OR	4.445
L17U18	-12.2	-44.5 1.30	29.6	1.30	IR	-113.7	153.4	IR	2.284	U38L39	247.5	0.0 1.00	102.0 1.24	IR	-454.7	509.4	IR	2.569	
					OR	-141.8	208.7	OR	2.916						OR	-567.1	693.4	OR	4.372
U18L19	-11.9	-44.4 1.30	29.7	1.30	IR	-113.7	153.4	IR	2.296	L39U40	-185.6	-95.1 1.25	0.0 1.00	IR	-454.7	509.4	IR	2.829	
					OR	-141.8	208.7	OR	2.929						OR	-567.1	693.4	OR	4.010
L19U20	74.8	-15.7 1.30	59.7	1.29	IR	-178.2	227.7	IR	2.562	U40L41	180.5	0.0 1.00	92.2 1.25	IR	-342.7	371.5	IR	2.072	
					OR	-222.2	309.9	OR	3.940						OR	-427.4	505.7	OR	3.527
U20L21	-98.9	-74.3 1.28	7.0	1.30	IR	-263.4	320.8	IR	2.214	L41U42	-110.7	-77.9 1.27	7.2 1.30	IR	-262.3	320.8	IR	1.945	
					OR	-328.5	436.6	OR	3.090						OR	-327.1	436.6	OR	2.776
L21U22	162.2	0.0 1.00	89.0	1.26	IR	-343.6	371.5	IR	2.353	U42L43	83.0	-17.2 1.30	62.8 1.29	IR	-176.9	227.7	IR	2.303	
					OR	-428.5	505.7	OR	3.860						OR	-220.6	309.9	OR	3.611
U22L23	-173.9	-94.8 1.25	0.0	1.00	IR	-454.7	509.4	IR	2.963	L43U44	-13.7	-47.3 1.30	31.6 1.30	IR	-112.1	153.4	IR	2.081	
					OR	-567.1	693.4	OR	4.148						OR	-139.8	208.7	OR	2.667
L23U24	235.2	0.0 1.00	101.9	1.24	IR	-454.7	509.4	IR	2.692	U44L45	-14.0	-47.4 1.30	31.5 1.30	IR	-112.1	153.4	IR	2.072	
					OR	-567.1	693.4	OR	4.497						OR	-139.8	208.7	OR	2.657
U24L25	-263.4	-107.0 1.23	0.0	1.00	IR	-601.3	681.5	IR	3.159	L45U46	83.2	-17.1 1.30	62.9 1.29	IR	-176.9	227.7	IR	2.298	
					OR	-749.8	927.6	OR	4.547						OR	-220.6	309.9	OR	3.605
L25U26	-251.5	-95.3L 1.23	21.5	1.23	IR	-526.0	589.3	IR	2.880	U46L47	-110.9	-78.0 1.27	7.2 1.30	IR	-262.3	320.8	IR	1.942	
					OR	-655.9	802.1	OR	4.243						OR	-327.1	436.6	OR	2.774
U26L27	230.6	-23.6 1.30	86.6L 1.23	IR	-453.5	509.4	IR	3.218	L47U48	180.1	0.0 1.00	92.8 1.26	IR	-342.7	371.5	IR	2.063		
					OR	-565.6	693.4	OR	5.342						OR	-427.4	505.7	OR	3.509
L27U28	-161.1	-75.8 1.23	23.6	1.30	IR	-382.8	430.0	IR	2.925	U 1L 1	-55.0	-41.8L 1.30	0.0 1.00	IR	-148.8	201.1	IR	2.245	
					OR	-477.4	585.3	OR	4.173						OR	-185.5	273.7	OR	3.124
U28L29	131.8	-25.8L 1.30	66.8 1.24	IR	-284.7	360.0	IR	3.419	U 3L 3	-57.0	-41.7L 1.30	0.0 1.00	IR	-148.8	201.1	IR	2.200		
					OR	-355.1	490.0	OR	5.366						OR	-185.5	273.7	OR	3.082
L29U30	-61.6	-57.6 1.26	32.6L 1.30	IR	-208.6	277.9	IR	2.550	U 5L 5	-57.0	-41.7L 1.30	0.0 1.00	IR	-148.8	201.1	IR	2.200		
					OR	-260.1	378.3	OR	3.444						OR	-185.5	273.7	OR	3.082
U30L31	32.2	-39.8L 1.29	48.4 1.27	IR	-117.8	174.6	IR	2.942	U 7L 7	-57.0	-41.7L 1.30	0.0 1.00	IR	-148.8	201.1	IR	2.200		
					OR	-146.9	237.6	OR	4.244						OR	-185.5	273.7	OR	3.082
										U 9L 9	-57.0	-41.7L 1.30	0.0 1.00	IR	-148.8	201.1	IR	2.200	

U10L10	0.0	0.0	0.0	1.00	0.0	1.00	IR	-185.5	273.7	OR	3.082
U11L11	-50.0	-41.6	1.30	0.0	1.00	IR	IR	-148.8	201.1	IR	999.999
U12L12	0.0	0.0	1.00	0.0	1.00	IR	IR	-148.8	201.1	IR	999.999
U13L13	-50.0	-41.6	1.30	0.0	1.00	IR	IR	-148.8	201.1	IR	2.374
U15L15	-47.0	-41.6	1.30	0.0	1.00	IR	IR	-141.8	174.6	IR	3.258
U17L17	-47.0	-41.6	1.30	0.0	1.00	IR	IR	-141.8	174.6	IR	2.280
U19L19	-47.0	-41.6	1.30	0.0	1.00	IR	IR	-176.9	237.6	OR	3.122
U21L21	-47.0	-41.6	1.30	0.0	1.00	IR	IR	-141.8	174.6	IR	2.280
U23L23	-50.0	-41.6	1.30	0.0	1.00	IR	IR	-141.8	174.6	IR	3.122
U24L24	0.0	0.0	1.00	0.0	1.00	IR	IR	-148.8	201.1	IR	999.999
U25L25	-50.0	-41.6	1.30	0.0	1.00	IR	IR	-148.8	201.1	IR	2.374
U26L26	0.0	0.0	1.00	0.0	1.00	IR	IR	-148.8	201.1	IR	999.999
U27L27	-55.0	-41.6	1.30	0.0	1.00	IR	IR	-148.8	201.1	IR	2.254
U29L29	-55.0	-41.6	1.30	0.0	1.00	IR	IR	-185.5	273.7	OR	3.138
U31L31	-55.0	-41.6	1.30	0.0	1.00	IR	IR	-148.8	201.1	IR	2.254
U33L33	-55.0	-41.6	1.30	0.0	1.00	IR	IR	-148.8	201.1	IR	3.138
U35L35	-55.0	-41.6	1.30	0.0	1.00	IR	IR	-148.8	201.1	IR	2.254
U37L37	-50.0	-41.6	1.30	0.0	1.00	IR	IR	-148.8	201.1	IR	3.138
U39L39	-50.0	-41.6	1.30	0.0	1.00	IR	IR	-148.8	201.1	IR	2.374
U41L41	-50.0	-41.6	1.30	0.0	1.00	IR	IR	-141.8	174.6	IR	2.208
U43L43	-50.0	-41.6	1.30	0.0	1.00	IR	IR	-176.9	237.6	OR	3.050
U45L45	-50.0	-41.6	1.30	0.0	1.00	IR	IR	-141.8	174.6	IR	2.208
								-176.9	237.6	OR	3.050

U47L47	-50.0	-41.6	1.30	0.0	1.00	IR	IR	-141.8	174.6	IR	2.208
								-176.9	237.6	OR	3.050
MEMBER L18L19 IS CRITICAL. SEE DETAILS BELOW.											
	DL	LL + IMPACT FORCE	ALLOWABLE FORCE								
	FORCE	COMP I.F.	TENS I.F.	COMP TENS						RATING	
231.6	0.0	1.21	117.4	1.21	IR	-350.3	382.1	IR	1.282	46.15	
					OR	-436.9	520.1	OR	2.457	88.46	
SUPPORT REACTIONS											
	SUPPORT	D.L.	LL+I	I.F.							
	U 0	182.4	79.8	1.22							
	L11	503.2	132.1	L 1.21							
	L25	468.7	133.1	L 1.21							
	L37	483.2	136.5	L 1.22							
	U48	150.1	76.5	1.23							

* TRUSS - LIVE LOAD TRS27 *											

MEMBER FORCES AND RATINGS											
I.F. = IMPACT FACTOR											
MEMBER	DL	LL + IMPACT FORCE	ALLOWABLE FORCE								
ID	FORCE	COMP I.F.	TENS I.F.	COMP TENS						RATING	FACTORS
U 0U 1	-132.0	-67.4	1.22	17.0	1.21	IR	-391.5	430.0	IR	3.849	
						OR	-488.2	585.3	OR	5.284	
U 1U 2	-132.0	-67.4	1.22	17.0	1.21	IR	-391.7	430.0	IR	3.853	
						OR	-488.5	585.3	OR	5.288	
U 2U 3	-283.2	-155.5	1.22	50.5	1.21	IR	-713.8	790.9	IR	2.769	
						OR	-890.2	1076.5	OR	3.903	
U 3U 4	-283.2	-155.5	1.22	50.5	1.21	IR	-713.8	790.9	IR	2.769	
						OR	-890.2	1076.5	OR	3.903	
U 4U 5	-304.2	-189.0	1.22	84.0	1.21	IR	-794.8	881.3	IR	2.596	
						OR	-991.1	1199.5	OR	3.635	
U 5U 6	-304.2	-189.0	1.22	84.0	1.21	IR	-794.8	881.3	IR	2.596	
						OR	-991.1	1199.5	OR	3.635	
U 6U 7	-194.8	-177.3	1.22	117.5	1.21	IR	-794.8	881.3	IR	3.383	
						OR	-991.1	1199.5	OR	4.490	
U 7U 8	-194.8	-177.3	1.22	117.5	1.21	IR	-794.8	881.3	IR	3.383	
						OR	-991.1	1199.5	OR	4.490	
U 8U 9	44.7	-112.4	1.22	151.0	1.21	IR	-642.7	713.3	IR	4.429	
						OR	-801.5	970.9	OR	6.135	
U 9U10	44.7	-112.4	1.22	151.0	1.21	IR	-642.7	713.3	IR	4.429	
						OR	-801.5	970.9	OR	6.135	
U10U11	391.6	0.0	1.22	182.5	1.21	IR	-797.1	881.3	IR	2.683	

U11U12	391.6	0.0	1.21	182.5	1.21	IR	-994.0	1199.5	OR	4.427
U12U13	101.7	0.0	1.21	60.5	1.21	IR	-797.2	881.3	IR	2.683
U13U14	101.7	0.0	1.21	60.5	1.21	IR	-994.1	1199.5	OR	4.427
U14U15	-107.8	-63.8	1.21	0.0	1.21	IR	-393.2	430.0	IR	5.423
U15U16	-107.8	-63.8	1.21	0.0	1.21	IR	-490.3	585.3	OR	7.989
U16U17	-223.5	-130.3	1.21	0.0	1.21	IR	-393.2	430.0	IR	5.423
U17U18	-223.5	-130.3	1.21	0.0	1.21	IR	-490.3	585.3	OR	7.989
U18U19	-223.7	-130.4	1.21	0.0	1.21	IR	-393.2	430.0	IR	5.616
U19U20	-223.7	-130.4	1.21	0.0	1.21	IR	-465.9	509.4	IR	7.422
U20U21	-108.3	-64.1	1.21	0.0	1.21	IR	-581.0	693.4	OR	7.422
U21U22	-108.3	-64.1	1.21	0.0	1.21	IR	-466.0	509.4	IR	5.618
U22U23	101.0	0.0	1.21	60.2	1.21	IR	-581.1	693.4	OR	7.423
U23U24	101.0	0.0	1.21	60.2	1.21	IR	-466.0	509.4	IR	1.861
U24U25	390.7	0.0	1.21	181.9	1.21	IR	-581.1	693.4	OR	2.744
U25U26	390.7	0.0	1.21	181.9	1.21	IR	-465.9	509.4	IR	1.861
U26U27	101.4	-112.5	1.21	152.6	1.21	IR	-581.1	693.4	OR	2.744
U27U28	101.4	-112.5	1.21	152.6	1.21	IR	-465.9	509.4	IR	1.858
U28U29	-80.5	-181.3	1.21	121.8	1.21	IR	-581.1	693.4	OR	2.741
U29U30	-80.5	-181.3	1.21	121.8	1.21	IR	-465.9	509.4	IR	1.858
U30U31	-138.8	-199.5	1.21	91.7	1.21	IR	-581.1	693.4	OR	2.741
U31U32	-138.8	-199.5	1.21	92.4	1.22	IR	-466.0	509.4	IR	5.581
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U32U33	-73.4	-181.4	1.21	123.7	1.22	IR	-872.3	966.8	IR	4.405
U33U34	-73.4	-181.4	1.21	123.7	1.22	IR	-1087.8	1315.9	OR	5.593
U34U35	115.6	-112.5	1.21	155.0	1.22	IR	-872.3	966.8	IR	4.405
U35U36	115.6	-112.5	1.21	155.0	1.22	IR	-1087.8	1315.9	OR	5.593
U36U37	411.8	0.0	1.21	184.7	1.22	IR	-643.2	713.3	IR	3.856
U37U38	411.8	0.0	1.23	184.7	1.22	IR	-802.1	970.9	OR	5.518
U39U39	107.8	0.0	1.23	61.2	1.22	IR	-643.3	713.3	IR	3.856
U39U40	107.8	0.0	1.23	61.2	1.22	IR	-802.2	970.9	OR	5.518
U40U41	-124.9	-71.0	1.23	0.0	1.22	IR	-797.1	881.3	IR	2.541
U41U42	-124.9	-71.0	1.23	0.0	1.22	IR	-994.0	1199.5	OR	4.264
U42U43	-259.0	-145.4	1.23	0.0	1.22	IR	-797.2	881.3	IR	2.541
U43U44	-259.0	-145.4	1.23	0.0	1.22	IR	-994.1	1199.5	OR	4.264
U44U45	-258.8	-145.3	1.23	0.0	1.22	IR	-393.2	430.0	IR	5.262
U45U46	-258.8	-145.3	1.23	0.0	1.22	IR	-490.3	585.3	OR	7.798
U46U47	-124.6	-70.8	1.23	0.0	1.22	IR	-393.2	430.0	IR	5.262
U47U48	-124.6	-70.8	1.23	0.0	1.22	IR	-490.3	585.3	OR	7.798
L 1L 2	217.0	-33.8	1.21	118.5	1.22	IR	-537.9	589.3	IR	5.817
L 2L 3	217.0	-33.8	1.21	118.5	1.22	IR	-670.8	802.1	OR	7.689
L 3L 4	303.0	-67.3	1.21	179.8	1.22	IR	-537.9	589.3	IR	5.817
L 4L 5	303.0	-67.3	1.21	179.8	1.22	IR	-670.8	802.1	OR	7.689
L 5L 6	258.9	-100.7	1.21	188.6	1.22	IR	-537.9	589.3	IR	1.918
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L 6L 7	258.9	-100.7	1.21	188.6	1.22	IR	-537.1	593.3	IR	1.773	OR	2.909
L 7L 8	84.4	-134.2	1.21	151.1	1.22	IR	-483.7	535.0	IR	2.983	OR	4.261
L 8L 9	84.4	-134.2	1.21	151.1	1.22	IR	-483.7	535.0	IR	2.983	OR	4.261
L 9L10	-220.3	-167.7	1.21	59.3	1.22	IR	-483.7	535.0	IR	1.571	OR	2.283
L10L11	-220.3	-167.7	1.21	59.3	1.22	IR	-484.8	535.0	IR	1.577	OR	2.291
L11L12	-238.5	-120.8	1.21	0.0	1.21	IR	-484.9	535.0	IR	2.039	OR	3.030
L12L13	-238.5	-120.8	1.21	0.0	1.21	IR	-484.9	535.0	IR	2.039	OR	3.030
L15L16	173.7	0.0	1.21	105.3	1.21	IR	-350.3	382.1	IR	1.980	OR	3.290
L16L17	173.7	0.0	1.21	105.3	1.21	IR	-350.3	382.1	IR	1.980	OR	3.290
L17L18	231.6	0.0	1.21	133.1	1.21	IR	-350.3	382.1	IR	1.130	OR	2.167
L18L19	231.6	0.0	1.21	133.1	1.21	IR	-350.3	382.1	IR	1.130	OR	2.167
L19L20	174.0	0.0	1.21	105.5	1.21	IR	-350.3	382.1	IR	1.973	OR	3.280
L20L21	174.0	0.0	1.21	105.5	1.21	IR	-350.3	382.1	IR	1.973	OR	3.280
L23L24	-237.7	-120.3	1.21	0.0	1.21	IR	-484.9	535.0	IR	2.055	OR	3.050
L24L25	-237.7	-120.3	1.21	0.0	1.21	IR	-484.9	535.0	IR	2.055	OR	3.050
L25L26	-244.0	-167.9	1.21	59.7	1.21	IR	-484.8	535.0	IR	1.434	OR	2.147
L26L27	-244.0	-167.9	1.21	59.7	1.21	IR	-484.0	535.0	IR	1.429	OR	2.142
L27L28	-1.3	-137.2	1.21	152.2	1.21	IR	-484.0	535.0	IR	3.518	OR	4.389
L28L29	-1.3	-137.2	1.21	152.2	1.21	IR	-484.0	535.0	IR	3.518	OR	4.389
L29L30	118.8	-106.4	1.21	137.1	1.21	IR	-537.4	593.3	IR	2.407	OR	3.494
L30L31	118.8	-106.4	1.21	137.1	1.21	IR	-537.4	593.3	IR	2.407	OR	3.494
L31L32	115.2	-108.0	1.22	137.1	1.21	IR	-537.4	593.3	IR	2.425	OR	

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L29U30	-61.6	-65.2	1.26	32.6	1.30	IR	-208.6	277.9	IR	2.255
						OR	-260.1	378.3	OR	3.047
U30L31	32.2	-43.8	1.29	54.6	1.27	IR	-117.8	174.6	IR	2.609
						OR	-146.9	237.6	OR	3.764
L31U32	37.9	-43.8	1.29	54.6	1.27	IR	-117.8	174.6	IR	2.505
						OR	-146.9	237.6	OR	3.660
U32L33	-67.3	-65.2	1.26	32.7	1.30	IR	-208.6	277.9	IR	2.169
						OR	-260.1	378.3	OR	2.960
L33U34	137.5	-26.6	1.30	75.6	1.24	IR	-284.7	360.0	IR	2.944
						OR	-355.1	490.0	OR	4.664
U34L35	-166.8	-85.9	1.23	26.6	1.30	IR	-382.8	430.0	IR	2.515
						OR	-477.4	585.3	OR	3.616
L35U36	236.4	-26.6	1.30	96.3	1.23	IR	-453.5	509.4	IR	2.835
						OR	-565.6	693.4	OR	4.746
U36L37	-257.0	-103.6	1.23	24.4	1.23	IR	-526.0	589.3	IR	2.596
						OR	-655.9	802.1	OR	3.850
L37U38	-275.7	-120.6	1.23	0.0	1.00	IR	-601.3	681.5	IR	2.699
						OR	-749.8	927.6	OR	3.931
U38L39	247.5	0.0	1.00	116.4	1.24	IR	-454.7	509.4	IR	2.250
						OR	-567.1	693.4	OR	3.830
L39U40	-185.6	-107.7	1.25	0.0	1.00	IR	-454.7	509.4	IR	2.500
						OR	-567.1	693.4	OR	3.543
U40L41	180.5	0.0	1.00	104.1	1.25	IR	-342.7	371.5	IR	1.834
						OR	-427.4	505.7	OR	3.123
L41U42	-110.7	-87.8	1.27	7.3	1.30	IR	-262.3	320.8	IR	1.727
						OR	-327.1	436.6	OR	2.465
U42L43	83.0	-19.8	1.30	70.5	1.29	IR	-176.9	227.7	IR	2.053
						OR	-220.6	309.9	OR	3.219
L43U44	-13.7	-52.7	1.30	35.9	1.30	IR	-112.1	153.4	IR	1.868
						OR	-139.8	208.7	OR	2.394
U44L45	-14.0	-52.8	1.30	35.8	1.30	IR	-112.1	153.4	IR	1.860
						OR	-139.8	208.7	OR	2.384
L45U46	83.2	-19.8	1.30	70.6	1.29	IR	-176.9	227.7	IR	2.048
						OR	-220.6	309.9	OR	3.214
U46L47	-110.9	-87.8	1.27	7.2	1.30	IR	-262.3	320.8	IR	1.724
						OR	-327.1	436.6	OR	2.462
L47U48	180.1	0.0	1.00	104.8	1.26	IR	-342.7	371.5	IR	1.826
						OR	-427.4	505.7	OR	3.107
U 1L 1	-55.0	-50.4	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.860
						OR	-185.5	273.7	OR	2.589
U 3L 3	-57.0	-50.2	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.829
						OR	-185.5	273.7	OR	2.561
U 5L 5	-57.0	-50.2	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.829

U43L43	-50.0	-47.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.926
						OR	-176.9	237.6	OR	2.661
U45L45	-50.0	-47.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.927
						OR	-176.9	237.6	OR	2.662
U47L47	-50.0	-47.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.928
						OR	-176.9	237.6	OR	2.663

MEMBER L18L19 IS CRITICAL. SEE DETAILS BELOW.

DL FORCE	LL + IMPACT FORCE		ALLOWABLE FORCE		RATING
	COMP	I.F.	TENS	I.F.	FACTOR TONS
231.6	0.0	1.21	133.1	1.21	IR -350.3 382.1 IR 1.130 45.22
					OR -436.9 520.1 OR 2.167 86.67

SUPPORT REACTIONS

SUPPORT	D.L.	LL+I	I.F.
U 0	182.4	90.7	1.22
L11	503.2	115.5	1.21
L25	468.7	114.4	1.21
L37	483.2	115.3	1.22
U48	150.1	86.6	1.23

 * TRUSS - LIVE LOAD ML80 *

MEMBER FORCES AND RATINGS I.F. = IMPACT FACTOR

MEMBER ID	DL FORCE	LL + IMPACT FORCE		ALLOWABLE FORCE		RATING FACTORS
		COMP	I.F.	COMP	TENS	
U 0U 1	-132.0	-64.3	1.22	16.5	1.21	IR 4.036 OR 5.540
U 1U 2	-132.0	-64.3	1.22	16.5	1.21	IR 4.039 OR 5.544
U 2U 3	-283.2	-149.3	1.22	48.9	1.21	IR 2.885 OR 4.066
U 3U 4	-283.2	-149.3	1.22	48.9	1.21	IR 2.885 OR 4.066
U 4U 5	-304.2	-185.0	1.22	81.3	1.21	IR 2.652 OR 3.714
U 5U 6	-304.2	-185.0	1.22	81.3	1.21	IR 2.652 OR 3.714
U 6U 7	-194.8	-171.4	1.22	113.8	1.21	IR 3.499 OR 4.644
U 7U 8	-194.8	-171.4	1.22	113.8	1.21	IR 3.499 OR 4.644
U 8U 9	44.7	-107.7	1.22	146.2	1.21	IR 4.574

U 7L 7	-57.0	-50.2	1.30	0.0	1.00	IR	-185.5	273.7	OR	2.561
						OR	-148.8	201.1	IR	1.829
U 9L 9	-57.0	-50.2	1.30	0.0	1.00	IR	-185.5	273.7	OR	2.561
						OR	-148.8	201.1	IR	1.829
U10L10	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U11L11	-50.0	-46.8	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.111
						OR	-185.5	273.7	OR	2.897
U12L12	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U13L13	-50.0	-46.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.115
						OR	-185.5	273.7	OR	2.902
U15L15	-47.0	-45.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	2.077
						OR	-176.9	237.6	OR	2.844
U17L17	-47.0	-45.5	1.30	0.0	1.00	IR	-141.8	174.6	IR	2.082
						OR	-176.9	237.6	OR	2.852
U19L19	-47.0	-45.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	2.082
						OR	-176.9	237.6	OR	2.851
U21L21	-47.0	-45.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	2.077
						OR	-176.9	237.6	OR	2.844
U23L23	-50.0	-46.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.115
						OR	-185.5	273.7	OR	2.902
U24L24	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U25L25	-50.0	-46.8	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.111
						OR	-185.5	273.7	OR	2.897
U26L26	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U27L27	-55.0	-49.4	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.897
						OR	-185.5	273.7	OR	2.641
U29L29	-55.0	-49.5	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.894
						OR	-185.5	273.7	OR	2.636
U31L31	-55.0	-49.5	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.894
						OR	-185.5	273.7	OR	2.636
U33L33	-55.0	-49.5	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.894
						OR	-185.5	273.7	OR	2.636
U35L35	-55.0	-49.4	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.897
						OR	-185.5	273.7	OR	2.641
U37L37	-50.0	-46.8	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.111
						OR	-185.5	273.7	OR	2.897
U39L39	-50.0	-46.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	2.115
						OR	-185.5	273.7	OR	2.902
U41L41	-50.0	-47.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.926
						OR	-176.9	237.6	OR	2.661

U 9U10	44.7	-107.7	1.22	146.2	1.21	IR	-801.5	970.9	OR	6.336	U30U31	-138.8	-195.6	1.21	88.5	1.21	IR	-872.3	966.8	IR	3.750
						OR	-642.7	713.3	IR	4.574						OR	-1087.8	1315.9	OR	4.851	
U10U11	391.6	0.0	1.22	176.7	1.21	IR	-797.1	881.3	IR	2.771	U31U32	-138.8	-195.6	1.21	89.2	1.22	IR	-872.3	966.8	IR	3.750
						OR	-994.0	1199.5	OR	4.572						OR	-1087.8	1315.9	OR	4.851	
U11U12	391.6	0.0	1.21	176.7	1.21	IR	-797.2	881.3	IR	2.771	U32U33	-73.4	-174.4	1.21	119.4	1.22	IR	-872.3	966.8	IR	4.581
						OR	-994.1	1199.5	OR	4.572						OR	-1087.8	1315.9	OR	5.817	
U12U13	101.7	0.0	1.21	59.0	1.21	IR	-393.2	430.0	IR	5.566	U33U34	-73.4	-174.4	1.21	119.4	1.22	IR	-872.3	966.8	IR	4.581
						OR	-490.3	585.3	OR	8.198						OR	-1087.8	1315.9	OR	5.817	
U13U14	101.7	0.0	1.21	59.0	1.21	IR	-393.2	430.0	IR	5.566	U34U35	115.6	-107.4	1.21	149.7	1.22	IR	-643.2	713.3	IR	3.993
						OR	-490.3	585.3	OR	8.198						OR	-802.1	970.9	OR	5.714	
U14U15	-107.8	-62.8	1.21	0.0	1.21	IR	-465.9	509.4	IR	5.703	U35U36	115.6	-107.4	1.21	149.7	1.22	IR	-643.3	713.3	IR	3.993
						OR	-581.0	693.4	OR	7.536						OR	-802.2	970.9	OR	5.714	
U15U16	-107.8	-62.8	1.21	0.0	1.21	IR	-466.0	509.4	IR	5.704	U36U37	411.8	0.0	1.21	178.4	1.22	IR	-797.1	881.3	IR	2.631
						OR	-581.1	693.4	OR	7.538						OR	-994.0	1199.5	OR	4.415	
U16U17	-223.5	-131.6	1.21	0.0	1.21	IR	-466.0	509.4	IR	1.843	U37U38	411.8	0.0	1.23	178.4	1.22	IR	-797.2	881.3	IR	2.631
						OR	-581.1	693.4	OR	2.718						OR	-994.1	1199.5	OR	4.415	
U17U18	-223.5	-131.6	1.21	0.0	1.21	IR	-465.9	509.4	IR	1.843	U39U39	107.8	0.0	1.23	59.4	1.22	IR	-393.2	430.0	IR	5.426
						OR	-581.1	693.4	OR	2.718						OR	-490.3	585.3	OR	8.042	
U18U19	-223.7	-131.7	1.21	0.0	1.21	IR	-465.9	509.4	IR	1.839	U39U40	107.8	0.0	1.23	59.4	1.22	IR	-393.2	430.0	IR	5.426
						OR	-581.1	693.4	OR	2.714						OR	-490.3	585.3	OR	8.042	
U19U20	-223.7	-131.7	1.21	0.0	1.21	IR	-465.9	509.4	IR	1.839	U40U41	-124.9	-69.5	1.23	0.0	1.22	IR	-537.9	589.3	IR	5.942
						OR	-581.1	693.4	OR	2.714						OR	-670.8	802.1	OR	7.854	
U20U21	-108.3	-63.1	1.21	0.0	1.21	IR	-466.0	509.4	IR	5.667	U41U42	-124.9	-69.5	1.23	0.0	1.22	IR	-537.9	589.3	IR	5.942
						OR	-581.1	693.4	OR	7.491						OR	-670.8	802.1	OR	7.854	
U21U22	-108.3	-63.1	1.21	0.0	1.21	IR	-465.9	509.4	IR	5.666	U42U43	-259.0	-145.7	1.23	0.0	1.22	IR	-537.9	589.3	IR	1.915
						OR	-581.0	693.4	OR	7.489						OR	-670.8	802.1	OR	2.827	
U22U23	101.0	0.0	1.21	58.6	1.21	IR	-393.2	430.0	IR	5.609	U43U44	-259.0	-145.7	1.23	0.0	1.22	IR	-537.9	589.3	IR	1.915
						OR	-490.3	585.3	OR	8.257						OR	-670.8	802.1	OR	2.827	
U23U24	101.0	0.0	1.21	58.6	1.21	IR	-393.2	430.0	IR	5.609	U44U45	-258.8	-145.6	1.23	0.0	1.22	IR	-537.9	589.3	IR	1.917
						OR	-490.3	585.3	OR	8.257						OR	-670.8	802.1	OR	2.830	
U24U25	390.7	0.0	1.21	176.2	1.21	IR	-797.2	881.3	IR	2.785	U45U46	-258.8	-145.6	1.23	0.0	1.22	IR	-537.9	589.3	IR	1.917
						OR	-994.1	1199.5	OR	4.591						OR	-670.8	802.1	OR	2.830	
U25U26	390.7	0.0	1.21	176.2	1.21	IR	-797.1	881.3	IR	2.785	U46U47	-124.6	-69.3	1.23	0.0	1.22	IR	-537.9	589.3	IR	5.965
						OR	-994.0	1199.5	OR	4.591						OR	-670.8	802.1	OR	7.882	
U26U27	101.4	-107.4	1.21	147.8	1.21	IR	-643.3	713.3	IR	4.139	U47U48	-124.6	-69.3	1.23	0.0	1.22	IR	-537.9	589.3	IR	5.965
						OR	-802.2	970.9	OR	5.882						OR	-670.8	802.1	OR	7.882	
U27U28	101.4	-107.4	1.21	147.8	1.21	IR	-643.2	713.3	IR	4.139	L 1L 2	217.0	-32.7	1.21	113.6	1.22	IR	-349.2	382.1	IR	1.454
						OR	-802.1	970.9	OR	5.882						OR	-435.4	520.1	OR	2.669	
U28U29	-80.5	-174.4	1.21	118.0	1.21	IR	-872.3	966.8	IR	4.541	L 2L 3	217.0	-32.7	1.21	113.6	1.22	IR	-349.2	382.1	IR	1.454
						OR	-1087.8	1315.9	OR	5.777						OR	-435.4	520.1	OR	2.669	
U29U30	-80.5	-174.4	1.21	118.0	1.21	IR	-872.3	966.8	IR	4.541	L 3L 4	303.0	-65.1	1.21	174.0	1.22	IR	-537.1	593.3	IR	1.668
						OR	-1087.8	1315.9	OR	5.777						OR	-669.8	807.5	OR	2.899	

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L 4L 5	303.0	-65.1	1.21	174.0	1.22	IR OR	-537.1 -669.8	593.3 807.5	IR OR	1.668 2.899
L 5L 6	258.9	-97.6	1.21	184.3	1.22	IR OR	-537.1 -669.8	593.3 807.5	IR OR	1.815 2.977
L 6L 7	258.9	-97.6	1.21	184.3	1.22	IR OR	-537.1 -669.8	593.3 807.5	IR OR	1.815 2.977
L 7L 8	84.4	-130.0	1.21	145.1	1.22	IR OR	-483.7 -603.2	535.0 728.1	IR OR	3.104 4.435
L 8L 9	84.4	-130.0	1.21	145.1	1.22	IR OR	-483.7 -603.2	535.0 728.1	IR OR	3.104 4.435
L 9L10	-220.3	-162.4	1.21	56.5	1.22	IR OR	-483.7 -603.2	535.0 728.1	IR OR	1.622 2.358
L10L11	-220.3	-162.4	1.21	56.5	1.22	IR OR	-484.8 -604.6	535.0 728.1	IR OR	1.629 2.366
L11L12	-238.5	-117.8	1.21	0.0	1.21	IR OR	-484.9 -604.6	535.0 728.1	IR OR	2.092 3.110
L12L13	-238.5	-117.8	1.21	0.0	1.21	IR OR	-484.9 -604.6	535.0 728.1	IR OR	2.092 3.110
L15L16	173.7	0.0	1.21	105.2	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	1.981 3.292
L16L17	173.7	0.0	1.21	105.2	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	1.981 3.292
L17L18	231.6	0.0	1.21	138.6	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	1.086 2.082
L18L19	231.6	0.0	1.21	138.6	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	1.086 2.082
L19L20	174.0	0.0	1.21	105.4	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	1.974 3.283
L20L21	174.0	0.0	1.21	105.4	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	1.974 3.283
L23L24	-237.7	-117.3	1.21	0.0	1.21	IR OR	-484.9 -604.6	535.0 728.1	IR OR	2.108 3.130
L24L25	-237.7	-117.3	1.21	0.0	1.21	IR OR	-484.9 -604.6	535.0 728.1	IR OR	2.108 3.130
L25L26	-244.0	-162.6	1.21	56.8	1.21	IR OR	-484.8 -604.6	535.0 728.1	IR OR	1.480 2.217
L26L27	-244.0	-162.6	1.21	56.8	1.21	IR OR	-484.0 -603.6	535.0 728.1	IR OR	1.476 2.211
L27L28	-1.3	-132.9	1.21	146.1	1.21	IR OR	-484.0 -603.5	535.0 728.1	IR OR	3.632 4.531
L28L29	-1.3	-132.9	1.21	146.1	1.21	IR OR	-484.0 -603.5	535.0 728.1	IR OR	3.632 4.531
L29L30	118.8	-103.0	1.21	191.1	1.21	IR OR	-537.4 -670.2	593.3 807.5	IR OR	2.483 3.603

U 6L 7	-101.4	-60.3	1.27	37.1	1.30	IR	-207.9	277.9	IR	1.767	L27U28	-161.1	-82.5	1.23	25.9	1.30	IR	-382.8	430.0	IR	2.687
						OR	-259.3	378.3	OR	2.620							OR	-477.4	585.3	OR	3.834
L 7U 8	174.9	-25.5	1.30	70.8	1.25	IR	-336.2	371.5	IR	2.778	U28L29	131.8	-25.9	1.30	73.1	1.24	IR	-284.7	360.0	IR	3.124
						OR	-419.2	505.7	OR	4.674							OR	-355.1	490.0	OR	4.904
U 8L 9	-204.6	-81.1	1.24	13.9	1.30	IR	-381.9	430.0	IR	2.186	L29U30	-61.6	-63.5	1.26	33.9	1.30	IR	-208.6	277.9	IR	2.314
						OR	-476.3	585.3	OR	3.350							OR	-260.1	378.3	OR	3.125
L 9U10	278.1	-2.3	1.30	91.7	1.23	IR	-453.1	509.4	IR	2.522	U30L31	32.2	-44.0	1.29	53.9	1.27	IR	-117.8	174.6	IR	2.643
						OR	-565.0	693.4	OR	4.527							OR	-146.9	237.6	OR	3.813
U10L11	-293.8	-98.1	1.23	0.0	1.00	IR	-601.2	681.5	IR	3.133	L31U32	37.9	-44.0	1.29	53.9	1.27	IR	-117.8	174.6	IR	2.537
						OR	-749.7	927.6	OR	4.647							OR	-146.9	237.6	OR	3.708
L11U12	-263.7	-114.0	1.23	0.0	1.00	IR	-601.3	681.5	IR	2.963	U32L33	-67.3	-63.5	1.26	33.9	1.30	IR	-208.6	277.9	IR	2.224
						OR	-749.8	927.6	OR	4.266							OR	-260.1	378.3	OR	3.036
U12L13	235.4	0.0	1.00	112.1	1.24	IR	-454.7	509.4	IR	2.443	L33U34	137.5	-25.8	1.30	73.0	1.24	IR	-284.7	360.0	IR	3.046
						OR	-567.1	693.4	OR	4.084							OR	-355.1	490.0	OR	4.826
L13U14	-175.0	-104.8	1.25	0.0	1.00	IR	-419.7	463.7	IR	2.334	U34L35	-166.8	-82.4	1.23	25.8	1.30	IR	-382.8	430.0	IR	2.620
						OR	-523.4	631.1	OR	3.323							OR	-477.4	585.3	OR	3.767
U14L15	161.4	0.0	1.00	98.3	1.26	IR	-343.6	371.5	IR	2.136	L35U36	236.4	-25.7	1.30	91.9	1.23	IR	-453.5	509.4	IR	2.969
						OR	-428.5	505.7	OR	3.500							OR	-565.6	693.4	OR	4.970
L15U16	-99.1	-83.7	1.28	6.8	1.30	IR	-263.4	320.8	IR	1.964	U36L37	-257.0	-98.5	1.23	23.6	1.23	IR	-526.0	589.3	IR	2.730
						OR	-328.5	436.6	OR	2.742							OR	-655.9	802.1	OR	4.049
U16L17	75.1	-20.1	1.30	68.3	1.29	IR	-178.2	227.7	IR	2.234	L37U38	-275.7	-113.5	1.23	0.0	1.00	IR	-601.3	681.5	IR	2.868
						OR	-222.2	309.9	OR	3.437							OR	-749.8	927.6	OR	4.177
L17U18	-12.2	-52.3	1.30	35.8	1.30	IR	-113.7	153.4	IR	1.940	U38L39	247.5	0.0	1.00	111.8	1.24	IR	-454.7	509.4	IR	2.343
						OR	-141.8	208.7	OR	2.477							OR	-567.1	693.4	OR	3.988
U18L19	-11.9	-52.2	1.30	35.9	1.30	IR	-113.7	153.4	IR	1.950	L39U40	-185.6	-104.4	1.25	0.0	1.00	IR	-454.7	509.4	IR	2.578
						OR	-141.8	208.7	OR	2.488							OR	-567.1	693.4	OR	3.654
L19U20	74.8	-20.2	1.30	68.3	1.29	IR	-178.2	227.7	IR	2.240	U40L41	180.5	0.0	1.00	102.0	1.25	IR	-342.7	371.5	IR	1.874
						OR	-222.2	309.9	OR	3.445							OR	-427.4	505.7	OR	3.190
U20L21	-98.9	-83.6	1.28	6.8	1.30	IR	-263.4	320.8	IR	1.970	L41U42	-110.7	-87.1	1.27	7.5	1.30	IR	-262.3	320.8	IR	1.741
						OR	-328.5	436.6	OR	2.749							OR	-327.1	436.6	OR	2.485
L21U22	162.2	0.0	1.00	98.9	1.26	IR	-343.6	371.5	IR	2.117	U42L43	83.0	-21.4	1.30	71.3	1.29	IR	-176.9	227.7	IR	2.029
						OR	-428.5	505.7	OR	3.474							OR	-220.6	309.9	OR	3.182
U22L23	-173.9	-104.4	1.25	0.0	1.00	IR	-454.7	509.4	IR	2.689	L43U44	-13.7	-55.0	1.30	38.2	1.30	IR	-112.1	153.4	IR	1.786
						OR	-567.1	693.4	OR	3.765							OR	-139.8	208.7	OR	2.289
L23U24	235.2	0.0	1.00	112.1	1.24	IR	-454.7	509.4	IR	2.446	U44L45	-14.0	-55.1	1.30	38.1	1.30	IR	-112.1	153.4	IR	1.779
						OR	-567.1	693.4	OR	4.087							OR	-139.8	208.7	OR	2.281
U24L25	-263.4	-114.0	1.23	0.0	1.00	IR	-601.3	681.5	IR	2.965	L45U46	83.2	-21.3	1.30	71.4	1.29	IR	-176.9	227.7	IR	2.025
						OR	-749.8	927.6	OR	4.269							OR	-220.6	309.9	OR	3.177
L25U26	-251.5	-99.0	1.23	23.6	1.23	IR	-526.0	589.3	IR	2.774	U46L47	-110.9	-87.1	1.27	7.4	1.30	IR	-262.3	320.8	IR	1.738
						OR	-655.9	802.1	OR	4.087							OR	-327.1	436.6	OR	2.482
U26L27	230.6	-25.9	1.30	92.3	1.23	IR	-453.5	509.4	IR	3.020	L47U48	180.1	0.0	1.00	102.6	1.26	IR	-342.7	371.5	IR	1.866
						OR	-565.6	693.4	OR	5.013							OR	-427.4	505.7	OR	3.173
U 1L 1	-55.0	-58.0	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.617											

U39L39	-50.0	-54.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.808
						OR	-185.5	273.7	OR	2.481
U41L41	-50.0	-55.5	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.654
						OR	-176.9	237.6	OR	2.286
U43L43	-50.0	-55.5	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.654
						OR	-176.9	237.6	OR	2.286
U45L45	-50.0	-55.5	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.655
						OR	-176.9	237.6	OR	2.286
U47L47	-50.0	-55.5	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.655
						OR	-176.9	237.6	OR	2.287

MEMBER L18L19 IS CRITICAL. SEE DETAILS BELOW.

DL FORCE	LL + IMPACT FORCE		ALLOWABLE FORCE		RATING	
	COMP	I.F.	TENS	I.F.	COMP	TENS
231.6	0.0	1.21	138.6	1.21	IR	-350.3 382.1
					OR	-436.9 520.1
					IR	1.086 39.80
					OR	2.082 76.29

SUPPORT REACTIONS

SUPPORT	D.L.	LL+I	I.F.
U 0	182.4	86.2	1.22
L11	503.2	108.6	1.21
L25	468.7	107.4	1.21
L37	483.2	108.3	1.22
U48	150.1	83.9	1.23

 * TRUSS - LIVE LOAD P-82 *

MEMBER FORCES AND RATINGS
 I.F. = IMPACT FACTOR

MEMBER ID	DL FORCE	LL + IMPACT FORCE		ALLOWABLE FORCE		RATING FACTORS	
		COMP	I.F.	TENS	I.F.	COMP	TENS
U 0U 1	-132.0	-140.8	1.22	33.5	1.21	IR	-391.5 430.0
						OR	-488.2 585.3
						IR	1.843
						OR	2.530
U 1U 2	-132.0	-140.8	1.22	33.5	1.21	IR	-391.7 430.0
						OR	-488.5 585.3
						IR	1.844
						OR	2.532
U 2U 3	-283.2	-318.8	1.22	99.4	1.21	IR	-713.8 790.9
						OR	-890.2 1076.5
						IR	1.351
						OR	1.904
U 3U 4	-283.2	-318.8	1.22	99.4	1.21	IR	-713.8 790.9
						OR	-890.2 1076.5
						IR	1.351
						OR	1.904
U 4U 5	-304.2	-371.5	1.22	165.3	1.21	IR	-794.8 881.3
						OR	-991.1 1199.5
						IR	1.321
						OR	1.849
U 5U 6	-304.2	-371.5	1.22	165.3	1.21	IR	-794.8 881.3
						OR	-991.1 1199.5
						IR	1.321
						OR	1.849
U 6U 7	-194.8	-357.4	1.22	231.2	1.21	IR	-794.8 881.3
						IR	1.678

U 3L 3	-57.0	-57.8	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.588
						OR	-185.5	273.7	OR	2.224
U 5L 5	-57.0	-57.8	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.588
						OR	-185.5	273.7	OR	2.224
U 7L 7	-57.0	-57.8	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.588
						OR	-185.5	273.7	OR	2.224
U 9L 9	-57.0	-57.8	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.588
						OR	-185.5	273.7	OR	2.224
U10L10	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U11L11	-50.0	-54.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.806
						OR	-185.5	273.7	OR	2.478
U12L12	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U13L13	-50.0	-54.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.808
						OR	-185.5	273.7	OR	2.481
U15L15	-47.0	-53.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.767
						OR	-176.9	237.6	OR	2.420
U17L17	-47.0	-53.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.770
						OR	-176.9	237.6	OR	2.424
U19L19	-47.0	-53.6	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.770
						OR	-176.9	237.6	OR	2.423
U21L21	-47.0	-53.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.767
						OR	-176.9	237.6	OR	2.420
U23L23	-50.0	-54.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.808
						OR	-185.5	273.7	OR	2.481
U24L24	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U25L25	-50.0	-54.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.806
						OR	-185.5	273.7	OR	2.478
U26L26	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U27L27	-55.0	-57.1	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.642
						OR	-185.5	273.7	OR	2.286
U29L29	-55.0	-57.2	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.640
						OR	-185.5	273.7	OR	2.283
U31L31	-55.0	-57.2	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.640
						OR	-185.5	273.7	OR	2.283
U33L33	-55.0	-57.2	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.640
						OR	-185.5	273.7	OR	2.283
U35L35	-55.0	-57.1	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.642
						OR	-185.5	273.7	OR	2.286
U37L37	-50.0	-54.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.806
						OR	-185.5	273.7	OR	2.478

U 7U 8	-194.8	-357.4	1.22	231.2	1.21	IR	OR	-991.1	1199.5	OR	2.228
						IR	OR	-794.8	881.3	IR	1.678
						OR	OR	-991.1	1199.5	OR	2.228
U 8U 9	44.7	-233.6	1.22	297.1	1.21	IR	OR	-642.7	713.3	IR	2.251
						OR	OR	-801.5	970.9	OR	3.118
U 9U10	44.7	-233.6	1.22	297.1	1.21	IR	OR	-642.7	713.3	IR	2.251
						OR	OR	-801.5	970.9	OR	3.118
U10U11	391.6	0.0	1.22	359.1	1.21	IR	OR	-797.1	881.3	IR	1.364
						OR	OR	-994.0	1199.5	OR	2.250
U11U12	391.6	0.0	1.21	359.1	1.21	IR	OR	-797.2	881.3	IR	1.364
						OR	OR	-994.1	1199.5	OR	2.250
U12U13	101.7	0.0	1.21	113.7	1.21	IR	OR	-393.2	430.0	IR	2.887
						OR	OR	-490.3	585.3	OR	4.252
U13U14	101.7	0.0	1.21	113.7	1.21	IR	OR	-393.2	430.0	IR	2.887
						OR	OR	-490.3	585.3	OR	4.252
U14U15	-107.8	-113.0	1.21	0.0	1.21	IR	OR	-465.9	509.4	IR	3.170
						OR	OR	-581.0	693.4	OR	4.188
U15U16	-107.8	-113.0	1.21	0.0	1.21	IR	OR	-466.0	509.4	IR	3.170
						OR	OR	-581.1	693.4	OR	4.189
U16U17	-223.5	-218.2	1.21	0.0	1.21	IR	OR	-466.0	509.4	IR	1.111
						OR	OR	-581.1	693.4	OR	1.639
U17U18	-223.5	-218.2	1.21	0.0	1.21	IR	OR	-465.9	509.4	IR	1.111
						OR	OR	-581.1	693.4	OR	1.639
U18U19	-223.7	-218.3	1.21	0.0	1.21	IR	OR	-465.9	509.4	IR	1.110
						OR	OR	-581.1	693.4	OR	1.637
U19U20	-223.7	-218.3	1.21	0.0	1.21	IR	OR	-465.9	509.4	IR	1.110
						OR	OR	-581.1	693.4	OR	1.637
U20U21	-108.3	-113.5	1.21	0.0	1.21	IR	OR	-466.0	509.4	IR	3.151
						OR	OR	-581.1	693.4	OR	4.165
U21U22	-108.3	-113.5	1.21	0.0	1.21	IR	OR	-465.9	509.4	IR	3.150
						OR	OR	-581.0	693.4	OR	4.164
U22U23	101.0	0.0	1.21	112.9	1.21	IR	OR	-393.2	430.0	IR	2.913
						OR	OR	-490.3	585.3	OR	4.288
U23U24	101.0	0.0	1.21	112.9	1.21	IR	OR	-393.2	430.0	IR	2.913
						OR	OR	-490.3	585.3	OR	4.288
U24U25	390.7	0.0	1.21	357.8	1.21	IR	OR	-797.2	881.3	IR	1.371
						OR	OR	-994.1	1199.5	OR	2.260
U25U26	390.7	0.0	1.21	357.8	1.21	IR	OR	-797.1	881.3	IR	1.371
						OR	OR	-994.0	1199.5	OR	2.260
U26U27	101.4	-236.1	1.21	300.3	1.21	IR	OR	-643.3	713.3	IR	2.038
						OR	OR	-802.2	970.9	OR	2.896
U27U28	101.4	-236.1	1.21	300.3	1.21	IR	OR	-643.2	713.3	IR	2.038
						OR	OR	-802.1	970.9	OR	2.896

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U28U29	-80.5	-372.6	1.21	239.6	1.21	IR	OR	-872.3	966.8	IR	2.125
						OR	OR	-1087.8	1315.9	OR	2.703
U29U30	-80.5	-372.6	1.21	239.6	1.21	IR	OR	-872.3	966.8	IR	2.125
						OR	OR	-1087.8	1315.9	OR	2.703
U30U31	-138.8	-393.6	1.21	183.6	1.21	IR	OR	-872.3	966.8	IR	1.864
						OR	OR	-1087.8	1315.9	OR	2.411
U31U32	-138.8	-393.6	1.21	185.0	1.22	IR	OR	-872.3	966.8	IR	1.864
						OR	OR	-1087.8	1315.9	OR	2.411
U32U33	-73.4	-372.7	1.21	247.7	1.22	IR	OR	-872.3	966.8	IR	2.144
						OR	OR	-1087.8	1315.9	OR	2.722
U33U34	-73.4	-372.7	1.21	247.7	1.22	IR	OR	-872.3	966.8	IR	2.144
						OR	OR	-1087.8	1315.9	OR	2.722
U34U35	115.6	-236.2	1.21	310.5	1.22	IR	OR	-643.2	713.3	IR	1.925
						OR	OR	-802.1	970.9	OR	2.755
U35U36	115.6	-236.2	1.21	310.5	1.22	IR	OR	-643.3	713.3	IR	1.925
						OR	OR	-802.2	970.9	OR	2.755
U36U37	411.8	0.0	1.21	370.1	1.22	IR	OR	-797.1	881.3	IR	1.269
						OR	OR	-994.0	1199.5	OR	2.129
U37U38	411.8	0.0	1.23	370.1	1.22	IR	OR	-797.2	881.3	IR	1.269
						OR	OR	-994.1	1199.5	OR	2.129
U39U39	107.8	0.0	1.23	117.9	1.22	IR	OR	-393.2	430.0	IR	2.732
						OR	OR	-490.3	585.3	OR	4.049
U39U40	107.8	0.0	1.23	117.9	1.22	IR	OR	-393.2	430.0	IR	2.732
						OR	OR	-490.3	585.3	OR	4.049
U40U41	-124.9	-130.1	1.23	0.0	1.22	IR	OR	-537.9	589.3	IR	3.176
						OR	OR	-670.8	802.1	OR	4.198
U41U42	-124.9	-130.1	1.23	0.0	1.22	IR	OR	-537.9	589.3	IR	3.176
						OR	OR	-670.8	802.1	OR	4.198
U42U43	-259.0	-251.5	1.23	0.0	1.22	IR	OR	-537.9	589.3	IR	1.109
						OR	OR	-670.8	802.1	OR	1.637
U43U44	-259.0	-251.5	1.23	0.0	1.22	IR	OR	-537.9	589.3	IR	1.109
						OR	OR	-670.8	802.1	OR	1.637
U44U45	-258.8	-251.5	1.23	0.0	1.22	IR	OR	-537.9	589.3	IR	1.110
						OR	OR	-670.8	802.1	OR	1.638
U45U46	-258.8	-251.5	1.23	0.0	1.22	IR	OR	-537.9	589.3	IR	1.110
						OR	OR	-670.8	802.1	OR	1.638
U46U47	-124.6	-129.7	1.23	0.0	1.22	IR	OR	-537.9	589.3	IR	3.187
						OR	OR	-670.8	802.1	OR	4.211
U47U48	-124.6	-129.7	1.23	0.0	1.22	IR	OR	-537.9	589.3	IR	3.187
						OR	OR	-670.8	802.1	OR	4.211
L 1L 2	217.0	-66.5	1.21	246.2	1.22	IR	OR	-349.2	382.1	IR	0.671
						OR	OR	-435.4	520.1	OR	1.231

L 2L 3	217.0	-66.5	1.21	246.2	1.22	IR OR	-349.2 -435.4	382.1 520.1	IR OR	0.671 1.231
L 3L 4	303.0	-132.4	1.21	361.0	1.22	IR OR	-537.1 -669.8	593.3 807.5	IR OR	0.804 1.397
L 4L 5	303.0	-132.4	1.21	361.0	1.22	IR OR	-537.1 -669.8	593.3 807.5	IR OR	0.804 1.397
L 5L 6	258.9	-198.3	1.21	371.9	1.22	IR OR	-537.1 -669.8	593.3 807.5	IR OR	0.899 1.475
L 6L 7	258.9	-198.3	1.21	371.9	1.22	IR OR	-537.1 -669.8	593.3 807.5	IR OR	0.899 1.475
L 7L 8	84.4	-264.1	1.21	310.2	1.22	IR OR	-483.7 -603.2	535.0 728.1	IR OR	1.452 2.075
L 8L 9	84.4	-264.1	1.21	310.2	1.22	IR OR	-483.7 -603.2	535.0 728.1	IR OR	1.452 2.075
L 9L10	-220.3	-330.0	1.21	124.3	1.22	IR OR	-483.7 -603.2	535.0 728.1	IR OR	0.798 1.160
L10L11	-220.3	-330.0	1.21	124.3	1.22	IR OR	-484.8 -604.6	535.0 728.1	IR OR	0.802 1.164
L11L12	-238.5	-234.7	1.21	0.0	1.21	IR OR	-484.9 -604.6	535.0 728.1	IR OR	1.050 1.560
L12L13	-238.5	-234.7	1.21	0.0	1.21	IR OR	-484.9 -604.6	535.0 728.1	IR OR	1.050 1.560
L15L16	173.7	0.0	1.21	185.4	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	1.125 1.869
L16L17	173.7	0.0	1.21	185.4	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	1.125 1.869
L17L18	231.6	0.0	1.21	211.1	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	0.713 1.366
L18L19	231.6	0.0	1.21	211.1	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	0.713 1.366
L19L20	174.0	0.0	1.21	185.7	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	1.121 1.864
L20L21	174.0	0.0	1.21	185.7	1.21	IR OR	-350.3 -436.9	382.1 520.1	IR OR	1.121 1.864
L23L24	-237.7	-233.6	1.21	0.0	1.21	IR OR	-484.9 -604.6	535.0 728.1	IR OR	1.058 1.571
L24L25	-237.7	-233.6	1.21	0.0	1.21	IR OR	-484.9 -604.6	535.0 728.1	IR OR	1.058 1.571
L25L26	-244.0	-330.4	1.21	126.5	1.21	IR OR	-484.8 -604.6	535.0 728.1	IR OR	0.729 1.091
L26L27	-244.0	-330.4	1.21	126.5	1.21	IR OR	-484.0 -603.6	535.0 728.1	IR OR	0.727 1.089
L 27L28	-1.3	-269.9	1.21	318.2	1.21	IR OR	-484.0 -603.5	535.0 728.1	IR OR	1.695 2.231

U35L35	-55.0	-79.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.178
						OR	-185.5	273.7	OR	1.640
U37L37	-50.0	-73.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.348
						OR	-185.5	273.7	OR	1.849
U39L39	-50.0	-73.0	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.353
						OR	-185.5	273.7	OR	1.857
U41L41	-50.0	-75.3	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.219
						OR	-176.9	237.6	OR	1.684
U43L43	-50.0	-75.3	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.219
						OR	-176.9	237.6	OR	1.684
U45L45	-50.0	-75.3	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.220
						OR	-176.9	237.6	OR	1.685
U47L47	-50.0	-75.2	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.220
						OR	-176.9	237.6	OR	1.686

MEMBER U36L37 IS CRITICAL. SEE DETAILS BELOW.

DL	LL	IMPACT	FORCE	ALLOWABLE	FORCE	RATING
FORCE	COMP	I.F.	TENS	I.F.	COMP	FACTOR
-261.9	-341.6	1.22	126.5	1.21	IR	0.653
					OR	1.003
						66.56
						102.32

SUPPORT REACTIONS

SUPPORT	D.L.	LL+I	I.F.
U 0	182.4	193.2	1.22
L11	503.2	259.9	1.21
L25	468.7	258.4	1.21
L37	483.2	260.9	1.22
U48	150.1	166.7	1.23

+
+ S T R I N G E R A N A L Y S I S +
+

THE STRINGER LOCATED AT 8.00 FT. FROM THE CENTER LINE OF GIRDER OR TRUSS IS CRITICAL.

DEAD LOADS ACTING ON STRINGER

STRINGER	SLAB	INPUT	TOTAL
WEIGHT	DL1	DL1	DL2
0.036	0.792	0.020	0.848
			0.000

* STRINGER - LIVE LOAD H20 *

MAXIMUM REACTIONS

REACTIONS MOMENTS

L47U48	180.1	0.0	1.00	192.0	1.26	IR	-327.1	436.6	OR	1.437
						OR	-342.7	371.5	IR	0.997
U 1L 1	-55.0	-82.1	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.143
						OR	-185.5	273.7	OR	1.591
U 3L 3	-57.0	-81.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.129
						OR	-185.5	273.7	OR	1.581
U 5L 5	-57.0	-81.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.129
						OR	-185.5	273.7	OR	1.581
U 7L 7	-57.0	-81.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.129
						OR	-185.5	273.7	OR	1.581
U 9L 9	-57.0	-81.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.129
						OR	-185.5	273.7	OR	1.581
U10L10	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U11L11	-50.0	-73.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.348
						OR	-185.5	273.7	OR	1.849
U12L12	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U13L13	-50.0	-73.0	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.353
						OR	-185.5	273.7	OR	1.857
U15L15	-47.0	-70.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.342
						OR	-176.9	237.6	OR	1.838
U17L17	-47.0	-70.3	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.350
						OR	-176.9	237.6	OR	1.848
U19L19	-47.0	-70.3	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.349
						OR	-176.9	237.6	OR	1.848
U21L21	-47.0	-70.7	1.30	0.0	1.00	IR	-141.8	174.6	IR	1.342
						OR	-176.9	237.6	OR	1.838
U23L23	-50.0	-73.0	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.353
						OR	-185.5	273.7	OR	1.857
U24L24	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U25L25	-50.0	-73.3	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.348
						OR	-185.5	273.7	OR	1.849
U26L26	0.0	0.0	1.00	0.0	1.00	IR	-148.8	201.1	IR	999.999
U27L27	-55.0	-79.6	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.178
						OR	-185.5	273.7	OR	1.640
U29L29	-55.0	-79.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.177
						OR	-185.5	273.7	OR	1.638
U31L31	-55.0	-79.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.177
						OR	-185.5	273.7	OR	1.638
U33L33	-55.0	-79.7	1.30	0.0	1.00	IR	-148.8	201.1	IR	1.177
						OR	-185.5	273.7	OR	1.638

SUPPORT	DL1	DL2	+ (LL+I)	+ I.F. - I.F.	+ I.F. - I.F.
1	4.0	0.0	29.0	0.0	1.30
2	8.1	0.0	26.9 L	0.0	1.30
3	8.1	0.0	26.9 L	0.0	1.30
4	4.0	0.0	29.0	0.0	1.30

NOTE: ALL SUPPORT REACTIONS AND END SHEARS IN EACH SPAN DUE TO A LIVE LOAD ARE CALCULATED BASED ON AASHTO ARTICLE 3.23.1 AS INTERPRETED IN SOL 431-93-05.

UNFACTORED MOMENTS AND SHEARS

SPAN 1 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X	DL1		DL2		+(LL+I)		-(LL+I)		DL1		DL2		+(LL+I)		-(LL+I)	
	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4.0	0.0	29.0	0.0	0.0	0.0	0.0	1.30
	SIMULT	SHEAR							SIMULT	MOM						
0.95	3.4	0.0	25.9	0.0	0.0	0.0	3.2	0.0	27.2	0.0	25.9	0.0	0.0	0.0	0.0	1.30
	SIMULT	SHEAR							SIMULT	MOM						
1.90	6.1	0.0	46.0	0.0	0.0	0.0	2.4	0.0	24.2	0.0	24.2	0.0	0.0	0.0	0.0	1.30
	SIMULT	SHEAR							SIMULT	MOM						
2.85	8.0	0.0	69.3	0.0	0.0	0.0	1.6	0.0	21.2	0.0	21.2	0.0	0.0	0.0	0.0	1.30
	SIMULT	SHEAR							SIMULT	MOM						
3.80	9.2	0.0	69.0	0.0	0.0	0.0	0.8	0.0	18.1	0.0	18.1	0.0	0.0	0.0	0.0	1.30
	SIMULT	SHEAR							SIMULT	MOM						
4.75	9.6	0.0	71.8	0.0	0.0	0.0	0.0	0.0	15.1	0.0	15.1	0.0	0.0	0.0	0.0	1.30
	SIMULT	SHEAR							SIMULT	MOM						
5.70	9.2	0.0	69.0	0.0	0.0	0.0	-0.8	0.0	12.1	0.0	12.1	0.0	0.0	0.0	0.0	1.30
	SIMULT	SHEAR							SIMULT	MOM						
6.65	8.0	0.0	60.3	0.0	0.0	0.0	-1.6	0.0	9.1	0.0	9.1	0.0	0.0	0.0	0.0	1.30
	SIMULT	SHEAR							SIMULT	MOM						
7.60	6.1	0.0	46.0	0.0	0.0	0.0	-2.4	0.0	6.0	0.0	6.0	0.0	0.0	0.0	0.0	1.30
	SIMULT	SHEAR							SIMULT	MOM						
8.55	3.4	0.0	25.9	0.0	0.0	0.0	-3.2	0.0	3.0	0.0	3.0	0.0	0.0	0.0	0.0	1.30
	SIMULT	SHEAR							SIMULT	MOM						
9.50	0.0	0.0	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.30
	SIMULT	SHEAR							SIMULT	MOM						

SPAN 2 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X	DL1		DL2		-(LL+I)		DL1		DL2		+(LL+I)		-(IL+I)		I.F.
	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	4.0	0.0	0.0	29.0	0.0	0.0	1.30	
0.95	SIMULT	SHEAR	0.0	0.0	0.0	0.0	0.0	SIMULT	MOM	0.0	0.0	0.0	0.0	0.0	
	3.4	0.0	25.9	0.0	0.0	0.0	0.0	3.2	0.0	27.2	0.0	25.9	0.0	1.30	
1.90	SIMULT	SHEAR	27.2	0.0	0.0	0.0	0.0	SIMULT	MOM	25.9	0.0	25.9	0.0	1.30	
	6.1	0.0	46.0	0.0	0.0	0.0	0.0	2.4	0.0	24.2	0.0	-6.0	0.0	1.30	
2.85	SIMULT	SHEAR	24.2	0.0	0.0	0.0	0.0	SIMULT	MOM	46.0	0.0	46.0	0.0	1.30	
	8.0	0.0	60.3	0.0	0.0	0.0	0.0	1.6	0.0	21.2	0.0	-9.1	0.0	1.30	
3.80	SIMULT	SHEAR	21.2	0.0	0.0	0.0	0.0	SIMULT	MOM	60.3	0.0	60.3	0.0	1.30	
	9.2	0.0	69.0	0.0	0.0	0.0	0.0	0.8	0.0	18.1	0.0	-12.1	0.0	1.30	
4.75	SIMULT	SHEAR	18.1	0.0	0.0	0.0	0.0	SIMULT	MOM	69.0	0.0	69.0	0.0	1.30	
	9.6	0.0	71.8	0.0	0.0	0.0	0.0	0.0	0.0	15.1	0.0	-15.1	0.0	1.30	
5.70	SIMULT	SHEAR	-15.1	0.0	0.0	0.0	0.0	SIMULT	MOM	71.8	0.0	71.8	0.0	1.30	
	9.2	0.0	69.0	0.0	0.0	0.0	0.0	-0.8	0.0	12.1	0.0	-18.1	0.0	1.30	
6.65	SIMULT	SHEAR	-18.1	0.0	0.0	0.0	0.0	SIMULT	MOM	69.0	0.0	69.0	0.0	1.30	
	8.0	0.0	60.3	0.0	0.0	0.0	0.0	-1.6	0.0	9.1	0.0	-21.2	0.0	1.30	
7.60	SIMULT	SHEAR	-21.2	0.0	0.0	0.0	0.0	SIMULT	MOM	60.3	0.0	60.3	0.0	1.30	
	6.1	0.0	46.0	0.0	0.0	0.0	0.0	-2.4	0.0	6.0	0.0	-24.2	0.0	1.30	
8.55	SIMULT	SHEAR	-24.2	0.0	0.0	0.0	0.0	SIMULT	MOM	46.0	0.0	46.0	0.0	1.30	
	3.4	0.0	25.9	0.0	0.0	0.0	0.0	-3.2	0.0	3.0	0.0	-27.2	0.0	1.30	

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NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

SPAN 1

X	NON-COMP		OVERLOAD		SHEAR	NON-COMPACT		COMPACT		COMPACT	
	MOMENT	STRENGTH	MOMENT	STRENGTH		RATING	FACTORS	RATING	FACTORS	IR	OR
0.00	154.9	B	123.9	85.8	85.8	1.28	V	2.14	V	1.28	V
0.95	154.9	B	123.9	85.8	85.8	1.38	V	2.31	V	1.38	V
1.90	154.9	B	123.9	85.8	85.8	1.47	T	2.46	T	1.54	O
2.85	154.9	B	123.9	85.8	85.8	1.10	T	1.84	T	1.15	O
3.80	154.9	B	123.9	85.8	85.8	0.96	T	1.59	T	1.00	O
4.75	154.9	B	123.9	85.8	85.8	0.92	T	1.53	T	0.96	O
5.70	154.9	B	123.9	85.8	85.8	0.96	T	1.59	T	1.00	O
6.65	154.9	B	123.9	85.8	85.8	1.10	T	1.84	T	1.15	O
7.60	154.9	B	123.9	85.8	85.8	1.47	T	2.46	T	1.54	O
8.55	154.9	B	123.9	85.8	85.8	1.38	V	2.31	V	1.38	V
9.50	154.9	B	123.9	85.8	85.8	1.28	V	2.14	V	1.28	V

SPAN 2

	X	NON-COMP STRENGTH	OVERLOAD MOMENT	SHEAR STRENGTH	NON-COMPACT RATING FACTORS IR OR	COMPACT MOMENT STRENGTH	COMPACT RATING FACTORS IR OR
0.00	154.9 B	123.9	85.8	1.28 V	2.14 V	172.2	1.28 V 2.14 V
0.95	154.9 B	123.9	85.8	1.38 V	2.31 V	172.2	1.38 V 2.31 V
1.90	154.9 B	123.9	85.8	1.47 T	2.46 T	172.2	1.54 O 2.56 O

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

SPAN 1

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT	
	MOMENT	STRENGTH	RATING	IR	MOMENT	RATING
0.00	154.9 B	123.9	85.8	1.24 V	172.2	1.24 V
0.95	154.9 B	123.9	85.8	1.38 V	172.2	1.38 V
1.90	154.9 B	123.9	85.8	1.47 T	172.2	1.54 O
2.85	154.9 B	123.9	85.8	1.10 T	172.2	1.15 O
3.80	154.9 B	123.9	85.8	0.96 T	172.2	1.00 O
4.75	154.9 B	123.9	85.8	0.92 T	172.2	0.96 O
5.70	154.9 B	123.9	85.8	0.96 T	172.2	1.00 O
6.65	154.9 B	123.9	85.8	1.10 T	172.2	1.15 O
7.60	154.9 B	123.9	85.8	1.47 T	172.2	1.54 O
8.55	154.9 B	123.9	85.8	1.38 V	172.2	1.38 V
9.50	154.9 B	123.9	85.8	1.24 V	172.2	1.24 V

SPAN 2

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT	
	MOMENT	STRENGTH	RATING	IR	MOMENT	RATING
0.00	154.9 B	123.9	85.8	1.24 V	172.2	1.24 V
0.95	154.9 B	123.9	85.8	1.38 V	172.2	1.38 V
1.90	154.9 B	123.9	85.8	1.47 T	172.2	1.54 O
2.85	154.9 B	123.9	85.8	1.10 T	172.2	1.15 O
3.80	154.9 B	123.9	85.8	0.96 T	172.2	1.00 O
4.75	154.9 B	123.9	85.8	0.92 T	172.2	0.96 O
5.70	154.9 B	123.9	85.8	0.96 T	172.2	1.00 O
6.65	154.9 B	123.9	85.8	1.10 T	172.2	1.15 O
7.60	154.9 B	123.9	85.8	1.47 T	172.2	1.54 O
8.55	154.9 B	123.9	85.8	1.38 V	172.2	1.38 V
9.50	154.9 B	123.9	85.8	1.25 V	172.2	1.25 V

SPAN 3

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT	
	MOMENT	STRENGTH	RATING	IR	MOMENT	RATING
0.00	154.9 B	123.9	85.8	1.24 V	172.2	1.24 V
0.95	154.9 B	123.9	85.8	1.38 V	172.2	1.38 V
1.90	154.9 B	123.9	85.8	1.47 T	172.2	1.54 O
2.85	154.9 B	123.9	85.8	1.10 T	172.2	1.15 O
3.80	154.9 B	123.9	85.8	0.96 T	172.2	1.00 O
4.75	154.9 B	123.9	85.8	0.92 T	172.2	0.96 O
5.70	154.9 B	123.9	85.8	0.96 T	172.2	1.00 O
6.65	154.9 B	123.9	85.8	1.10 T	172.2	1.15 O
7.60	154.9 B	123.9	85.8	1.47 T	172.2	1.54 O
8.55	154.9 B	123.9	85.8	1.38 V	172.2	1.38 V
9.50	154.9 B	123.9	85.8	1.43 V	172.2	1.43 V

* STRINGER - LIVE LOAD TK527 *

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MAXIMUM REACTIONS

SUPPORT	DL1	DL2	+(LL+I)	-(LL+I)	+I.F. -I.F.	MOMENTS
1	4.0	0.0	29.2	0.0	1.30	+I.F. -I.F.
2	8.1	0.0	33.7	0.0	1.30	
3	8.1	0.0	33.7	0.0	1.30	
4	4.0	0.0	29.2	0.0	1.30	

NOTE: ALL SUPPORT REACTIONS AND END SHEARS IN EACH SPAN DUE TO A LIVE LOAD
ARE CALCULATED BASED ON AASHTO ARTICLE 3.23.1
AS INTERPRETED IN SOL 431-93-05.

UNFACTORED MOMENTS AND SHEARS

SPAN 1 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X	DL1		DL2		+(LL+I)		-(LL+I)		+I.F. -I.F.		MOMENTS	
	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.95	3.4	0.0	25.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.90	6.1	0.0	43.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2.85	8.0	0.0	54.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3.80	9.2	0.0	57.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4.75	9.6	0.0	56.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5.70	9.2	0.0	57.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.65	8.0	0.0	54.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7.60	6.1	0.0	43.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.55	3.4	0.0	25.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.50	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

SPAN 2 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X	DL1		DL2		+(LL+I)		-(LL+I)		+I.F. -I.F.		MOMENTS	
	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
0.95	3.4	0.0	25.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1.90	6.1	0.0	43.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2.85	8.0	0.0	54.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3.80	9.2	0.0	57.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4.75	9.6	0.0	56.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
5.70	9.2	0.0	57.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
6.65	8.0	0.0	54.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
7.60	6.1	0.0	43.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
8.55	3.4	0.0	25.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
9.50	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

X	STRENGTH	STRENGTH	STRENGTH	IR	OR	STRENGTH	IR	OR
0.00	154.9 B	123.9	85.8	1.27 V	2.12 V	172.2	1.27 V	2.12 V
0.95	154.9 B	123.9	85.8	1.38 V	2.30 V	172.2	1.38 V	2.30 V
1.90	154.9 B	123.9	85.8	1.55 T	2.59 T	172.2	1.62 O	2.70 O
2.85	154.9 B	123.9	85.8	1.23 T	2.05 T	172.2	1.28 O	2.13 O
3.80	154.9 B	123.9	85.8	1.14 T	1.91 T	172.2	1.19 O	1.99 O
4.75	154.9 B	123.9	85.8	1.16 T	1.94 T	172.2	1.21 O	2.02 O
5.70	154.9 B	123.9	85.8	1.14 T	1.91 T	172.2	1.19 O	1.99 O
6.65	154.9 B	123.9	85.8	1.23 T	2.05 T	172.2	1.28 O	2.13 O
7.60	154.9 B	123.9	85.8	1.55 T	2.59 T	172.2	1.62 O	2.70 O
8.55	154.9 B	123.9	85.8	1.38 V	2.30 V	172.2	1.38 V	2.30 V
9.50	154.9 B	123.9	85.8	1.27 V	2.12 V	172.2	1.27 V	2.12 V

SPAN 3

X	NON-COMP		NON-COMPACT		COMPACT		COMPACT	
	MOMENT	OVERLOAD	SHEAR	RATING FACTORS	MOMENT	RATING FACTORS	IR	OR
	STRENGTH	STRENGTH	STRENGTH	IR	OR	STRENGTH	IR	OR
0.00	154.9 B	123.9	85.8	1.27 V	2.12 V	172.2	1.27 V	2.12 V
0.95	154.9 B	123.9	85.8	1.38 V	2.30 V	172.2	1.38 V	2.30 V
1.90	154.9 B	123.9	85.8	1.55 T	2.59 T	172.2	1.62 O	2.70 O
2.85	154.9 B	123.9	85.8	1.23 T	2.05 T	172.2	1.28 O	2.13 O
3.80	154.9 B	123.9	85.8	1.14 T	1.91 T	172.2	1.19 O	1.99 O
4.75	154.9 B	123.9	85.8	1.16 T	1.94 T	172.2	1.21 O	2.02 O
5.70	154.9 B	123.9	85.8	1.14 T	1.91 T	172.2	1.19 O	1.99 O
6.65	154.9 B	123.9	85.8	1.23 T	2.05 T	172.2	1.28 O	2.13 O
7.60	154.9 B	123.9	85.8	1.55 T	2.59 T	172.2	1.62 O	2.70 O
8.55	154.9 B	123.9	85.8	1.38 V	2.30 V	172.2	1.38 V	2.30 V
9.50	154.9 B	123.9	85.8	1.27 V	2.12 V	172.2	1.27 V	2.12 V

* STRINGER - LIVE LOAD ML80 *

MAXIMUM REACTIONS

SUPPORT	DL1	DL2	REACTIONS		MOMENTS	
			+ (LL+I)	- (LL+I)	+ I.F. - I.F.	- I.F. - I.F.
1	4.0	0.0	31.1	0.0	1.30	
2	8.1	0.0	39.3	0.0	1.30	
3	8.1	0.0	39.3	0.0	1.30	
4	4.0	0.0	31.1	0.0	1.30	

NOTE: ALL SUPPORT REACTIONS AND END SHEARS IN EACH SPAN DUE TO A LIVE LOAD ARE CALCULATED BASED ON AASHTO ARTICLE 3.23.1 AS INTERPRETED IN SOL 431-93-05

UNFACTORED MOMENTS AND SHEARS

SPAN 1 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X	DL1		DL2 + (LL+I)		- (LL+I)		DL1		DL2 + (LL+I)		- (LL+I)	
	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	31.1	0.0	1.30
	SIMULT	SHEAR	0.0	0.0	0.0	SIMULT	MOM	0.0	0.0	28.0	0.0	1.30
0.95	3.4	0.0	26.6	0.0	0.0	3.2	0.0	28.0	0.0	28.0	-1.9	1.30
	SIMULT	SHEAR	28.0	0.0	0.0	SIMULT	MOM	0.0	2.4	0.0	23.0	-3.9
1.90	6.1	0.0	43.6	0.0	0.0	0.0	0.0	43.6	0.0	43.6	29.6	1.30
	SIMULT	SHEAR	23.0	0.0	0.0	SIMULT	MOM	0.0	1.6	0.0	19.1	-5.8
2.85	8.0	0.0	54.3	0.0	0.0	0.0	0.0	54.3	0.0	54.3	38.8	1.30
	SIMULT	SHEAR	19.1	0.0	0.0	SIMULT	MOM	0.0	0.0	0.0	0.0	0.0

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

SPAN	X	NON-COMP		OVERLOAD		NON-COMPACT		COMPACT		COMPACT			
		MOMENT		MOMENT		RATING FACTORS		MOMENT		RATING FACTORS			
		STRENGTH	B	STRENGTH	B	IR	OR	STRENGTH	IR	IR	OR		
0.00	154.9	B	123.9	85.8	1.27	V	2.12	V	172.2	1.27	V	2.12	V
0.95	154.9	B	123.9	85.8	1.38	V	2.30	V	172.2	1.38	V	2.30	V
1.90	154.9	B	123.9	85.8	1.55	T	2.59	T	172.2	1.62	O	2.70	O
2.85	154.9	B	123.9	85.8	1.23	T	2.05	T	172.2	1.28	O	2.13	O
3.80	154.9	B	123.9	85.8	1.14	T	1.91	T	172.2	1.19	O	1.99	O
4.75	154.9	B	123.9	85.8	1.16	T	1.94	T	172.2	1.21	O	2.02	O
5.70	154.9	B	123.9	85.8	1.14	T	1.91	T	172.2	1.19	O	1.99	O
6.65	154.9	B	123.9	85.8	1.23	T	2.05	T	172.2	1.28	O	2.13	O
7.60	154.9	B	123.9	85.8	1.55	T	2.59	T	172.2	1.62	O	2.70	O
8.55	154.9	B	123.9	85.8	1.38	V	2.30	V	172.2	1.38	V	2.30	V
9.50	154.9	B	123.9	85.8	1.27	V	2.12	V	172.2	1.27	V	2.12	V

SPAN 2

[illegible]

8.55	3.4	0.0	26.6	0.0	-3.2	0.0	1.9	-28.0	1.30
	SIMULT	SHEAR	-28.0	0.0	SIMULT	MOM	16.6	26.6	
9.50	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	-31.1	1.30
	SIMULT	SHEAR	0.0	0.0	SIMULT	MOM	0.0	0.0	

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

SPAN 1									
=====									
X	NON-COMP OVERLOAD			NON-COMPACT			COMPACT		
	MOMENT	STRENGTH	SHEAR	RATING	IR	OR	MOMENT	STRENGTH	RATING
0.00	154.9 B	123.9	85.8	1.20 V	1.20 V	1.99 V	172.2	1.20 V	1.99 V
0.95	154.9 B	123.9	85.8	1.35 V	1.35 V	2.24 V	172.2	1.35 V	2.24 V
1.90	154.9 B	123.9	85.8	1.55 T	1.55 T	2.59 T	172.2	1.62 O	2.70 O
2.85	154.9 B	123.9	85.8	1.23 T	1.23 T	2.05 T	172.2	1.28 O	2.13 O
3.80	154.9 B	123.9	85.8	1.14 T	1.14 T	1.91 T	172.2	1.19 O	1.99 O
4.75	154.9 B	123.9	85.8	1.08 T	1.08 T	1.80 T	172.2	1.13 O	1.88 O
5.70	154.9 B	123.9	85.8	1.14 T	1.14 T	1.91 T	172.2	1.19 O	1.99 O
6.65	154.9 B	123.9	85.8	1.23 T	1.23 T	2.05 T	172.2	1.28 O	2.13 O
7.60	154.9 B	123.9	85.8	1.55 T	1.55 T	2.59 T	172.2	1.62 O	2.70 O
8.55	154.9 B	123.9	85.8	1.35 V	1.35 V	2.24 V	172.2	1.35 V	2.24 V
9.50	154.9 B	123.9	85.8	1.20 V	1.20 V	1.99 V	172.2	1.20 V	1.99 V

SPAN 2									
=====									
X	NON-COMP OVERLOAD			NON-COMPACT			COMPACT		
	MOMENT	STRENGTH	SHEAR	RATING	IR	OR	MOMENT	STRENGTH	RATING
0.00	154.9 B	123.9	85.8	1.20 V	1.20 V	1.99 V	172.2	1.20 V	1.99 V
0.95	154.9 B	123.9	85.8	1.35 V	1.35 V	2.24 V	172.2	1.35 V	2.24 V
1.90	154.9 B	123.9	85.8	1.55 T	1.55 T	2.59 T	172.2	1.62 O	2.70 O
2.85	154.9 B	123.9	85.8	1.23 T	1.23 T	2.05 T	172.2	1.28 O	2.13 O
3.80	154.9 B	123.9	85.8	1.14 T	1.14 T	1.91 T	172.2	1.19 O	1.99 O
4.75	154.9 B	123.9	85.8	1.08 T	1.08 T	1.80 T	172.2	1.13 O	1.88 O
5.70	154.9 B	123.9	85.8	1.14 T	1.14 T	1.91 T	172.2	1.19 O	1.99 O
6.65	154.9 B	123.9	85.8	1.23 T	1.23 T	2.05 T	172.2	1.28 O	2.13 O
7.60	154.9 B	123.9	85.8	1.55 T	1.55 T	2.59 T	172.2	1.62 O	2.70 O
8.55	154.9 B	123.9	85.8	1.35 V	1.35 V	2.24 V	172.2	1.35 V	2.24 V
9.50	154.9 B	123.9	85.8	1.20 V	1.20 V	1.99 V	172.2	1.20 V	1.99 V

SPAN 3									
=====									
X	NON-COMP OVERLOAD			NON-COMPACT			COMPACT		
	MOMENT	STRENGTH	SHEAR	RATING	IR	OR	MOMENT	STRENGTH	RATING
0.00	154.9 B	123.9	85.8	1.20 V	1.20 V	1.99 V	172.2	1.20 V	1.99 V
0.95	154.9 B	123.9	85.8	1.35 V	1.35 V	2.24 V	172.2	1.35 V	2.24 V
1.90	154.9 B	123.9	85.8	1.55 T	1.55 T	2.59 T	172.2	1.62 O	2.70 O
2.85	154.9 B	123.9	85.8	1.23 T	1.23 T	2.05 T	172.2	1.28 O	2.13 O
3.80	154.9 B	123.9	85.8	1.14 T	1.14 T	1.91 T	172.2	1.19 O	1.99 O
4.75	154.9 B	123.9	85.8	1.08 T	1.08 T	1.80 T	172.2	1.13 O	1.88 O
5.70	154.9 B	123.9	85.8	1.14 T	1.14 T	1.91 T	172.2	1.19 O	1.99 O
6.65	154.9 B	123.9	85.8	1.23 T	1.23 T	2.05 T	172.2	1.28 O	2.13 O
7.60	154.9 B	123.9	85.8	1.55 T	1.55 T	2.59 T	172.2	1.62 O	2.70 O
8.55	154.9 B	123.9	85.8	1.35 V	1.35 V	2.24 V	172.2	1.35 V	2.24 V
9.50	154.9 B	123.9	85.8	1.20 V	1.20 V	1.99 V	172.2	1.20 V	1.99 V

SPAN 2 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30									
=====									
X	DL1		DL2		DL1		DL2		I.F.
	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.30
0.95	3.4	0.0	26.6	0.0	3.2	0.0	28.0	0.0	1.30
1.90	6.1	0.0	43.6	0.0	2.4	0.0	23.0	-1.9	1.30
2.85	8.0	0.0	54.3	0.0	1.6	0.0	19.1	-5.8	1.30
3.80	9.2	0.0	57.6	0.0	0.8	0.0	15.2	-7.8	1.30
4.75	9.6	0.0	60.8	0.0	0.0	0.0	11.3	-11.3	1.30
5.70	9.2	0.0	57.6	0.0	0.0	0.0	7.8	-15.2	1.30
6.65	8.0	0.0	54.3	0.0	-1.6	0.0	5.8	-19.1	1.30
7.60	6.1	0.0	43.6	0.0	-2.4	0.0	3.9	-23.0	1.30
8.55	3.4	0.0	26.6	0.0	-3.2	0.0	1.9	-28.0	1.30
9.50	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	-31.1	1.30

SPAN 3 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30									
=====									
X	DL1		DL2		DL1		DL2		I.F.
	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.30
0.95	3.4	0.0	26.6	0.0	3.2	0.0	28.0	0.0	1.30
1.90	6.1	0.0	43.6	0.0	2.4	0.0	23.0	-1.9	1.30
2.85	8.0	0.0	54.3	0.0	1.6	0.0	19.1	-5.8	1.30
3.80	9.2	0.0	57.6	0.0	0.8	0.0	15.2	-7.8	1.30
4.75	9.6	0.0	60.8	0.0	0.0	0.0	11.3	-11.3	1.30
5.70	9.2	0.0	57.6	0.0	-0.8	0.0	7.8	-15.2	1.30
6.65	8.0	0.0	54.3	0.0	-1.6	0.0	5.8	-19.1	1.30
7.60	6.1	0.0	43.6	0.0	-2.4	0.0	3.9	-23.0	1.30
8.55	3.4	0.0	26.6	0.0	-3.2	0.0	1.9	-28.0	1.30
9.50	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	-31.1	1.30

SPAN 3 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30									
=====									
X	DL1		DL2		DL1		DL2		I.F.
	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	MOMENT	SHEAR	
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.30
0.95	3.4	0.0	26.6	0.0	3.2	0.0	28.0	0.0	1.30
1.90	6.1	0.0	43.6	0.0	2.4	0.0	23.0	-1.9	1.30
2.85	8.0	0.0	54.3	0.0	1.6	0.0	19.1	-5.8	1.30
3.80	9.2	0.0	57.6	0.0	0.8	0.0	15.2	-7.8	1.30
4.75	9.6	0.0	60.8	0.0	0.0	0.0	11.3	-11.3	1.30
5.70	9.2	0.0	57.6	0.0	-0.8	0.0	7.8	-15.2	1.30
6.65	8.0	0.0	54.3	0.0	-1.6	0.0	5.8	-19.1	1.30
7.60	6.1	0.0	43.6	0.0	-2.4	0.0	3.9	-23.0	1.30
	SIMULT	SHEAR	-23.0	0.0	SIMULT	MOM	29.6	43.6	

* STRINGER - LIVE LOAD P-82 *									

MAXIMUM REACTIONS									
REACTIONS									
+I.F. -I.F. +I.F. -I.F.									
SUPPORT	DL1	DL2	+(LL+I)	-(LL+I)	+(LL+I)	-(LL+I)	+(LL+I)	-(LL+I)	+(LL+I)
1	4.0	0.0	40.7	0.0	1.30	0.0	1.30	0.0	1.30
2	8.1	0.0	55.5	0.0	1.30	0.0	1.30	0.0	1.30
3	8.1	0.0	55.5	0.0	1.30	0.0	1.30	0.0	1.30
4	4.0	0.0	40.7	0.0	1.30	0.0	1.30	0.0	1.30

SPAN 3 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X									
0.00	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)	I.F.
	MOMENT	MOMENT	MOMENT	MOMENT	SHEAR	SHEAR	SHEAR	SHEAR	
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.30
0.95	SIMULT SHEAR	0.0	0.0	0.0	SIMULT MOM	0.0	0.0	0.0	1.30
	3.4	0.0	34.8	0.0	3.2	0.0	36.7	-2.6	1.30
1.90	SIMULT SHEAR	36.7	0.0	0.0	SIMULT MOM	34.8	21.8	-5.1	1.30
	6.1	0.0	57.2	0.0	2.4	0.0	30.1	-38.8	1.30
2.85	SIMULT SHEAR	30.1	0.0	0.0	SIMULT MOM	57.2	38.8	-10.2	1.30
	8.0	0.0	71.2	0.0	1.6	0.0	25.0	-7.7	1.30
3.80	SIMULT SHEAR	25.0	0.0	0.0	SIMULT MOM	71.2	50.9	-14.8	1.30
	9.2	0.0	75.5	0.0	0.8	0.0	19.9	-70.2	1.30
4.75	SIMULT SHEAR	19.9	0.0	0.0	SIMULT MOM	75.5	58.2	-30.1	1.30
	9.6	0.0	79.7	0.0	0.0	0.0	14.8	-75.5	1.30
5.70	SIMULT SHEAR	12.8	0.0	0.0	SIMULT MOM	70.2	75.5	-40.7	1.30
	9.2	0.0	75.5	0.0	-0.8	0.0	10.2	-25.0	1.30
6.65	SIMULT SHEAR	-19.9	0.0	0.0	SIMULT MOM	58.2	34.8	-40.7	1.30
	8.0	0.0	71.2	0.0	-1.6	0.0	7.7	0.0	1.30
7.60	SIMULT SHEAR	-25.0	0.0	0.0	SIMULT MOM	50.9	71.2	0.0	1.30
	6.1	0.0	57.2	0.0	-2.4	0.0	5.1	0.0	1.30
8.55	SIMULT SHEAR	-30.1	0.0	0.0	SIMULT MOM	38.8	57.2	0.0	1.30
	3.4	0.0	34.8	0.0	-3.2	0.0	2.6	0.0	1.30
9.50	SIMULT SHEAR	-36.7	0.0	0.0	SIMULT MOM	21.8	34.8	0.0	1.30
	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	0.0	1.30
	SIMULT SHEAR	0.0	0.0	0.0	SIMULT MOM	0.0	0.0	0.0	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

SPAN 1									

X									
0.00	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)	I.F.
	MOMENT	MOMENT	MOMENT	MOMENT	SHEAR	SHEAR	SHEAR	SHEAR	
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.30
0.95	SIMULT SHEAR	0.0	0.0	0.0	SIMULT MOM	0.0	0.0	0.0	1.30
	3.4	0.0	34.8	0.0	3.2	0.0	36.7	-2.6	1.30
1.90	SIMULT SHEAR	36.7	0.0	0.0	SIMULT MOM	34.8	21.8	-5.1	1.30
	6.1	0.0	57.2	0.0	2.4	0.0	30.1	-38.8	1.30
2.85	SIMULT SHEAR	30.1	0.0	0.0	SIMULT MOM	57.2	38.8	-10.2	1.30
	8.0	0.0	71.2	0.0	1.6	0.0	25.0	-7.7	1.30
3.80	SIMULT SHEAR	25.0	0.0	0.0	SIMULT MOM	71.2	50.9	-14.8	1.30
	9.2	0.0	75.5	0.0	0.8	0.0	19.9	-70.2	1.30
4.75	SIMULT SHEAR	19.9	0.0	0.0	SIMULT MOM	75.5	58.2	-30.1	1.30
	9.6	0.0	79.7	0.0	0.0	0.0	14.8	-75.5	1.30
5.70	SIMULT SHEAR	12.8	0.0	0.0	SIMULT MOM	70.2	75.5	-40.7	1.30
	9.2	0.0	75.5	0.0	-0.8	0.0	10.2	-25.0	1.30
6.65	SIMULT SHEAR	-19.9	0.0	0.0	SIMULT MOM	58.2	34.8	-40.7	1.30
	8.0	0.0	71.2	0.0	-1.6	0.0	7.7	0.0	1.30
7.60	SIMULT SHEAR	-25.0	0.0	0.0	SIMULT MOM	50.9	71.2	0.0	1.30
	6.1	0.0	57.2	0.0	-2.4	0.0	5.1	0.0	1.30
8.55	SIMULT SHEAR	-30.1	0.0	0.0	SIMULT MOM	38.8	57.2	0.0	1.30
	3.4	0.0	34.8	0.0	-3.2	0.0	2.6	0.0	1.30
9.50	SIMULT SHEAR	-36.7	0.0	0.0	SIMULT MOM	21.8	34.8	0.0	1.30
	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	0.0	1.30
	SIMULT SHEAR	0.0	0.0	0.0	SIMULT MOM	0.0	0.0	0.0	1.30

SPAN 2 - LIVE LOAD IMPACT FACTORS : POS MOM 1.30

X									
0.00	DL1	DL2	+(LL+I)	-(LL+I)	DL1	DL2	+(LL+I)	-(LL+I)	I.F.
	MOMENT	MOMENT	MOMENT	MOMENT	SHEAR	SHEAR	SHEAR	SHEAR	
0.00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.30
0.95	SIMULT SHEAR	0.0	0.0	0.0	SIMULT MOM	0.0	0.0	0.0	1.30
	3.4	0.0	34.8	0.0	3.2	0.0	36.7	-2.6	1.30
1.90	SIMULT SHEAR	36.7	0.0	0.0	SIMULT MOM	34.8	21.8	-5.1	1.30
	6.1	0.0	57.2	0.0	2.4	0.0	30.1	-38.8	1.30
2.85	SIMULT SHEAR	30.1	0.0	0.0	SIMULT MOM	57.2	38.8	-10.2	1.30
	8.0	0.0	71.2	0.0	1.6	0.0	25.0	-7.7	1.30
3.80	SIMULT SHEAR	25.0	0.0	0.0	SIMULT MOM	71.2	50.9	-14.8	1.30
	9.2	0.0	75.5	0.0	0.8	0.0	19.9	-70.2	1.30
4.75	SIMULT SHEAR	19.9	0.0	0.0	SIMULT MOM	75.5	58.2	-30.1	1.30
	9.6	0.0	79.7	0.0	0.0	0.0	14.8	-75.5	1.30
5.70	SIMULT SHEAR	12.8	0.0	0.0	SIMULT MOM	70.2	75.5	-40.7	1.30
	9.2	0.0	75.5	0.0	-0.8	0.0	10.2	-25.0	1.30
6.65	SIMULT SHEAR	-19.9	0.0	0.0	SIMULT MOM	58.2	34.8	-40.7	1.30
	8.0	0.0	71.2	0.0	-1.6	0.0	7.7	0.0	1.30
7.60	SIMULT SHEAR	-25.0	0.0	0.0	SIMULT MOM	50.9	71.2	0.0	1.30
	6.1	0.0	57.2	0.0	-2.4	0.0	5.1	0.0	1.30
8.55	SIMULT SHEAR	-30.1	0.0	0.0	SIMULT MOM	38.8	57.2	0.0	1.30
	3.4	0.0	34.8	0.0	-3.2	0.0	2.6	0.0	1.30
9.50	SIMULT SHEAR	-36.7	0.0	0.0	SIMULT MOM	21.8	34.8	0.0	1.30
	0.0	0.0	0.0	0.0	-4.0	0.0	0.0	0.0	1.30
	SIMULT SHEAR	0.0	0.0	0.0	SIMULT MOM	0.0	0.0	0.0	1.30

X	NON-COMP		OVERLOAD		SHEAR STRENGTH	NON-COMPACT		COMPACT MOMENT STRENGTH	COMPACT		
	MOMENT STRENGTH	MOMENT	MOMENT	RATING FACTORS		IR	RATING FACTORS		IR		
0.00	154.9	B	123.9	85.8	85.8	0.91	V	1.52	V	1.52	V
0.95	154.9	B	123.9	85.8	85.8	1.03	V	1.71	V	1.71	V
1.90	154.9	B	123.9	85.8	85.8	1.19	T	1.98	T	1.24	O
2.85	154.9	B	123.9	85.8	85.8	0.94	T	1.56	T	1.24	O
3.80	154.9	B	123.9	85.8	85.8	0.87	T	1.46	T	1.24	O
4.75	154.9	B	123.9	85.8	85.8	0.82	T	1.37	T	1.24	O
5.70	154.9	B	123.9	85.8	85.8	0.87	T	1.46	T	1.24	O
6.65	154.9	B	123.9	85.8	85.8	0.94	T	1.56	T	1.24	O
7.60	154.9	B	123.9	85.8	85.8	1.19	T	1.98	T	1.24	O
8.55	154.9	B	123.9	85.8	85.8	1.03	V	1.71	V	1.24	O
9.50	154.9	B	123.9	85.8	85.8	0.91	V	1.52	V	1.24	O

SPAN 3

X	NON-COMP		OVERLOAD		SHEAR	NON-COMPACT		COMPACT		
	MOMENT	STRENGTH	MOMENT	STRENGTH		IR	OR	IR	OR	
0.00	154.9	B	123.9	85.8	0.91	V	1.52	V	1.52	V
0.95	154.9	B	123.9	85.8	1.03	V	1.71	V	1.03	V
1.90	154.9	B	123.9	85.8	1.19	T	1.98	T	1.24	O
2.85	154.9	B	123.9	85.8	0.94	T	1.56	T	0.98	O
3.80	154.9	B	123.9	85.8	0.87	T	1.46	T	0.91	O
4.75	154.9	B	123.9	85.8	0.82	T	1.37	T	0.86	O
5.70	154.9	B	123.9	85.8	0.87	T	1.46	T	0.91	O
6.65	154.9	B	123.9	85.8	0.94	T	1.56	T	0.98	O
7.60	154.9	B	123.9	85.8	1.19	T	1.98	T	1.24	O
8.55	154.9	B	123.9	85.8	1.03	V	1.71	V	1.03	V
9.50	154.9	B	123.9	85.8	0.91	V	1.52	V	0.91	V

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP		OVERLOAD		NON-COMPACT				COMPACT				
	MOMENT	STRENGTH	MOMENT	STRENGTH	RATING FACTORS		IR	MOMENT	RATING FACTORS		IR	MOMENT	
0.00	154.9	B	123.9	85.8	0.90	V	1.51	V	172.2	0.90	V	1.51	V
1.60	154.9	B	123.9	85.8	1.27	V	2.12	V	172.2	1.27	V	2.12	V
3.20	154.9	B	123.9	85.8	0.80	T	1.34	T	172.2	0.84	O	1.40	O
4.80	154.9	B	123.9	85.8	0.57	T	0.95	T	172.2	0.60	O	1.00	O
6.40	154.9	B	123.9	85.8	0.42	T	0.70	T	172.2	0.44	O	0.74	O
8.00	154.9	B	123.9	85.8	0.40	T	0.67	T	172.2	0.42	O	0.70	O

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* FLOORBEAM - LIVE LOAD HS20 *
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LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 32.00
LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
0.00	0.0	0.0	0.0	3.3	0.0	41.6	1.30
1.60	5.1	0.0	47.4	3.1	0.0	29.6	1.30

FACTOR CODES: M - MOMENT, V - SHEAR

3.20 9.8 0.0 81.5 2.8 0.0 25.5 1.30
 4.80 14.1 0.0 110.2 2.5 0.0 21.3 1.30
 6.40 18.0 0.0 143.5 2.3 0.0 20.8 1.30
 8.00 21.4 0.0 145.6 -2.0 0.0 13.0 1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
 UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
 SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	RATING FACTORS
0.00	154.9 B	123.9	85.8	0.90 V	1.51 V	0.90 V
1.60	154.9 B	123.9	85.8	1.27 V	2.12 V	1.27 V
3.20	154.9 B	123.9	85.8	0.80 T	1.34 T	0.84 O
4.80	154.9 B	123.9	85.8	0.57 T	0.95 T	0.60 O
6.40	154.9 B	123.9	85.8	0.42 T	0.70 T	0.44 O
8.00	154.9 B	123.9	85.8	0.40 T	0.67 T	0.42 O

 * FLOORBEAM - LIVE LOAD TK527 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 33.83
 LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1		DL2		LL+I	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	RATING FACTORS
0.00	0.0	0.0	0.0	0.0	44.0	1.30
1.60	5.1	0.0	50.1	3.1	31.3	1.30
3.20	9.8	0.0	86.2	2.8	26.9	1.30
4.80	14.1	0.0	116.5	2.5	22.5	1.30
6.40	18.0	0.0	151.7	2.3	22.0	1.30
8.00	21.4	0.0	153.9	-2.0	13.7	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
 UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
 SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	RATING FACTORS
0.00	154.9 B	123.9	85.8	1.42 V	172.2	0.85 V
1.60	154.9 B	123.9	85.8	1.21 V	2.01 V	1.21 V
3.20	154.9 B	123.9	85.8	0.76 T	1.27 T	0.79 O
4.80	154.9 B	123.9	85.8	0.50 T	0.90 T	0.57 O
6.40	154.9 B	123.9	85.8	0.40 T	0.67 T	0.42 O
8.00	154.9 B	123.9	85.8	0.38 T	0.63 T	0.40 O

 * FLOORBEAM - LIVE LOAD ML80 *

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LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 35.78
 LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1		DL2		LL+I	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	RATING FACTORS
0.00	0.0	0.0	0.0	3.3	0.0	46.5
1.60	5.1	0.0	53.0	3.1	0.0	33.1
3.20	9.8	0.0	91.2	2.8	0.0	28.5
4.80	14.1	0.0	123.3	2.5	0.0	23.8
6.40	18.0	0.0	160.5	2.3	0.0	23.3
8.00	21.4	0.0	162.8	-2.0	0.0	14.5

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
 UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
 SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	RATING FACTORS
0.00	154.9 B	123.9	85.8	0.81 V	1.35 V	0.81 V
1.60	154.9 B	123.9	85.8	0.72 T	1.20 T	0.75 O
3.20	154.9 B	123.9	85.8	0.51 T	0.85 T	0.53 O
4.80	154.9 B	123.9	85.8	0.38 T	0.63 T	0.40 O
6.40	154.9 B	123.9	85.8	0.36 T	0.60 T	0.38 O
8.00	154.9 B	123.9	85.8	0.36 T	0.60 T	0.38 O

 * FLOORBEAM - LIVE LOAD P-82 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 46.89
 LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1		DL2		LL+I	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	RATING FACTORS
0.00	0.0	0.0	0.0	3.3	0.0	61.0
1.60	5.1	0.0	69.5	3.1	0.0	43.4
3.20	9.8	0.0	119.5	2.8	0.0	37.3
4.80	14.1	0.0	161.6	2.5	0.0	31.2
6.40	18.0	0.0	210.3	2.3	0.0	30.5
8.00	21.4	0.0	213.4	-2.0	0.0	19.1

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
 UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
 SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT	
	MOMENT	STRENGTH	SHEAR	RATING FACTORS	MOMENT	RATING FACTORS
0.00	154.9 B	123.9	85.8	0.62 V	1.03 V	0.62 V
						1.03 V

1.60 154.9 B 123.9 85.8 0.87 V 1.45 V 172.2 0.87 V 1.45 V
 3.20 154.9 B 123.9 85.8 0.55 T 0.91 T 172.2 0.57 O 0.95 O
 4.80 154.9 B 123.9 85.8 0.39 T 0.65 T 172.2 0.41 O 0.68 O
 6.40 154.9 B 123.9 85.8 0.29 T 0.48 T 172.2 0.30 O 0.50 O
 8.00 154.9 B 123.9 85.8 0.27 T 0.46 T 172.2 0.29 O 0.48 O

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 < FLOORBEAM 2 >
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DEAD LOADS ACTING ON FLOORBEAM

UNIFORM LOAD INPUT CONCENTRATED LOADS
 FLOORBEAM DL1 DL2
 WEIGHT DL1 DL2
 0.036 0.130 8.000 8.055 0.000

 * FLOORBEAM - LIVE LOAD H20 *

LIVE LOAD REACTION FROM DECK (ONE LANE) : 32.08
 LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1	DL2	LL+I	DL1	DL2	LL+I	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	41.7	1.30
1.60	8.4	0.0	47.5	5.1	0.0	29.7	1.30
3.20	16.3	0.0	81.7	4.8	0.0	25.5	1.30
4.80	23.8	0.0	110.5	4.6	0.0	21.4	1.30
6.40	30.9	0.0	143.9	4.3	0.0	20.9	1.30
8.00	37.5	0.0	146.0	-4.0	0.0	13.0	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
 UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
 SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD			NON-COMPACT			COMPACT		
	MOMENT	STRENGTH	RATING	IR	OR	STRENGTH	IR	OR	
0.00	154.9 B	123.9	85.8	0.87 V	1.45 V	172.2	0.87 V	1.45 V	
1.60	154.9 B	123.9	85.8	1.23 V	2.05 V	172.2	1.23 V	2.05 V	
3.20	154.9 B	123.9	85.8	0.75 T	1.26 T	172.2	0.79 O	1.32 O	
4.80	154.9 B	123.9	85.8	0.52 T	0.86 T	172.2	0.54 O	0.91 O	
6.40	154.9 B	123.9	85.8	0.37 T	0.61 T	172.2	0.39 O	0.65 O	
8.00	154.9 B	123.9	85.8	0.34 T	0.56 T	172.2	0.35 O	0.59 O	

 * FLOORBEAM - LIVE LOAD HS20 *

LIVE LOAD REACTION FROM DECK (ONE LANE) : 32.08
 LIVE LOAD IMPACT FACTORS : POS MOM 1.30

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UNFACTORED MOMENTS AND SHEARS

X	DL1	DL2	LL+I	DL1	DL2	LL+I	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	41.7	1.30
1.60	8.4	0.0	47.5	5.1	0.0	29.7	1.30
3.20	16.3	0.0	81.7	4.8	0.0	25.5	1.30
4.80	23.8	0.0	110.5	4.6	0.0	21.4	1.30
6.40	30.9	0.0	143.9	4.3	0.0	20.9	1.30
8.00	37.5	0.0	146.0	-4.0	0.0	13.0	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
 UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
 SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD			NON-COMPACT			COMPACT		
	MOMENT	STRENGTH	RATING	IR	OR	STRENGTH	IR	OR	
0.00	154.9 B	123.9	85.8	0.87 V	1.45 V	172.2	0.87 V	1.45 V	
1.60	154.9 B	123.9	85.8	1.23 V	2.05 V	172.2	1.23 V	2.05 V	
3.20	154.9 B	123.9	85.8	0.75 T	1.26 T	172.2	0.79 O	1.32 O	
4.80	154.9 B	123.9	85.8	0.52 T	0.86 T	172.2	0.54 O	0.91 O	
6.40	154.9 B	123.9	85.8	0.37 T	0.61 T	172.2	0.39 O	0.65 O	
8.00	154.9 B	123.9	85.8	0.34 T	0.56 T	172.2	0.35 O	0.59 O	

 * FLOORBEAM - LIVE LOAD TKS27 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 38.60
 LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1	DL2	LL+I	DL1	DL2	LL+I	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	50.2	1.30
1.60	8.4	0.0	57.2	5.1	0.0	35.8	1.30
3.20	16.3	0.0	98.3	4.8	0.0	30.7	1.30
4.80	23.8	0.0	133.0	4.6	0.0	25.7	1.30
6.40	30.9	0.0	173.1	4.3	0.0	25.1	1.30
8.00	37.5	0.0	175.6	-4.0	0.0	15.7	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
 UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
 SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD			NON-COMPACT			COMPACT		
	MOMENT	STRENGTH	RATING	IR	OR	STRENGTH	IR	OR	
0.00	154.9 B	123.9	85.8	0.73 V	1.21 V	172.2	0.73 V	1.21 V	
1.60	154.9 B	123.9	85.8	1.02 V	1.70 V	172.2	1.02 V	1.70 V	
3.20	154.9 B	123.9	85.8	0.63 T	1.05 T	172.2	0.66 O	1.09 O	
4.80	154.9 B	123.9	85.8	0.43 T	0.72 T	172.2	0.45 O	0.75 O	
6.40	154.9 B	123.9	85.8	0.31 T	0.51 T	172.2	0.32 O	0.54 O	

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* FLOORBEAM - LIVE LOAD HS20 *

LIVE LOAD REACTION FROM DECK (ONE LANE) : 32.08
LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	41.7	1.30
1.60	8.4	0.0	47.5	5.1	0.0	29.7	1.30
3.20	16.3	0.0	81.7	4.8	0.0	25.5	1.30
4.80	23.8	0.0	110.5	4.6	0.0	21.4	1.30
6.40	30.9	0.0	143.9	4.3	0.0	20.9	1.30
8.00	37.5	0.0	146.0	-4.0	0.0	13.0	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT		RATING FACTORS	
	MOMENT	MOMENT	SHEAR	RATING	MOMENT	STRENGTH	IR	OR
0.00	154.9 B	123.9	85.8	0.87 V	1.45 V	172.2	0.87 V	1.45 V
1.60	154.9 B	123.9	85.8	1.23 V	2.05 V	172.2	1.23 V	2.05 V
3.20	154.9 B	123.9	85.8	0.75 T	1.26 T	172.2	0.75 T	1.32 O
4.80	154.9 B	123.9	85.8	0.52 T	0.86 T	172.2	0.52 T	0.91 O
6.40	154.9 B	123.9	85.8	0.37 T	0.61 T	172.2	0.37 T	0.65 O
8.00	154.9 B	123.9	85.8	0.34 T	0.56 T	172.2	0.34 T	0.59 O

* FLOORBEAM - LIVE LOAD TK527 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 38.60
LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	50.2	1.30
1.60	8.4	0.0	57.2	5.1	0.0	35.8	1.30
3.20	16.3	0.0	98.3	4.8	0.0	30.7	1.30
4.80	23.8	0.0	133.0	4.6	0.0	25.7	1.30
6.40	30.9	0.0	173.1	4.3	0.0	25.1	1.30
8.00	37.5	0.0	175.6	-4.0	0.0	15.7	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

NON-COMP OVERLOAD	NON-COMPACT		COMPACT	
	MOMENT	SHEAR	MOMENT	RATING FACTORS

X	STRENGTH	STRENGTH	STRENGTH	IR	OR	STRENGTH	IR	OR
0.00	154.9 B	123.9	85.8	0.73 V	1.21 V	172.2	0.73 V	1.21 V
1.60	154.9 B	123.9	85.8	1.02 V	1.70 V	172.2	1.02 V	1.70 V
3.20	154.9 B	123.9	85.8	0.63 T	1.05 T	172.2	0.66 O	1.09 O
4.80	154.9 B	123.9	85.8	0.43 T	0.72 T	172.2	0.45 O	0.75 O
6.40	154.9 B	123.9	85.8	0.31 T	0.51 T	172.2	0.32 O	0.54 O
8.00	154.9 B	123.9	85.8	0.28 T	0.46 T	172.2	0.30 O	0.49 O

* FLOORBEAM - LIVE LOAD ML80 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 44.45
LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	57.8	1.30
1.60	8.4	0.0	65.9	5.1	0.0	41.2	1.30
3.20	16.3	0.0	113.3	4.8	0.0	35.4	1.30
4.80	23.8	0.0	153.1	4.6	0.0	29.6	1.30
6.40	30.9	0.0	199.4	4.3	0.0	28.9	1.30
8.00	37.5	0.0	202.3	-4.0	0.0	18.1	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR
UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER
SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT			
	MOMENT	SHEAR	MOMENT	RATING FACTORS	MOMENT	RATING FACTORS		
	STRENGTH	STRENGTH	IR	OR	IR	OR		
0.00	154.9 B	123.9	85.8	0.63 V	1.05 V	172.2	0.63 V	1.05 V
1.60	154.9 B	123.9	85.8	0.89 V	1.48 V	172.2	0.89 V	1.48 V
3.20	154.9 B	123.9	85.8	0.54 T	0.91 T	172.2	0.57 O	0.95 O
4.80	154.9 B	123.9	85.8	0.37 T	0.62 T	172.2	0.39 O	0.65 O
6.40	154.9 B	123.9	85.8	0.27 T	0.44 T	172.2	0.28 O	0.47 O
8.00	154.9 B	123.9	85.8	0.24 T	0.40 T	172.2	0.26 O	0.43 O

* FLOORBEAM - LIVE LOAD P-82 *

LIVE LOAD REACTION FROM DECK (ONE TRUCK) : 62.53
LIVE LOAD IMPACT FACTORS : POS MOM 1.30

UNFACTORED MOMENTS AND SHEARS

X	DL1 MOMENT	DL2 MOMENT	LL+I MOMENT	DL1 SHEAR	DL2 SHEAR	LL+I SHEAR	I.F.
0.00	0.0	0.0	0.0	5.4	0.0	81.3	1.30
1.60	8.4	0.0	92.7	5.1	0.0	57.9	1.30
3.20	16.3	0.0	159.3	4.8	0.0	49.8	1.30
4.80	23.8	0.0	215.4	4.6	0.0	41.7	1.30
6.40	30.9	0.0	280.4	4.3	0.0	40.6	1.30
8.00	37.5	0.0	284.5	-4.0	0.0	25.4	1.30

NOTE: THE SHEAR CAPACITIES CALCULATED HEREIN ARE BASED ON STIFFENED OR UNSTIFFENED EQUATIONS AS SPECIFIED BY INPUT REGARDLESS OF THE STIFFENER SPACINGS INPUT AND ARE NOT CHECKED AGAINST AASHTO CRITERIA.

STRENGTHS AND LOAD FACTOR RATINGS

X	NON-COMP OVERLOAD		NON-COMPACT		COMPACT	
	MOMENT	STRENGTH	SHEAR	RATING	MOMENT	STRENGTH
0.00	154.9 B	123.9	85.8	0.45 V	0.75 V	0.75 V
1.60	154.9 B	123.9	85.8	0.63 V	1.05 V	1.05 V
3.20	154.9 B	123.9	85.8	0.39 T	1.72.2	0.41 O
4.80	154.9 B	123.9	85.8	0.27 T	0.44 T	0.28 O
6.40	154.9 B	123.9	85.8	0.19 T	0.31 T	0.20 O
8.00	154.9 B	123.9	85.8	0.17 T	0.29 T	0.18 O

+++++
+ R A T I N G S U M M A R Y +
+++++

MEMBER: TRUSS

ALLOWABLE STRESS RATING		MEMBER	
LOAD	FACTOR	TONS	SPAN
H20 IR (CRITICAL)	1.78	35.6	L36L37
OR (CRITICAL)	2.74	54.7	L36L37
HS20 IR (CRITICAL)	1.28	46.2	L18L19
OR (CRITICAL)	2.28	82.2	L36L37
TK527 IR (CRITICAL)	1.13	45.2	L18L19
OR (CRITICAL)	2.01	80.4	L36L37
ML80 IR (CRITICAL)	1.09	39.8	L18L19
OR (CRITICAL)	2.08	76.2	L36L37
P-82 IR (CRITICAL)	0.65	66.6	L36L37
OR (CRITICAL)	1.00	102.3	L36L37

USE FOR TRUSS RATINGS

MEMBER: STRINGER AT 8.00

LOAD FACTOR RATING		X	
LOAD	FACTOR	TONS	SPAN
H20 IR (CRITICAL)	0.96 O	19.1	4.75 3
OR (CRITICAL)	1.59 O	31.8	4.75 3
IR (POS MOM)	0.96 O	19.1	4.75 3
OR (POS MOM)	1.59 O	31.8	4.75 3
HS20 IR (CRITICAL)	0.96 O	34.4	4.75 3
OR (CRITICAL)	1.59 O	57.3	4.75 3
IR (POS MOM)	0.96 O	34.4	4.75 3
OR (POS MOM)	1.59 O	57.3	4.75 3
TK527 IR (CRITICAL)	1.19 O	47.8	5.70 3
OR (CRITICAL)	1.99 O	79.6	5.70 3
IR (POS MOM)	1.19 O	47.8	5.70 3
OR (POS MOM)	1.99 O	79.6	5.70 3
ML80 IR (CRITICAL)	1.13 O	41.3	4.75 3
OR (CRITICAL)	1.88 O	68.9	4.75 3
IR (POS MOM)	1.13 O	41.3	4.75 3
OR (POS MOM)	1.88 O	68.9	4.75 3
P-82 IR (CRITICAL)	0.86 O	87.7	4.75 3
OR (CRITICAL)	1.43 O	146.2	4.75 3
IR (POS MOM)	0.86 O	87.7	4.75 3
OR (POS MOM)	1.43 O	146.2	4.75 3

SEE HAND CAROS FOR FLOORBEAM RATINGS

MEMBER: FLOORBEAM

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LOAD FACTOR RATING		X	
LOAD	FACTOR	TONS	FLBM
H20 IR (CRITICAL)	0.35 O	7.1	8.00 3
OR (CRITICAL)	0.59 O	11.8	8.00 3
IR (POS MOM)	0.35 O	7.1	8.00 3
OR (POS MOM)	0.59 O	11.8	8.00 3
HS20 IR (CRITICAL)	0.35 O	12.8	8.00 3
OR (CRITICAL)	0.59 O	21.3	8.00 3
IR (POS MOM)	0.35 O	12.8	8.00 3
OR (POS MOM)	0.59 O	21.3	8.00 3
TK527 IR (CRITICAL)	0.30 O	11.8	8.00 3
OR (CRITICAL)	0.49 O	19.7	8.00 3
IR (POS MOM)	0.30 O	11.8	8.00 3
OR (POS MOM)	0.49 O	19.7	8.00 3
ML80 IR (CRITICAL)	0.26 O	9.4	8.00 3
OR (CRITICAL)	0.43 O	15.6	8.00 3
IR (POS MOM)	0.26 O	9.4	8.00 3
OR (POS MOM)	0.43 O	15.6	8.00 3
P-82 IR (CRITICAL)	0.18 O	18.6	8.00 3
OR (CRITICAL)	0.30 O	31.0	8.00 3
IR (POS MOM)	0.18 O	18.6	8.00 3
OR (POS MOM)	0.30 O	31.0	8.00 3

RATING FACTOR CODES:

- T - TOP STEEL STRESS/STRENGTH GOVERNS
- B - BOTTOM STEEL STRESS/STRENGTH GOVERNS
- C - CONCRETE STRESS/STRENGTH GOVERNS
- R - REINFORCEMENT STRESS/STRENGTH GOVERNS
- V - SHEAR STRESS/STRENGTH GOVERNS
- blank - COMPACT MOMENT STRENGTH GOVERNS
- O - OVERLOAD PROVISIONS GOVERN
- I - MOMENT-SHEAR INTERACTION GOVERNS
- F - SECTION DOES NOT MEET FLANGE PROJECTION/THICKNESS RATIO CRITERIA
- W - SECTION DOES NOT MEET WEB DEPTH/THICKNESS RATIO CRITERIA

NON-COMPACT MOMENT STRENGTH CODES:

- B - SECTION IS BRACED
- U - SECTION IS UNBRACED

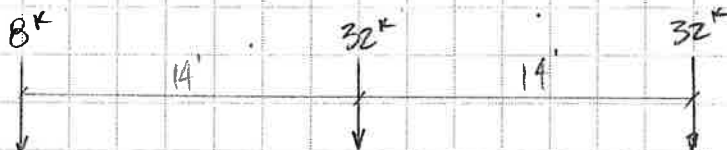
NOTE: ALL RATINGS ARE BASED ON THE NUMBER OF DESIGN LANES OR THE ACTUAL TRAFFIC LANES AS DEFINED BY "D" OR "L" ENTERED FOR LANES IN THE PROJECT IDENTIFICATION.

FLOORBEAM RATING ANALYSIS

- AS PER CALCS BY HARRY L. DOVELL, ANALYZE F.B AS BEING CONTINUOUS IF NEEDED
- AS PER CALCS BY HARRY L. DOVELL, DL ON F.B = 550 #/ft.

LIVE LOAD: FIND THE AXLE LOAD ON EACH FLOORBEAM ASSUMING THE SLAB TO ACT AS A SIMPLE SPAN BTWN. FLOORBEAMS.

MAX FLOORBEAM SPACING = 9.5'



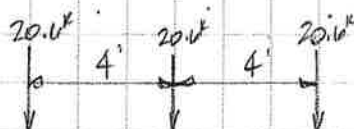
HS20 $R_{FB} = 32^k$



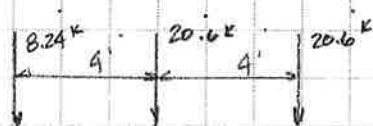
I.M.L $R_{FB} = 30^k + 30^k \times \frac{5.5'}{9.5'} = 47.4^k$



H20 $R_{FB} = 32^k$



ML-BD $R_{FB} = 20.6^k + 2 \times 20.6^k \times \frac{5.5'}{9.5'} = 44.4^k$



TK527 $R_{FB} = 20.6^k + 8.24^k \times \frac{5.5'}{9.5'} + 20.6^k \times \frac{5.5'}{9.5'} = 37.3^k$



PB2 $R_{FB} = 27^k + 2 \times 27^k \times \frac{5.5'}{9.5'} = 58.2^k$

Chapter 2 Program Description

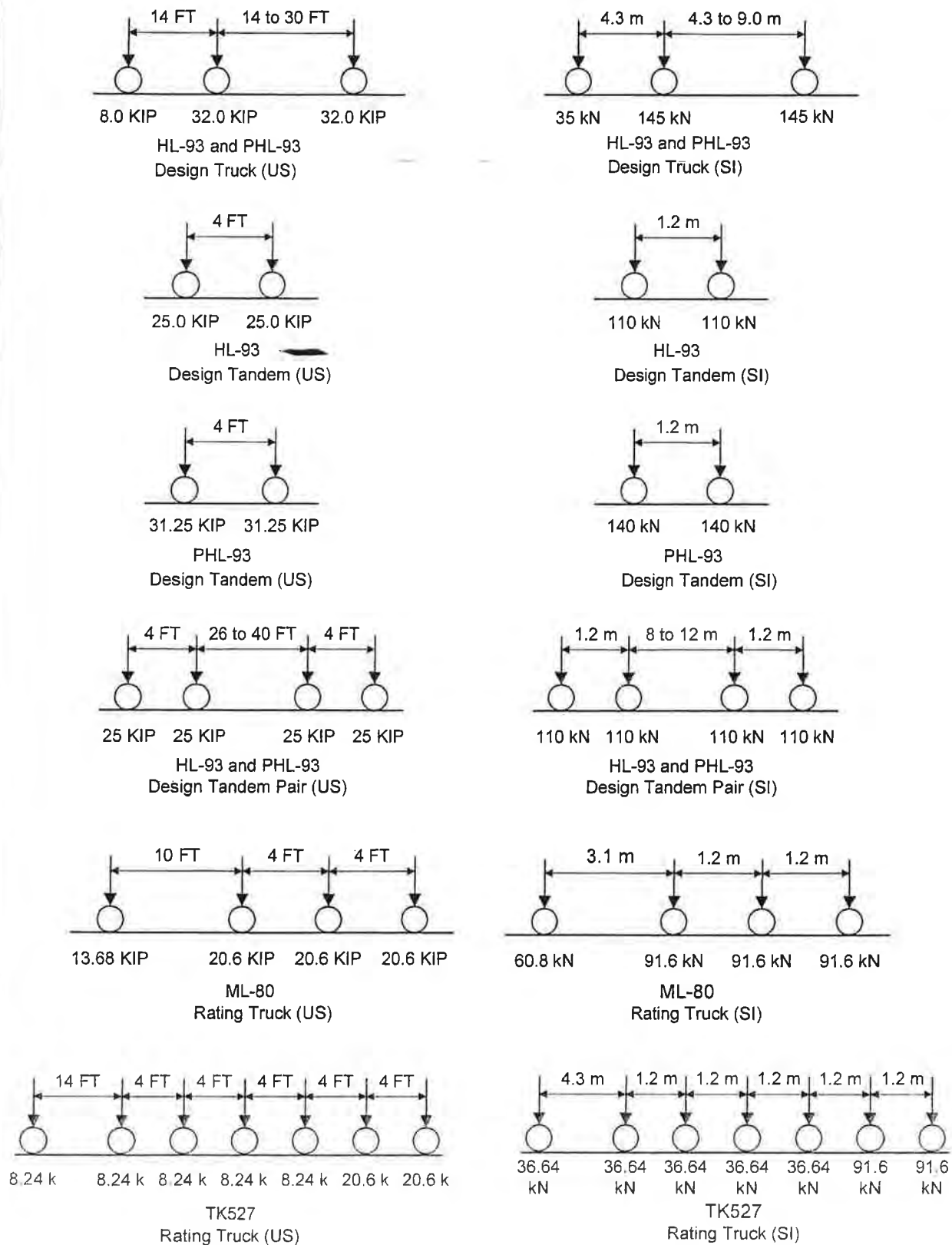
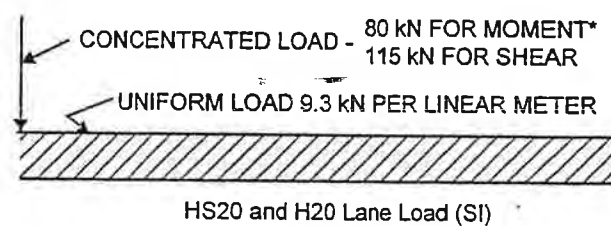
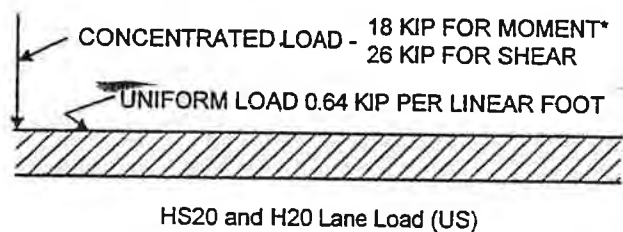
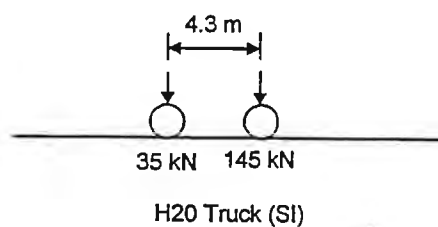
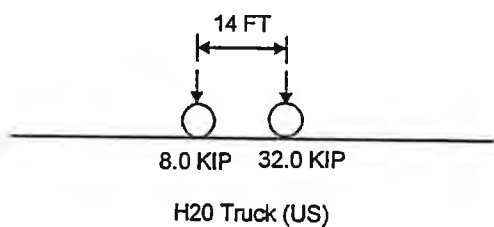
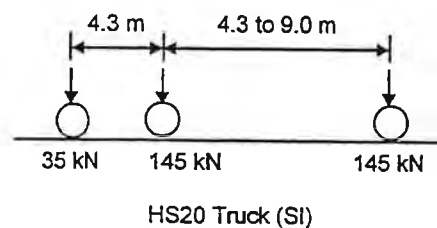
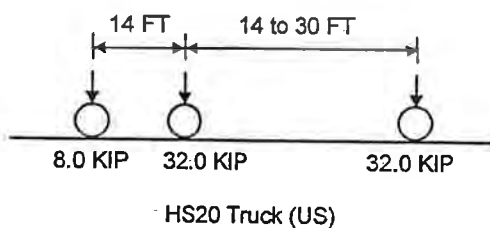
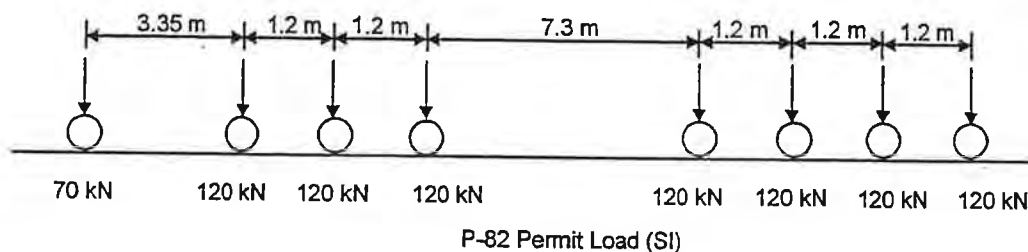
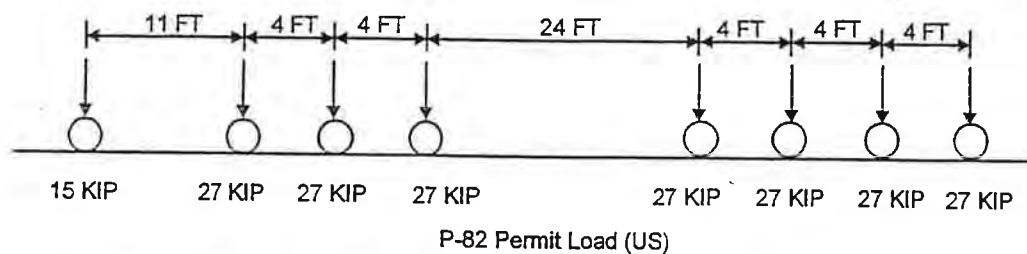


Figure 2.5-1 LRFD, ML-80, and TK527 Live Loading



* use two concentrated loads for negative moment

Figure 2.5-2 P-82, HS20, and H20 Live Loading

FLOORBEAM RATING ANALYSIS

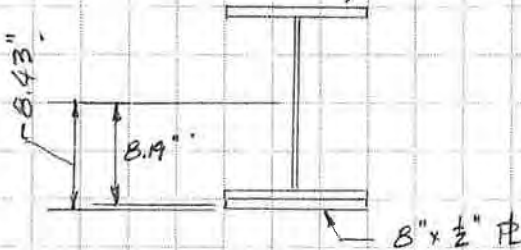
DEAD LOAD

SPAN = 7.5'

F.B. SPAN LENGTH: I.F. TO I.F. OF MAIN TRUSSES = 15'

$$M_{DL} = 550 \frac{\text{#}}{\text{ft}} \times 7.5'^2 \div 8 = 3867 \text{ #} \cdot \text{ft}$$

$$V_{DL} = 550 \frac{\text{#}}{\text{ft}} \times 7.5' \div 2 = 2,063 \text{ #}$$



$$I = 448 \text{ in}^4 + (8 \text{ in} \times \frac{1}{2} \text{ in}) \times 8.19^2 = 716 \text{ in}^4$$

$$f_b = 3867 \text{ #} \cdot \text{ft} \times 12 \text{ in} \times 8.43 \text{ in} \div 716 \text{ in}^4 = 546 \text{ PSI}$$

$$f_v = 2063 \text{ #} \div (13.625 \text{ in} \times .295 \text{ in}) = 513 \text{ PSI}$$

LIVE LOAD -

• I.M.L.

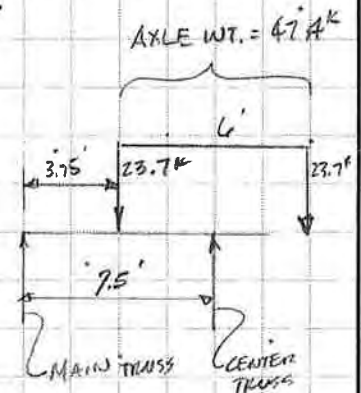
$$M_{LL+I} = (23.7' \times 7.5' \div 4) \times 1000 \times 1.3 = 59,768 \text{ #} \cdot \text{ft}$$

$$V_{LL+I} = 23.7' \times 1000 \times 1.3 = 30,810 \text{ #}$$

$$f_b = 59,768 \text{ #} \cdot \text{ft} \times 12 \times 8.43 \text{ in} \div 716 \text{ in}^4 = 8,162 \text{ PSI}$$

$$f_v = 30,810 \text{ #} \div (13.625 \text{ in} \times .295 \text{ in}) = 7,665 \text{ PSI}$$

IMPACT ?



INVENTRY RATING: I.M.L.

$$\text{MOMENT: } (18,000 \text{ PSI} - 546 \text{ PSI}) \div 8,162 \text{ PSI} = 2.13$$

$$\text{SHEAR: } (11,000 \text{ PSI} - 513 \text{ PSI}) \div 7,665 \text{ PSI} = 1.36$$

SEE SPREADSHEET
FOR RATINGS OF
OTHER VEHICLES.

OPERATING RATING: I.M.L.

$$\text{MOMENT: } (24,500 \text{ PSI} - 546 \text{ PSI}) \div 8,162 \text{ PSI} = 2.93$$

$$\text{SHEAR: } (15,000 \text{ PSI} - 513 \text{ PSI}) \div 7,665 \text{ PSI} = 1.89$$



JOB: Monaca Bridge Rating
SHEET NO.: 95 OF 97
CALCULATED BY: AMK DATE: 5/20/2008
CHECKED BY: DFL DATE: 5-20-08
JOB NUMBER: 4396.001

Floorbeam Rating

DL = 550 lb./ft
Span = 7.5 ft
I = 716 in⁴
c = 8.43 in.
Web depth = 13.625 in.
Web Thick = 0.295 in.

Dead Load

M_{DL} = 3867 ft-lb
V_{DL} = 2063 lb

Live Load

Wheel reactions on floorbeam:

	axle wt	wheel wt.	Impact	LL + I
HS20	32	16	1.3	20.8
H20	32	16	1.3	20.8
TK527	37.3	18.65	1.3	24.245
ML80	44.4	22.2	1.3	28.86
P-82	58.2	29.1	1.3	37.83

Live Load Moments

$$M = P \times L / 4$$

M_{HS20} = 39000 ft-lb
M_{H20} = 39000 ft-lb
M_{TK527} = 45459 ft-lb
M_{ML80} = 54113 ft-lb
M_{P-82} = 70931 ft-lb

Live Load Shear

$$V = R$$

V_{HS20} = 20800 Lb.
V_{H20} = 20800 Lb.
V_{TK527} = 24245 Lb.
V_{ML80} = 28860 Lb.
V_{P-82} = 37830 Lb.

96 / 97

Calculate Stresses

Bending Stresses

$Fb_{DL} = 546 \text{ psi}$
 $Fb_{HS20} = 5510 \text{ psi}$
 $Fb_{H20} = 5510 \text{ psi}$
 $Fb_{TK527} = 6423 \text{ psi}$
 $Fb_{ML80} = 7645 \text{ psi}$
 $Fb_{P-82} = 10022 \text{ psi}$

Shear Stresses

$Fv_{DL} = 513$
 $Fv_{HS20} = 5175$
 $Fv_{H20} = 5175$
 $Fv_{TK527} = 6032$
 $Fv_{ML80} = 7180$
 $Fv_{P-82} = 9412$

Ratings

Inventory Stress Levels

Bending 18000 psi
 Shear 11000 psi

Operating Stress Levels

Bending 24500 psi
 Shear 15000 psi

Rating = (Allow. Stress - DL stress) / LL Stress

FLOOR BEAM RATINGS					
	HS20	H20	TK527	ML80	P-82
Inventory Rating - Moment	3.17	3.17	2.72	2.28	1.74
Inventory Rating - Shear	2.03	2.03	1.74	1.46	1.11
Operating Rating - Moment	4.35	4.35	3.73	3.13	2.39
Operating Rating - Shear	2.80	2.80	2.40	2.02	1.54



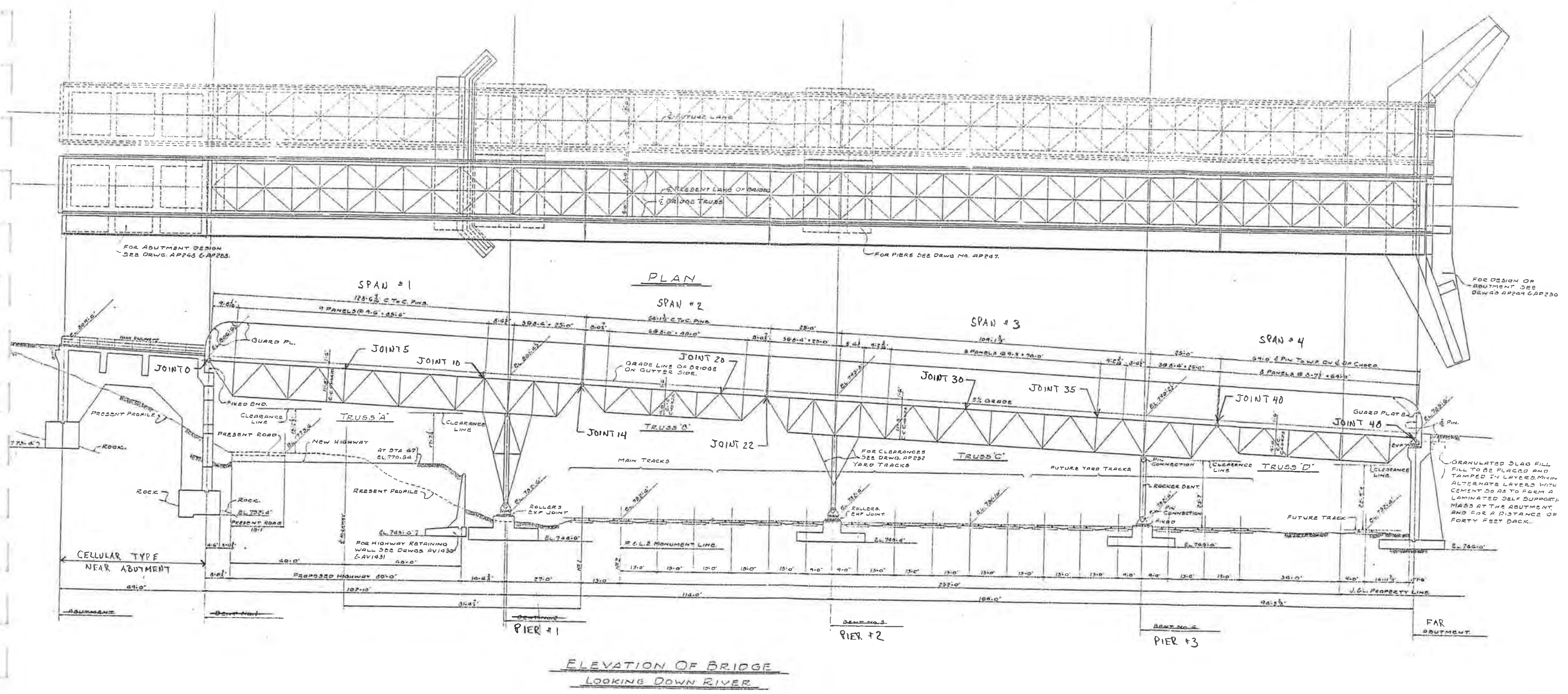
Herbert, Rowland & Grubic, Inc.
Engineering & Related Services

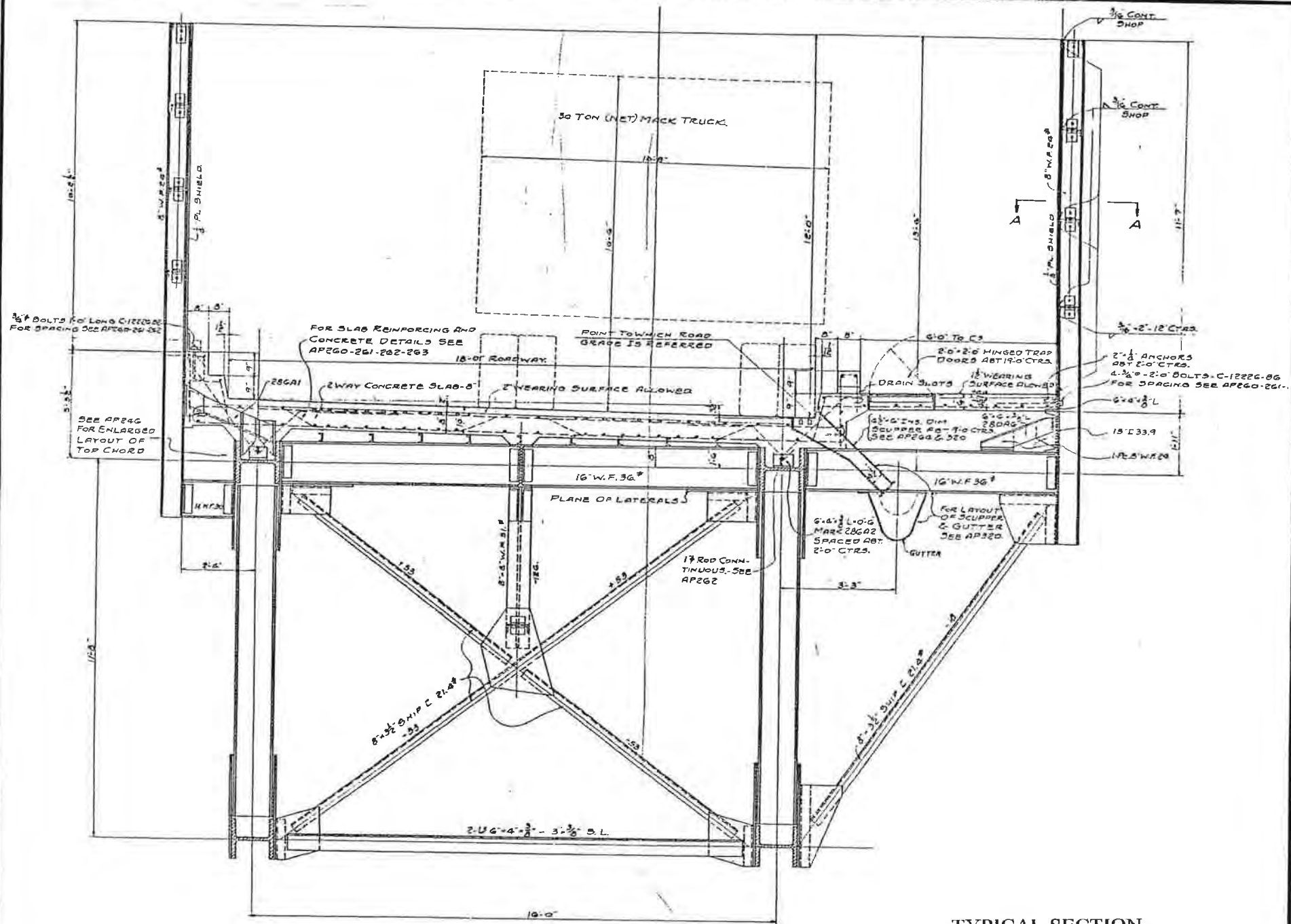
JOB MONACA BRIDGE RATING
SHEET NO. 97 OF 97
CALCULATED BY AMK DATE 5.20.08
CHECKED BY _____ DATE _____
SCALE 4396.001

RATING SUMMARY

	H520 (36T)		H20 (20T)		TK527 (40 T)		ML80 (37.7T)		PB2 (102T)	
	FACTOR	TONS	FACTOR	TONS	FACTOR	TONS	FACTOR	TONS	FACTOR	TONS
INVENTORY	1.28	46.2	1.78	35.6	1.13	45.2	1.09	39.8	—	—
OPERATING	2.28	82.2	2.74	54.7	2.01	80.4	2.02*	76.2*	1.00	102.3

* FLOORBEAM SHEAR CONTROLS RATING. ALL OTHER RATINGS CONTROLLED BY TRUSS MEMBERS





TYPICAL SECTION
N.T.S